

ANATOMY, PHYSIOLOGY & PATHOLOGY NOTES OF THE **ENDOCRINE** SYSTEM

FOURTH EDITION

PRE-SUMMARIZED
READY-TO-STUDY
HIGH-YIELD NOTES

FOR THE TIME-POOR
MEDICAL, PRE-MED,
USMLE OR PA STUDENT



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A Message From Our Team

Studying medicine or any health-related degree can be stressful; believe us, we know from experience! The human body is an incredibly complex organism, and finding a way to streamline your learning is crucial to succeeding in your exams and future profession. Our goal from the outset has been to create the greatest educational resource for the next generation of medical students, and to make them as affordable as possible.

In this fourth edition of our notes we have made a number of text corrections, formatting updates, and figure updates which we feel will enhance your study experience. We have also endeavoured to use only open-source images and/or provide attribution where possible.

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Anatomy & Physiology Notes:

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OVERVIEW OF THE ENDOCRINE SYSTEM

OVERVIEW OF THE ENDOCRINE SYSTEM

Endocrinology:

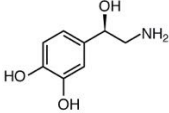
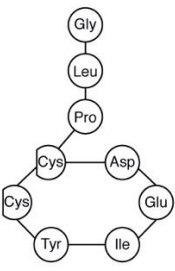
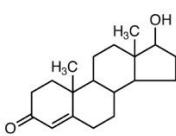
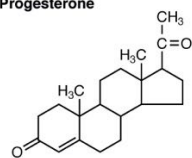
- **Endocrinology:** The scientific study of **Hormones (Chemical Messengers)** and the endocrine organs.
- Endocrine system is critical for **maintaining Homeostasis**

What is a Hormone?

- Chemical signalling molecules secreted by endocrine glands into the extracellular fluids to exert an effect elsewhere in the body
- Hormones travel in blood or lymph throughout the body
- **BIOLOGICAL SPECIFICITY:** Hormones Interact with **specific receptors** of **specific cells** of **specific organs**.

Hormones Are Either Steroidal or Non-Steroidal:

- **Steroids Hormones:**
 - Sex Hormones – Eg: Testosterone (Androgens), Oestrogens, Progestogens.
 - Adrenal Hormones – Eg: Glucocorticoids, Mineralocorticoids
 - (Derived from Cholesterol)
- **Non-Steroid Hormones:**
 - **Amino Acid Derivatives:**
 - Catecholamines (Eg: Adrenaline, Nor-Adrenaline & Dopamine) (Derived from Tyrosine)
 - Histamine (Derived from Histidine)
 - All Thyroid Hormones (Derived from Tyrosine)
 - **Proteins:**
 - All Pituitary Hormones
 - **Fatty-Acid Derivatives:**
 - Eg: Prostaglandins
 - Eg: Thromboxanes
 - **Purines:**
 - Eg: Adenosine
 - **Dissolved Gases:**
 - Eg: Nitric Oxide

Hormone Class	Components	Example(s)
Amine Hormone	Amino acids with modified groups (e.g. norepinephrine's carboxyl group is replaced with a benzene ring)	Norepinephrine 
Peptide Hormone	Short chains of linked amino acids	Oxytocin 
Protein Hormone	Long chains of linked amino acids	Human Growth Hormone 
Steroid Hormones	Derived from the lipid cholesterol	Testosterone  Progesterone 

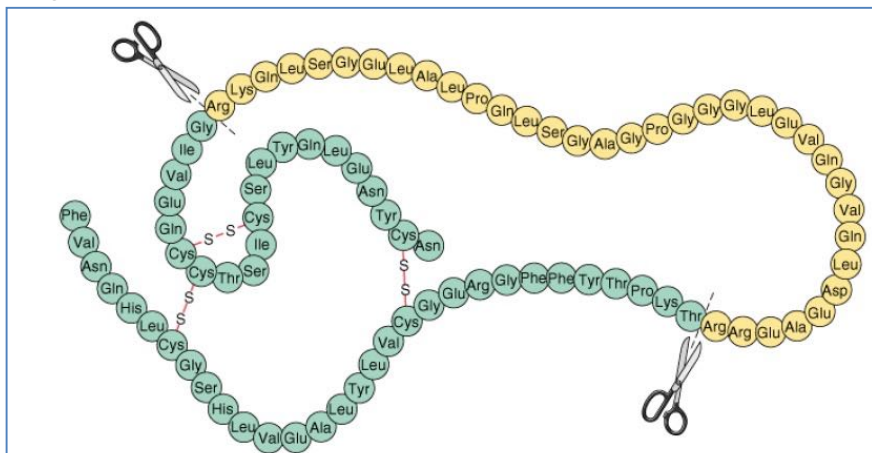
Synthesis of Chemical Messengers:

- Steroidogenesis:

- All steroid hormones are derived from Cholesterol
- **There are 5 Families of Steroids, each with their main physiological member:**
 - Progestogens (Progesterone)
 - Androgens (Testosterone)
 - Mineralocorticoids (Aldosterone)
 - Glucocorticoids (Cortisol)
 - Oestrogens (Oestrogen)

- Protein/Peptide Synthesis & Processing:

- Synthesis of polypeptide hormones can be more complex than Transcription & Translation.
- Some Protein Hormones are initially synthesised as longer **Pre-Prohormones**
- These **Pre-Prohormones** are then cleaved, leaving **Prohormones**
- These **Prohormones** are then cleaved again, leaving active **Hormones**
 - Eg: Insulin:



'Reactive' Properties of Chemical Messengers:

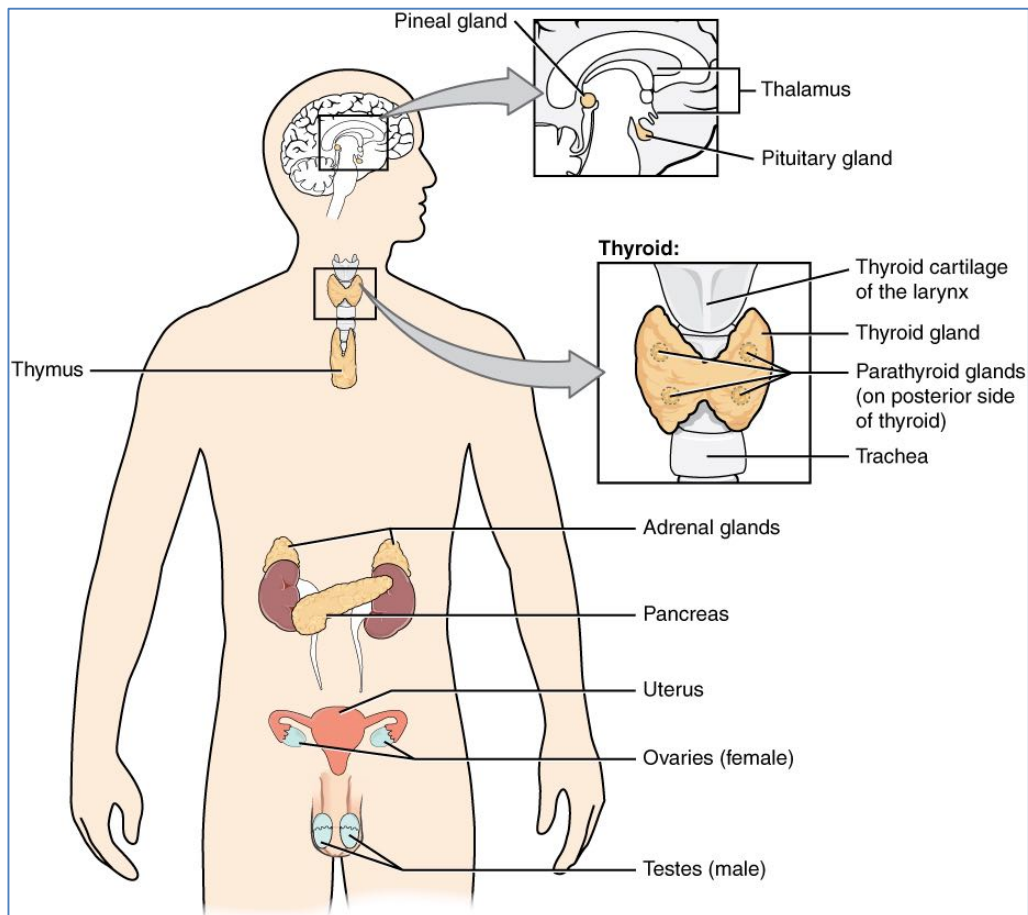
- **Biological Specificity:** Certain Chemical Messengers will only fit into certain receptors
- **Affinity:** The degree to which a chemical is attracted towards a receptor
- **Efficacy:** The degree of effectiveness of the binding of the messenger to the receptor
- **'Agonists':** Chemical Messengers with **High Affinity & High Efficacy**
- **'Antagonists':** Chemical Messengers with **High Affinity** but **Low Efficacy**
Note: There are no Endogenous *Receptor*-Antagonists, Only Exogenous (Drugs)
- **Hormone Binding Proteins:** Proteins that inactivate hormones by binding to them, limiting Bioactivity
- **Epitope:** An Immunologically active binding site on a protein to which an antibody can attach

Endocrine Glands:

- **Endocrine Glands** are **Ductless** and secrete by **Exocytosis** into the **Extracellular Fluid** → Diffuses into Blood

Classical Endocrine Glands:

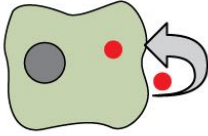
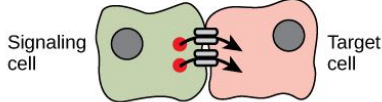
- Hypothalamus
- Pituitary gland
- Pineal gland
- Thyroid gland
- Parathyroid glands (dorsal aspect of thyroid gland)
- Thymus
- Adrenal glands
- Pancreas (has exocrine parts for digestion) (endocrine part secretes insulin)
- Gonads: Testes/Ovaries (also exocrine)



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Long or Short-Range Signalling?

- **Endocrine:** Some signals are “broadcasted” throughout the entire body via bloodstream. → **Hormones** (produced by endocrine cells) [TV]
- **Autocrine:** Signals that affect only cells of the same cell type as the emitting cell. [doctor conference]
- **Paracrine:** Signals (aka local mediators) that act on cells in the vicinity of the emitting cell but on different cell types than the emitting cell. [Lecture]

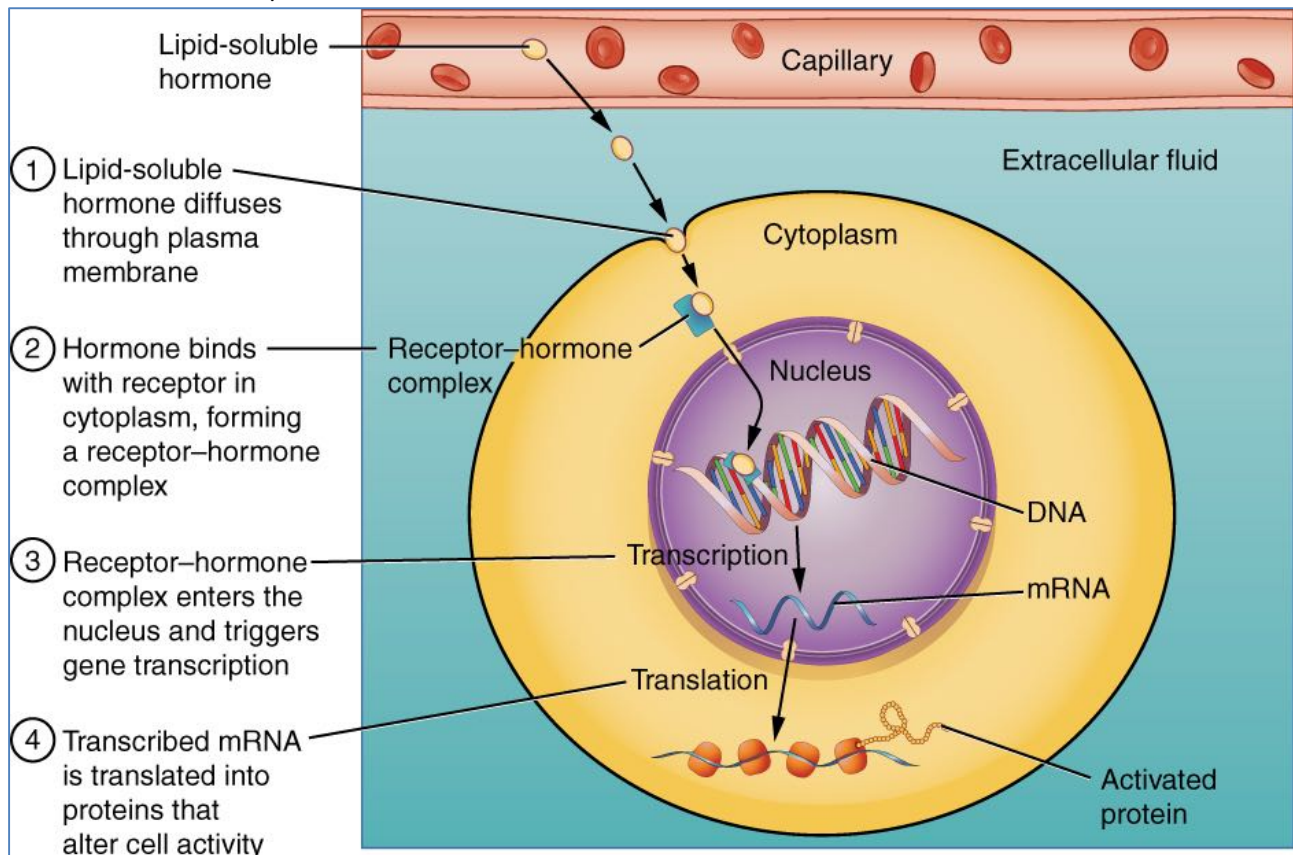
Forms of Chemical Signaling	
Autocrine	A cell targets itself.
	
Signaling across gap junctions	A cell targets a cell connected by gap junctions.
	
Paracrine	A cell targets a nearby cell.
	
Endocrine	A cell targets a distant cell through the bloodstream.
	

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2 Main Receptor Types: (Intracellular & Membrane-bound Receptors)

• Intracellular Receptors:

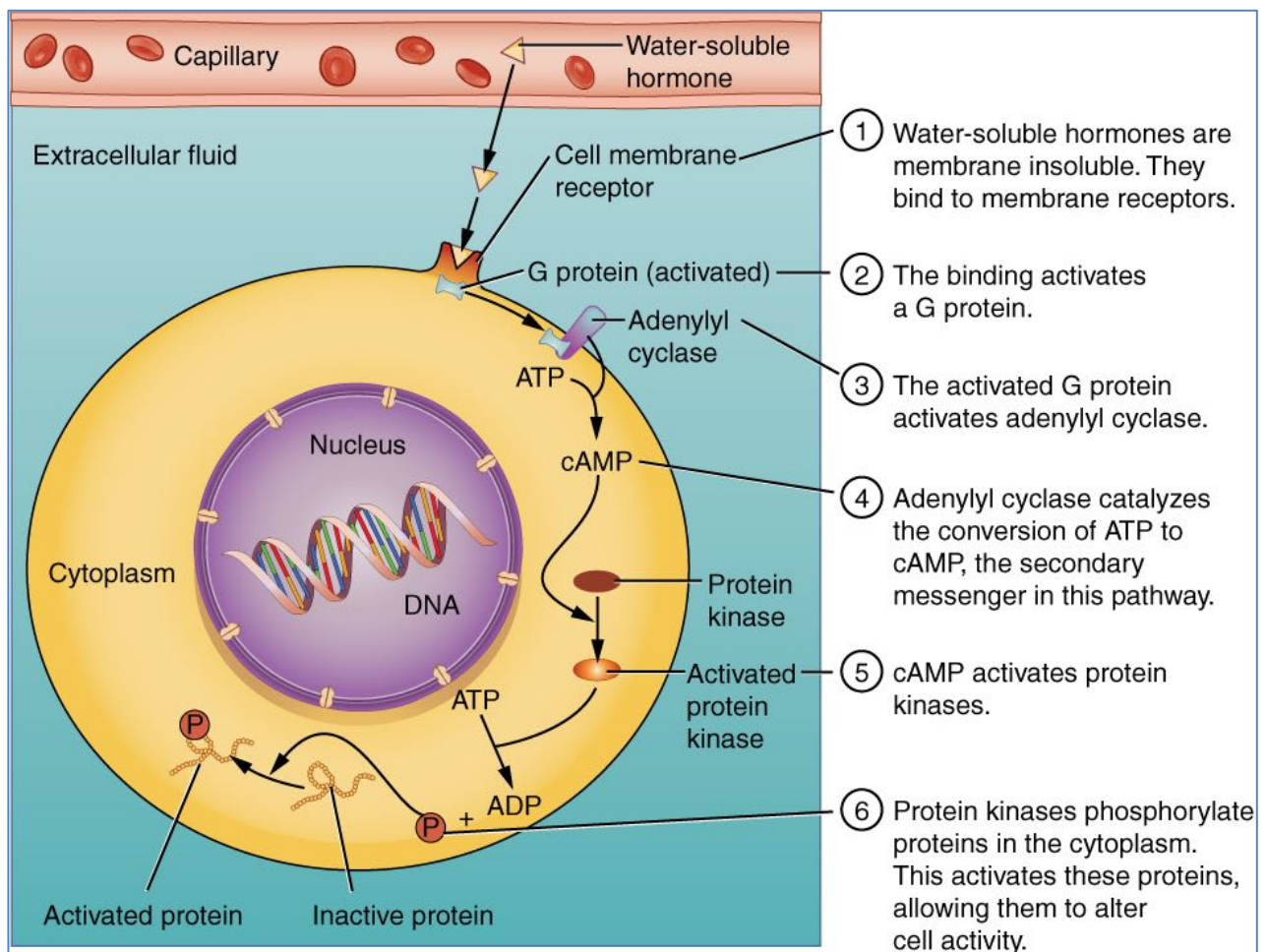
- Lipid-soluble hormones (steroid/thyroid hormones) & even gasses (nitric oxide-blood vessel dilation)
 - **Steroid hormones** bind to receptor proteins in the cytosol or the nucleus that regulate gene expression.



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• Plasma-Membrane-Bound-Receptors:

- Most signal molecules can't cross the plasma membrane of the target cell.
- Most intracellular signalling proteins act as **molecular switches** activated by either **phosphorylation** OR **GTP-Binding** (swapping a GDP for a GTP)
- **3 Types:**
 - **Ion-Channel-Linked Receptors**
 - Resulting signal is a flow of ions across the membrane – produces an electric current.
 - **Enzyme-Linked Receptors**
 - When activated – act as enzymes or are associated with enzymes inside the cell.
 - **G-Protein-Linked Receptors (more common)**
 - Binds to a class of membrane-bound **GTP-Binding-protein (G-Protein)** → becomes activated and released to migrate across the membrane, initiating a cascade of other effects.
 - Some G-Proteins directly regulate ion channels in the plasma membrane.
 - Other G-Proteins activate membrane-bound enzymes. Eg: **adenylyl-cyclase** → increases the [**second messenger (cyclic-AMP)**] → activates an intracellular **signalling protein** (eg: A protein kinase) OR turns on **genes via activated Protein Kinase 'A' (PKA)**.



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Tissue Responsiveness:

- Receptor Downregulation:

- Where a **decreased receptor density** in the membrane decreases the responsiveness of that cell to that receptor's stimuli.
- This is achieved **by Internalising** the **receptor-ligand complex**, dissociating the ligand, and recycling the receptor back to the surface.

- Receptor Desensitisation:

- Where a **change in receptor structure** decreases the responsiveness of that cell to that receptor's stimuli.
- Why? To prevent multiple, rapid stimulations.

Receptor Regulation

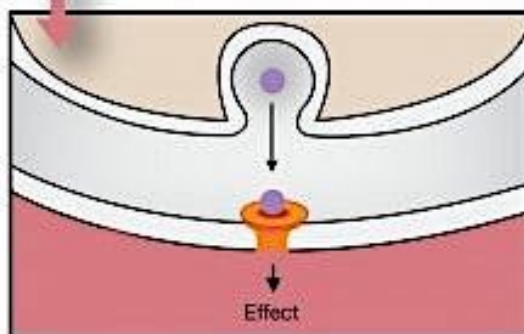
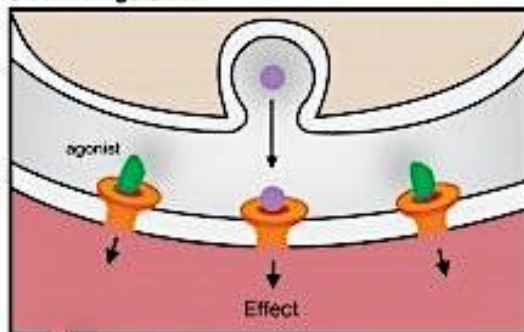
Receptor Downregulation (Desensitization)

Sustained Agonist Activity
↓
Endocytosis
↓
Recycling or Degradation
↓
Synthesis of New Protein
↓
Incorporation into Membrane

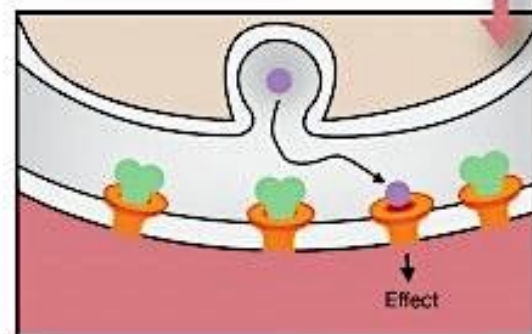
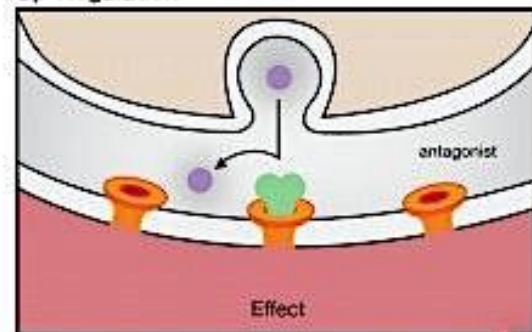
Receptor Upregulation (Supersensitization)

Receptor Downregulation
↓
Increase total number of receptors
↓
Decrease down regulation
↓
Withdrawal of antagonist
↓
elevated number of receptors can
produce exaggerated response

Down-Regulation



Up-Regulation



Source: Unattributable

Regulation of Hormone Release:

- 3 Mechanisms:

- **1- Humoral:**
 - Where the concentration of a solute in the blood (Eg: High Glucose/Low Calcium) is detected by a specific gland, stimulating hormone release (Eg: Insulin/Parathyroid Hormone)
- **2- Neural:**
 - Where the Nervous System Directly stimulates hormone release.
 - Eg: Sympathetic NS Activated → Stimulates Adrenal Medulla → Secretes Catecholamines.
- **3- Hormonal:**
 - Where one hormone stimulates the release of another hormone from a different cell.
 - Eg: The Hypothalamus secretes hormones → Stimulate Ant. Pituitary → Secretes Hormones.
 - Eg: The Ant. Pituitary secretes Hormones → Stimulate other organs to secrete hormones.

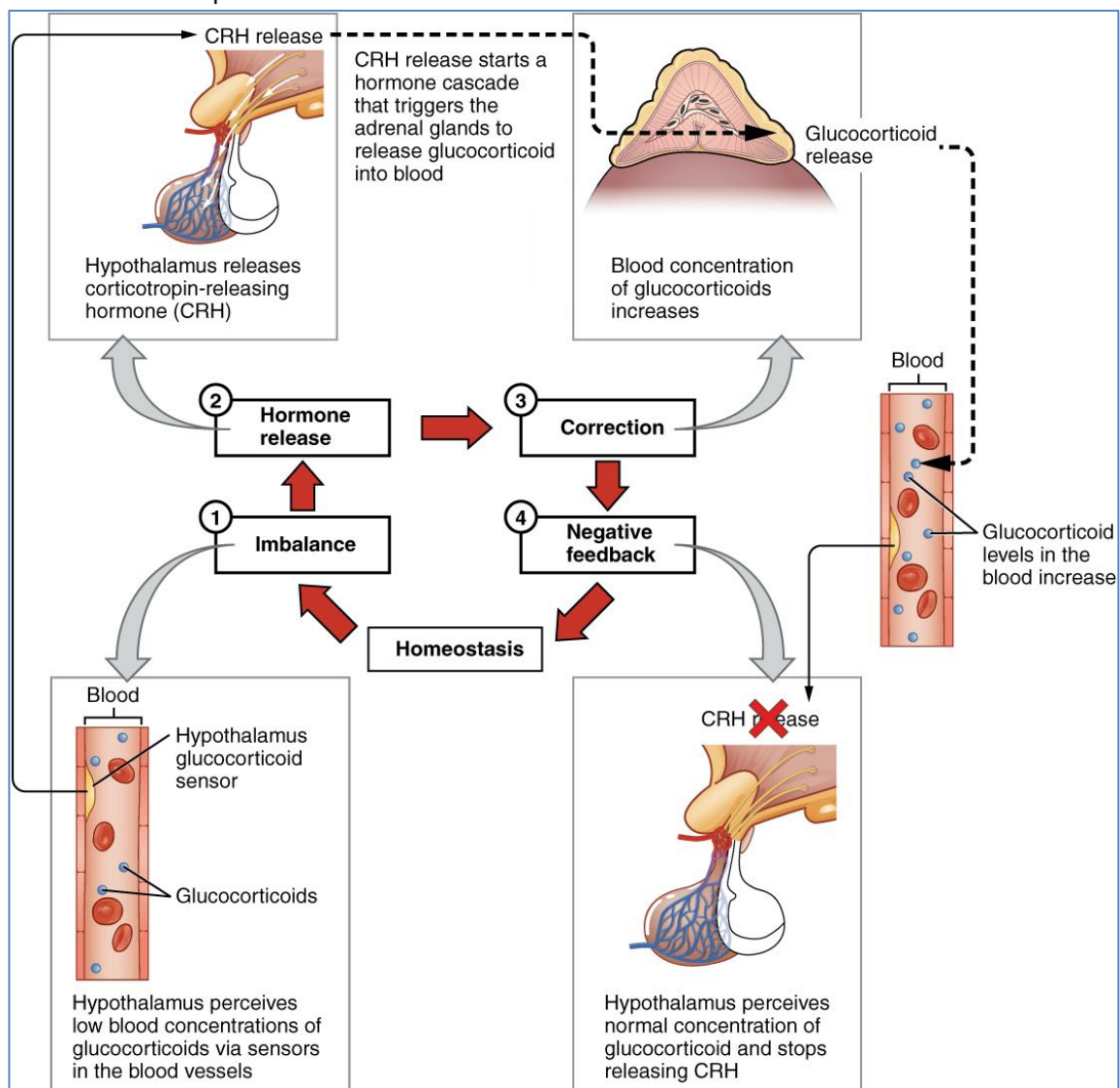
FEEDBACK:

• Negative:

- Most common
- Where the **Biological Response** causes a **Decreased Hormone Release**.
- Maintains levels around a stable intrinsic/pre-set level.
- Involved in homeostatic control.

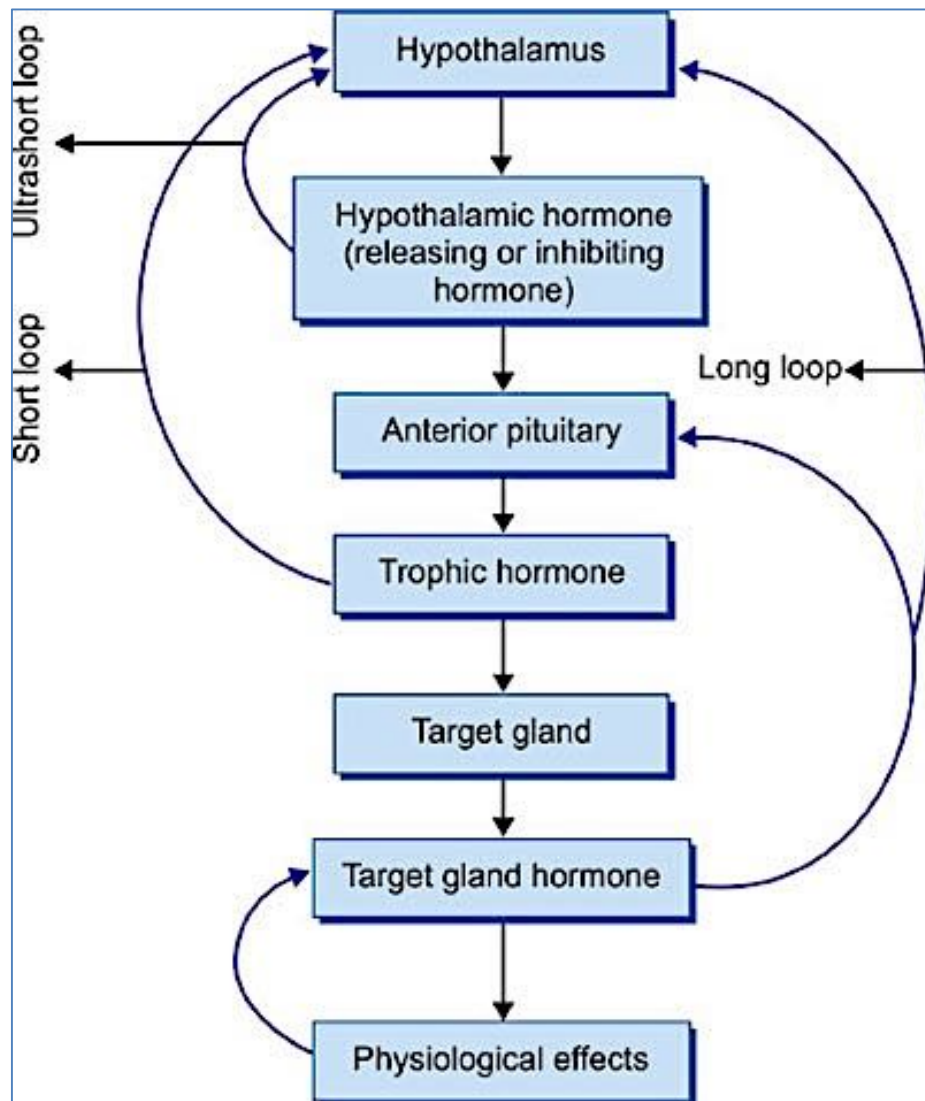
• Positive:

- Uncommon (Lactation & Parturition)
- Where the **Biological Response** causes an **Increased Hormone Release**
- Are therefore Unstable mechanisms
- Terminate upon removal of initial stimulus.



Levels of Feedback Loops:

- Feedback may occur at many different levels within an endocrine axis.
 - **Ultra-Short Loop:**
 - The secreted hormone feeds back to the same tissue that secreted it.
 - Eg: A Hypothalamic Hormone feeds back to the Hypothalamus.
 - **Short Loop:**
 - The secreted hormone feeds back to the tissue that stimulated its secretion.
 - Eg: The Hormone secreted by the Target Organ feeds back to the Pituitary.
 - Or. The Hormone secreted by the Pituitary feeds back to the Hypothalamus.
 - **Long Loop:**
 - The hormone secreted by the target organ feeds directly back to the Hypothalamus.



Source: Unattributable

Major Hormones & Their Functions:

Endocrine gland	Associated hormones	Chemical class	Effect
Pituitary (anterior)	Growth hormone (GH)	Protein	Promotes growth of body tissues
Pituitary (anterior)	Prolactin (PRL)	Peptide	Promotes milk production
Pituitary (anterior)	Thyroid-stimulating hormone (TSH)	Glycoprotein	Stimulates thyroid hormone release
Pituitary (anterior)	Adrenocorticotrophic hormone (ACTH)	Peptide	Stimulates hormone release by adrenal cortex
Pituitary (anterior)	Follicle-stimulating hormone (FSH)	Glycoprotein	Stimulates gamete production
Pituitary (anterior)	Luteinizing hormone (LH)	Glycoprotein	Stimulates androgen production by gonads
Pituitary (posterior)	Antidiuretic hormone (ADH)	Peptide	Stimulates water reabsorption by kidneys
Pituitary (posterior)	Oxytocin	Peptide	Stimulates uterine contractions during childbirth
Thyroid	Thyroxine (T ₄), triiodothyronine (T ₃)	Amine	Stimulate basal metabolic rate
Thyroid	Calcitonin	Peptide	Reduces blood Ca ²⁺ levels
Parathyroid	Parathyroid hormone (PTH)	Peptide	Increases blood Ca ²⁺ levels
Adrenal (cortex)	Aldosterone	Steroid	Increases blood Na ⁺ levels
Adrenal (cortex)	Cortisol, corticosterone, cortisone	Steroid	Increase blood glucose levels
Adrenal (medulla)	Epinephrine, norepinephrine	Amine	Stimulate fight-or-flight response
Pineal	Melatonin	Amine	Regulates sleep cycles
Pancreas	Insulin	Protein	Reduces blood glucose levels
Pancreas	Glucagon	Protein	Increases blood glucose levels
Testes	Testosterone	Steroid	Stimulates development of male secondary sex characteristics and sperm production
Ovaries	Oestrogens and progesterone	Steroid	Stimulate development of female secondary sex characteristics and prepare the body for childbirth

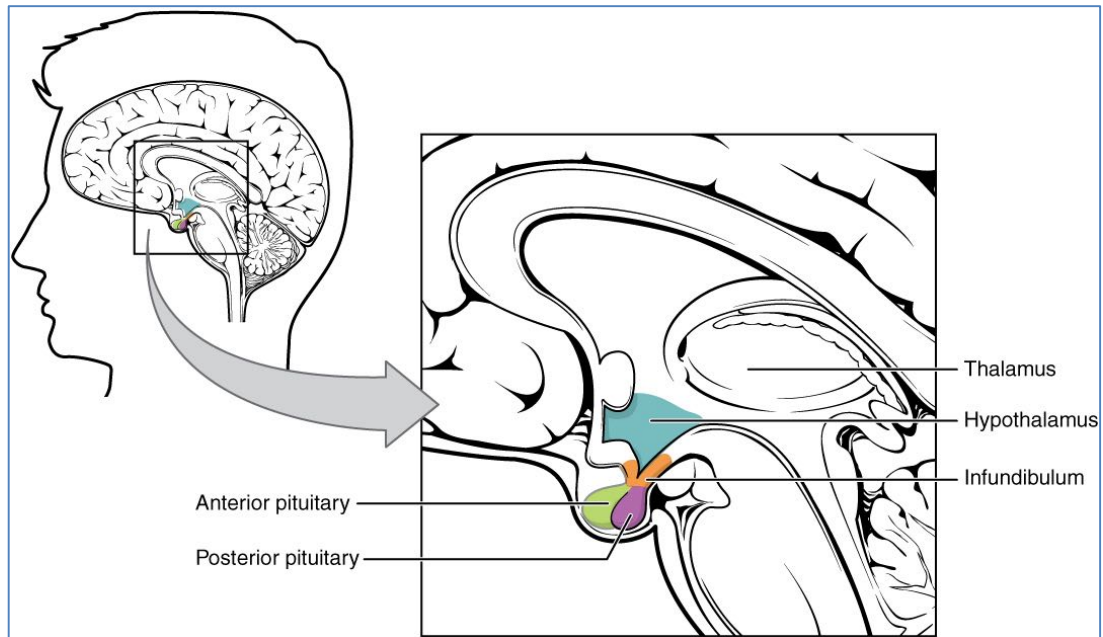
THE HYPOTHALAMUS & PITUITARY GLAND

THE HYPOTHALAMUS & PITUITARY GLAND

The Hypothalamus:

- Location:

- In the diencephalon @ base of the brain, just below the thalamus.
- 'Hypo'-Thalamus = Below the Thalamus



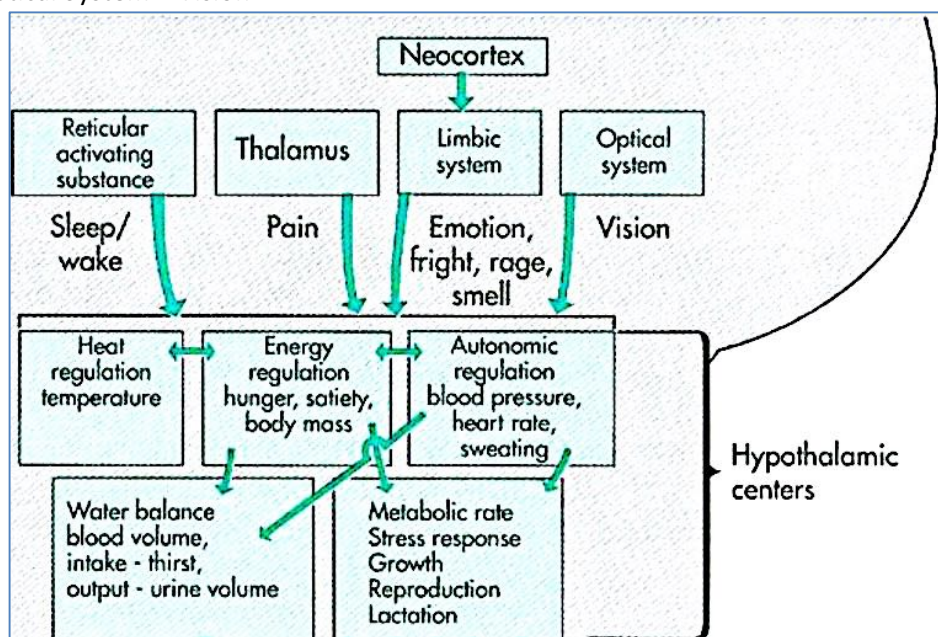
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- Functions:

- The Hypothalamus receives information from multiple higher brain centres, integrates it, decides on a response, and orders the pituitary to secrete specific hormones to elicit the response.
 - I.e: Links the nervous system to the endocrine system via the pituitary gland.
- Controls body temperature, hunger, thirst, fatigue, anger, and cycles. Synthesizes & secretes **neurohormones** (hypothalamic-releasing hormones) which stimulate or inhibit the secretion of pituitary hormones.

- Inputs:

- RAS (Reticular Activating System/Substance) – Regulates drowsiness by releasing Serotonin.
- Thalamus – Plays a role in Pain Perception
- Neocortex & Limbic System – Emotional Centre
- Optical System – Vision

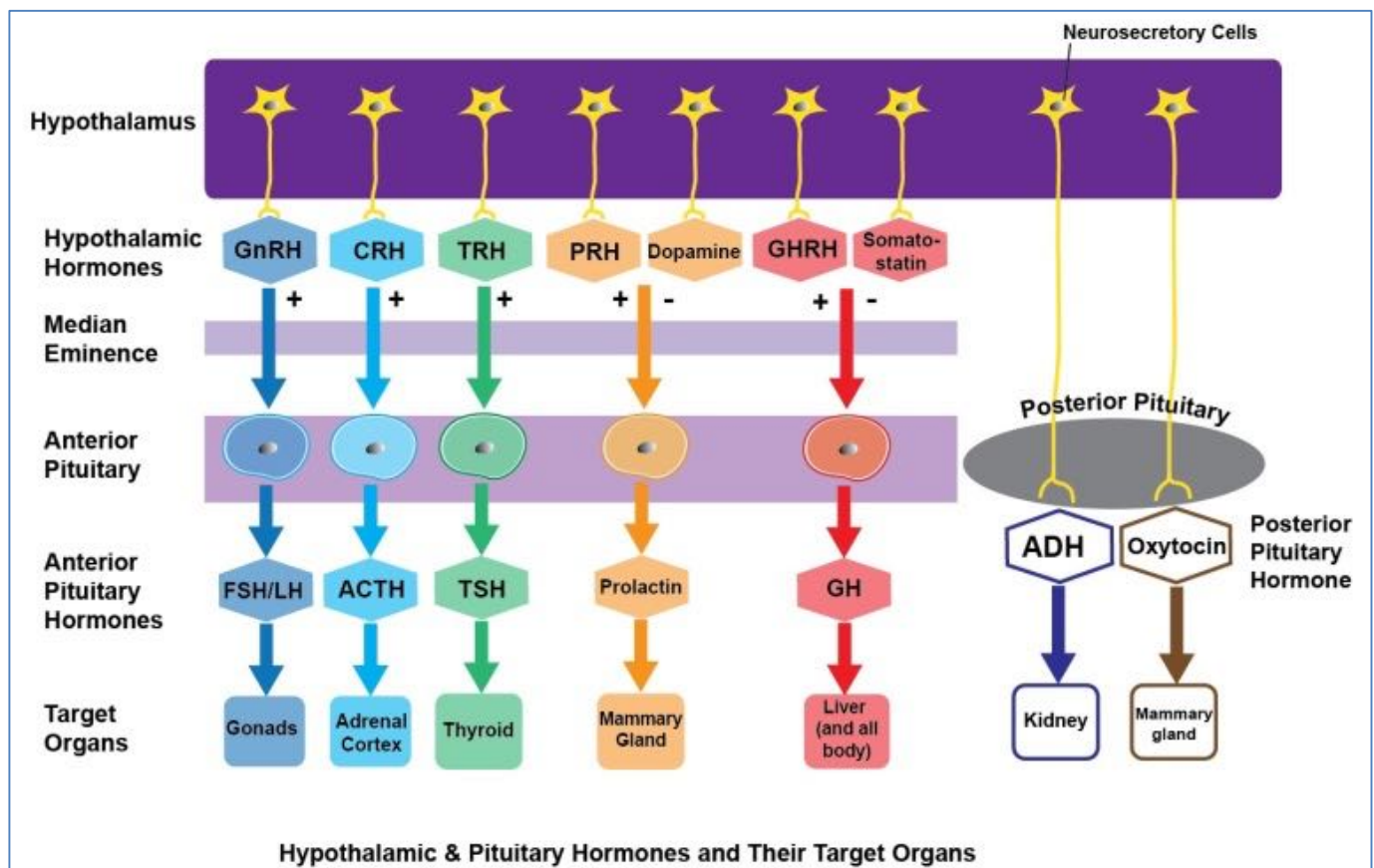


Source: Unattributable

- **Outputs:**
 - o Anterior Pituitary
 - o Posterior Pituitary
 - o Brain-Stem (Autonomic NS)

- **Hypothalamic Regulatory Hormones:**

Abbreviation	Full Name	Stimulated/Inhibited Hormone
GHRH	Growth-Hormone Releasing Hormone	Stimulates Release of Growth Hormone
SS	Somatostatin	Inhibits Release of Growth Hormone & TSH
TRH	Thyrotropin Releasing Hormone	Stimulates Release of TSH & Prolactin
PRH	Prolactin Releasing Hormone	Stimulates Release of Prolactin
GnRH	Gonadotropin Releasing Hormone	Stimulates Release of Gonadotropins; FSH & LH
CRH	Corticotrophin Releasing Hormone	Stimulates Release of ACTH
PIH	Prolactin Inhibiting Hormone	Inhibits Release of Prolactin



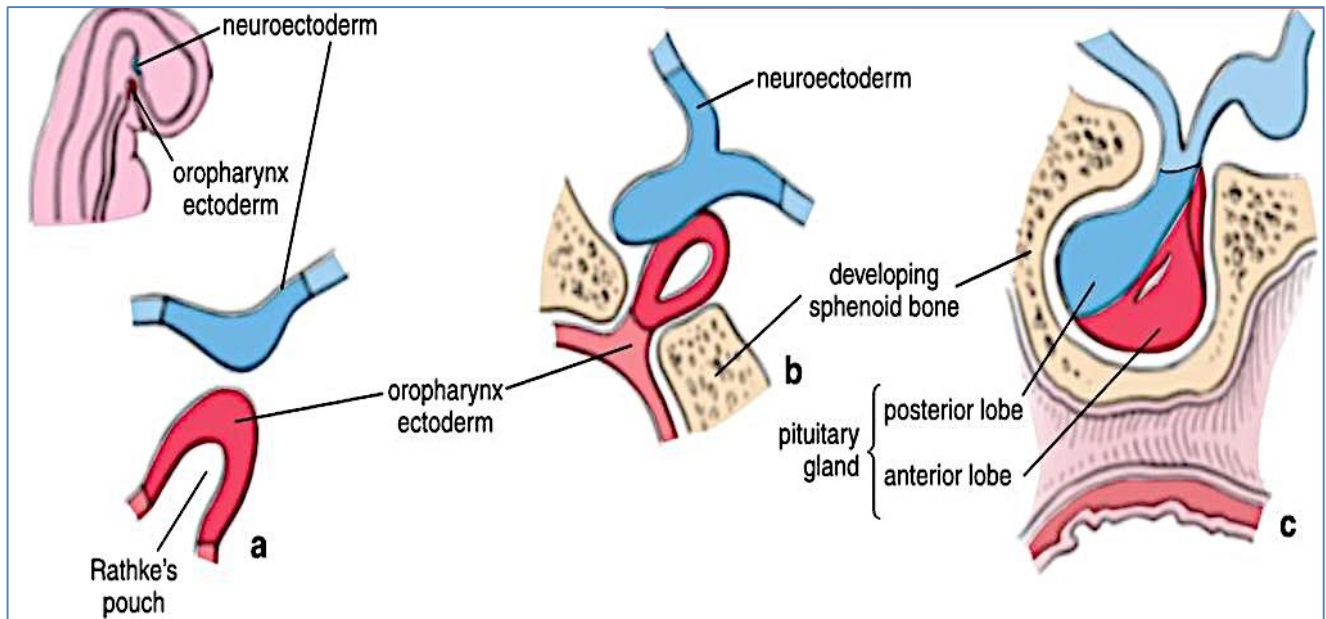
<https://www.ncbi.nlm.nih.gov/books/NBK551529/figure/article-692.image.f1/>

The Pituitary Gland:

- **Location:**
 - o Just Anterior to Pons of the Brainstem
 - o Just Posterior to the Optic Chiasma
 - o Just Inferior to the Hypothalamus
 - o Connects superiorly to the hypothalamus (above it) via the **Infundibulum**
- **Function:**
 - o Secretes at least 9 hormones.
 - o Function is controlled by hypothalamus (which secretes releasing/inhibiting hormones)

Embryology of the Pituitary Gland:

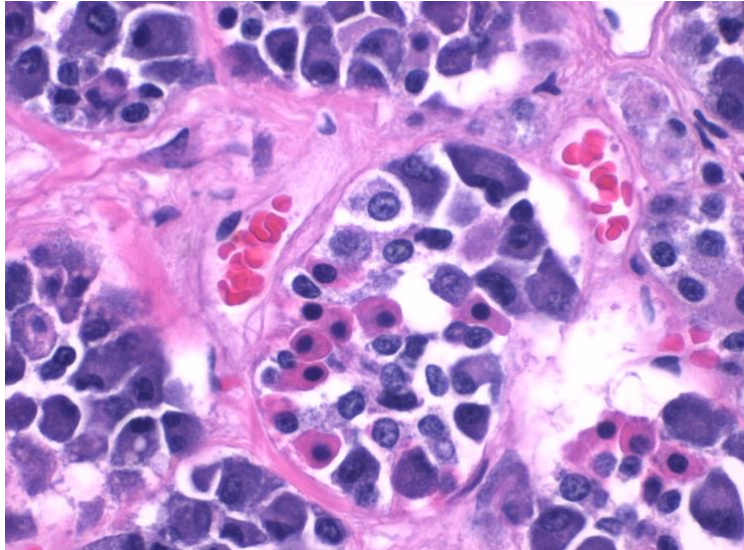
- **Q:** Why is the Anterior Pituitary **Endocrine**, but the Posterior Pituitary **Neuronal**?
- **A:** Because they have different embryonic origins.
 - **Anterior Pituitary:**
 - Arises from an upward out-pouching of the **Oral-Ectoderm** from the roof of the oral cavity called **Rathke's Pouch**. This pouch pinches off from the oral cavity and is later separated by the sphenoid bone.
 - Consists of **Epithelial/Glandular Tissue**, & therefore **Manufactures & Secretes Hormones**.
 - **Posterior Pituitary:**
 - Originates from a downward out-pouching of **Neuro-Ectoderm** from the brain in the floor of the 3rd ventricle.
 - Consists of **Neural Tissue**, & therefore **Secretes Neurohormones**.



Source: Unattributable

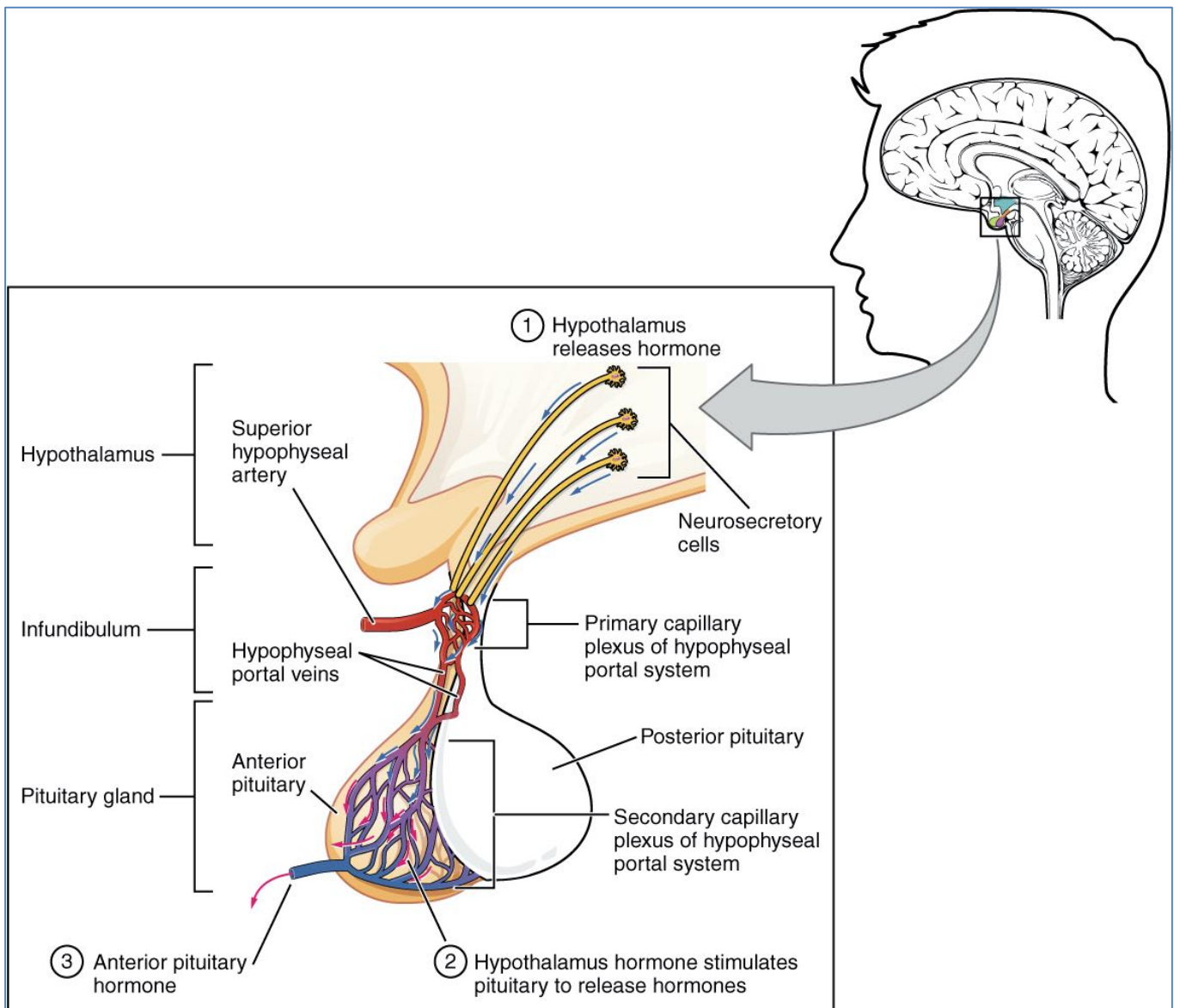
Anterior Pituitary: (Adenohypophysis)

- Glandular Tissue (adeno = gland)
- **Responsible for Producing:**
 - o **TSH** (Thyroid Stimulating Hormone)
 - o **FSH/LH** (Follicle Stimulating Hormone; Luteinizing Hormone)
 - o **PRL** (Prolactin)
 - o **ACTH** (Adreno CorticoTropic Hormone)
 - o **GH** (Growth Hormone)
- **Different Cell Types in the Anterior Pituitary Secrete Specific Hormones:**
 - o **Gonadotrophs:** FSH & LH
 - o **Corticotrophs:** ACTH (Adreno-Cortico-Tropic Hormone)
 - o **Thyrotrophs:** TSH
 - o **Mammotrophs:** Prolactin
 - o **Somatotrophs:** Growth Hormone & Somatotropin
- **Histology – Glandular structure:**
 - o Clusters of acini surrounded by blood vessels
 - o Acini - mosaics of different cells:
 - (acidophils – red, basophils – dark blue, chromophobes - colourless)
 - Note: Pituitary Tumours may be from any of the 3 cells
 - o **PLENTY of blood vessels (neither arteries or veins; but 'Portal Vessels' – ie: Blood comes *only* from the hypothalamus → carries the hypothalamic hormones.)**



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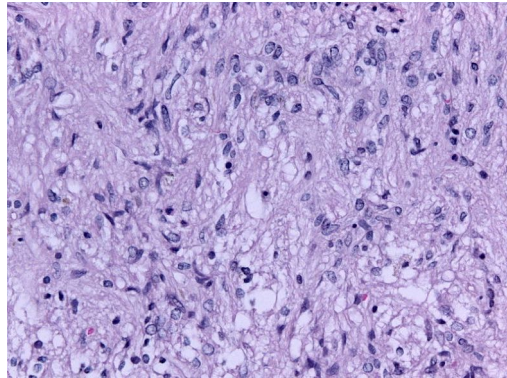
- **Blood Supply:**
 - o Arterial blood enters via **Hypophyseal Branches** of the **Internal Carotid Arteries**.
- **Venous Drainage:**
 - o Venous blood leaves via **venules** which drain into the **Dural Sinuses**.



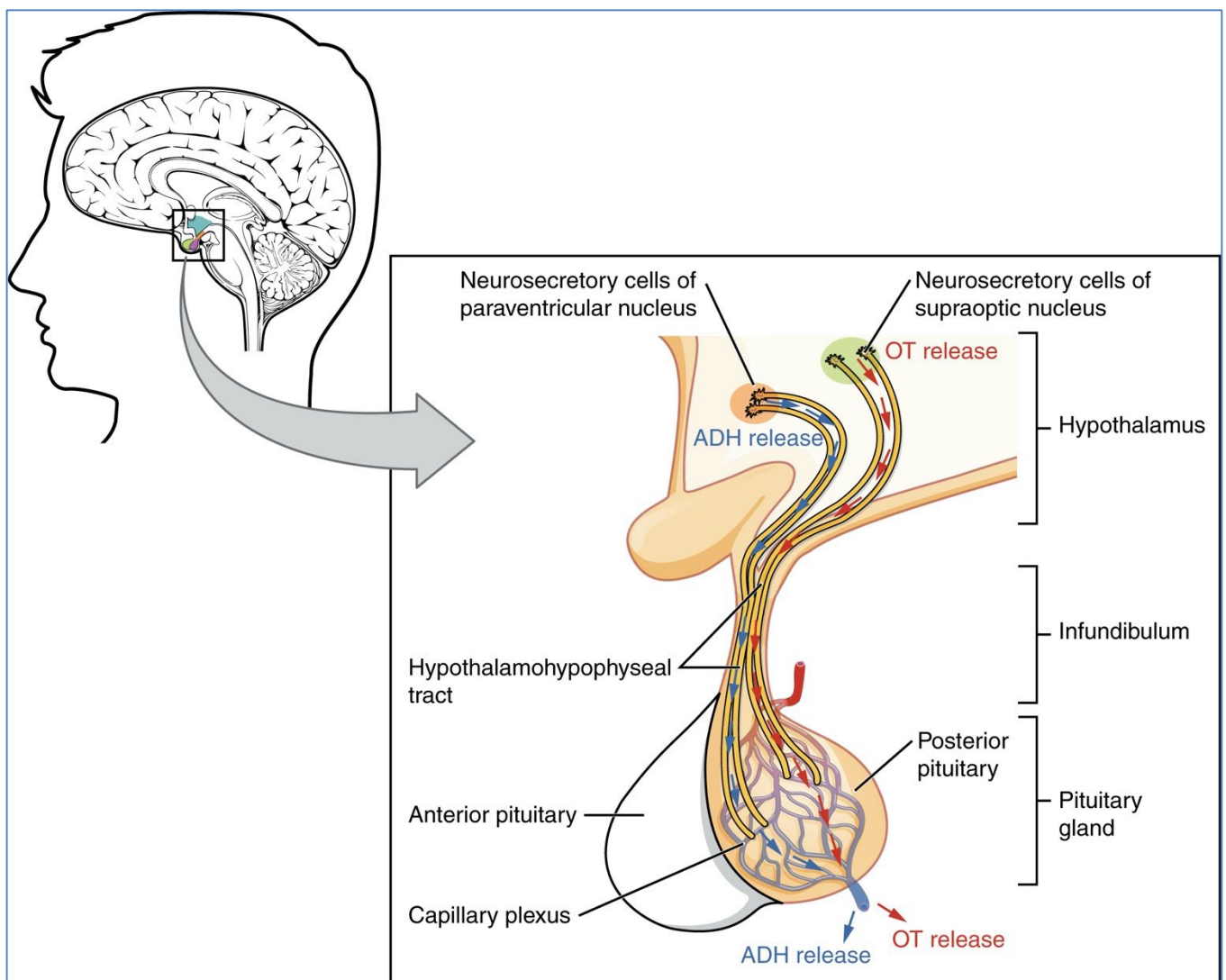
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Posterior Pituitary: (Neurohypophysis)

- Made of **Nervous Tissue**
- Is essentially an **Extension of the Hypothalamus**
- **Supraoptic & Paraventricular Nuclei** in the hypothalamus synthesize Oxytocin & ADH → Transport them to their axon terminals in the **Posterior Pituitary**.
 - Hormones released as needed via exocytosis in Post-Pituitary
 - ADH
 - Oxytocin
- **Histology – Just like normal brain tissue. (Neural Origin)**
 - Note: NO neurones, but plenty of axons.
 - Many supporting cells (Astrocytes, oligodendrocytes)
 - **Plus Blood Vessels (neither arteries or veins; but 'Portal Vessels' – i.e: Blood comes *only* from the hypothalamus → carries the hypothalamic hormones.)**



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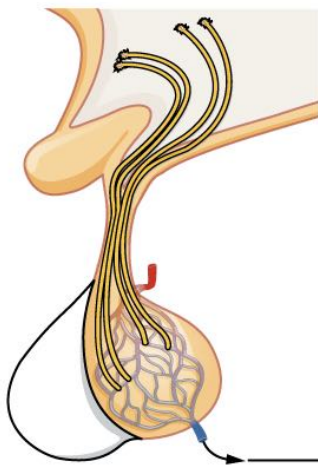
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Pituitary Hormones:

Pituitary lobe	Associated hormones	Chemical class	Effect
Anterior	Growth hormone (GH)	Protein	Promotes growth of body tissues
Anterior	Prolactin (PRL)	Peptide	Promotes milk production from mammary glands
Anterior	Thyroid-stimulating hormone (TSH)	Glycoprotein	Stimulates thyroid hormone release from thyroid
Anterior	Adrenocorticotrophic hormone (ACTH)	Peptide	Stimulates hormone release by adrenal cortex
Anterior	Follicle-stimulating hormone (FSH)	Glycoprotein	Stimulates gamete production in gonads
Anterior	Luteinizing hormone (LH)	Glycoprotein	Stimulates androgen production by gonads
Posterior	Antidiuretic hormone (ADH)	Peptide	Stimulates water reabsorption by kidneys
Posterior	Oxytocin	Peptide	Stimulates uterine contractions during childbirth
Intermediate zone	Melanocyte-stimulating hormone	Peptide	Stimulates melanin formation in melanocytes

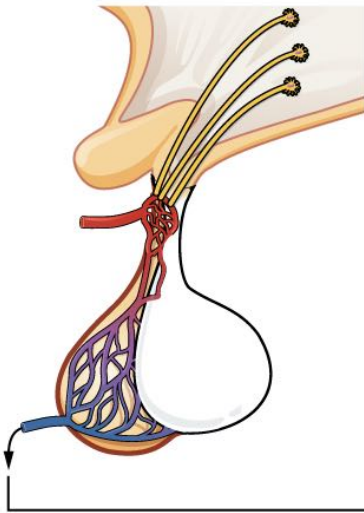
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Posterior Pituitary Hormones



Releasing hormone (hypothalamus)	Pituitary hormone	Target	Effects
ADH	Stores ADH	Kidneys, sweat glands, circulatory system	Water balance
-	OT	Female reproductive system	Triggers uterine contractions during childbirth

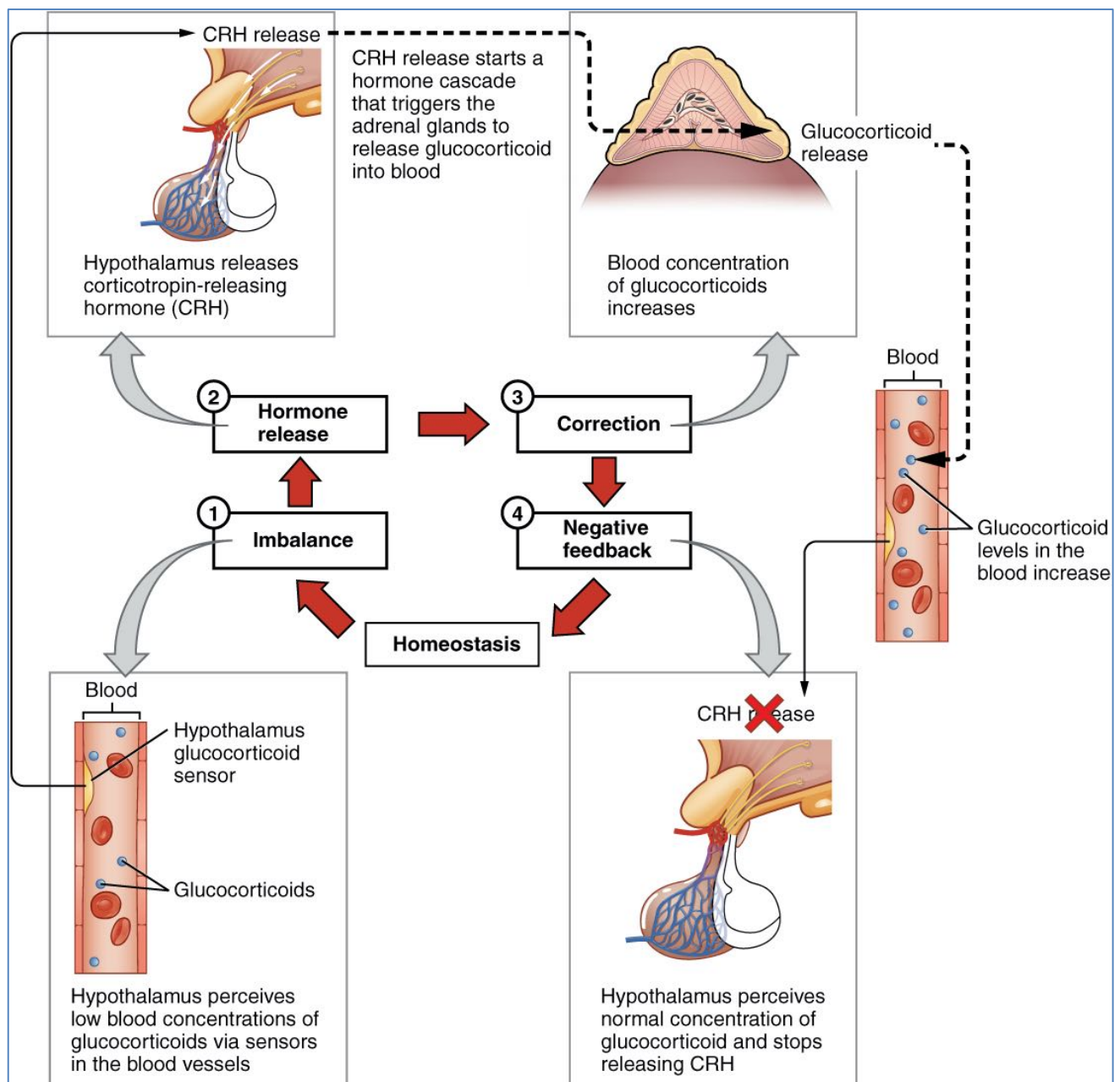
Anterior Pituitary Hormones



Releasing hormone (hypothalamus)	Pituitary hormone	Target	Effects
GnRH	LH	Reproductive system	Stimulates production of sex hormones by gonads
GnRH	FSH	Reproductive system	Stimulates production of sperm and eggs
TRH	TSH	Thyroid gland	Stimulates the release of thyroid hormone (TH). TH regulates metabolism.
PRH (inhibited by PIH)	PRL	Mammary glands	Promotes milk production
GHRH (inhibited by GHIH)	GH	Liver, bone, muscles	Induces targets to produce insulin-like growth factors (IGF). IGFs stimulate body growth and a higher metabolic rate.
CRH	ACTH	Adrenal glands	Induces targets to produce glucocorticoids, which regulate metabolism and the stress response

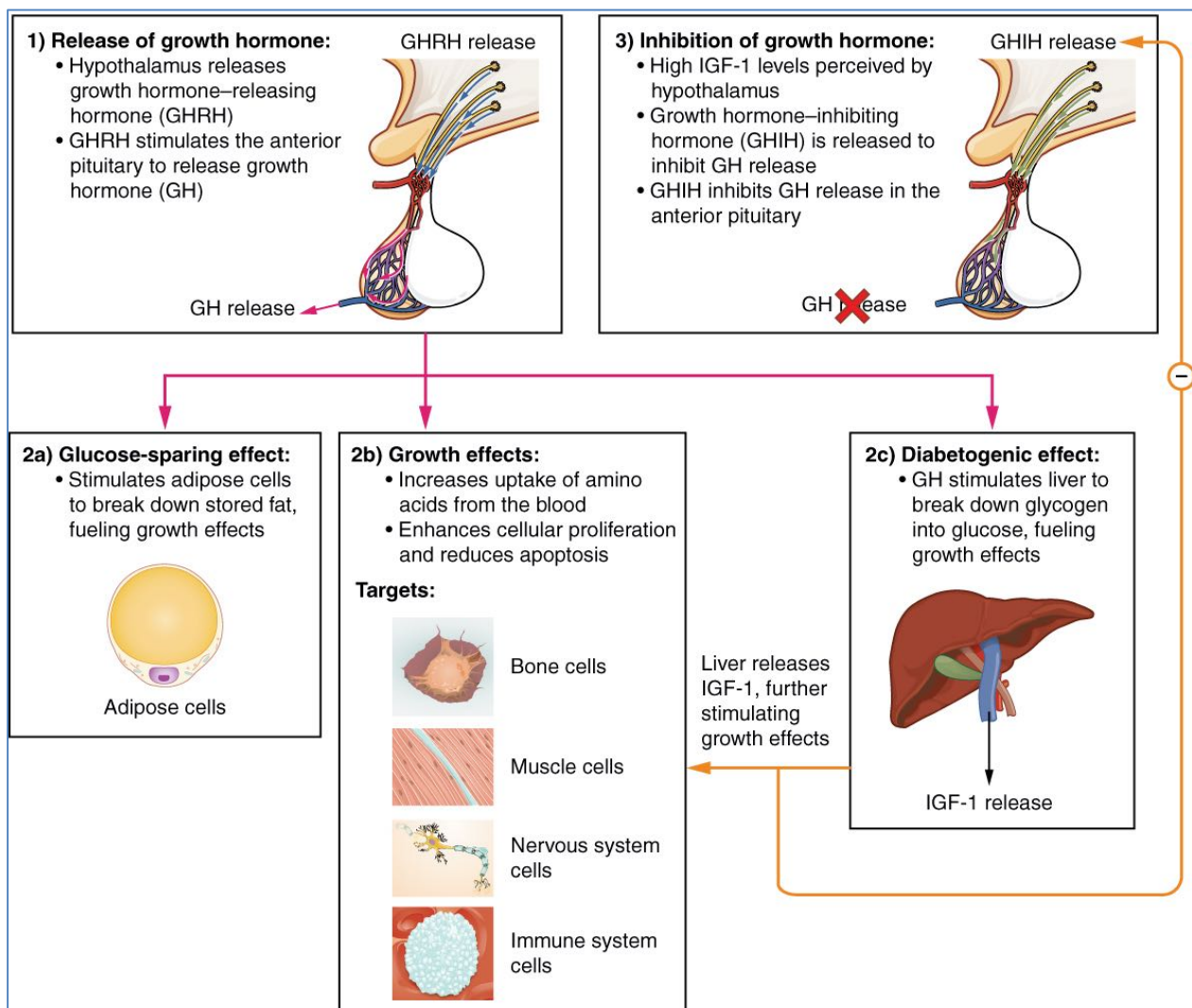
ACTH – Adrenocorticotrophic Hormone:

- **Secreted By:**
 - Corticotrophs in the Anterior Pituitary
- **Primary Action:**
 - Stimulates adrenocortical cells in the **Zona Fasciculata** of the **Adrenal Cortex** to secrete **Glucocorticoids**.
 - To reduce inflammation
 - To increase blood glucose levels
 - To increase lipolysis & proteolysis
- **Release is Stimulated By:**
 - Corticotrophin Releasing Hormone (CRH)
 - CRH is secreted by the Hypothalamus in response to:
 - Stress
 - Low blood glucose
 - Low glucocorticoid levels
 - Increased Sympathetic Activity
 - Normal Diurnal Rhythm
- **Release is Regulated By:**
 - Negative feedback loop between Hypothalamus-Pituitary-Adrenal glands



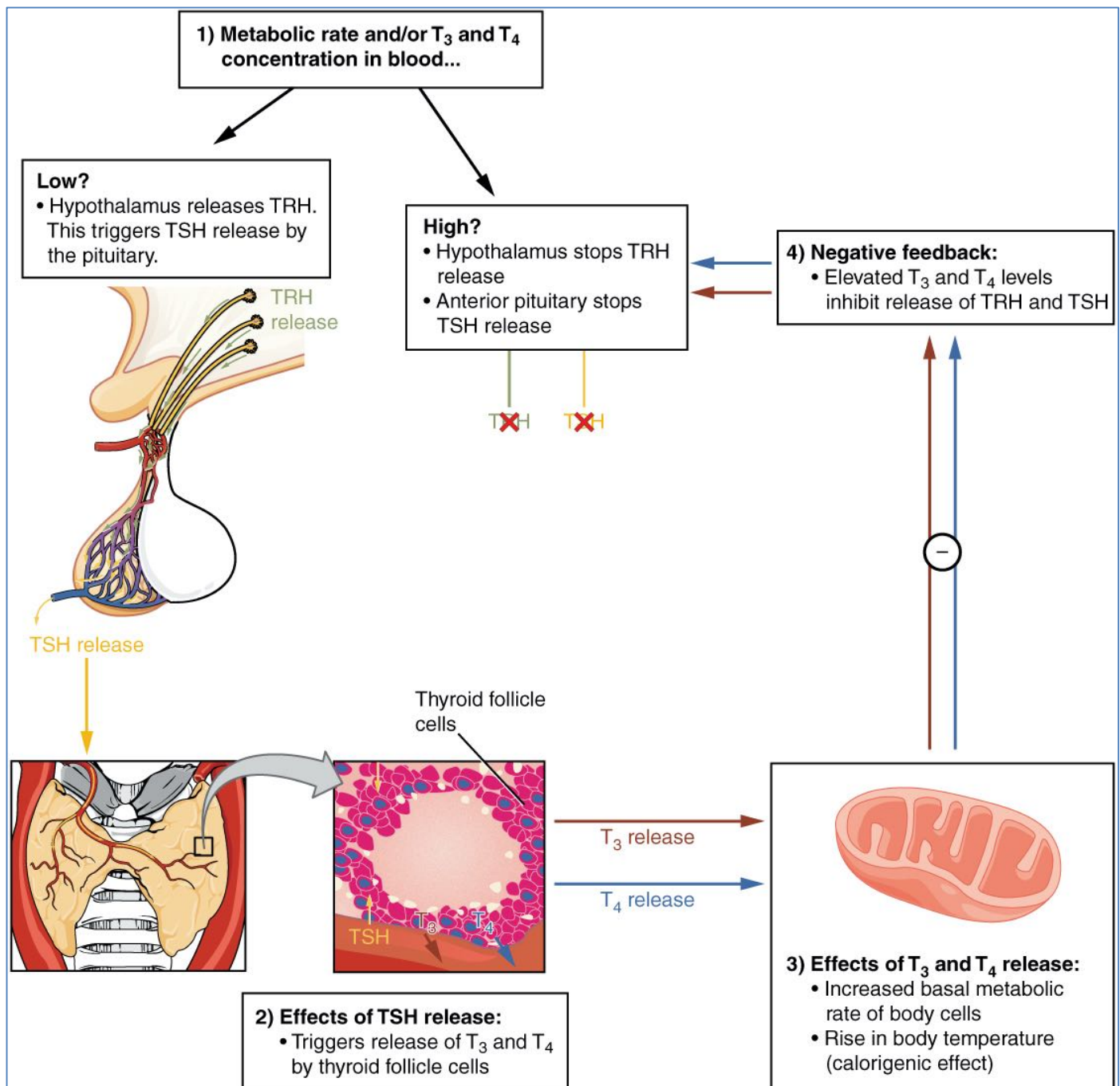
GH – Growth Hormone:

- **Secreted By:**
 - Somatotrophs in the Anterior Pituitary
- **Primary Actions:**
 - **↑Proportion of LEAN BODY MASS:**
 - ↑Muscle Mass
 - ↓Adipose Tissue
 - **Stimulates cell metabolism, growth and division throughout the body**
 - Stimulates Protein Synthesis
 - Decreases Protein Breakdown
 - Stimulates Bone Osteoblast activity
 - **Also has anti-insulin-like effects:**
 - Stimulates Gluconeogenesis (Liver)
 - Stimulates Glycogenolysis (Liver)
 - Increases tissue insulin resistance
 - Stimulates Lipolysis (Adipose tissue)
 - **Indirect Actions (Via Somatomedins/IGF's):**
 - **Linear Bone Growth** (↑Collagen & Protein Synthesis)
 - **Tissue Growth** (↑Protein Synthesis & Gene Replication/Transcription)
- **Release is Stimulated By:**
 - Growth Hormone Releasing Hormone (GHRH)
 - GHRH is secreted by the Hypothalamus in a diurnal manner & in response to low GH levels
 - Also stimulated by Ghrelin from the stomach
- **Release is Regulated By:**
 - Negative feedback loop between Hypothalamus & Pituitary gland



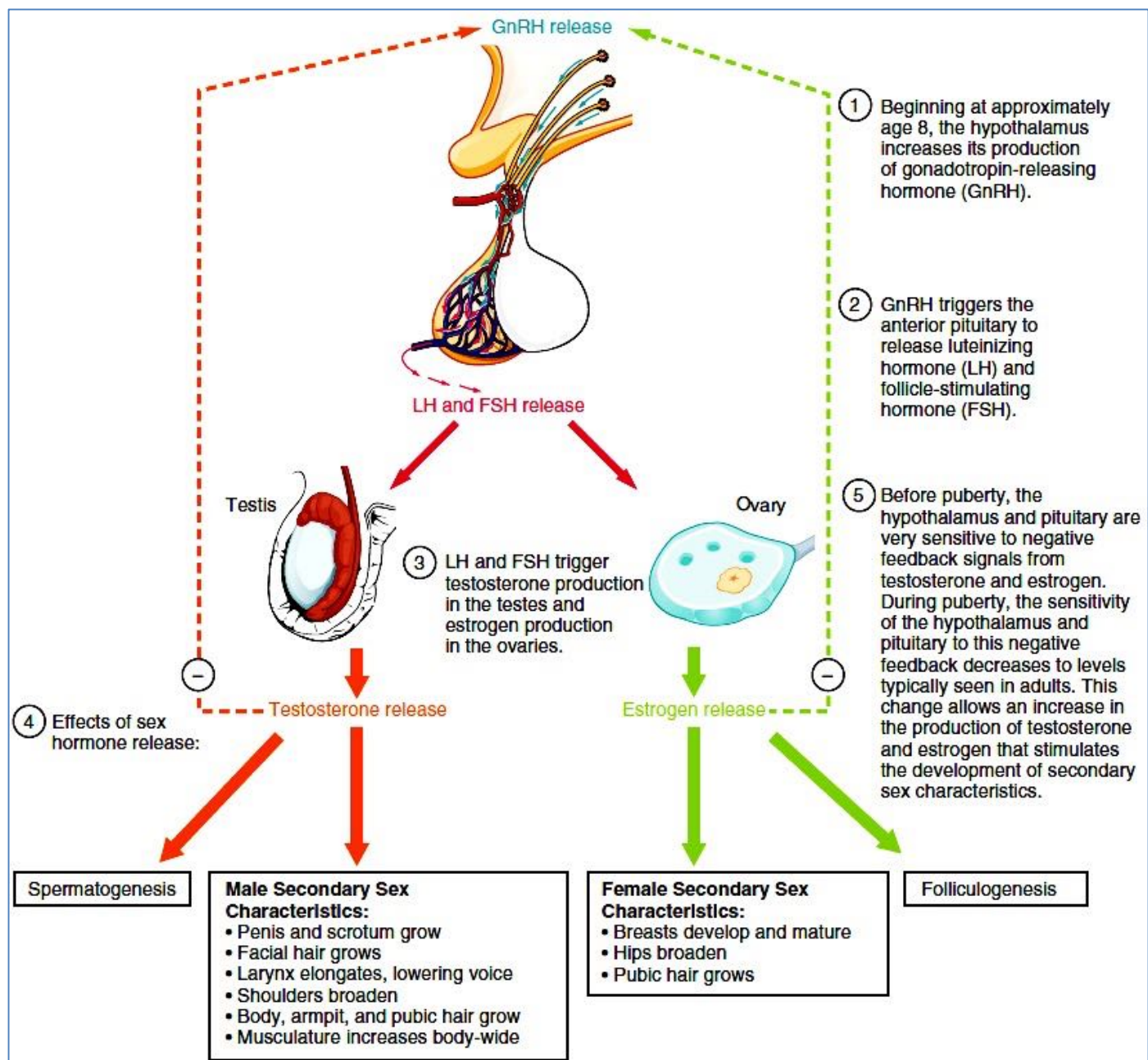
TSH - Thyroid Stimulating Hormone:

- **Secreted By:**
 - o Thyrotrophs of the Anterior Pituitary
- **Primary Action:**
 - o Stimulates Thyroid Gland Growth
 - o Stimulates Thyroid Hormone Synthesis
 - o Stimulates Thyroid Hormone Release
- **Release is Stimulated By:**
 - o Thyrotropin-Releasing Hormone (TRH)
 - o TRH is secreted by the Hypothalamus in response to:
 - Low T₃/T₄ Blood Levels
 - Decreased metabolism
 - Cold stress
 - High-energy-use situations
- **Release is Regulated By:**
 - o Negative feedback loop between Hypothalamus-Pituitary-Thyroid glands:



FSH/LH – Follicle Stimulating Hormone & Luteinizing Hormone:

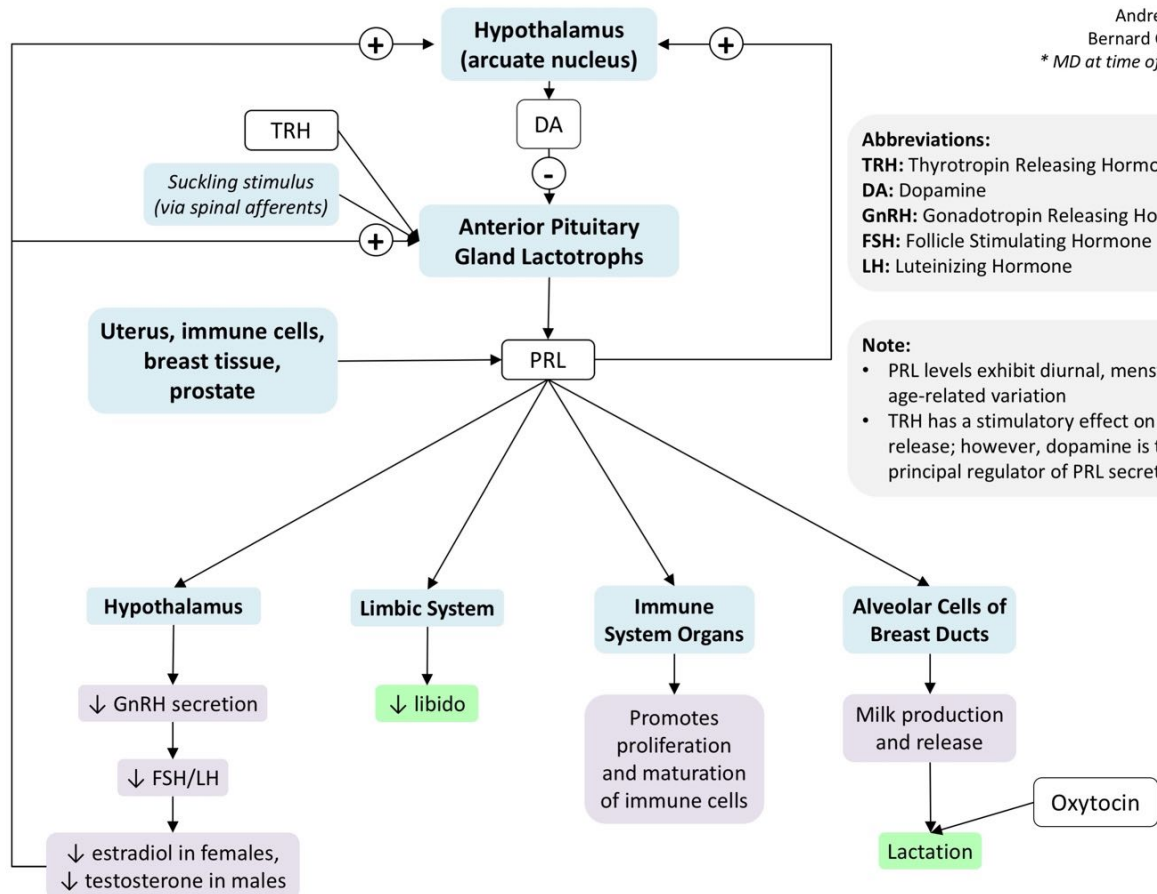
- **Secreted By:**
 - o Gonadotrophs of the Anterior Pituitary
- **Primary Action:**
 - o Important in Puberty → Development of Sexual Characteristics:
 - Primary Sex Characteristics (Genital Development)
 - Secondary Sex Characteristics (Eg: Pubic hair, voice changes, breast development)
 - o Important for Gamete Production
 - Males: LH → Leydig Cells → Produces Testosterone
 - Males: FSH → Sertoli Cells → Produce Sperm
 - Females: LH → Ovarian Follicles & Progesterone/Oestrogen Synthesis
 - Females: FSH → Recruits Ovarian Follicles early in menstrual cycle
- **Release is Stimulated By:**
 - o Gonadotropin-Releasing Hormone (GnRH)
- **Release is Regulated By:**
 - o Negative feedback loop between Hypothalamus, Anterior Pituitary & Gonads:



PRL – Prolactin:

- **Secreted By:**
 - Lactotrophs in the Anterior Pituitary
- **Primary Action:**
 - Targets Breast Tissue:
 - → Stimulates growth of glandular breast tissue during pregnancy
 - → Stimulates Breast Milk Production after birth
- **Release is Stimulated By:**
 - Prolactin-Releasing Hormone (PRH)
- **Release is Regulated By:**
 - Complex feedback loop involving dopamine, suckling stimuli, and other sex hormones

Feedback Loop: Prolactin (PRL)



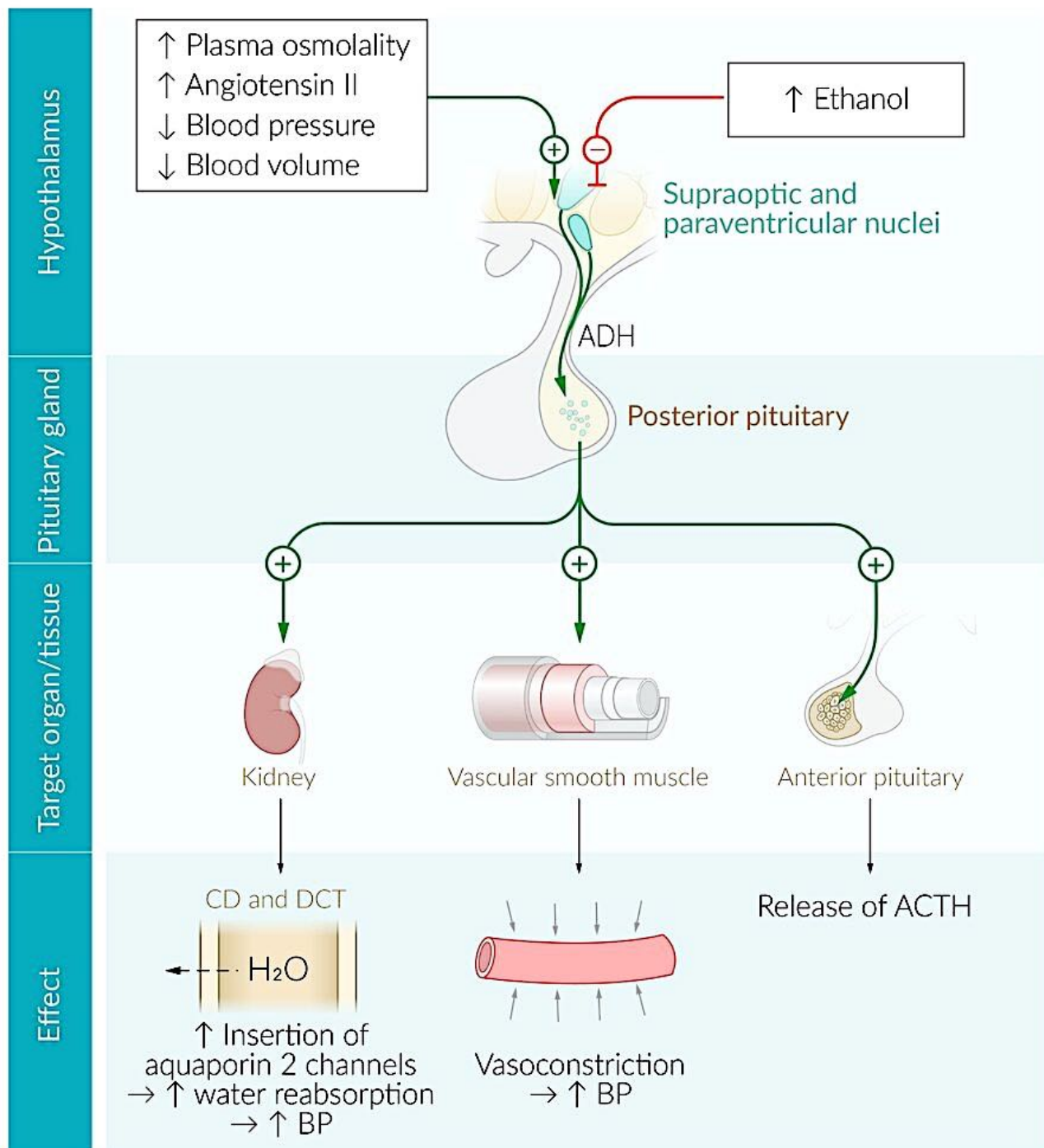
Legend: Pathophysiology Mechanism Sign/Symptom/Lab Finding Complications Published November 19, 2018 on www.thecalgaryguide.com



<https://calgaryguide.ucalgary.ca/feedback-loop-prolactin-prl/>

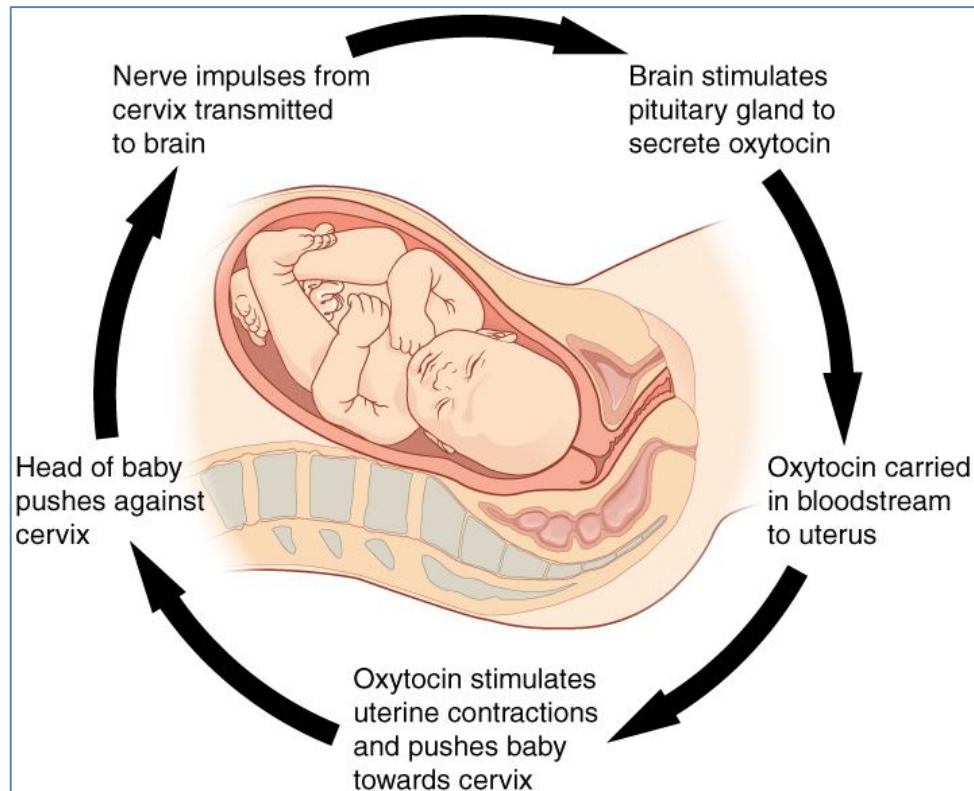
ADH – Anti-Diuretic Hormone (AKA: “Vasopressin”):

- **Secreted By:**
 - Posterior Pituitary
- **Primary Action:**
 - Regulates Blood Pressure & Blood Volume
 - V1 Receptors → Arteriole Constriction → Increased Systemic Vascular Resistance
 - V2 Receptors → Increases water reabsorption in the Renal Collecting Ducts
- **Release is Stimulated By:**
 - Low blood volume
 - Increase of serum osmolality
 - Angiotensin-II
- **Release is Regulated By:**
 - Feedback mechanism between Plasma osmolality & Osmoreceptors in Hypothalamus, Total body water, and the Renin-Angiotensin System



Oxytocin:

- **Secreted By:**
 - Posterior Pituitary
- **Primary Action:**
 - Progression of Labour
 - Let-Down Reflex in Breastfeeding
- **Release is Stimulated By:**
 - Baby Suckling (Breastfeeding)
 - Fetal head pushing against cervix (During Labour)
- **Release is Regulated By:**
 - Positive Feedback Mechanisms:
 - **Birth:**

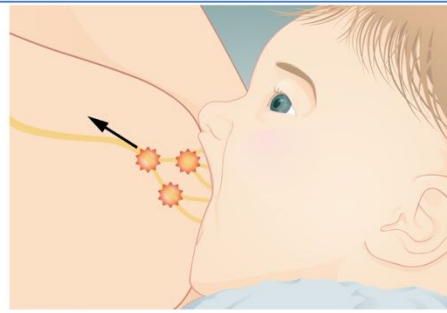


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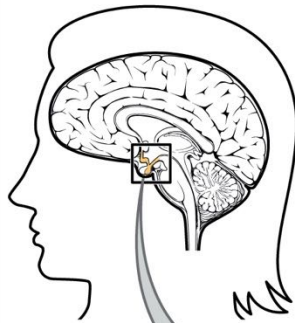
- **Breastfeeding:**

Increased milk production triggers increased suckling by infant (positive feedback loop).

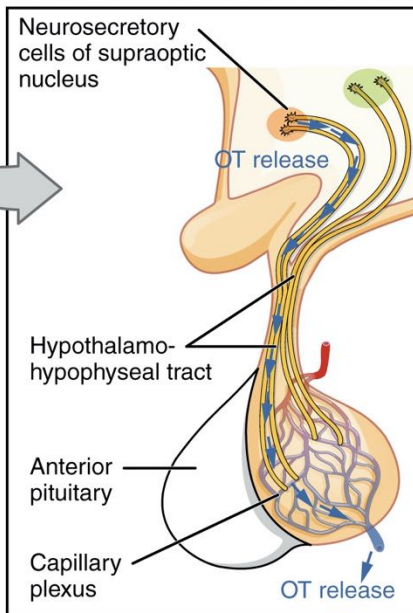
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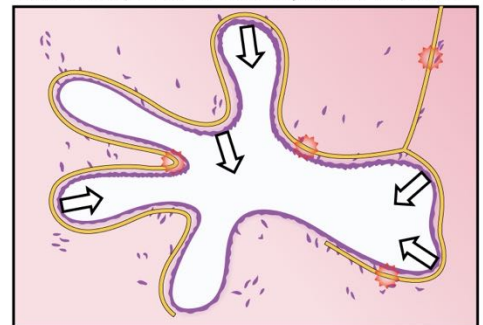
Suckling triggers sensory nerve impulses in the areola.



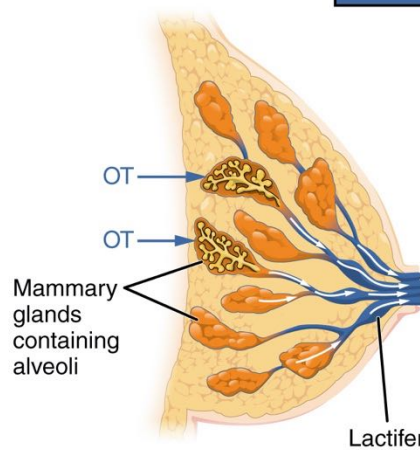
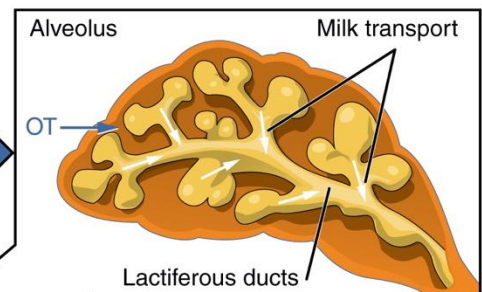
Brain receives sensory impulses from the areola and releases oxytocin (OT) from the hypothalamus and posterior pituitary.



Lactocytes in mammary alveoli produce milk in response to sensory nerve impulses.



Oxytocin (OT) triggers myoepithelial cells to squeeze milk from alveoli so it drains into lactiferous ducts.



Milk is pooled in lactiferous sinus before being discharged through nipple pores.

Lactiferous sinus

THE PINEAL GLAND

THE PINEAL GLAND

Location:

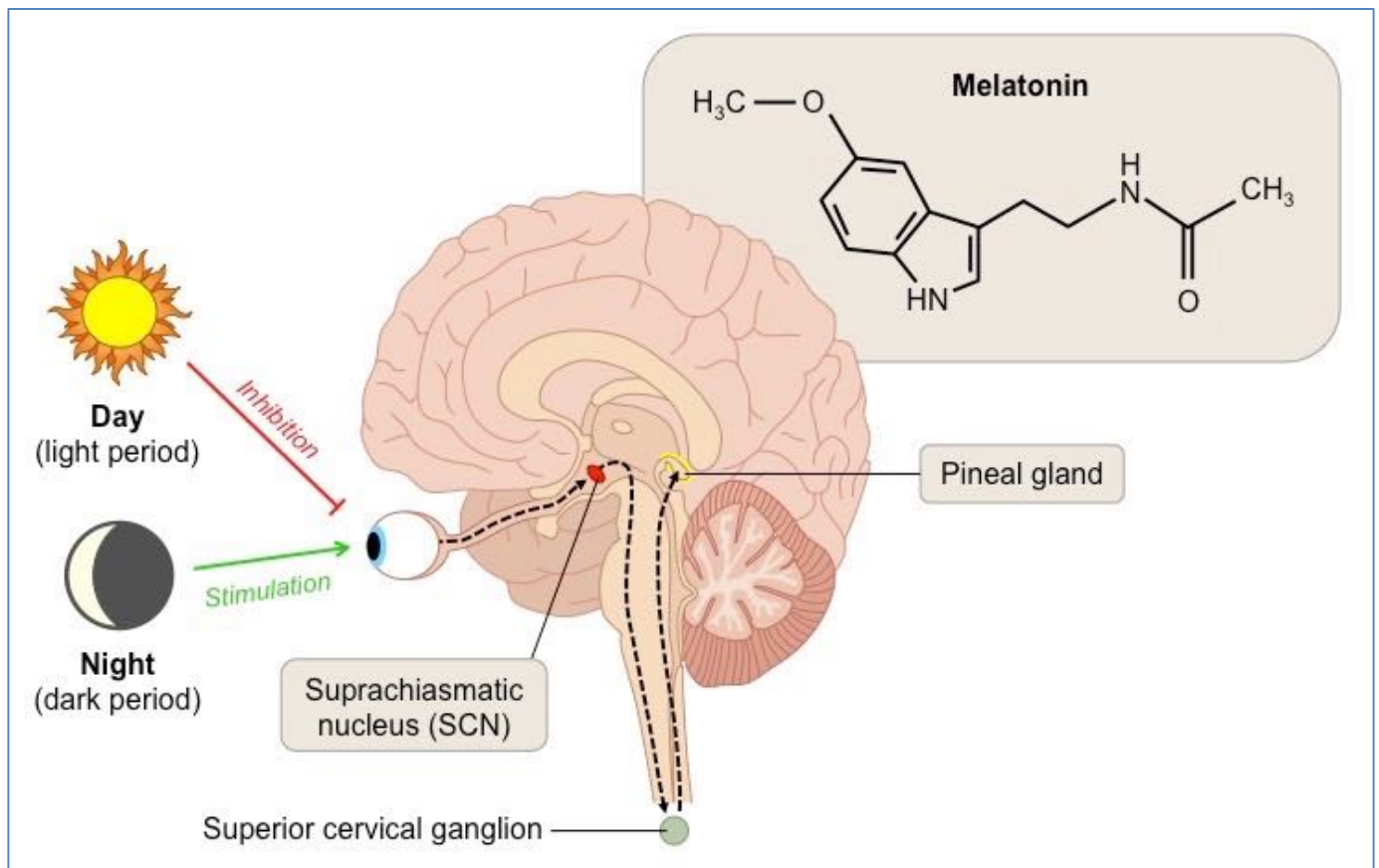
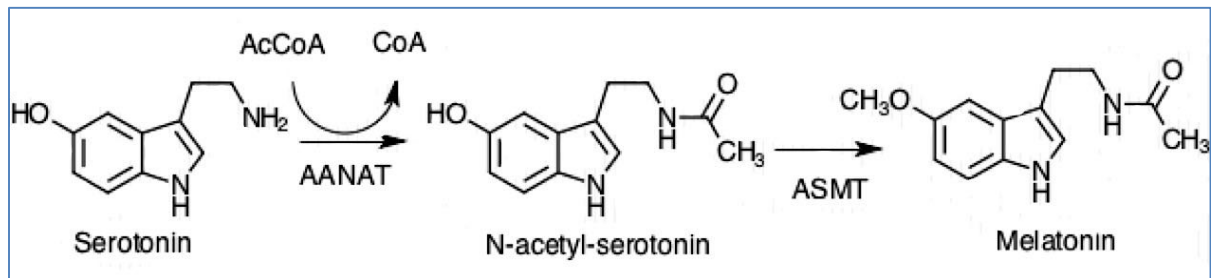
- In the 'Epithalamus'
- Behind the Hypothalamus & Pituitary gland

Function:

- Synthetises Melatonin
- Releases Melatonin at night due to **Decreasing Light Levels**
- Regulates body's Circadian Rhythm (Sleep/wake cycles)

Melatonin:

- A hormone derived from serotonin
- Modulates sleep patterns (both circadian and seasonal)



Source: Unattributable

THE THYROID GLAND

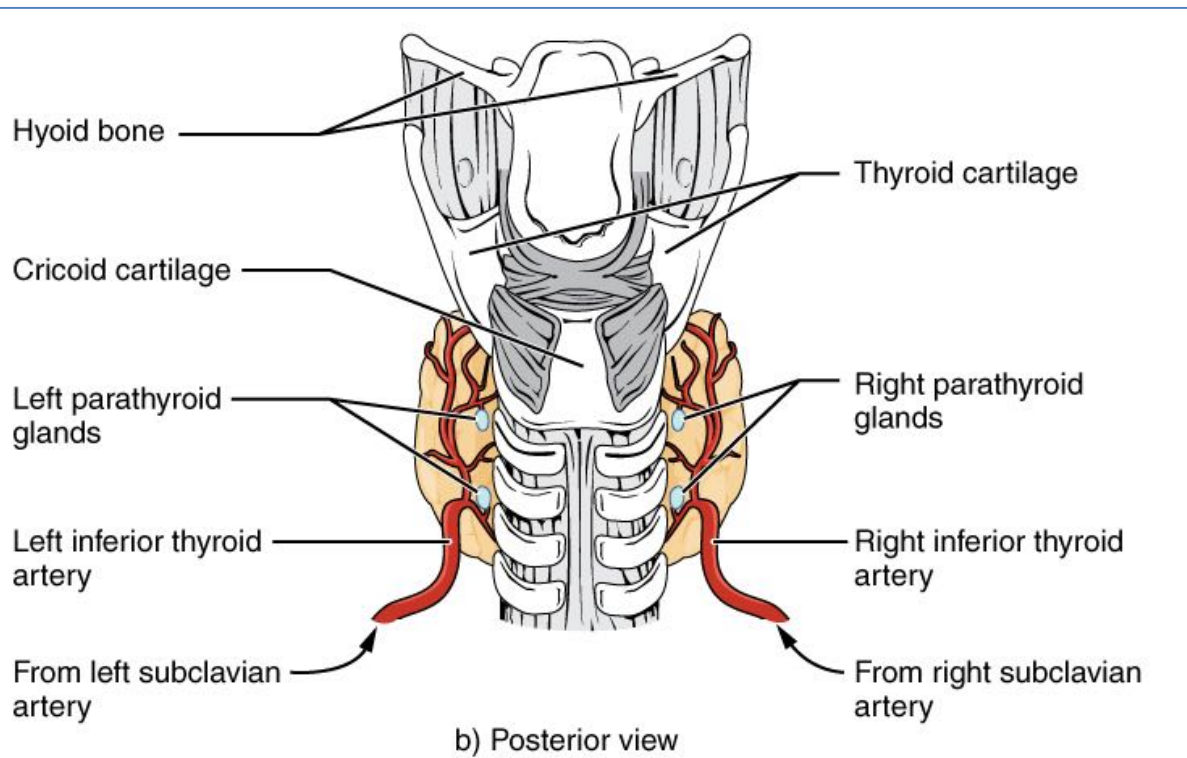
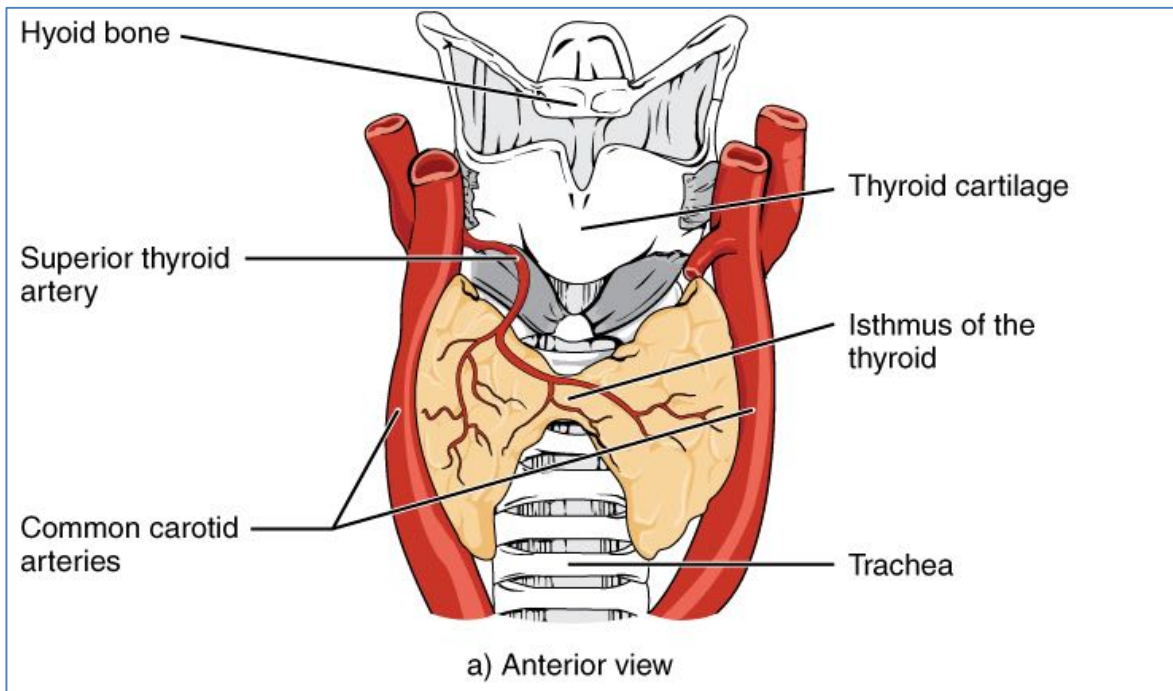
THE THYROID GLAND

General Info:

- There are Thyroid-Hormone Receptors everywhere in the body.
- 75-100µg of Thyroid Hormone is secreted per day

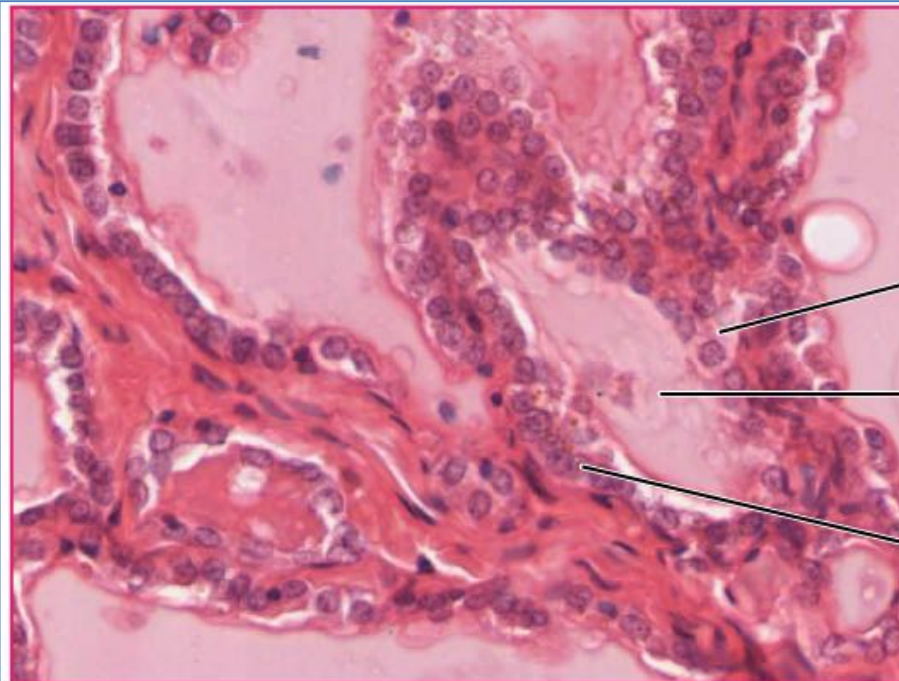
Anatomy of Thyroid Gland:

- A **Bilobar** Gland (2 Lobes – L&R) connected in the middle by the “**Isthmus**” of the Thyroid.
- **Location:**
 - o Immediately below the Larynx on each side of, and anterior to, the Trachea.
- **Rich Blood Supply:**
 - o Required for Building Blocks of Hormones.
 - o Required for Quick Release of Hormones.
 - o Flow - Regulated by the Sympathetic NS.



- **Composed of Millions of "Follicles":**

- Each contains a pool of Thyroid "**Colloid**" of stored Thyroid Hormones bound to **Thyroglobulin**.
 - Allows for a 2-3mth reserve of thyroid hormones.
- Each pool is lined by a layer of "**Principal/Follicle Cells**" that secrete **Thyroid Hormones** (T_3 & T_4).
- There are also patches of "**Parafollicular Cells**" that secrete **Calcitonin**.



Parafollicular cell

Colloid-containing follicle

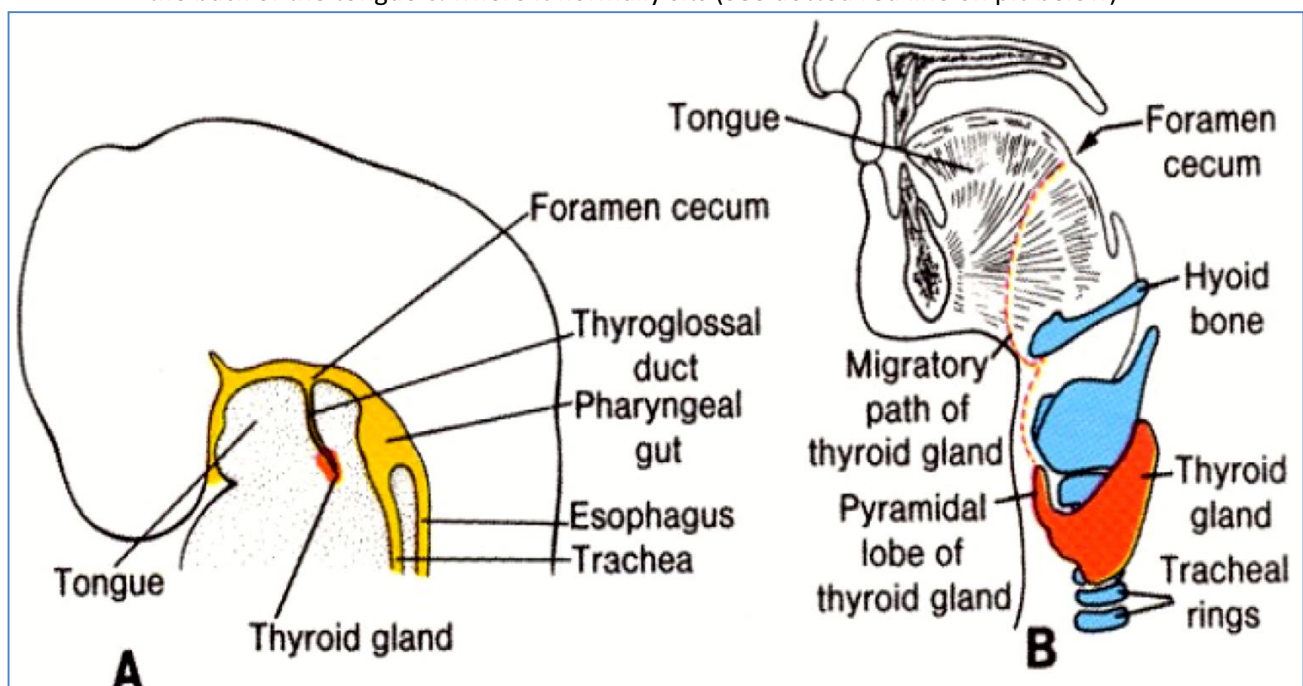
Follicle cells (cuboidal epithelium)

c) Thyroid follicle cells

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Thyroid Embryology:

- Forms from **Pharyngeal Pouches** (@4-5wks)
- Forms from the **Endoderm** germ layer.
- Once formed, it migrates downwards & becomes Bi-lobed.
 - Note: Sometimes things go wrong during this migration, leaving a person's thyroid gland between the back of the tongue & where it normally sits (See dotted red line on pic below)



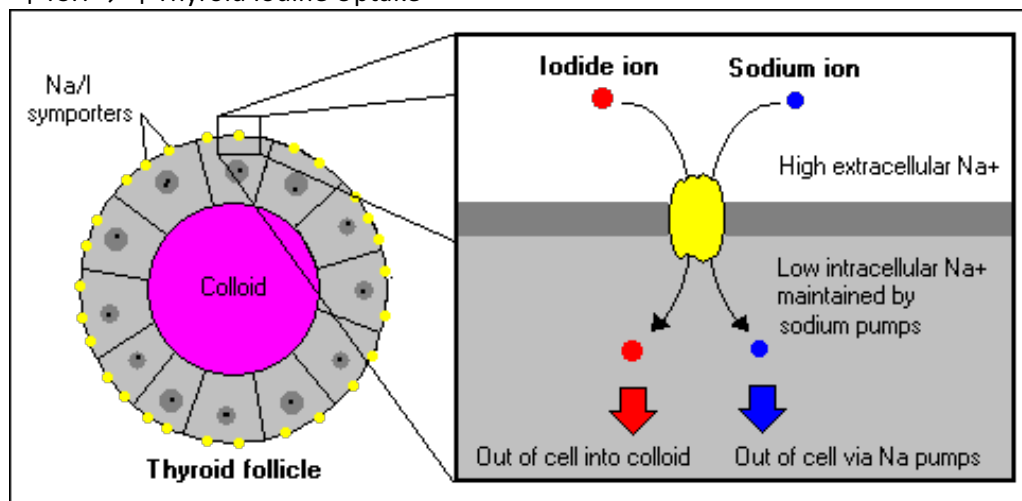
Source: Unattributable

Major Thyroid Hormones Produced (And Proportions):

- **T₃ – Triiodothyronine (7%)** (Iodinated Derivative of Tyrosine – 3x Iodines) (Most Biologically Active)
- **T₄ – Thyroxine (93%)** (Iodinated Derivative of Tyrosine – 4x Iodines) (Less Biologically Active)
 - o (Since these hormones are stored in the 'Colloid', there is ≈2-3mths of 'backup' Reserve)
- **Calcitonin** (Polypeptide)
 - o **Secreted By** – The Parafoollicular Cells of the Thyroid Gland
 - o **Function:** ↓ **Plasma-Ca⁺ levels** (By ↓Osteoclast Activity & ↑Osteoblast Activity)
 - o **Stimulated By:** ↑Extracellular [Ca⁺] (Note: Opposite of PTH)

Iodine Balance:

- **Iodine is an essential component of the 2 major Thyroid Hormones & Is a Dietary Requirement.**
- **Of the Iodine ingested;**
 - o 20% is Selectively Removed from blood by Thyroid Gland & used in Thyroid Hormone Synthesis.
 - o 80% is Excreted by the Kidneys
- Iodine is **Actively** taken up by Thyroid Gland via **"Iodine Trapping"**.
- The **Rate** of Iodine Uptake (Trapping) depends on **TSH Concentration**.
 - o ↑TSH → ↑Thyroid Iodine Uptake



Source: Unattributable

How TSH Stimulates Thyroid Hormone Synthesis/Secretion:

- **Binding of TSH to Follicle Cell → Activates AdenylylCyclase → ↑cAMP → Activates pKa (Protein Kinase A) → Phosphorylates Various Enzymes in Follicle Cell → Changes Activity of:**
 1. ↑Cleavage of Thyroglobulin in Lysosomes (Ie: ↑Release of T₃ & T₄)
 2. ↑Activity of Iodine Pump (The Rate Limiting Step) → ↑Iodine Available for Synthesis
 3. ↑Iodination of Tyrosine → ↑Synthesis of DIT's & MIT's → ↑Thyroid Hormone Synthesis
 4. ↑Size & Secretory Activity of Follicle Cells
 5. ↑# of Follicle Cells
- **Effects of TSH on the Thyroid:**
 - o Thyroid Follicle Hyperplasia
 - o ↑Iodine Uptake from the Blood (*Iodine Trapping*)
 - o ↑Thyroid Hormone Synthesis
 - o ↑Release of T₃ & T₄

Synthesis of Thyroid Hormone: (Stimulated by TSH)

1. **Active Uptake of Iodide by the Principal/Follicle Cells (Iodide "Trapping"):**
 - o Active Iodide uptake against massive Electrochemical Gradient.
 - o **Note: This is the Rate-Limiting Step of TH Synthesis.**
2. **Iodide Oxidation (by Peroxidase):**
 - o Oxidation of Iodide Ions (I⁻) → Iodine Molecules (I₂).
3. **Secretion of Active Iodine into Lumen of Colloid:**
4. **Synthesis of Thyroglobulin by Rough-ER+Golgi & Secretion into Lumen of Colloid:**
 - o Tyrosines – the basis of Thyroglobulin (A large poly-peptide of ≈70 Tyrosines)
5. **Iodines stick to the Tyrosines on the Thyroglobulin in Colloid → DIT or MIT (Di/Mono-Iodo Tyrosine)**

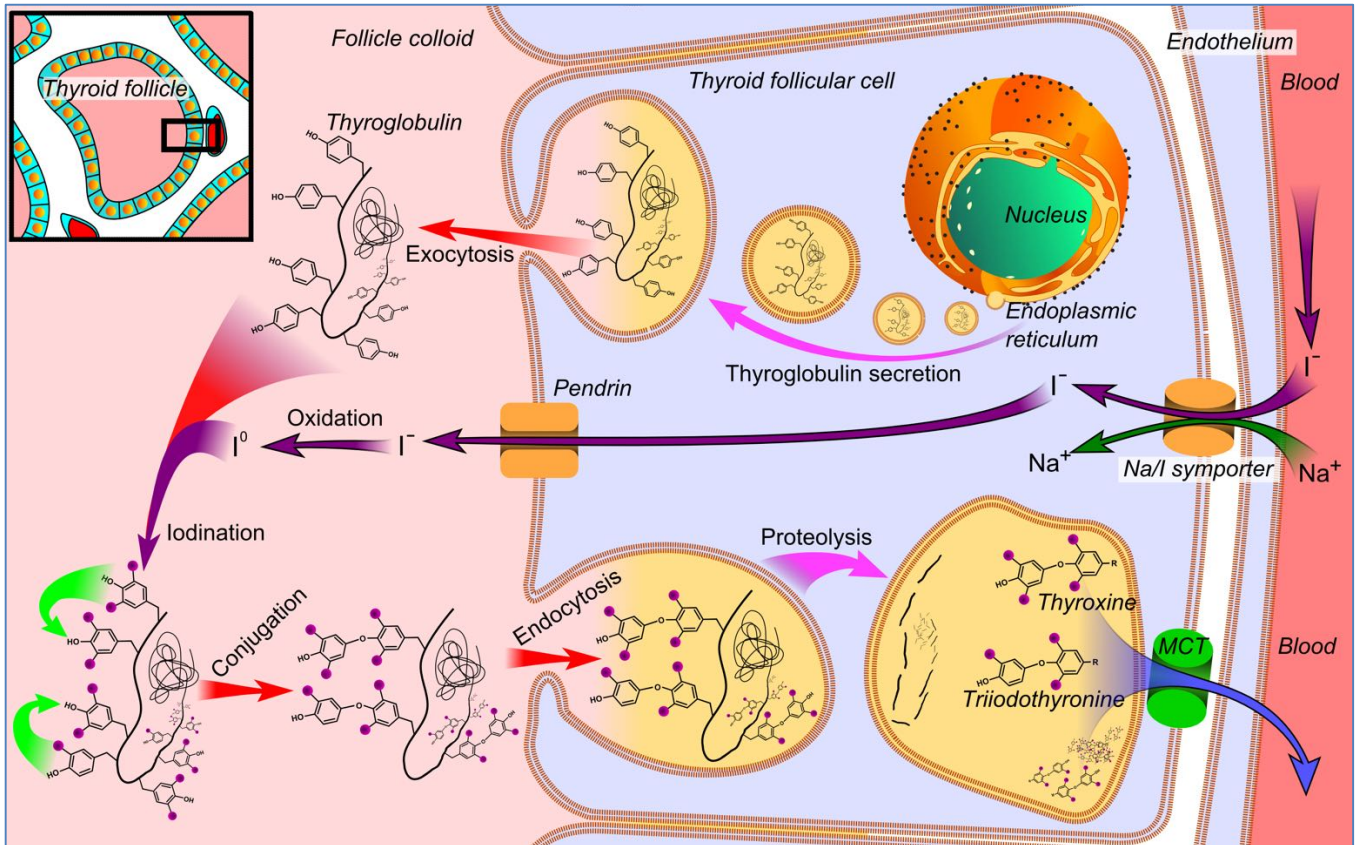
Hormone Release Mechanism:

6. Some of the Thyroglobulin Colloid is Endocytosed + Combined with Lysosome:

- Lysosomal Enzymes cleave the T_3 & T_4 from the Thyroglobulin.
- Note: In the process, many of the unpaired DIT's/MIT's are also released. These are De-Iodinated by **Deiodinase**. Both the freed Iodines & Tyrosines are recycled.

7. Vesicle of Cleaved T_3 & T_4 Breaks Down, Releasing Hormones into Cytosol

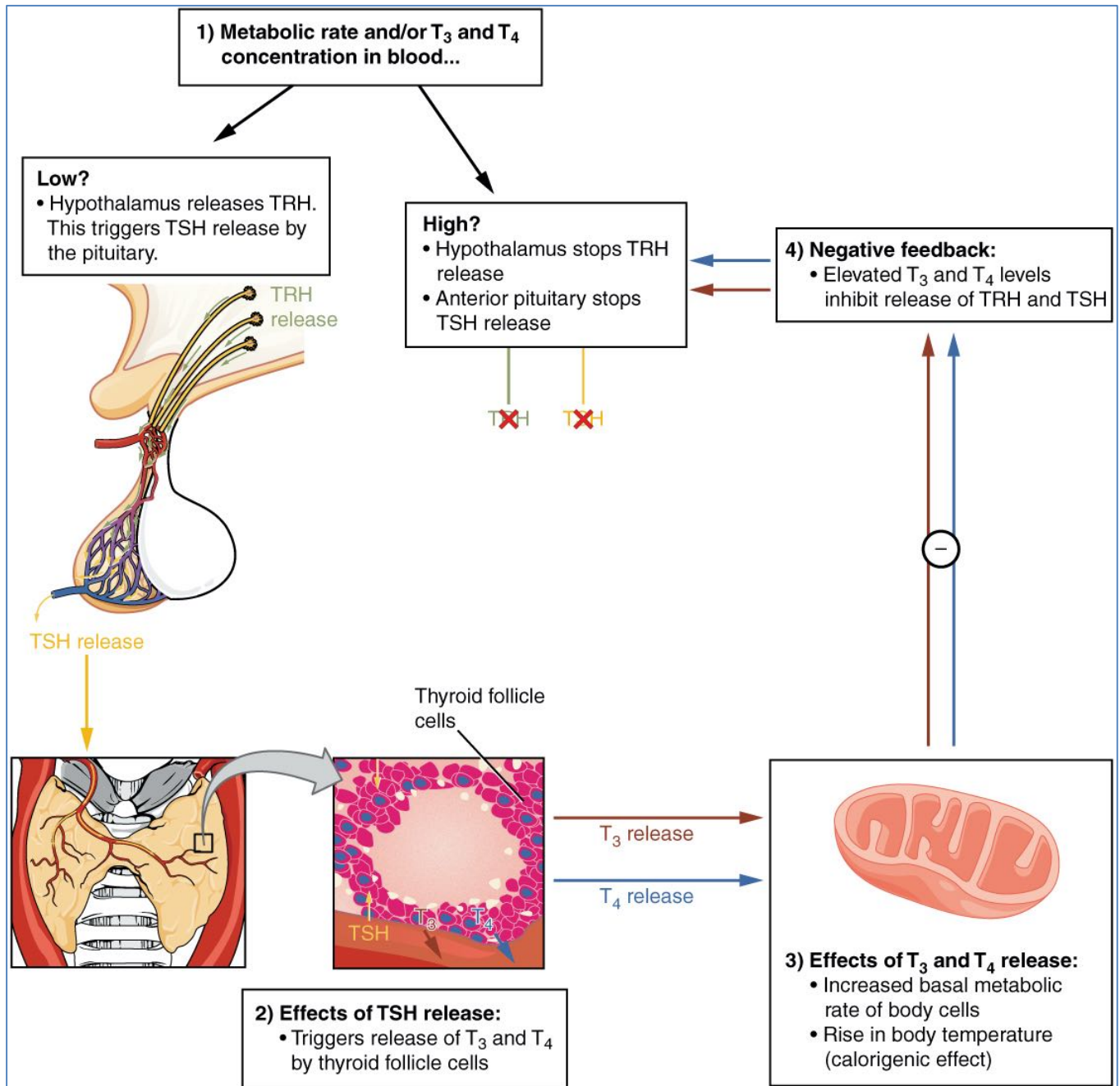
8. Hormones in Cytosol Diffuse through Basement Membrane → Combine with Binding Proteins in the Blood Stream (Thyroxine-Binding Protein/Albumin)



Häggström, Mikael (2014). "Medical gallery of Mikael Häggström 2014". WikiJournal of Medicine 1 (2). Public Domain. CC0, via Wikimedia Commons

Regulation of Thyroid Hormone Production/Release:

1. Hypothalamus Secretes "**Thyrotropin-Releasing Hormone (TRH)**" into portal circulation of Pituitary.
2. TRH Stimulates Anterior Pituitary to Secrete "**Thyroid Stimulating Hormone (TSH)**".
3. TSH Stimulates Thyroid Gland to Secrete:
 - *Primarily Thyroxine (T_4) (The relatively inactive Thyroid Hormone → converted to T_3)
 - And Some Triiodothyronine (T_3) (The most active Thyroid Hormone)
4. T_3 & T_4 Circulate in the Bloodstream Eliciting their effects + Provide Neg:Feedback to Ant. Pituitary



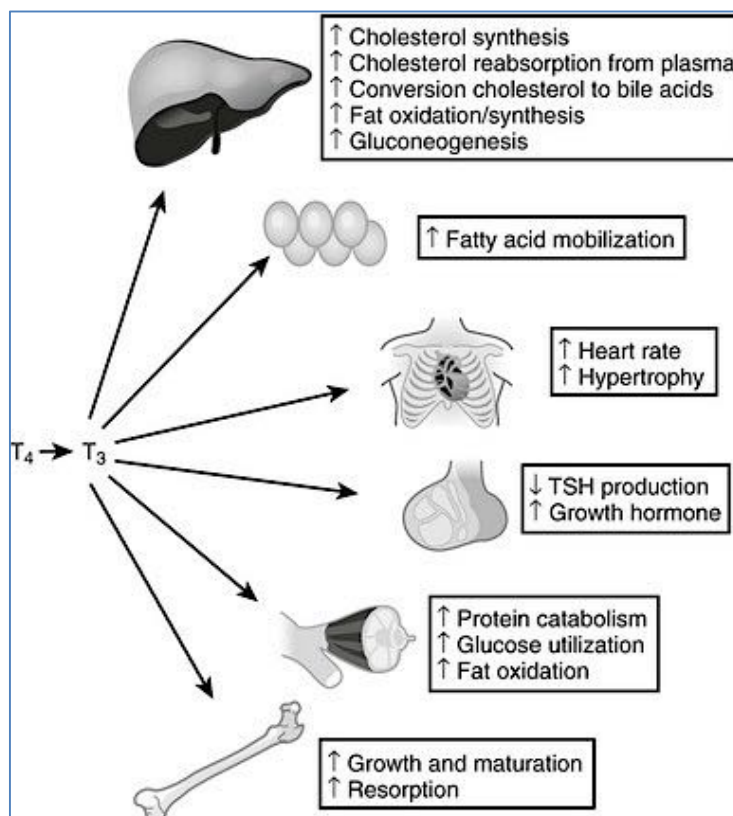
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Transport of Thyroid Hormones (Binding Proteins):

- Note: 75-100 μ g of Thyroid Hormone is secreted per day
- Thyroid hormone must be bound to carrier proteins when in bloodstream to avoid filtration by kidneys.
- **Common Thyroid-Hormone Carrier Proteins:**
 - 70% - "**Thyroxine-Binding Globulin (TBG)**"
 - 30% - Albumin
 - Note: The minute %age of unbound Thyroid Hormones are those eliciting their effects
 - I.e: TH must be unbound to be able to enter cells & bind to Intracellular Receptors.

Mechanism of Action of Thyroid Hormone:

1. Thyroxine (T_4) reaches target cell
 2. Binding Protein releases Thyroxine (T_4)
 3. Thyroxine (T_4) diffuses into cytosol → Converts to T_3
 4. T_3 (The most active form) enters Nucleus → Binds to Nuclear Receptor on DNA → Alters Gene Transcription.
 5. Activating Different Genes → leads to Change in Cell's Protein/Enzyme profile → Change in Activity.
- **Cellular Changes:**
 - ↑Carbohydrate/Fat Metabolism
 - ↑Glucose Uptake
 - ↑Protein Synthesis + Catabolism
 - ↑Mitochondrial Activity & Number
 - ↑Na/K-ATPase Activity
 - **Bodily Changes:**
 - **CVS:** ↑ O_2 Consumption → ↑Cardiac Output, HR & Respiration
 - **GIT:**
 - ↑Food Intake (↑Glucose Absorption from GIT)
 - ↑Secretion of Digestive Juices
 - ↑GIT Motility
 - **Metabolism:**
 - ↑Metabolism (Basal Metabolic Rate)
 - ↑Insulin Secretion
 - ↑Lipolysis → Higher [FFA] in Plasma
 - ↑Body Temp → Sweating
 - ↑Vitamin Requirements due to ↑Quantities of Enzymes (Of which vitamins are a vital component)
 - **Bone:** ↑Bone Turnover & ↑Resorption
 - **Muscle:** ↑Speed of Muscle Contraction & ↑Speed of Relaxation
 - **Sympathetic NS:** ↑Catecholamine Sensitivity of Heart, Muscle, Fat & Lymphocytes
 - **Note:** Because Thyroid Hormones act by Gene Activation, they are said to have a **Long 'Latent Period'**, during which they seem to have no discernible effect.
 - **Thyroxine:** 2-3 Days
 - **Triiodothyronine:** 6-12 Hours



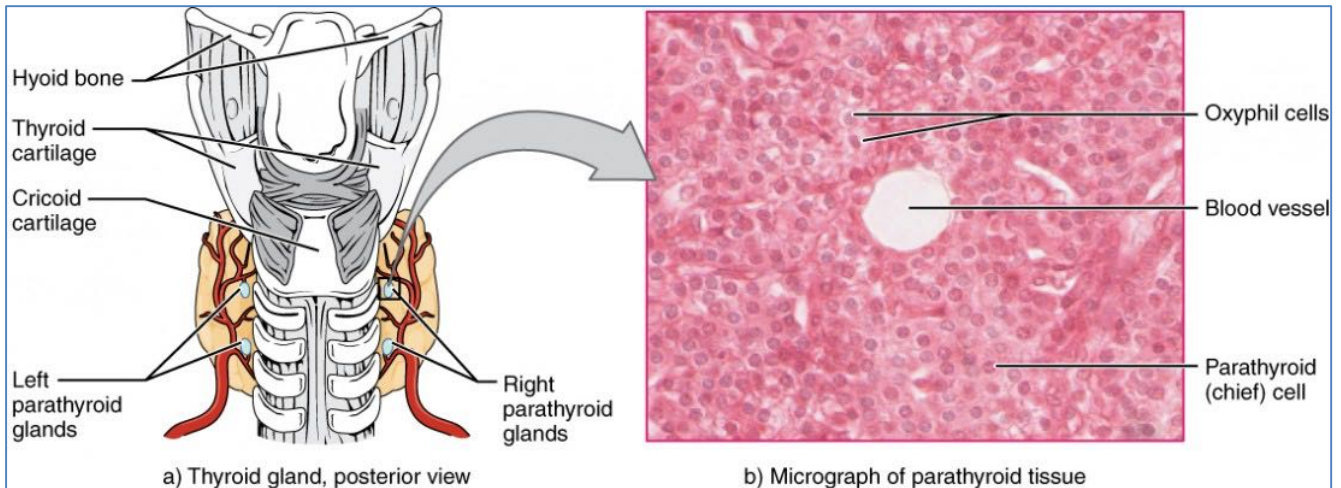
Source: Unattributable

PARATHYROID GLANDS

PARATHYROID GLANDS

Parathyroid Anatomy:

- **Macro Structures:**
 - 4x small Endocrine Glands on the Posterior Surface of the Thyroid Gland
 - 2x on Left; 2x on Right
 - Size of a grain of rice.
- **Micro Structures:**
 - Densely packed cells (As opposed to follicle structure of Thyroid Gland)
 - **2 Cell Types:**
 - **Parathyroid Chief Cells:**
 - Secrete Parathyroid Hormone (PTH)
 - **Oxyphil Cells:**
 - Unknown function.



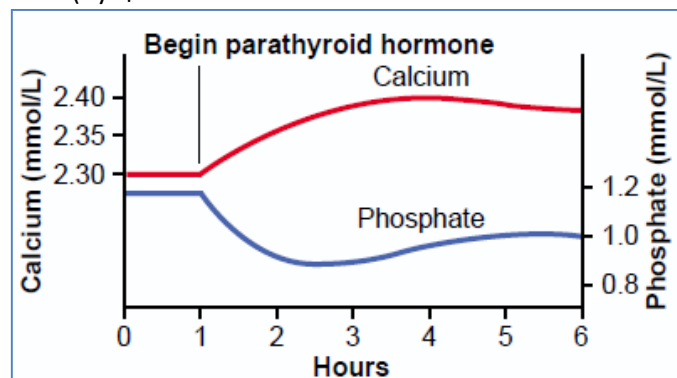
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Parathyroid Physiology:

- **Secretes Parathyroid Hormone → Important role in Calcium Homeostasis in Blood & Bones**
 - (Important for Excitable Tissues)
 - (Important for Bone Integrity)
- **(Also has effects on Phosphate)**
- **Note: Parathyroid Gland is NOT under Hypothalamic Control!!! – Functions Autonomously**

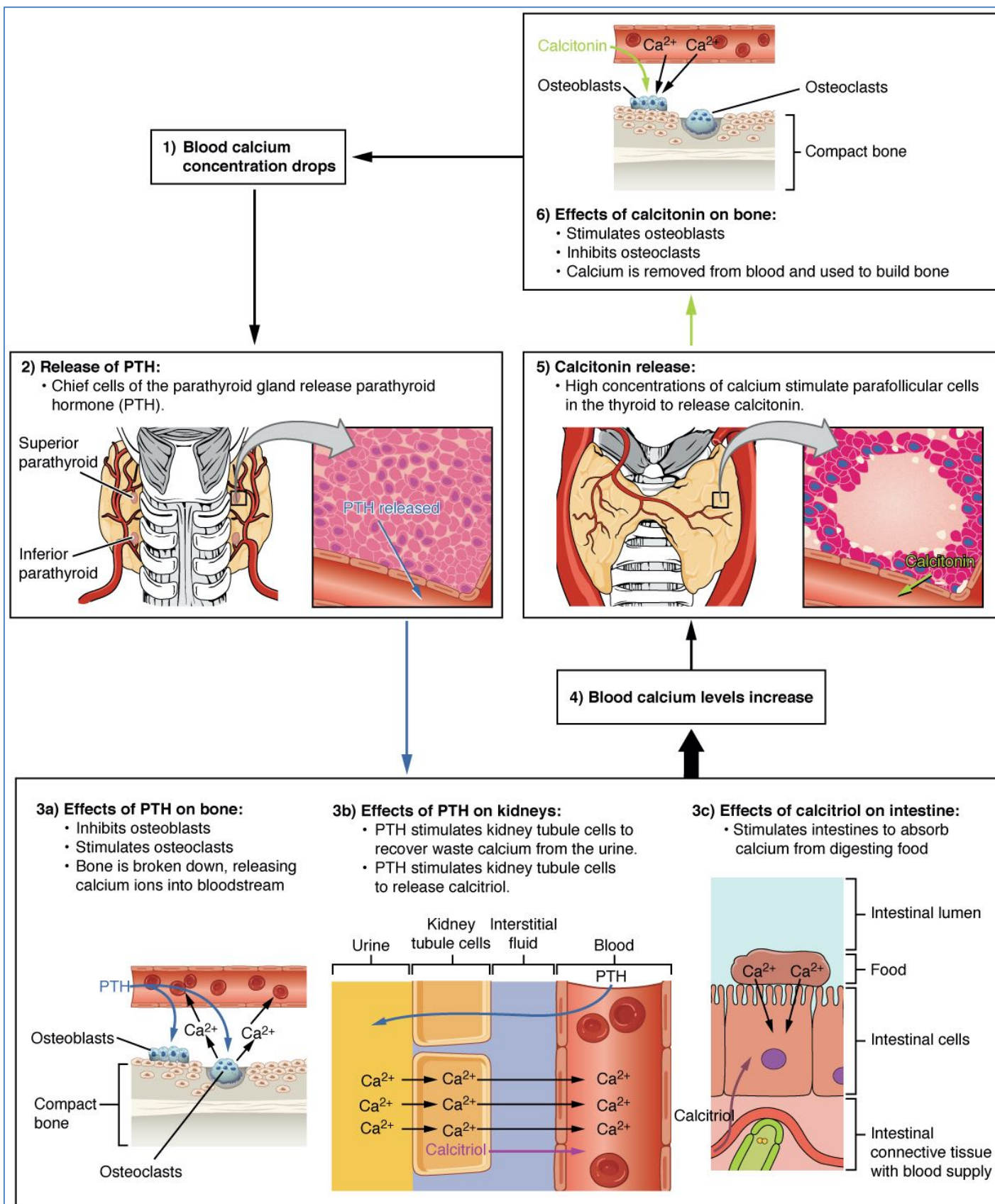
Parathyroid Hormone (PTH):

- **Secreted by** – The Chief Cells of the Parathyroid Glands
- **Release Stimulated By:**
 - ↓ Extracellular $[Ca^{+}]$ – Very Sensitive
- **Release Inhibited By:**
 - ↑ Extracellular $[Ca^{+}]$ – Very Sensitive
- **Aims to:**
 - ↑ **Plasma- Ca^{+} levels** (By Increasing Bone Ca^{+}/P^{-} Resorption & ↓ Renal Ca^{+} Excretion)
 - ↓ **Plasma- P^{-} levels** (By ↑ Renal P^{-} Excretion so that it exceeds Bone P^{-} Resorption)



Primary Effects:

- Stimulates Osteoclasts → Mobilises Ca from Bone Matrix → ↑Calcium in Blood
- Activates Vit-D in Kidneys → ↑GI Absorption of Ca^{2+} → ↑Calcium in Blood
- ↑Renal Calcium Reabsorption → ↓Renal Excretion → ↑Calcium in Blood
- (Increases Renal Excretion of Phosphate → ↓Phosphate in Blood)

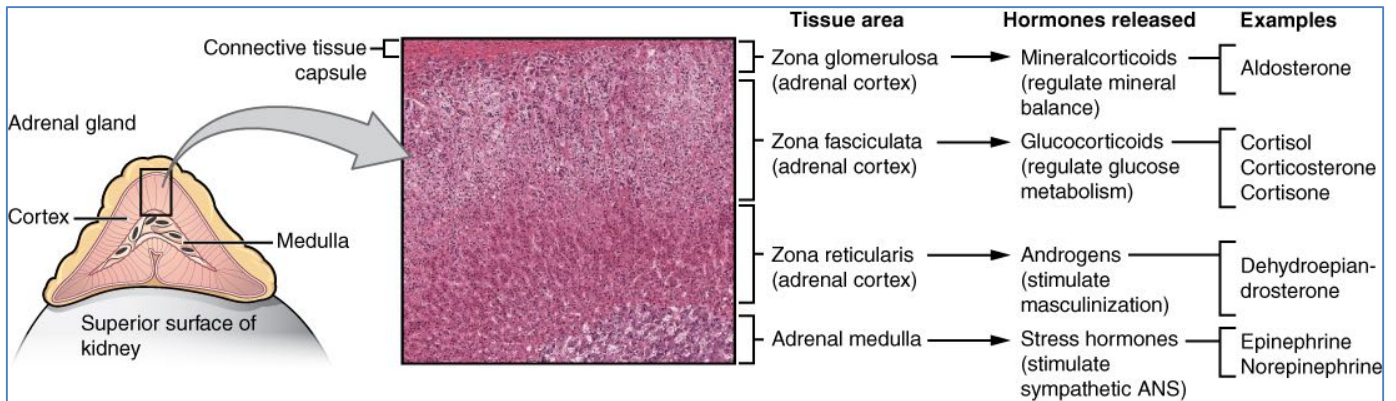


THE ADRENAL GLANDS

THE ADRENAL GLANDS

Anatomy of the Adrenal Glands:

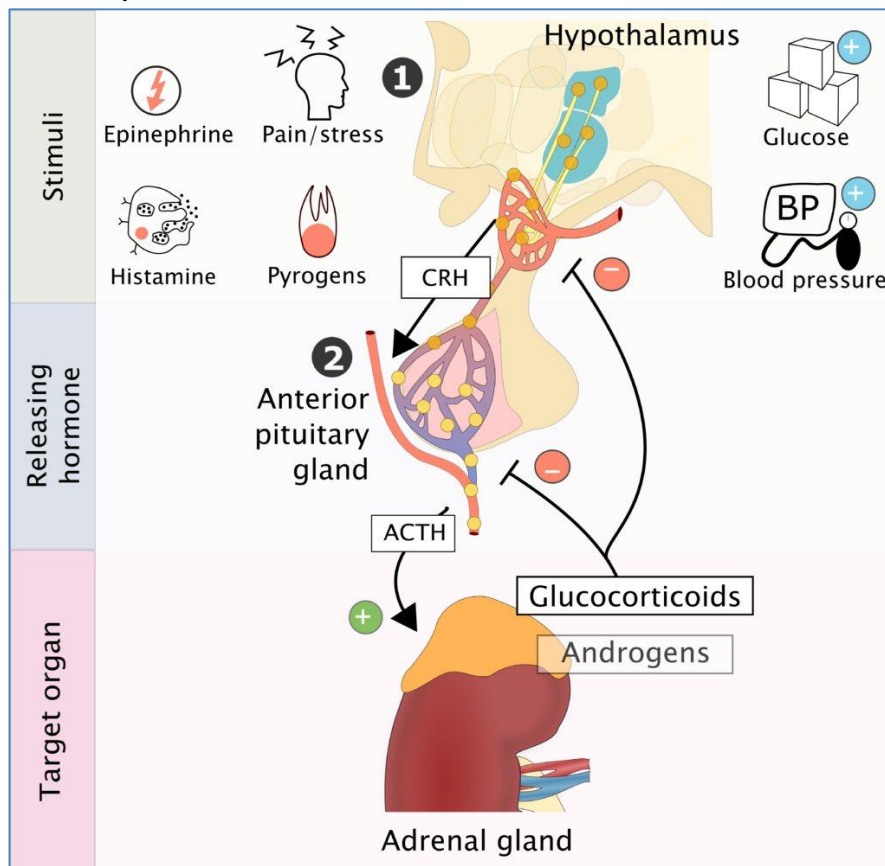
- Endocrine Glands that sit on top of the Kidneys
- Retroperitoneal
- Two Layers:
 - 1- Cortex (3 Zones) **(Remember GFR)**
 - 1- Zona **G**lomerulosa (Outer) → Mineralocorticoids (**Aldosterone**)
 - 2- Zona **F**asciculata → Glucocorticoids (**Cortisol**)
 - 3- Zona **R**eticularis (Inner) → Adrenal **A**ndrogens (DHEA)
 - 2- Medulla (Middle)
 - Chromaffin Cells → Catecholamines (**Adrenaline, Noradrenaline**)



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Regulation of the Adrenal Glands:

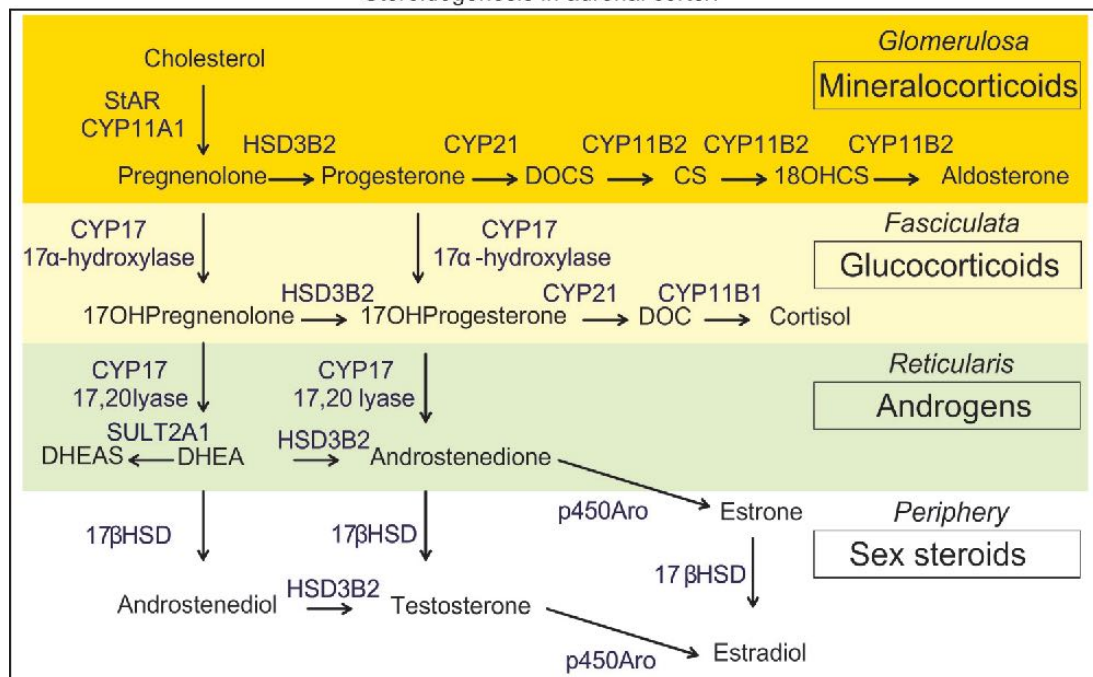
- Adrenal Cortex Secretions are controlled by ACTH (Adreno-Corticotrophic Hormone) by Anterior Pituitary
- Anterior Pituitary releases ACTH in response to CRH (Corticotrophin Releasing Factor/Hormone)
- **Negative feedback loop:**



Charmandari E, Nicolaides NC, Chrousos GP. Adrenal insufficiency. *Lancet*. 2014; 383 (9935): p.2152-2167. doi: [10.1016/s0140-6736\(13\)61684-0](https://doi.org/10.1016/s0140-6736(13)61684-0) . | [Open in Read by QxMD](#)

Adrenocortical Hormones:

- **All Adrenocortical Hormones are Steroidal** (Derived from cholesterol)
 - o Adrenal cortex cells convert Cholesterol into → '**Prognenolone**'
 - Prognenolone → Zona Glomerulosa → Aldosterone
 - Prognenolone → Zona Fasciculata → Cortisol
 - Prognenolone → Zona Reticularis → Testosterone & Oestrogen

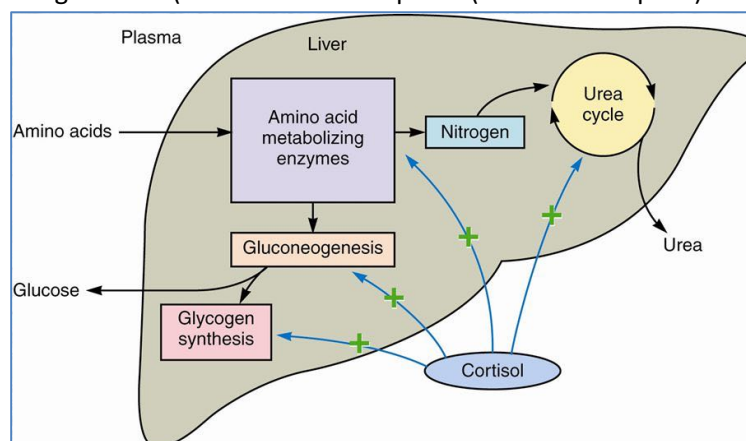


Lefèvre, Lucile et al. "Adrenocortical growth and cancer." *Comprehensive Physiology* 5 1 (2015): 293-326 .

- **Stress Hormones:**
 - o **Corticosteroids (Cortisol)** → ↑ Blood Glucose, Immunosuppression, ↓ Bone Formation.
 - o **Catecholamines (Adrenalines)** → ↑ Blood Glucose, ↑ HR & BP, ↓ Parasympathetics.
- **Electrolyte Balance:**
 - o **Aldosterone (A Mineralocorticoid)** → ↑ Na⁺ Retention, ↑ K⁺ Excretion, ↑ BP

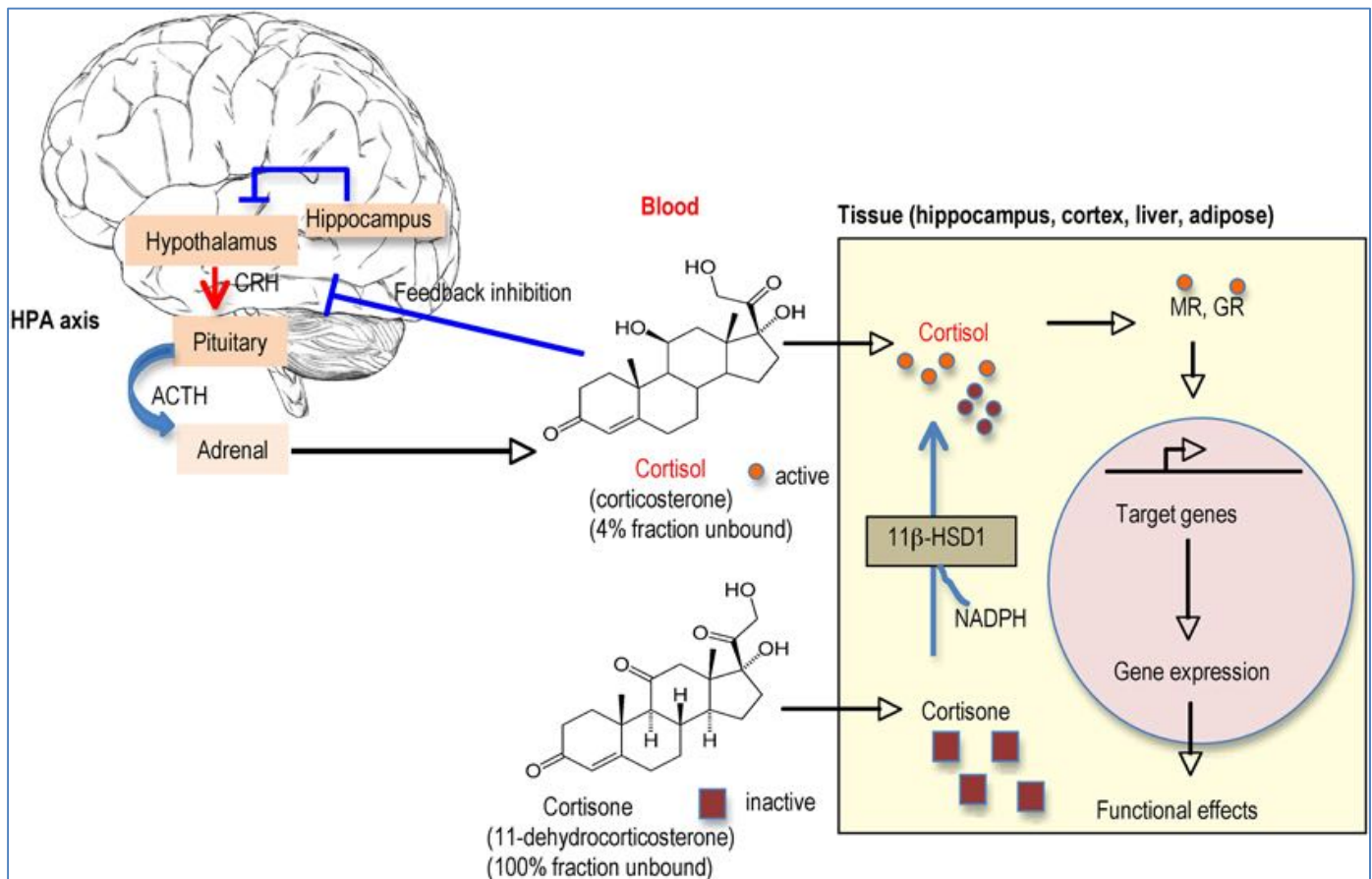
Cortisol:

- **Secreted By:**
 - o Cells of the Zona Fasciculata of the Adrenal Cortex
 - o Derived from Cholesterol
- **Metabolic Actions:**
 - o Increases Blood [Glucose] ; Stimulates Gluconeogenesis (Liver) → Glucose Output by Liver
 - o Proteolysis (Muscle) } ↑ Availability of Gluconeogenic Substrates in Blood (Glycerol + AA's)
 - o Lipolysis (Adipose) }
 - o Stimulates the Urea Cycle (Required to handle Wastes from Amino-Acid metabolism → Glucose)
 - o Glucose-Sparing Effect – (Inhibits Glucose Uptake (Muscle & Adipose))

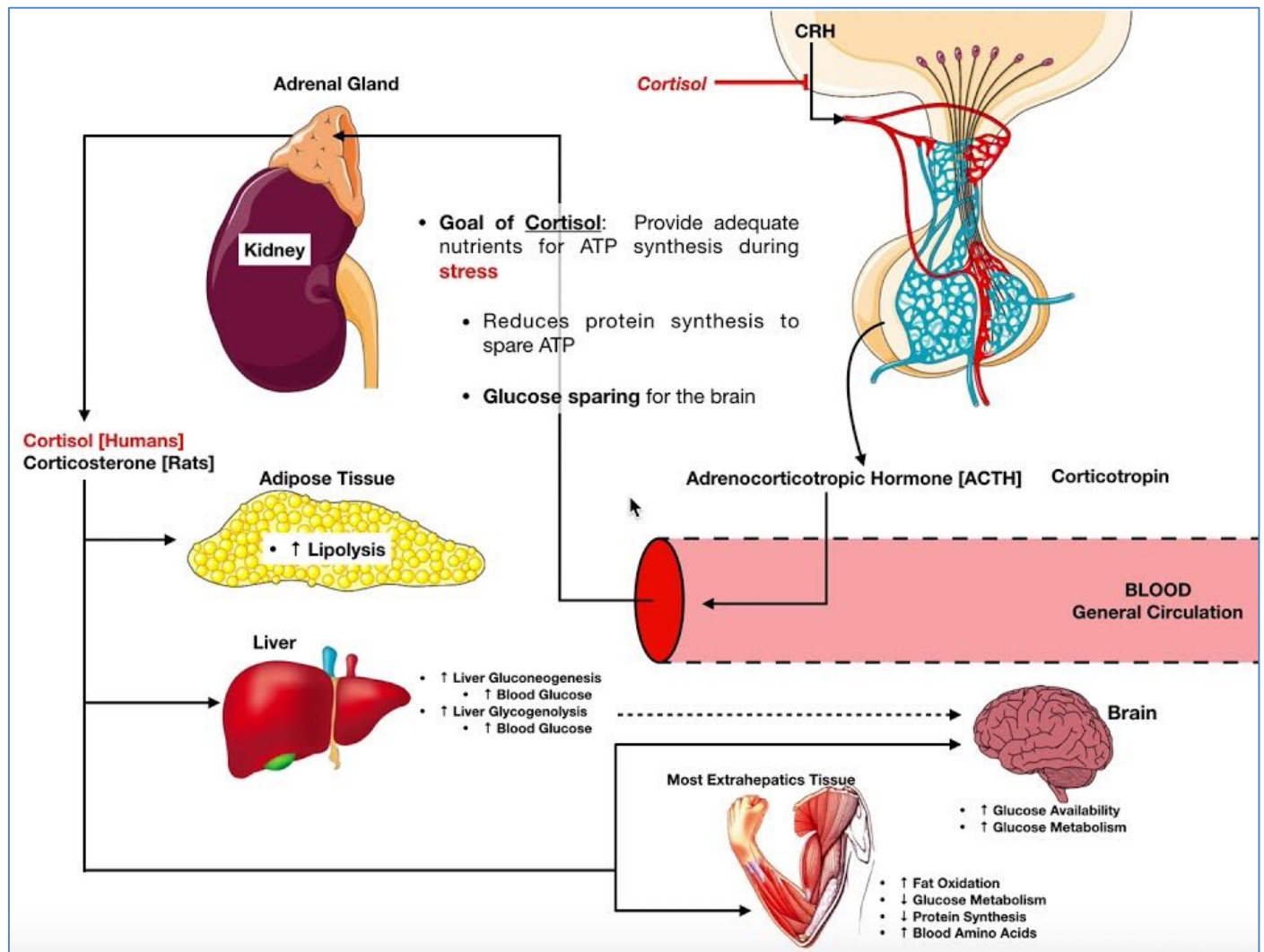


Source: Unattributable

- **Immune Actions:**
 - Reduces intensity of immune/inflammatory responses.
 - By reducing the production of Inflammatory mediators
 - Eg: Prostaglandins, Thromboxanes, Leukotrienes
 - Eg: Interleukins, Interferon, TNF
 - By reducing T-Cell Proliferation
 - By reducing Neutrophil Phagocytosis
- **Other Actions:**
 - Assists in maintaining blood pressure
 - Reduces bone formation
 - Increases renal filtration
 - Increases EPO (Erythropoietin) release
- **Release is Stimulated By Hypothalamus in Response to:**
 - Stress (Infection, Trauma, Sympathetic overdrive, Psychological stressors)
 - Physical Activity
 - Low Blood Glucose
- **Release is Regulated By:**
 - Hypothalamus Secretes Corticotrophin-Releasing Hormone (CRH) → ACTH by Anterior Pituitary
 - Negative Feedback Mechanism between Hypothalamus, Pituitary & Adrenal gland



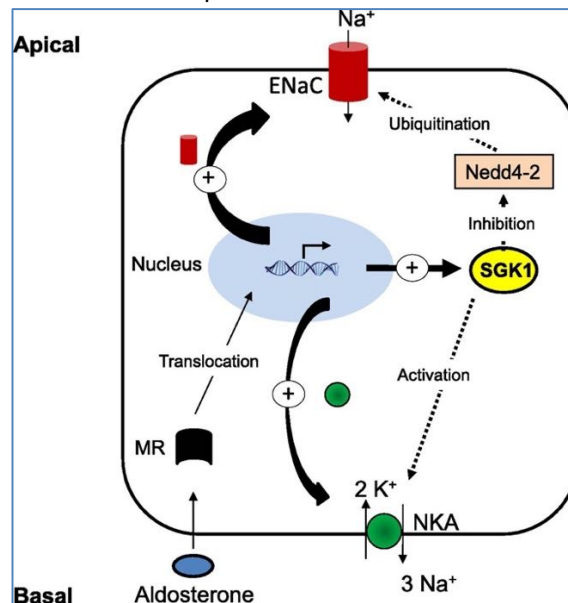
Yau JLW and Seckl JR, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0/>>, via Wikimedia Commons



<https://www.passeidireto.com/arquivo/93740033/cortisol>

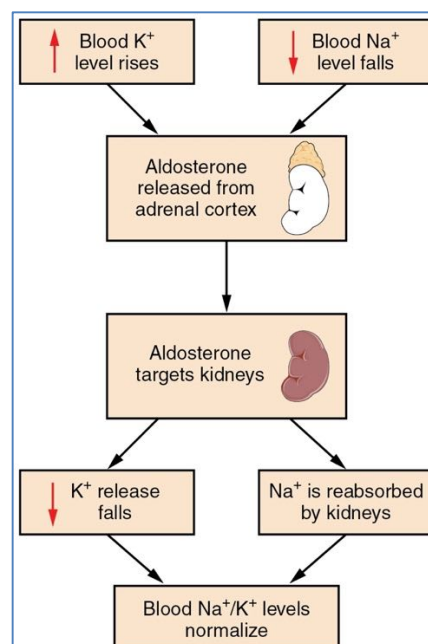
Aldosterone:

- **Secreted By:**
 - Zona Glomerulosa Cells of the Adrenal Glands
- **Primary Actions:**
 - **Sodium Homeostasis** → Causes Na Reabsorption in the Renal Distal Tubules & Collecting Ducts
 - **Regulates extracellular fluid volume** (Via increasing blood Na concentration & renal absorption)
 - **Potassium Homeostasis** → Increases K Secretion in the Renal Distal Tubules & Collecting Ducts
- **Works by:**
 - **ACTIVATING the Na/K-ATPases in the Principal Cells of Distal & Collecting Ducts:**
 - Increases Na^+ & Cl^- Reabsorption
 - Increases K^+ Secretion
 - **Promotes Na^+ -Channel (ENaC) Synthesis & Insertion into Luminal Membrane:**
 - Facilitates the Na^+ Reabsorption mentioned above.



<https://jasn.asnjournals.org/content/27/9/2554/tab-figures-data>

- **Release is Stimulated By Hypothalamus in Response to:**
 - *Angiotensin-II, Part of the Renin-Angiotensin System (Due to Renin Release by Kidneys)
 - *Hyponatraemia (Low Na^+ in Blood)
 - *Hyperkalaemia (High K^+ in Blood)
 - Stress
- **Release is Regulated By:**

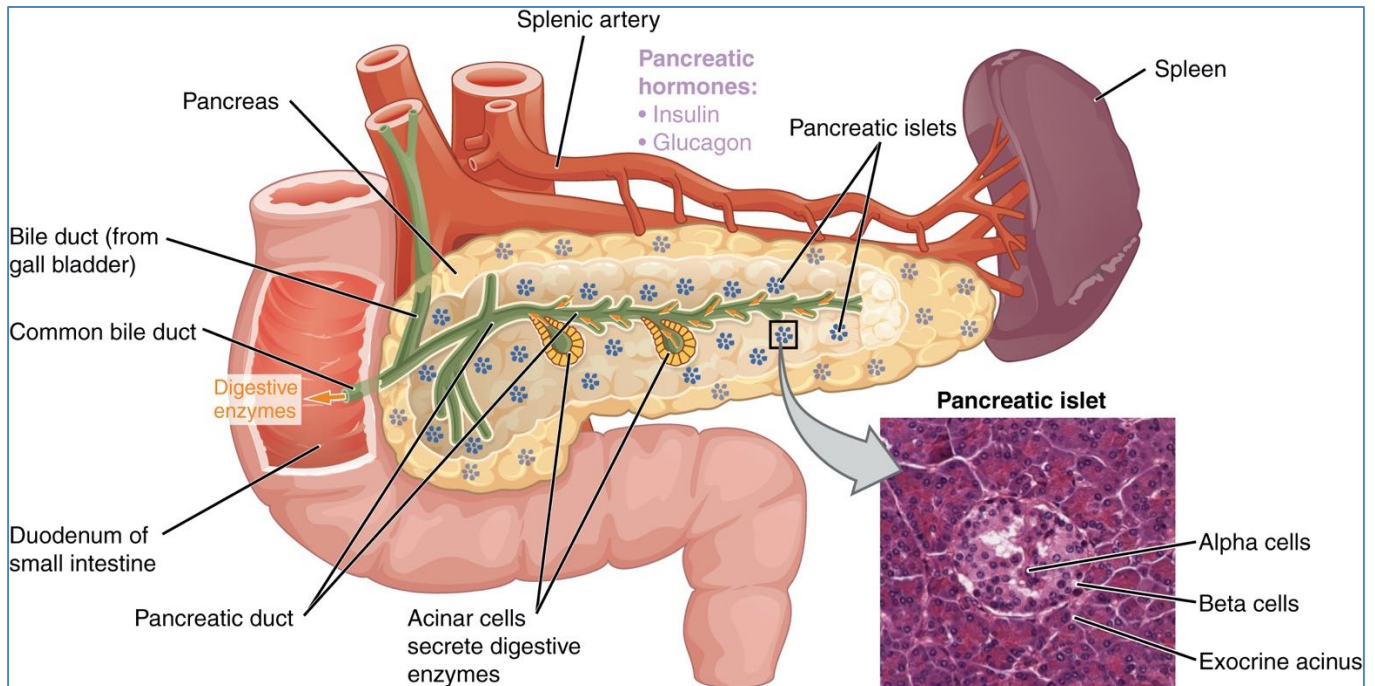


THE 'ENDOCRINE' PANCREAS

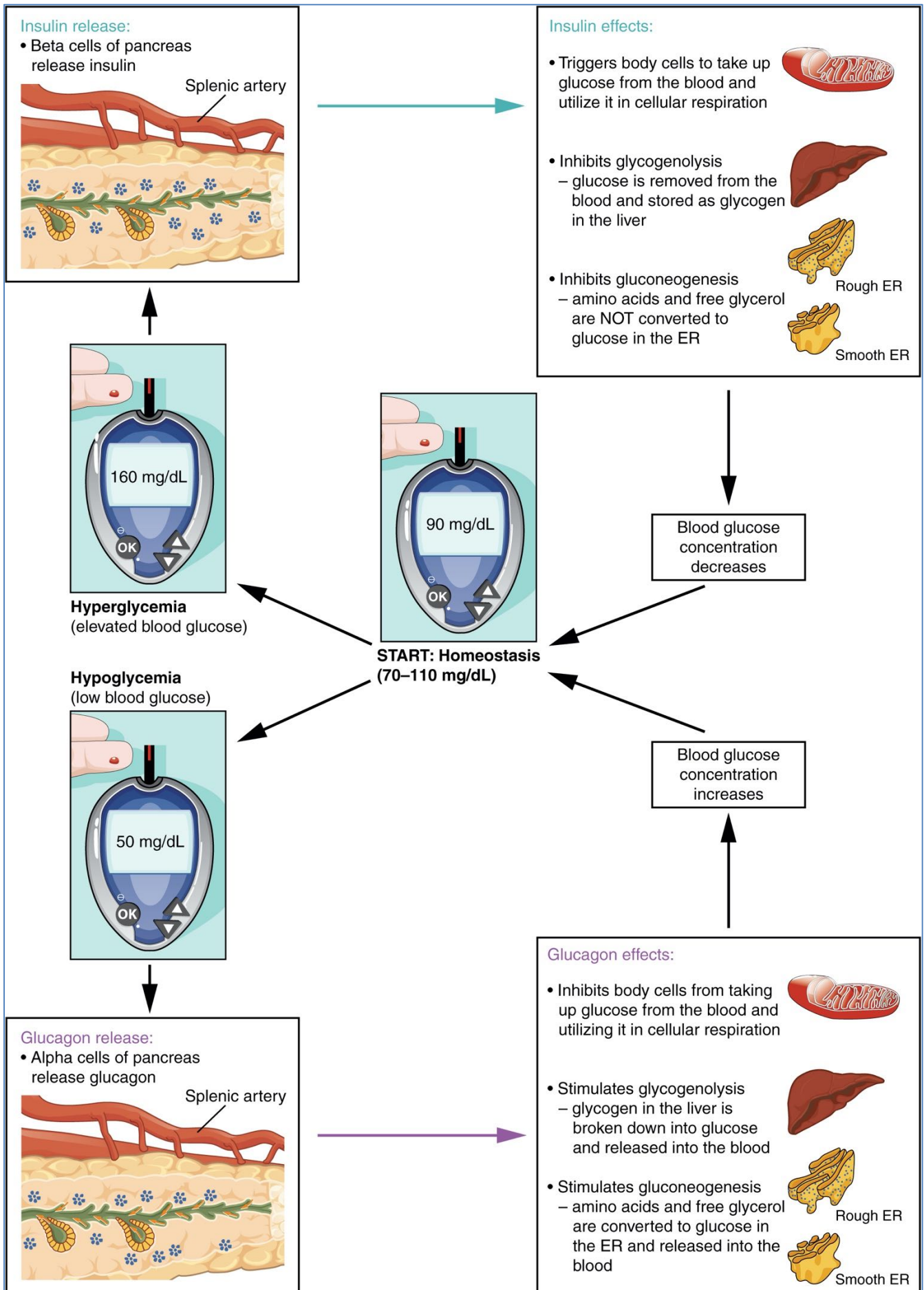
THE 'ENDOCRINE' PANCREAS

Anatomy of the Pancreas:

- Elongated, Horizontal, Retroperitoneal Organ
- Head –Cupped by the Duodenum
- Tail – abuts the spleen
- **Exocrine Component (99% of Cells)** – Acinar Epithelial Cells + Duct Network
 - o **Produces pancreatic juice** (incl. digestive enzymes)
- **Endocrine Component (1% of Cells)** – Islets of Langerhans + 4 Cell Types. *Of these:*
 - o 25% are **Alpha (α) Cells** – Secrete **Glucagon** ($\uparrow$ BSL)
 - o 60% are **Beta (β) Cells** – Secrete **Insulin** ($\downarrow$ BSL)
 - o 10% are **Delta (δ) Cells** – Secrete **Somatostatin**
 - o 5% are **PP Cells** – Secrete **Pancreatic Polypeptide** (Plays a role in Satiety)



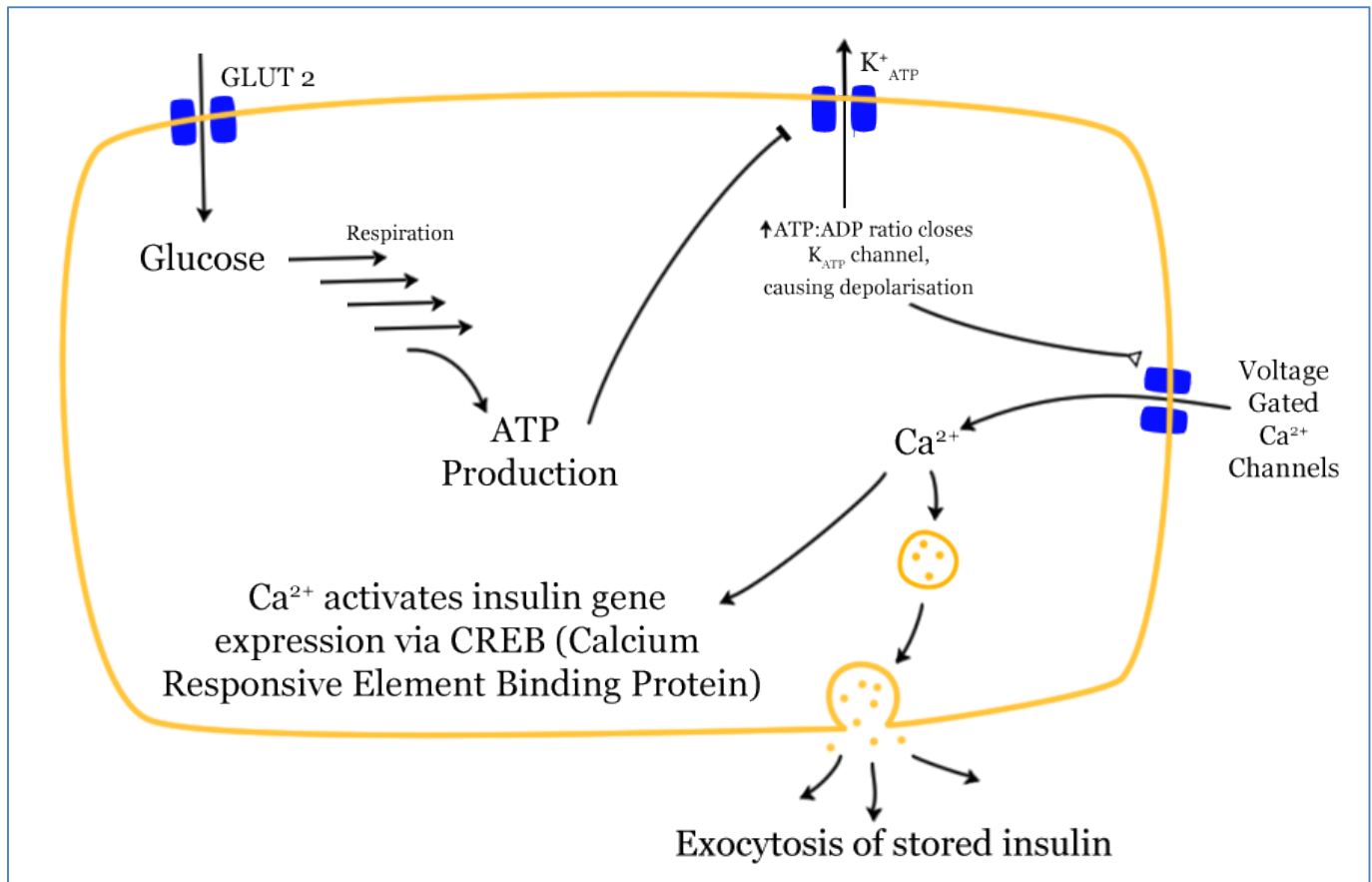
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Pancreatic Hormones:

Insulin:

- **Secreted By:**
 - o Beta (β) Cells of the Pancreas
- **Mechanism of Insulin Release from β -Cells of Pancreas:**
 1. $\uparrow$ Blood Glucose $\rightarrow \uparrow$ *Insulin-Independent* Uptake of Glucose into Pancreas (Via GLUT-2)
 2. $\rightarrow \uparrow$ ATP Production in β -Cell.
 3. $\uparrow$ ATP Closes the ATP-Gated- K^+ Channels in β -Cell Membrane $\rightarrow$ Depolarises the β -Cell
 4. Depolarisation $\rightarrow$ opens Voltage-Gated Ca^{2+} Channels $\rightarrow$ Influx of Ca^{2+}
 5. Influx of $Ca^{2+} \rightarrow Ca^{2+}$ Mediated Exocytosis of Insulin Vesicles (Similar to ACh Release in Muscles)

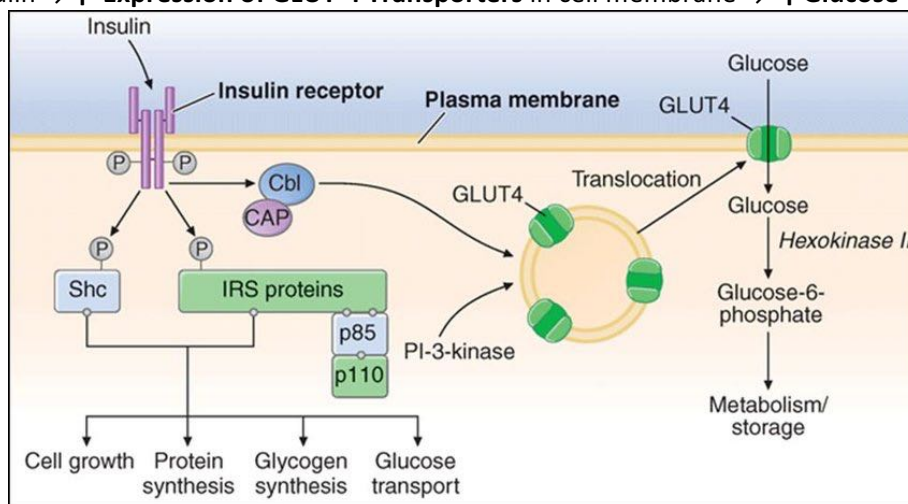


Prisonblues at the English-language Wikipedia, CC BY-SA 3.0 <<http://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

- **Insulin only affects *Insulin-Sensitive* Tissues (Ie: Those that expresses GLUT-4 Transporters):**
 - o Liver
 - o Muscle
 - o Adipose Tissue
- **Insulin doesn't affect *Insulin-Independent* Tissues:**
 - o Blood Vessels (Endothelium)
 - o Myocardium of Heart
 - o Nervous System
 - o Red Blood Cells
 - o Kidneys
 - o Eyes

- **Primary Actions:**

- Insulin → ↑ **Expression of GLUT-4 Transporters** in cell membrane → ↑ **Glucose Uptake** in tissues



Current Trends in Pharmacological Treatment of Type II Diabetes Mellitus - Scientific Figure on ResearchGate.
Available from: https://www.researchgate.net/figure/Mechanism-of-action-of-insulin-17_fig1_325007874

- **Fasted State:**

- There will be some GLUT-4 Transporters expressed in the Plasma Membrane.
- However, most will be found in the membranes of Cytoplasmic Vesicles within the cell.

- **Fed State:**

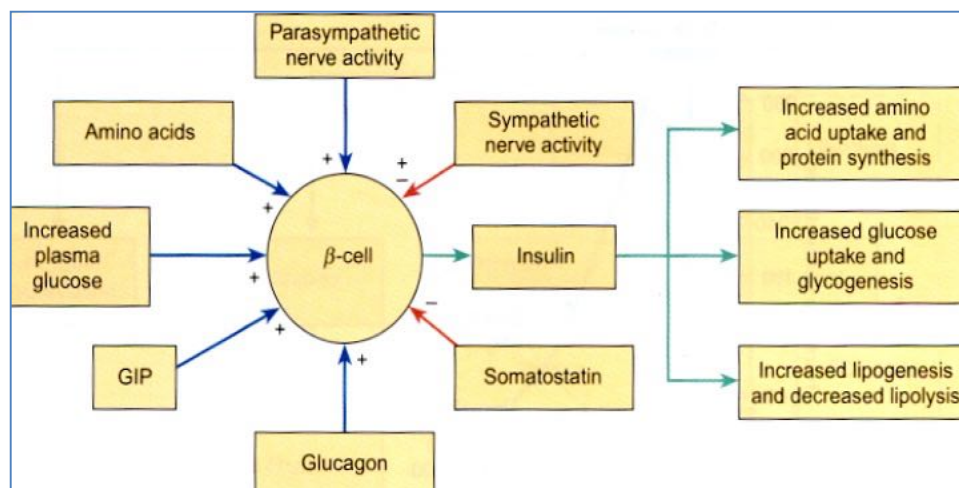
- Binding of Insulin to Receptors → Initiates a Signalling Cascade → Movement of GLUT-4 Laden Vesicles to the Cell Surface
- Upon reaching the Plasma Membrane, the vesicles fuse with it → ↑ Plasma Membrane-[GLUT-4].
- ↑ Plasma Membrane-[GLUT-4] → ↑ Glucose Uptake
- Signalling Cascade also Causes →
 - Glucose Metabolism
 - Protein Synthesis
 - Glycogen Synthesis
 - Inhibition of Gluconeogenesis
 - Lipid Synthesis.

- **Insulin Release is Stimulated By:**

- Parasympathetic NS (Rest & Digest)
- ↑ Blood [Amino Acids]
- ↑ Blood [Glucose]
- Gastrointestinal Peptide (GIP)
- Glucagon – (Weak Stimulator)

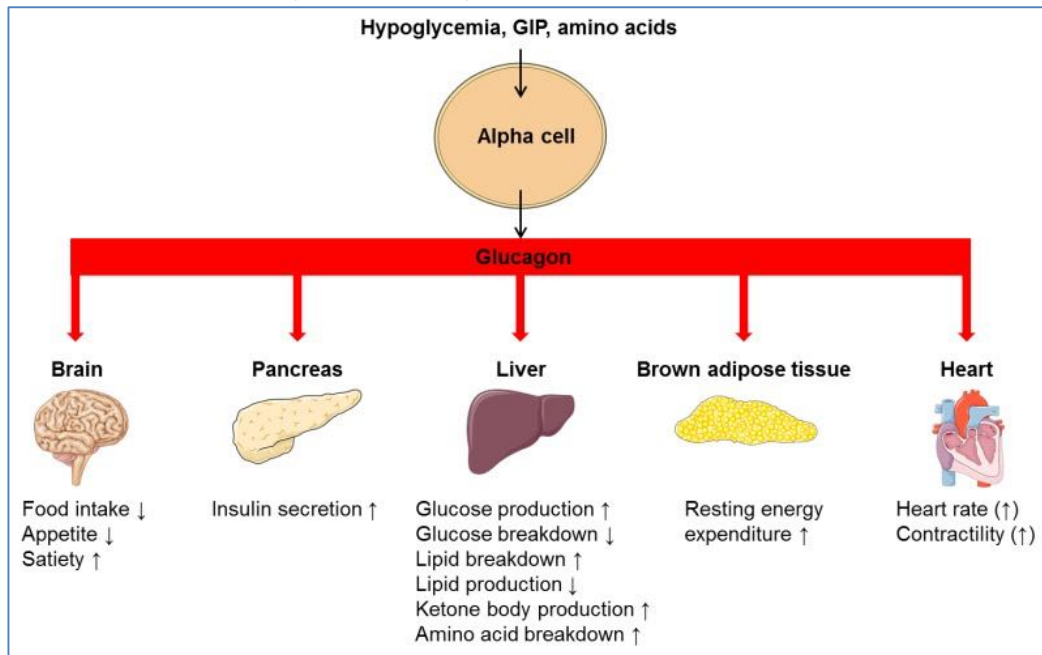
- **Insulin Release is Regulated By:**

- Sympathetic NS (Acts to ↑ Blood [Glucose] for Fight/Flight Response)
- Somatostatin



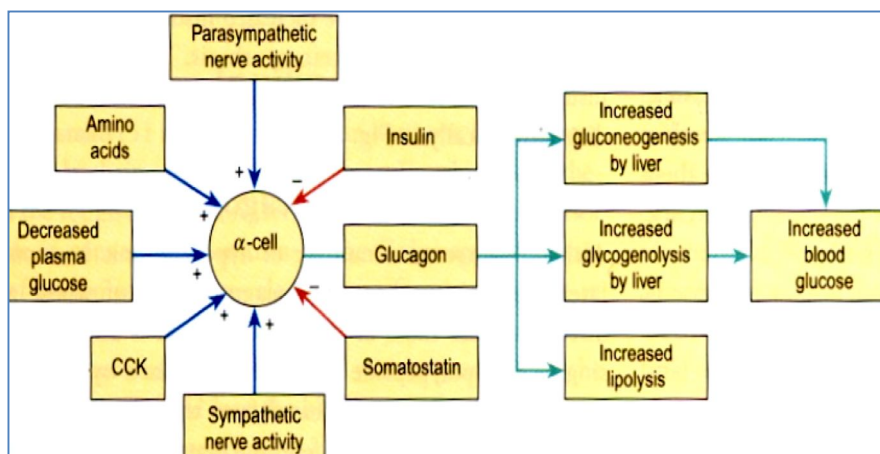
Glucagon:

- **Secreted By:**
 - Pancreatic Alpha Cells
- **Primary Actions:**
 - Acts to Increase Blood Glucose; by:
 - Stimulating Hepatic Glycogenolysis
 - Stimulating Hepatic Gluconeogenesis
 - Stimulates Lipolysis
 - Stimulates Ketogenesis
 - Inhibits Hepatic Glycolysis
 - Inhibits Fatty Acid Synthesis in Liver
 - Inhibits Fat Deposition in Adipose Tissue



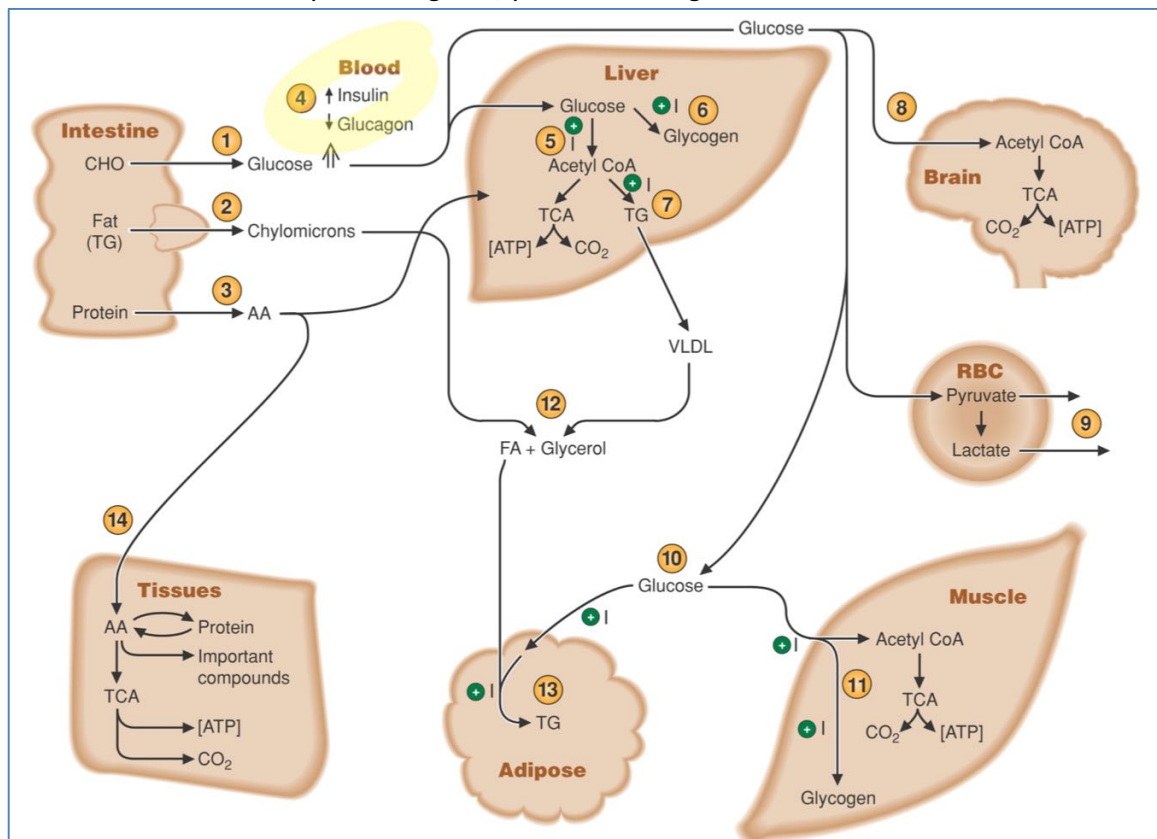
<https://www.ncbi.nlm.nih.gov/books/NBK279127/>

- **Release is Stimulated By:**
 - Low Glucose Levels (Eg: Between meals or while sleeping)
 - Low Insulin levels
 - Increased Sympathetic Drive (Alpha adrenergic receptors)
 - Cholecystokinin (CCK)
 - Exercise
- **Release is Regulated By:**
 - Elevated Blood Glucose levels
 - Elevated Insulin Levels
 - Increased Parasympathetic Drive (M3 muscarinic receptors)
 - Amino Acids Alanine & Arginine (Elevated following high protein meal)
 - Somatostatin

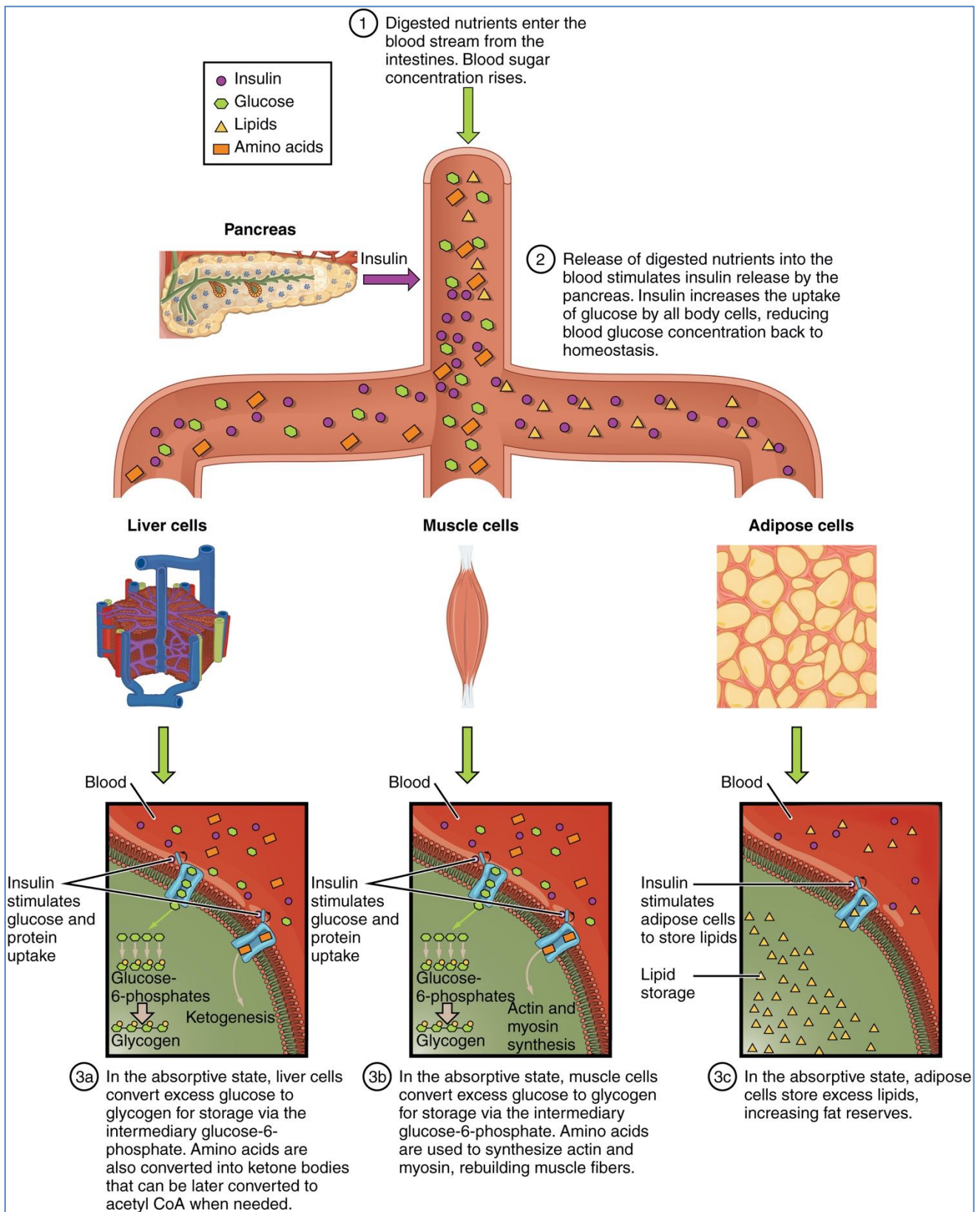


The “Fed State” – Directly After a Meal:

- **↑INSULIN:**
 - **Stimulates:**
 - **Nutrient Uptake from the Blood:**
 - **Glucose** (Liver, Muscle & Adipose)
 - Via ↑GLUT-4 Receptors (**Muscle & Adipose**)
 - Via ↑Glucose Utilisation (Liver)
 - **Amino Acids** (Liver, Muscle & Adipose)
 - **Fatty Acids** (Liver & Adipose)
 - Via ↑*Lipoprotein Lipase (LPL)* Activity in Adipose Tissue.
 - **Macromolecular Synthesis (& Storage):**
 - **Glycogenesis** (Liver & Muscle) – (Note: Glucose → Triglycerides in Adipose)
 - **Proteingenesi**s (Liver, Muscle & Adipose)
 - **Lipogenesis** (Liver & Adipose)
 - **Glycolysis – In all body cells**
 - **Inhibits:**
 - **Gluconeogenesis** (Liver)
 - **Ketogenesis** (Liver)
 - **Macromolecular Breakdown:**
 - **Lipolysis** (Liver & Adipose)
 - **Glycogenolysis** (Liver & Muscle)
 - **Proteolysis** (Liver, Muscle & Adipose)
- **+ the “Incretin Effect”:**
 - **Incretins (Released by GIT after a meal) Further Stimulates Insulin Release from Pancreas.**
 - Hence → The Insulin Response to Oral Glucose is much Greater & Quicker than IV Glucose.
 - ∴ **New Avenue for Diabetes Management:**
 - **Incretins:**
 - *Intestinal* glucose intake → Intestines release **Incretins (glucagon-like peptide-1 [GLP-1] and Glucose-dependent Insulinotropic Polypeptide [GIP])** → Stimulate β Cells to ↑Insulin Release and Suppress α-Cells and ↓Glucagon.
 - **Note: Incretins are Destroyed by Dipeptidyl Peptidase-4 (DPP-4)**
 - ∴ By Inhibiting DPP, you can Prolong the Action of Incretins

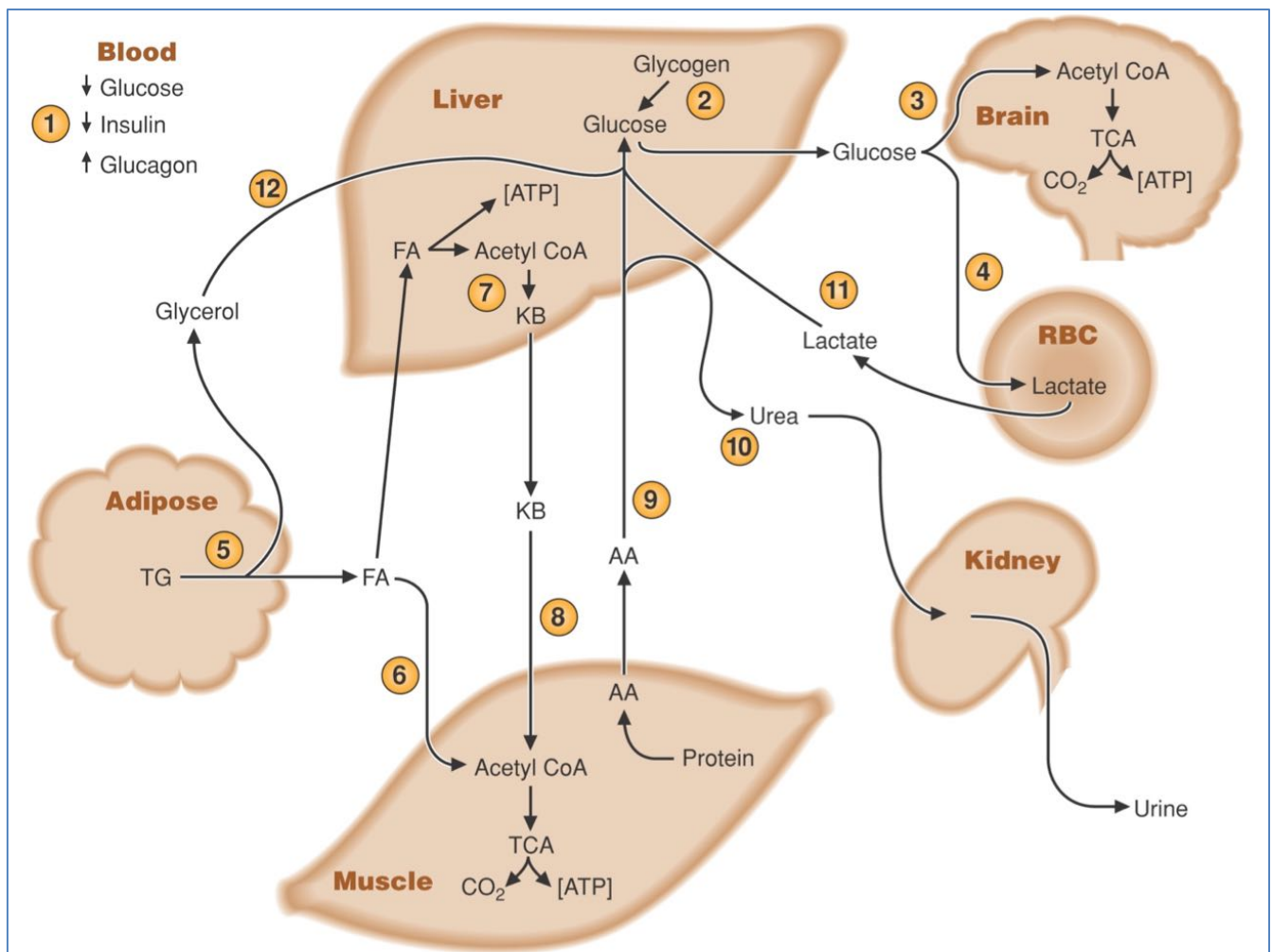


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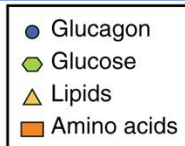


The "Fasted State" - ~3hrs After a Meal:

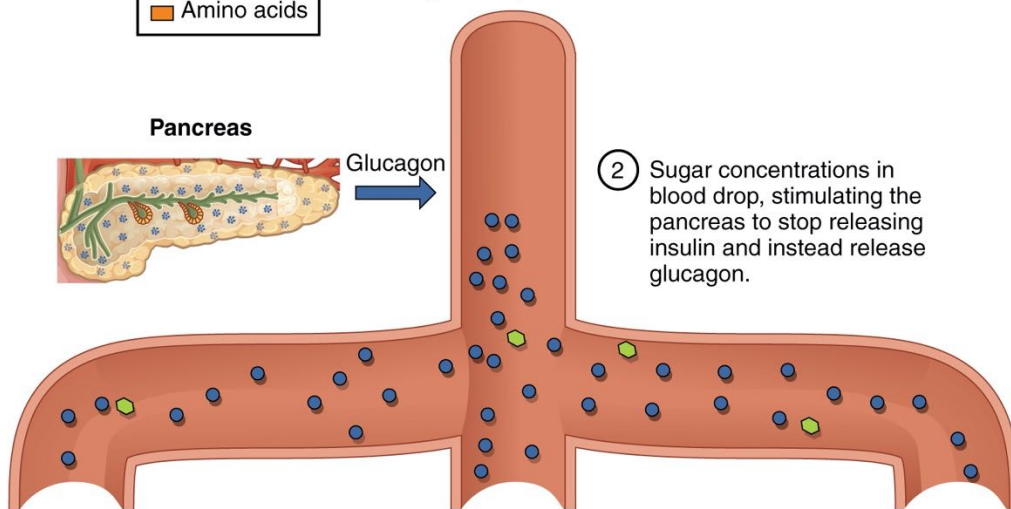
- **↑GLUCAGON**
 - *Activates Glycogenolysis (Liver) → ↑Blood [Glucose] (Note: Glucagon = Powerful Glycogenolytic)
 - Activates Gluconeogenesis (Liver) → ↑Blood [Glucose]
 - Stimulates Amino Acid Uptake (Liver) → ↑Gluconeogenesis → ↑Blood Glucose
 - Activates Lipolysis (Adipose) → ↑Blood [FA's] (Note: Glucagon = Powerful Lipolytic)
 - Stimulates Ketogenesis (Liver)
- **↓INSULIN → "Glucose-Sparing" Effect:**
 - **Increased Availability of Gluconeogenic Substrates:**...due to:
 - ↓Inhibition of Gluconeogenesis – (↑Level of Gluconeogenesis)
 - ↓Inhibition of Lipolysis – (↑Level of Lipolysis)
 - ↓Inhibition of Proteolysis – (↑Level of Proteolysis)
 - **↓Glucose Uptake by:**
 - Muscle
 - Liver
 - Adipose
 - **Glucose-Sparing →** More Glucose for Brain & Nerves (Glucose = 1^o Fuel)
- **Note: Insulin is Low, but is still high enough to prevent:**
 - Massive Lipolysis (As Glucagon is a powerful lipolytic)
 - Massive Ketogenesis (Normally, ↑Insulin Inhibits Ketogenesis) – Therefore Low Insulin allows some Ketogenesis but Prevents Ketoacidosis.)
 - Massive Proteolysis



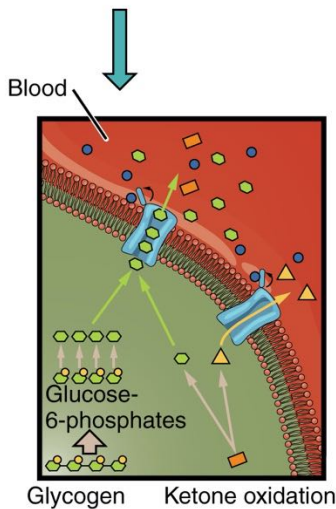
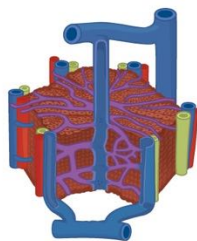
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① **Postabsorptive state:**
no nutrients enter the bloodstream from the digestive system

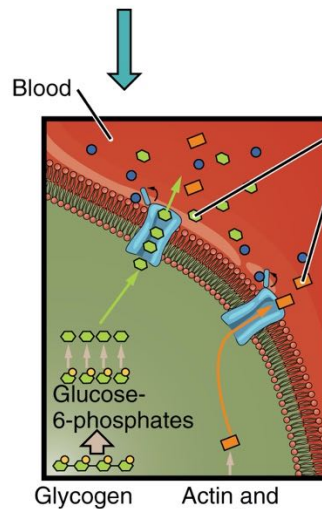
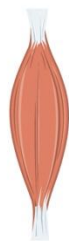


Liver cells



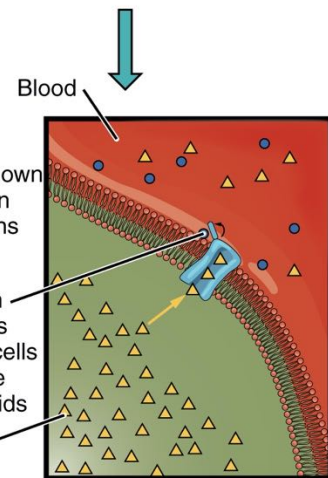
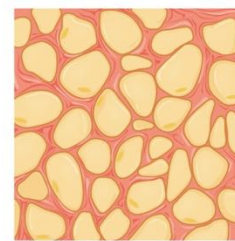
Glycogen Ketone oxidation

Muscle cells



Glycogen Actin and myosin catabolism

Adipose cells



Glucagon stimulates the breakdown of glycogen and proteins

Glucagon stimulates adipose cells to release stored lipids

Lipid storage

③a In the postabsorptive state glycogenolysis releases glucose into the blood, returning blood glucose concentrations to homeostasis. Ketone oxidation also releases lipids and additional glucose, which body cells can use to generate ATP.

③b In the postabsorptive state, muscle cells release glucose into the blood, returning blood glucose concentrations to homeostasis. Catabolized amino acids from muscle proteins can also generate ATP after undergoing ketogenesis and ketone oxidation in the liver.

③c In the postabsorptive state, adipose cells release stored lipids, which can be used to generate glucose, ketone bodies, or ATP.

CALCIUM & PHOSPHATE HOMEOSTASIS

CALCIUM & PHOSPHATE HOMEOSTASIS

Functions of Calcium & Phosphate:

- **Calcium:**
 - **Structural Purposes:**
 - Development & Maintenance of Skeleton
 - **Biochemical Purposes:**
 - Mediates exchange between Intracellular & Extracellular Compartments (eg: ACh Release)
 - Responsible for many Endocrine & Exocrine Secretions.
 - Can act as a secondary messenger (Like cAMP).
 - Role in Muscle Contraction & Nerve Impulses
 - Role in Blood Clotting
- **Phosphate:**
 - **Structural Purposes:**
 - Development & Maintenance of Skeleton
 - Phospholipids are a major structural component of Plasma Membrane
 - **Biochemical Purposes:**
 - Phosphate Release from Nucleotides (eg: ATP → ADP) is the Major Source of Cellular Energy.
 - The Phosphodiester-Bond Provides the backbone for RNA & DNA.
 - Phosphorylation provides a basis for Receptor Activation & Signal Transduction.
- **(Magnesium - Also important):**
 - **Structural Purposes:**
 - Important in Integrity of Bone
 - Note: 50% of body-Mg is stored in bone.
 - **Biochemical Purposes:**
 - Sufficient Mg^{+} is required to have Normal Ca^{+}
 - **Mg^{+} Regulation:**
 - Mg^{+} Levels controlled by modulating Mg^{+} Reabsorption in the Distal Tubules of the Kidney:
 - Mg^{+} Reabsorption stimulated by:
 - ↓ Extracellular Fluid Volume
 - ↓ Plasma Mg^{+}
 - Hypocalcaemia
 - Mechanism is unknown.

Bone Chemistry:

- **Bone Consists of 2 Things:**
 - **30% (By Weight) = Organic Bone Matrix:**
 - 90% of which is Collagen Fibres
 - 10% of which is Ground Substance
 - **70% (By Weight) = Bone Salts:**
 - The major salt = *Hydroxyapatite* ($Ca_{10}(PO_4)_6(OH)_2$)
 - Consists of mainly Calcium & Phosphate.
 - Also contains other *amorphous calcium salts* used for rapid exchange (See below)

Ca^{+} Exchange Between Plasma & Bone:

- **Experiment:** An Intravenous Ca^{+} injection dissipates within an hour. Likewise, large-scale removal of Ca is just as quickly compensated for.
- **How?** – The bone acts as a Calcium Exchanger, quickly mobilizing/depositing calcium salts to buffer the Extracellular-Fluid Ca^{+} Concentration.

Serum Concentrations:

- Calcium:

- Levels depend on 3 Processes:
 - Intestinal Absorption (Ie: To ↑ Serum Ca⁺)
 - Renal Excretion (Ie: To ↓ Serum Ca⁺)
 - Resorption/Deposition of Bone (Ie: To ↑ Serum Ca⁺)
- The Above Processes are Regulated by 3 Hormones:
 - PTH - Parathyroid Hormone
 - Calcitonin
 - Vitamin D (The Active Form)
- Calcium levels are tightly regulated - @ ≈ 9.4mg/dl OR 2.4mmol/L.
- Note: Only ≈1% of the Body's Ca⁺ is Extracellular. The Rest is Stored in Bones.
 - Hence, the Bones = Ca⁺ Reservoir.
- Extracellular Ca⁺ exists in 3 Forms:
 - 50% Ionized = Ca⁺ NOT Bound to Anything (Diffusible)
 - (Note: This is the functionally important form.)
 - 10% In Covalent Compounds (Diffusible)
 - 40% Bound to Plasma Proteins (Eg: Albumin) (Non-Diffusible)

- Phosphorus:

- Levels depend on:
 - Age
 - Gender
 - Dietary Intake.
 - Calcium-Controlling Hormones.
- Note: Only ≈1% of the Body's Phosphate is Extracellular. The Rest is Stored in Bones.
- Phosphorus levels are loosely regulated - @ ≈ 2.4 - 4.1 mg/dl.

Regulation of Plasma Ca⁺ & Phos. Levels:

- Intestinal Absorption:

- Calcium:
 - Normally, Ca⁺ is poorly absorbed by the Intestines.
 - **Vitamin D** Increases Ca⁺ Absorption by the Intestines (POTENT)
 - **PTH** indirectly promotes Intestinal Ca⁺ Absorption by ↑ Vit-D Activation by the Kidneys.
- Phosphate:
 - Absorption occurs very easily
 - (Ie: Almost all dietary Phosphate is absorbed into the blood, and later excreted in urine)

- Renal Excretion:

- Calcium:
 - Normally, 99% of Filtered Ca⁺ is Reabsorbed...
 - 90% happens in PCT, Loop of Henle & early DCT.
 - 10% happens in the late DCT – **and is Very Selective** (Depending on Blood-Ca⁺)
 - If Blood-Ca⁺ is Above Normal – All remaining Ca⁺ is expelled in urine.
 - Note: **Calcitonin** weakly ↑ Calcium Excretion
 - If Blood-Ca⁺ is Below Normal – All remaining Ca⁺ is reabsorbed
 - Note: **PTH** Greatly ↓ Calcium Excretion in the Kidneys (Ie: ↑ Reabsorption)
- Phosphate:
 - Renal Phosphate excretion is via an 'Overflow Mechanism':
 - If Blood-Phosphate is Below 1mmol/L – All filtered Phosphate is Reabsorbed
 - If Blood-Phosphate is Above 1mmol/L – Phosphate is excreted @ a rate relative to its conc.
 - Note: **PTH** Greatly ↑ Phosphate Excretion in the Kidneys.

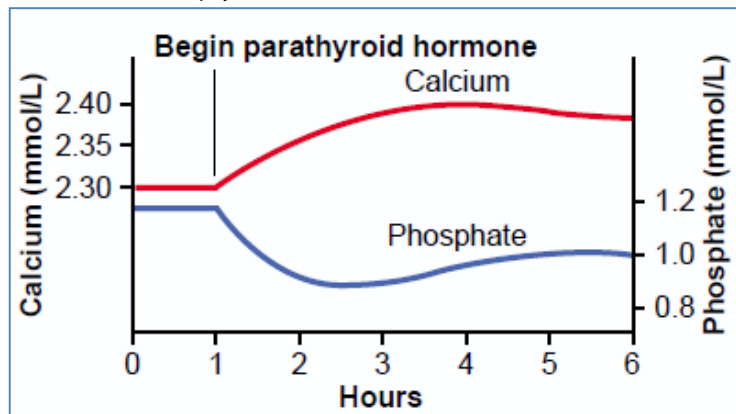
- Resorption/Deposition of Mineralized Bone:

- **PTH** promotes Osteoclast Activity (Bone Resorption)
- **Vitamin D** promotes Bone Calcification (Deposition) (Mechanism Unknown)
- **Calcitonin** promotes Bone Calcification (Deposition) (By Inhibiting Osteoclast Activity)

The 3 Major Hormones:

1- Parathyroid Hormone (PTH):

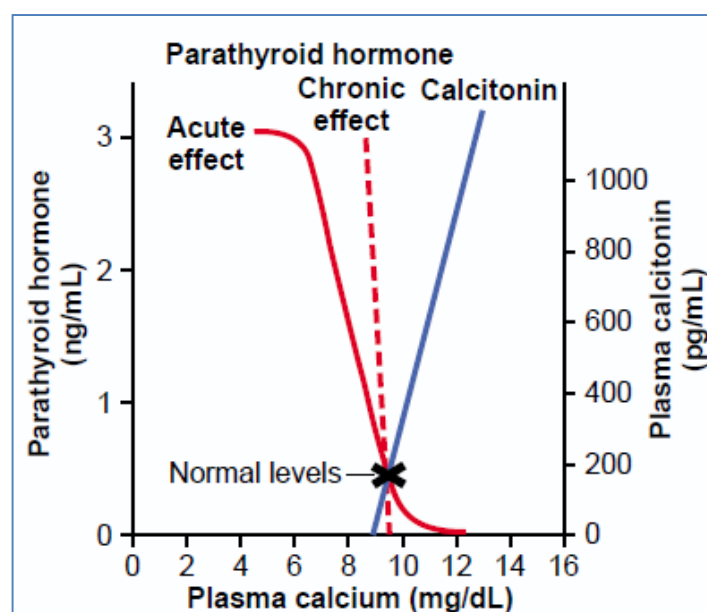
- **Secreted by** – The Chief Cells of the Parathyroid Glands
- **Aims to:**
 - **↑Plasma- Ca^+ levels** (By Increasing Bone Ca^+ / P^- Resorption & ↓ Renal Ca^+ Excretion)
 - **↓Plasma- P^- levels** (By ↑Renal P^- Excretion so that it exceeds Bone P^- Resorption)



- **Primary Effects:**
 - Mobilises Ca & Phos from bone Matrix (Bone Resorption) (By Stimulating Osteoclast Activity)
 - Stimulates Osteoblast & Osteoclast Proliferation → Promotes bone turnover.
 - Decreases Renal Excretion of Calcium (By Increasing Calcium Reabsorption in DCT)
 - Note: **PTH** is essential here to prevent excess loss of Calcium & therefore prevent calcium depletion in ECF & Bone.
 - Increases Renal Excretion of Phosphate (By Preventing Phosphate Reabsorption in PCT)
 - Increases Activation of Vit-D in Kidneys→ Indirectly increases intestinal absorption of Ca^+ / P^- .
- **Stimulated By:**
 - ↓ Extracellular $[\text{Ca}^+]$ – Very Sensitive
- **Inhibited By:**
 - ↑ Extracellular $[\text{Ca}^+]$ – Very Sensitive

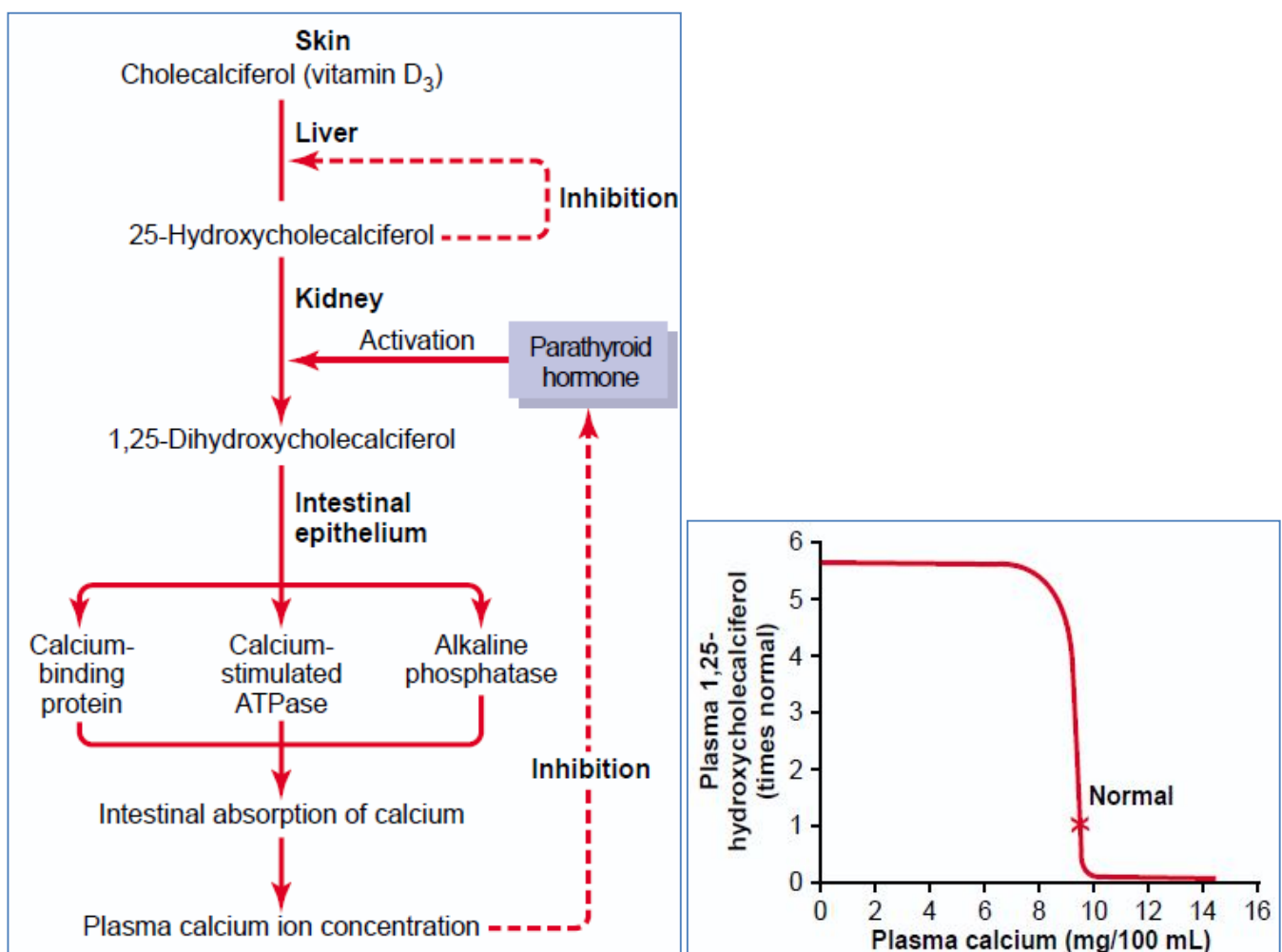
○ Regulators:

Stimulated By:	Inhibited By:
↓ Calcium	↑ Calcium
↑ Phosphate (Indirect)	Vit D ₃
↓ Magnesium	
Cortisol	



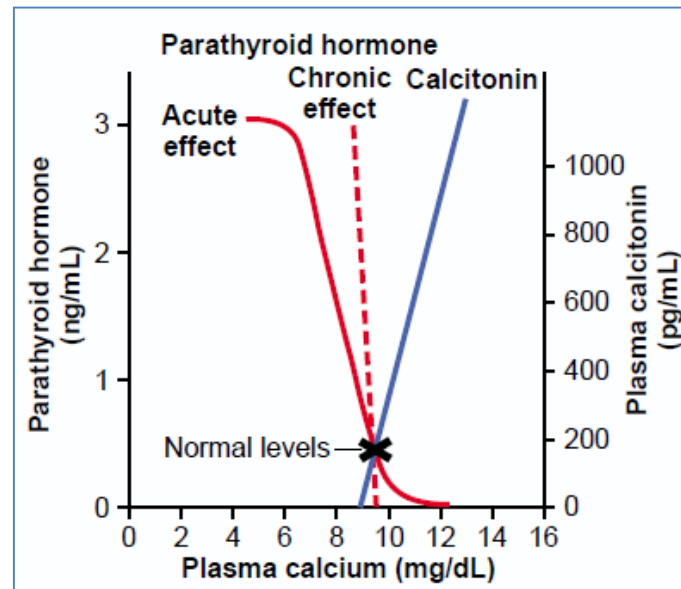
- 2- Vitamin D:

- **Aims to:**
 - $\uparrow$ **Plasma- Ca^+ / P^- levels** (by Increasing intestinal Ca^+ / P^- absorption)
- **Primary Effects:**
 - $\uparrow$ Intestinal Calcium Absorption
 - $\uparrow$ Intestinal Phosphate Absorption (Even better than usual)
 - Aids **PTH** in mobilizing Ca & Phos from bone Matrix.
 - (In Small Quantities, it can $\uparrow$ Bone Mineralization (Mechanism Unknown))
- **Vit-D Activation:**
 - Vit-D *itself* is not the active form that causes the above effects. It must first be **Activated**.
 - Vit-D is converted through a series of reactions in the **Skin**, **Liver** & the **Kidneys** to produce the final active product = **1,25-dihydroxycholecalciferol** aka. **1,25(OH) $_2$ D $_3$** .
 - See Below for Steps:
 - **Note:** The conversion in the Liver has Neg:Feedback for 2 Important Reasons:
 - 1- Prevents excessive 25-Hydroxycholecalciferol in the plasma, which in turn prevents excessive activation by kidneys $\rightarrow$ maintains Ca^+ ion concentration.
 - 2- Conserves the Vit-D $_3$ stored in the Liver for future use. (Because the converted forms only last a few weeks, whereas Vit-D $_3$ lasts for months)
 - **Note:** The conversion in the Kidneys is controlled by **PTH**:
 - Without PTH, none of the 1,25(OH) $_2$ D $_3$ is formed.
 - Therefore, PTH has a huge influence on the levels of body's functional Vit-D.
 - **Furthermore**, since Plasma- Ca^+ levels determine PTH levels, Plasma- Ca^+ has an Indirect, but **STRONG Negative Feedback Effect** as well. (**Even a slight increase in $[\text{Ca}^+]$ above 10mg/dL, sharply suppresses PTH secretion $\rightarrow$ $\downarrow$ 25-Hydroxycholecalciferol – See Diagram)**

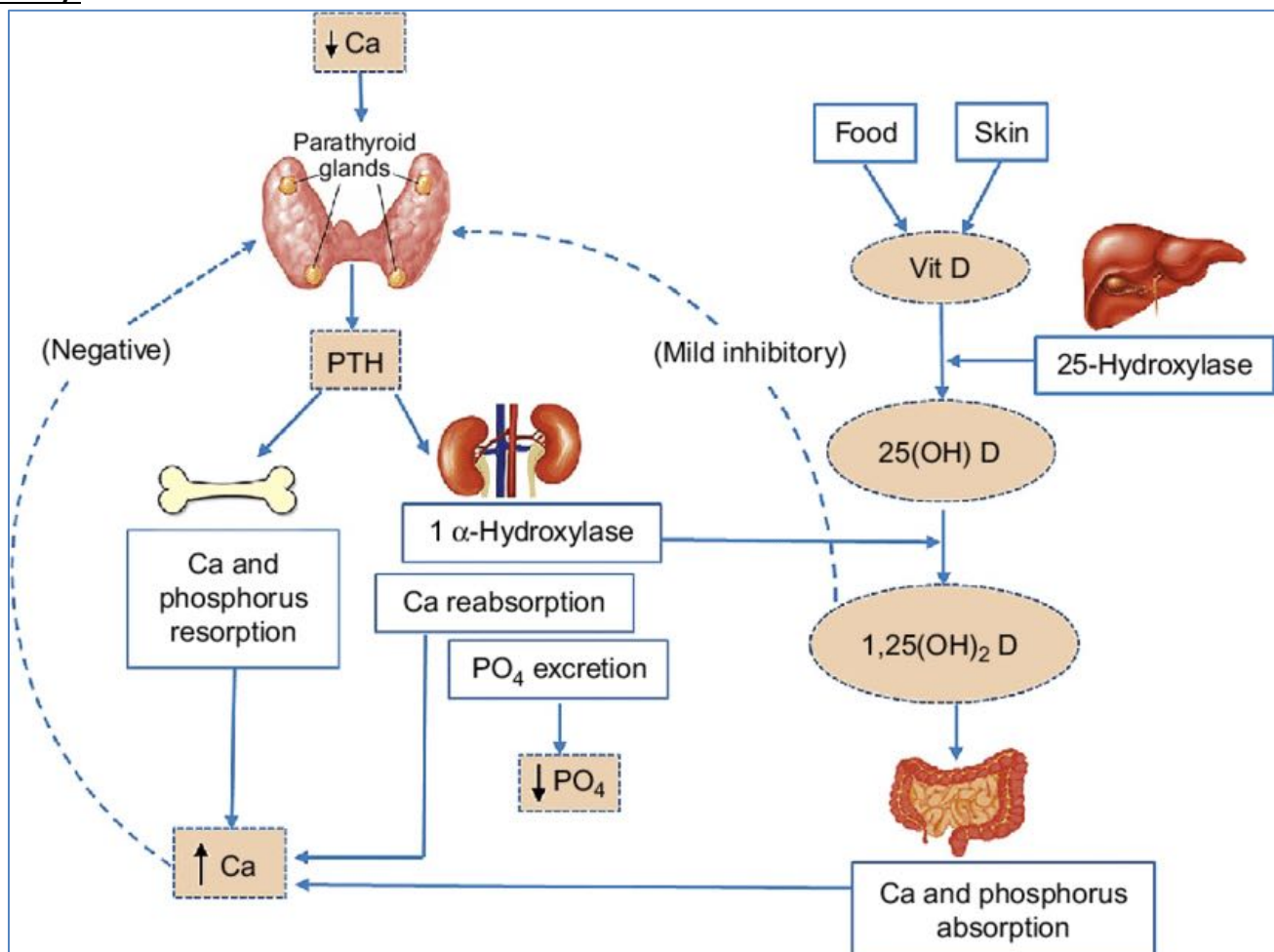


3- Calcitonin:

- **Secreted By** – The Parafollicular Cells of the Thyroid Gland
- **Aims to:**
 - **↓ Plasma- Ca^{2+} levels** (By ↓ Osteoclast Activity so that Bone Deposition is Favoured)
 - This effect is much greater in children due to rapid remodelling.
- **Primary Effects:**
 - Decreases the Activity & Proliferation of Osteoclasts → Favours Bone-Salt Deposition.
- **Stimulated By:**
 - ↑ Extracellular [Ca^{2+}] (Note: Opposite of PTH) (See Below Diagrams)



Summary:



FLUID & ELECTROLYTE HOMEOSTASIS

FLUID & ELECTROLYTE HOMEOSTASIS

Why Maintain Fluid & Electrolyte Balance?:

- Critical for **Normal Cell Function**
- Critical for Chemical Stability (**Homeostasis**) of Surrounding Fluids
- *Electrolyte Balance (Particularly Na^+ & K^+) – Critical for **function of Excitable Tissues**
- Critical for **Blood Pressure Homeostasis**

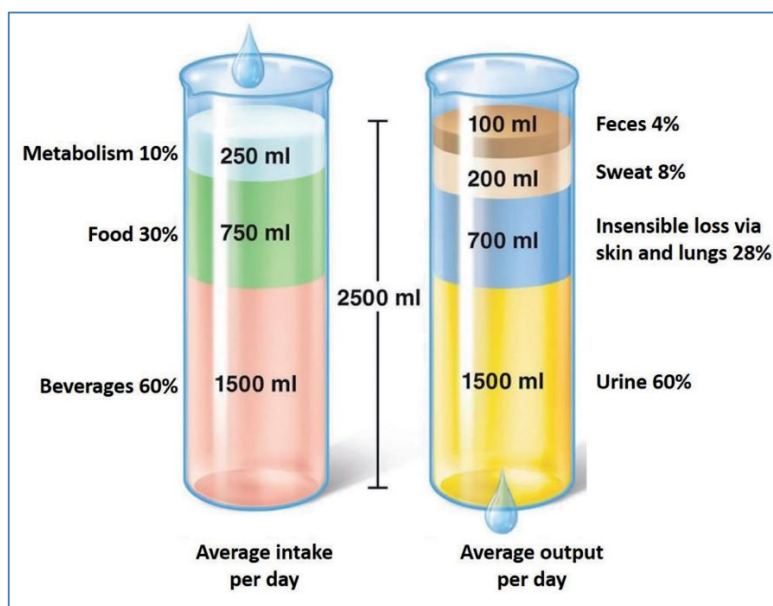
FLUID BALANCE:

Normal Adult Fluid Volume $\approx$ 40 Liters:

- **Extracellular** = 15 Litres
 - o 3 Litres = **Plasma**
 - o 12 Litres = **Interstitial Fluid**
- **Intracellular** = 25 Litres

Water Intake & Output:

- **Intake:**
 - o Produced in Metabolism
 - o Contained in Foods
 - o Consumed Fluids
- **Output:**
 - o Faeces (Obligatory)
 - o Sweat (Obligatory)
 - o Lungs (Obligatory)
 - o Urine



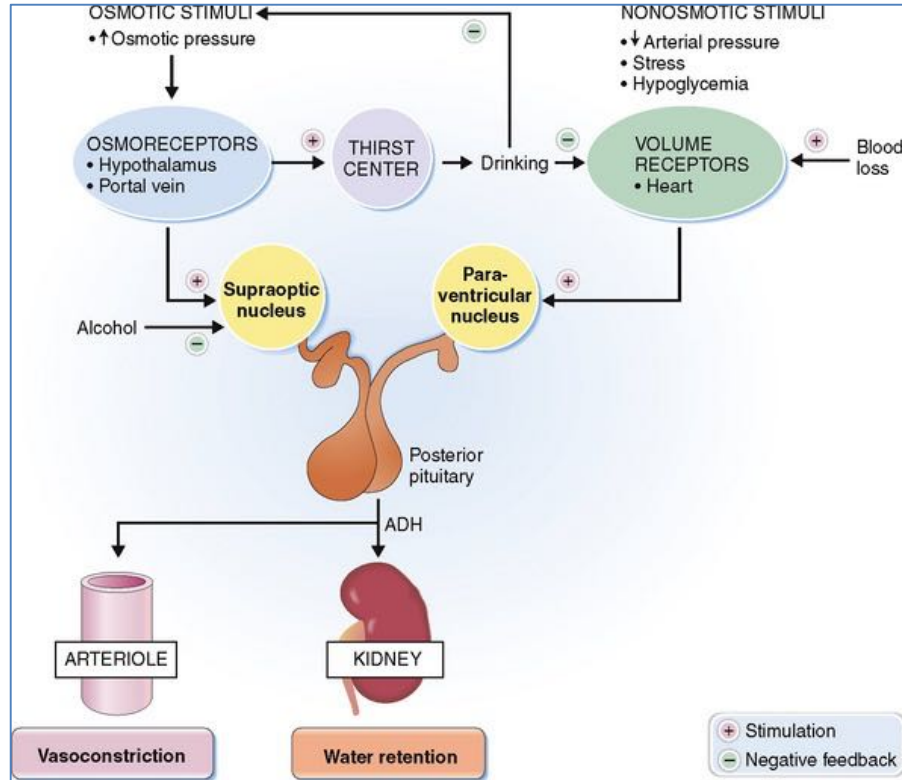
<https://repository.ubn.ru.nl/bitstream/handle/2066/191611/191611.pdf>

Regulation of Water Intake (Thirst) – Hypothalamic Triggers:

- **Thirst Triggered by 2 Things:**
 - o 1- A 10%+ Decrease in Plasma Volume....OR
 - o 2- A 1-2% Increase in Plasma Osmolality
- **1- Decreased Plasma Volume** → Reduced Blood Flow to Salivary Glands → Cellular Dehydration of Salivary Gland Cells → **"Dry Mouth"** → Triggers Thirst Centre in Hypothalamus.
- **2- Increased Plasma Osmolality** → Directly Causes Cellular Dehydration of **Osmoreceptors** in the Hypothalamus → Stimulates the Thirst Centre.
- **Note:** Extracellular Dehydration → Hypovolaemia → Baroreceptors stimulate Thirst Centre causes Renin Release from Kidneys.

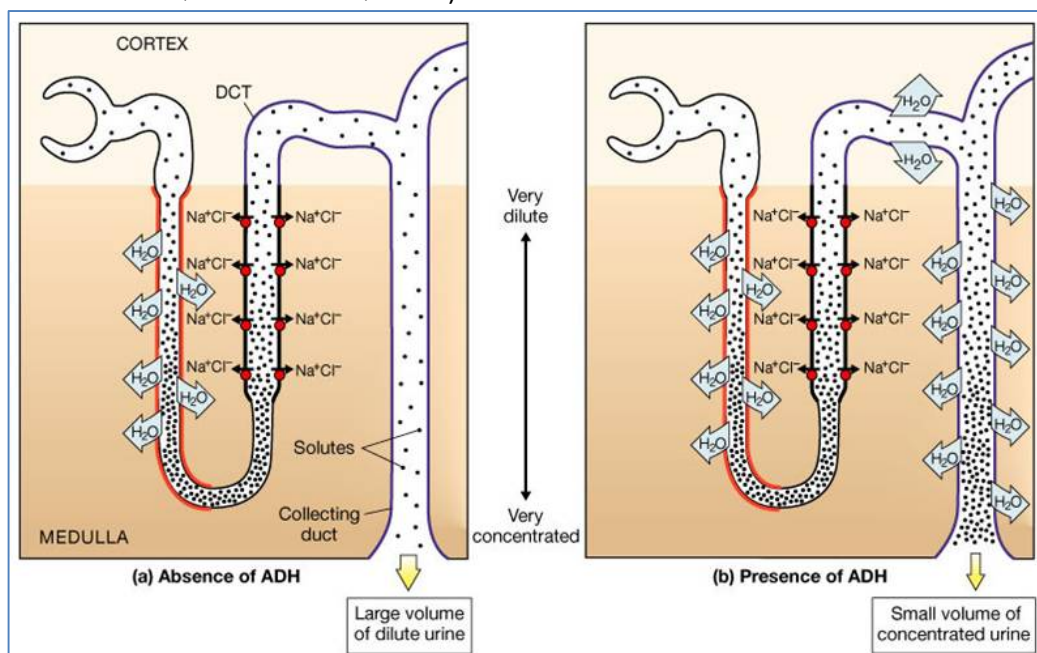
Regulation of Water Output:

- Water loss is unavoidable (Obligatory Water Loss – From Lungs, Skin, Faeces, Minimum Urine Output 500mL)
- Aside from Obligatory Loss, the rest (lost through Urine) is Regulated by **Anti-Diuretic Hormone (ADH)**.
- **Anti-Diuretic Hormone (ADH) → ↓ Water Output:**
 - Acts to increase Blood Volume.
 - Released from the Posterior Pituitary Gland
 - Released in response to:
 - ↑ Plasma Osmolality (↑ [Na⁺]) → Stimulation of Osmoreceptors in Hypothalamus
 - ↓ Plasma Volume.



<https://basicmedicalkey.com/the-endocrine-system-7/>

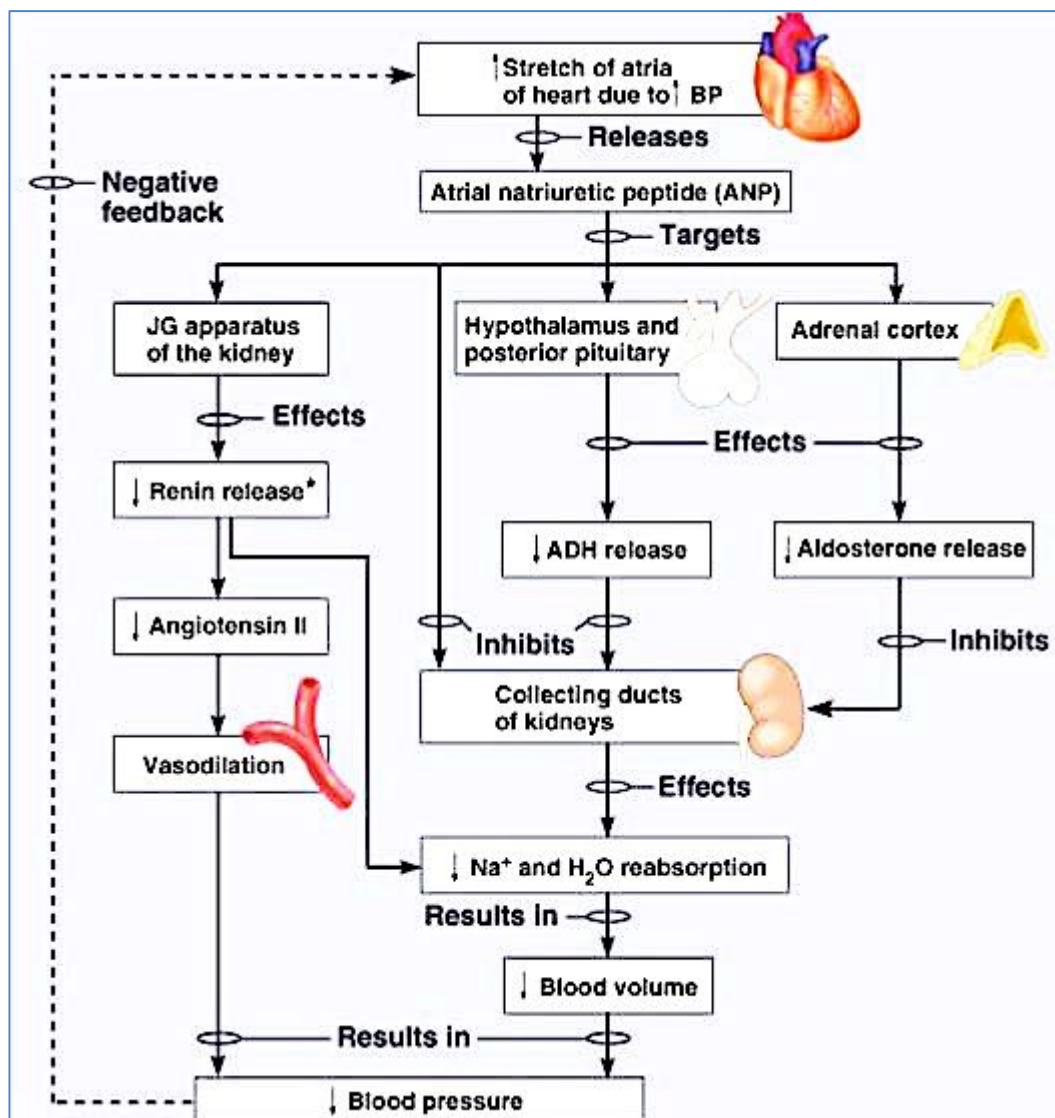
- **Works by INCREASING H₂O Permeability of Distal & Collecting Ducts:**
 - Distal Tubules & Collecting Ducts are **Normally Impermeable to H₂O**
 - **However**, the Presence of ADH → ↑ # of Aquaporins In Membrane → ↑ Permeability to H₂O
 - **This ↑ Permeability to H₂O + High [Solute] in Medulla → H₂O Reabsorption** (From Collecting Duct → Interstitium → Blood)



<https://www.austincc.edu/apreview/PhysText/Renal.html>

- Atrial Natriuretic Peptide (ANP) → ↑ Water Output:

- Acts to:
 - ↓ blood volume
 - ↓ Blood [Na]
- Secreted by Atrial Myocytes of the Heart
- Released in response to:
 - High Blood Pressure (Atrial Stretch)
- Works by:
 - Dilating Afferent Glomerular Arteriole
 - Constricting Efferent Glomerular Arteriole
 - ↑ Filtration Pressure → ↑ Filtration → ↑ H₂O & Na Excretion.
 - Inhibits Renin Secretion → Inhibits Renin-Angiotensin System
 - Inhibits Aldosterone Secretion from Adrenal Cortex.
 - Inhibits ADH Release from Post. Pituitary



<https://azkurs.org/bio2305-vascular-physiology-perfusion--blood-flow-through-tiss.html>

ELECTROLYTE BALANCE:

Significant Electrolytes:

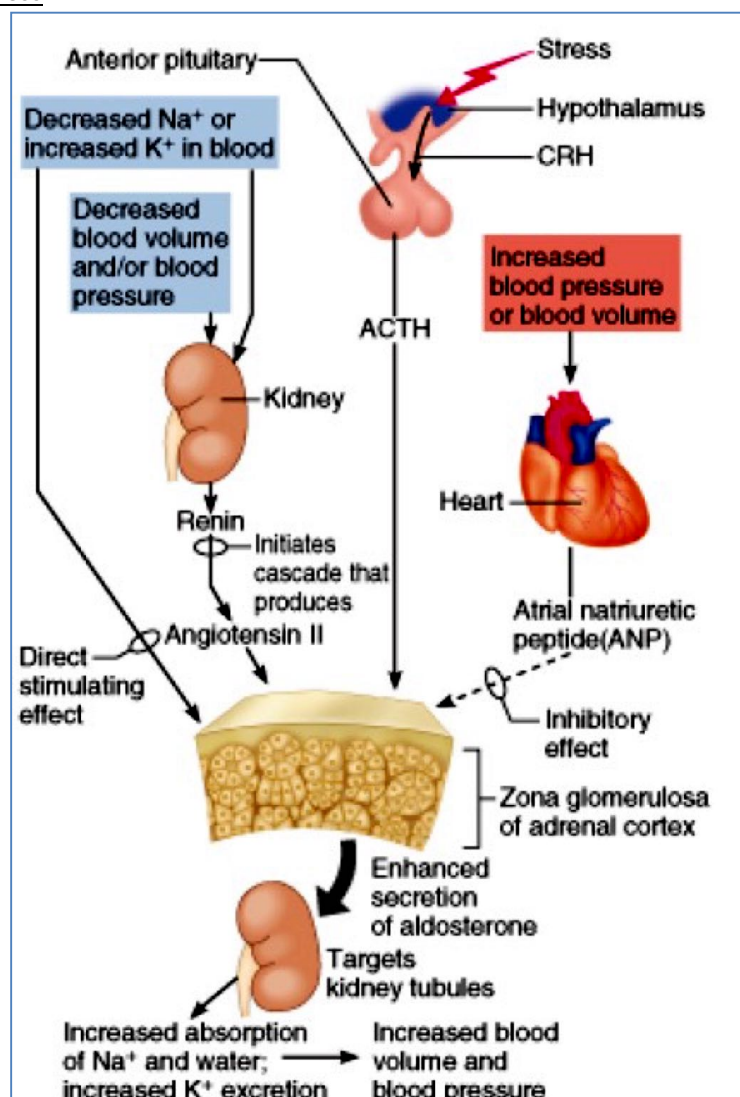
- Na^+ = Major Extracellular Cation
 - Cl^- = Major Extracellular Anion
 - K^+ = Major Intracellular Cation
- Account for 80% of Osmolality of Interstitial Fluid & Plasma.
- Accounts for 50% of Osmolality of Intracellular Fluid

Why Maintain Electrolytes

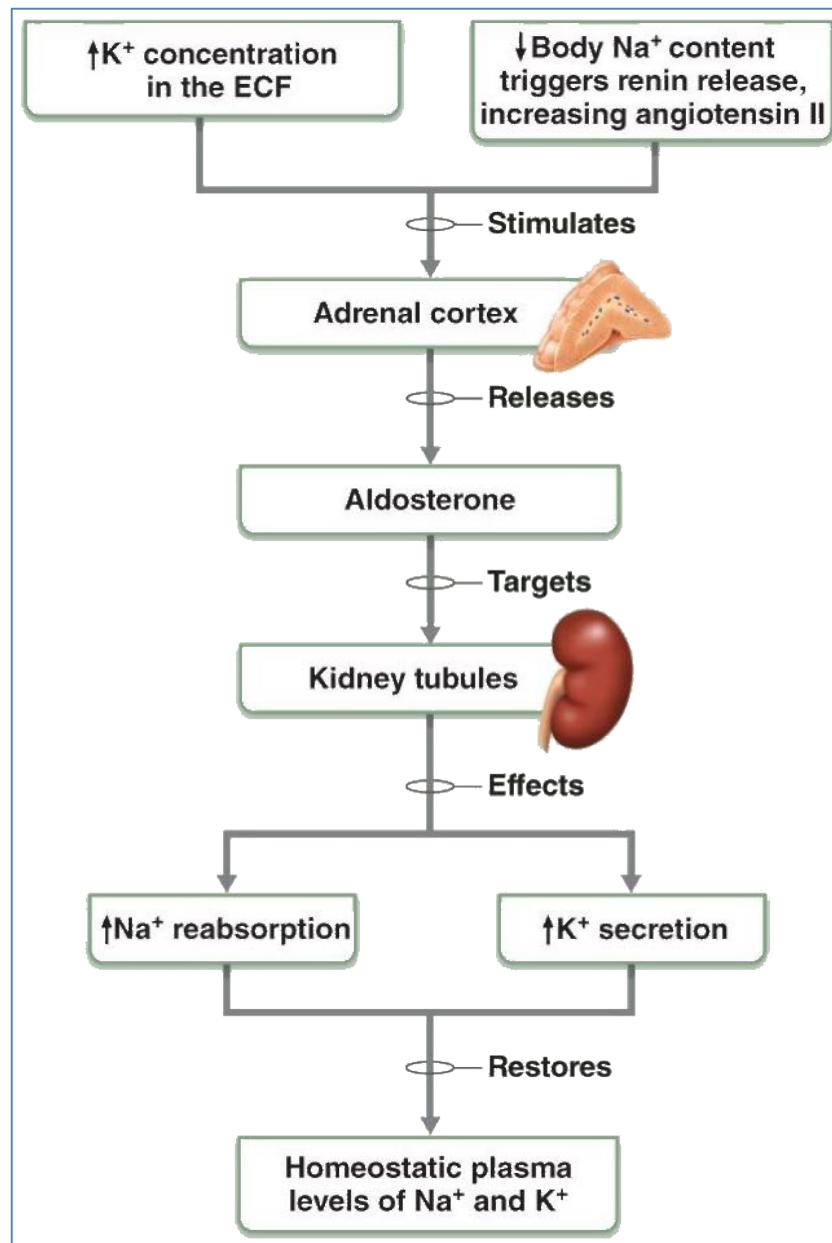
- Na^+ = Important for Heart & Nerve Function/Cellular Transport
- K^+ = Important for Heart Function/Cellular Transport
 - o (Note: too high Extracellular K^+ interferes with Cardiac Function = Fatal)
- Ca^{2+} = Important for Muscle, Heart & Nerve Function/Bone Formation
- Mg^{2+} = Important for Acetylcholine Release → Important for Neural & Cardiac Function
- HPO_4^{2-} = Important for Bone Formation (Bone salts – primarily calcium & phosphates)

Regulation of Na^+ - (The Main Extracellular Electrolyte):

- Primary role in Fluid & Electrolyte Balance (Because Water Follows Na^+ Movement)
- Extracellular $[\text{Na}^+]$ is normally stable & is **Regulated by levels of Aldosterone:**
- **Aldosterone → ↑Na Resorption:**
 - o Aldosterone = Steroid Hormone Released from The Adrenal Cortex.
 - o Released in response to:
 - *Angiotensin-II, Part of the Renin-Angiotensin System (Due to Renin Release by Kidneys)
 - *Hyponatraemia (Low Na^+ in Blood)
 - *Hyperkalaemia (High K^+ in Blood)
 - Stress



Source: Unattributable

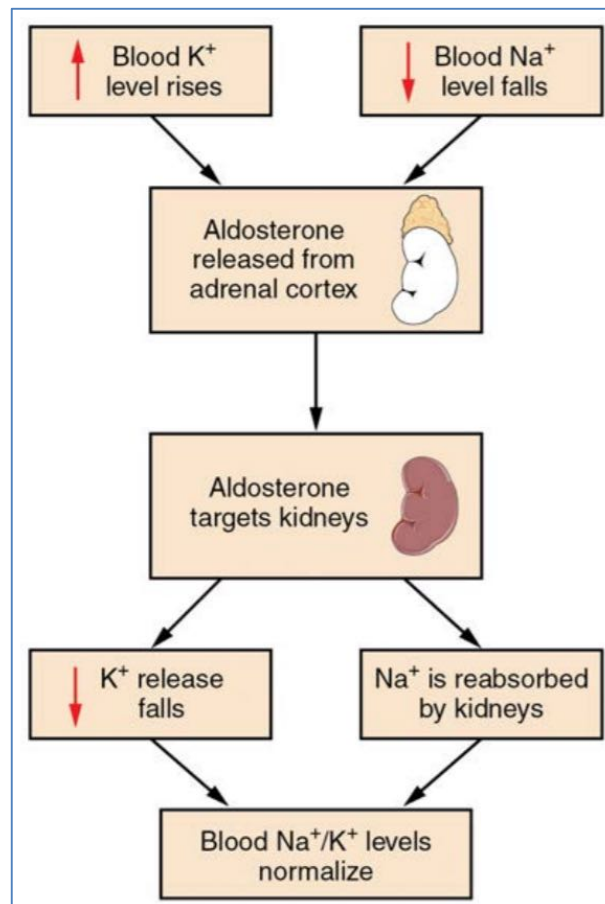


Case study notes, Acid base balance study; fundacionhenrydunant.org

- **Effects:**
 - Increases Na⁺ Reabsorption of the Principal Cells of the Distal & Collecting Ducts of the Nephron.
 - **If Aldosterone is High** – All Na in Filtrate is reabsorbed
 - **If Aldosterone is Low** – No Na in Filtrate is reabsorbed
- **Works by:**
 - **ACTIVATING the Na/K-ATPases in the Principal Cells of Distal & Collecting Ducts:**
 - Increases Na⁺ & Cl⁻ Reabsorption
 - Increases K⁺ Secretion
 - **Promotes Na⁺-Channel Synthesis & Insertion into Luminal Membrane:**
 - Facilitates the Na⁺ Reabsorption mentioned above.

Regulation of K^+ : The Primary Intracellular Electrolyte:

- Primary Roles in Normal Neuromuscular Function, Membrane Potentials & Membrane Transport.
- **Deficient Intracellular K^+ :**
 - o Cell membrane will be more Negative than normal (Ie: *Hyperpolarised*)
 - o Therefore it'll be harder to initialize an action potential as it takes more to reach threshold.
- **Excess Intracellular K^+ :**
 - o Cell membrane will be more Positive than normal (Ie: *Depolarised*)
 - o Therefore it'll be easier to initialize an action potential as it takes less to reach threshold.
- **Effect on the Heart:**
 - o The heart is particularly sensitive to K^+ Levels.
 - o Both Too High & Too Low K^+ Levels will Disrupt Electrical Conduction of the Heart → Can be Fatal.
- **Regulating K^+ Levels:**
 - o Relies solely on K^+ Secretion by the "**Principal Cells**" in the Collecting Ducts of the Kidneys.
 - o **Principal Cells Detect $[K^+]$ in the Blood:**
 - High Blood $[K^+]$ → K^+ Secretion is Increased
 - High Blood $[K^+]$ → K^+ Secretion is Decreased
 - o **Adrenal Glands Detect $[K^+]$ in the Blood:**
 - High Blood $[K^+]$ DIRECTLY Stimulates **Aldosterone** Release from Adrenal Cortex.
 - o **Aldosterone** → Activates Na^+/K^+ -ATPase's in the Distal Tubules & Collecting Ducts:
 - This Increases Reabsorption of Na^+ , Cl^- & H_2O from Distal Tubule → Interstitium
 - But ALSO causes Secretion of K^+ into the Filtrate.



<https://www.shivajicollege.ac.in/sPanel/uploads/econtent/0f09b17521b6e0f9ec6f25ab9c65f78d.pdf>

ENDOCRINE REGULATION OF GROWTH

ENDOCRINE REGULATION OF GROWTH

Influences on Growth:

- Genetics
- Nutrition
- **Endocrine Factors**
 - o (Growth Hormone)
 - o (IGF's – Insulin-like Growth Factors)

Phases of Growth:

- **Note:** These Differ in their *Rates of Growth* and *Regulators/Contributors*:

Phase of Growth	Major Regulators/Contributors		
	Nutrition	Hormonal	Genetics
Foetal (In Utero)	Yes - #1	Insulin (Acts as a growth factor in this phase) IGF-I	No
Infantile (Birth → 2yrs)	Yes - #1	GH & IGF is present, but in low amounts – NOT Imperative.	Yes – (Only after a few months after birth)
Pre-Pubertal (Childhood)	-Ve influence only if malnourished	- IGF Levels Increase - GF Receptors Increase	Yes - #1
	Note: Growth <i>Velocity</i> progressively declines during this phase (Transition from Infant → Child) Note: Body Proportions start to change.		
Pubertal (Early Teens)	-Ve influence only if malnourished	Sex Hormones: - →GH Release - →Epiphyseal Closure GH → Causes IGF Release GH + IGF → Bone Elongation	
Post-Pubertal (Late Teens)	Note: Growth <i>Velocity</i> peaks & then stays same for ≈6yrs. - (The last 3 years mainly concern the Trunk)		

Major Hormones involved with Growth:

- Growth Hormone, AKA: Somatotropin
- Insulin-like Growth Factors (Somatomedins) (IGF-I & IGF-II)
- Somatostatin (Inhibits secretion of GH from Ant. Pit.)
- Thyroid Hormone
- Cortisol – (Not Direct – has a 'permissive' role. Ie: Other growth hormones are more effective if it's present)
- Sex Hormones (Oestrogen/Testosterone)

Note: Somatostatin = The Primary Inhibitor of Growth:

- Produced By Numerous Organs, Including:
 - o Hypothalamus,
 - o Placenta,
 - o & Pancreas
 } Ie: In organs that Release other Hormones (Ie: Somatostatin = A safeguard to Mass-Release of Hormone) – (Paracrine)
- Primary Actions Include (but not limited to):
 - o (Mostly an 'Inhibitory Hormone' - Can be *Endocrine/Paracrine/Autocrine/Luminal/Neurohormone*)
 - o **In Pancreas:** Inhibits Insulin Release; Also Inhibits Glucagon Release; Also inhibits Exocrine Pancreas
 - o **In Anterior Pituitary:** Inhibits GH release, TSH release, Prolactin release, & ACTH release
 - o **In GI Tract:** Slows gastric emptying, Slows peristalsis, Suppresses pancreatic hormones
- Release is Stimulated By:
 - o Peptide 'Urocortin3' (Ucn3) released by Beta Cell to inhibit insulin release
 - o Cholecystokinin (CCK) released by Antral D-Cells
 - o Many other locally-relevant stimulants; (research ongoing)
- Release is Regulated By:
 - o Locally-relevant factors; (research ongoing)

Hypothalamic Hormone of Growth:

- **GHRH (Growth-Hormone Releasing Hormone):**
 - **Produced Mainly in:** Hypothalamus (But also in GIT, Pancreas & Placenta)
 - **GRF Release Stimulated By:**
 - Dopamine
 - GABA
 - α -Adrenergic Agonists
 - **Exerts Effects on:** Somatotropes (Anterior Pituitary) → ↑ Growth Hormone Release.

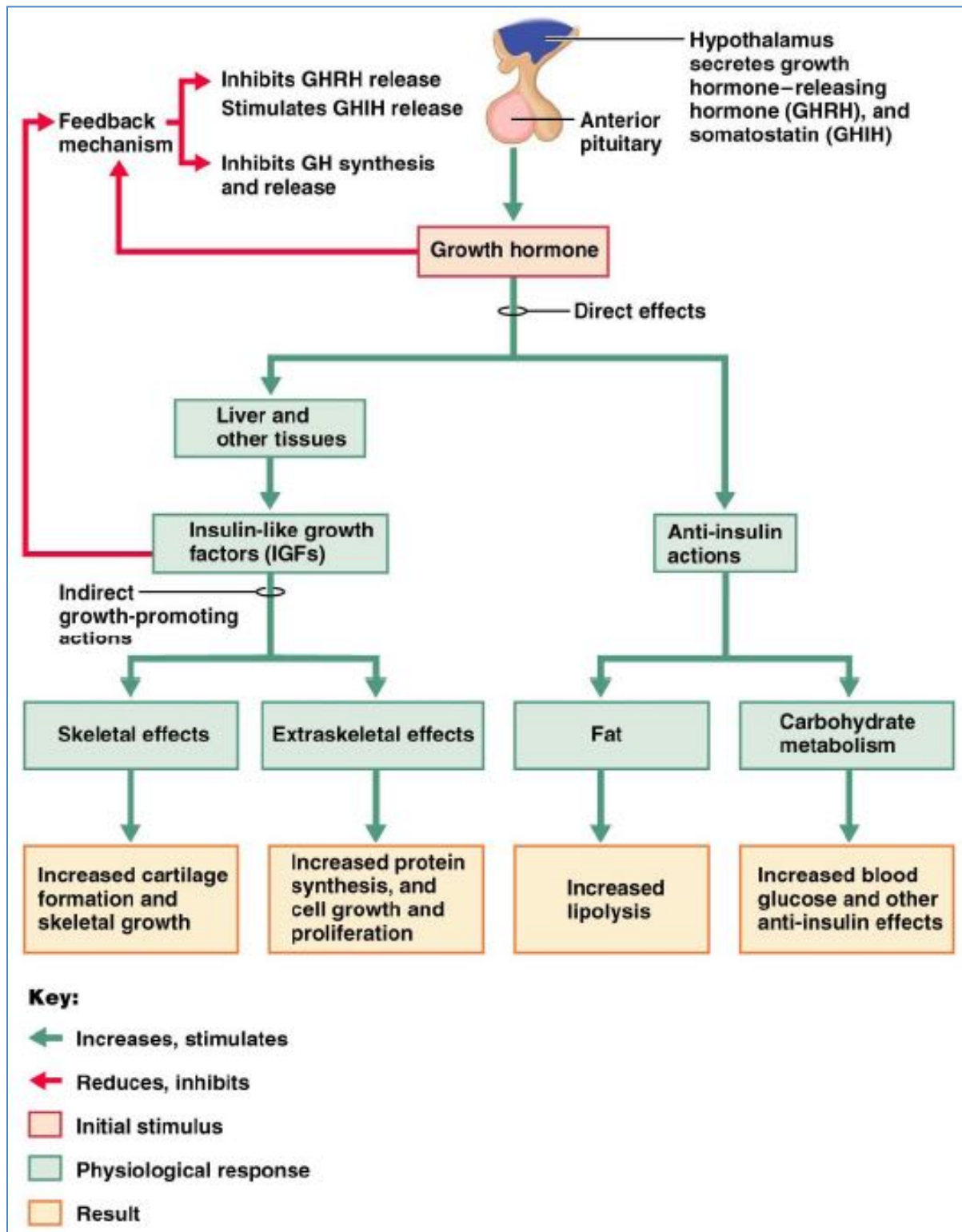
Anterior Pituitary Hormone of Growth:

- **Growth Hormone:**
 - **Produced by:** Anterior Pituitary (After ≈2mths old)
 - **Blood Transport:** ≈45% bound to Carrier Protein – (Prolongs ½ Life)
 - **Regulation of Release:**
 - **Stimulation:** GRH - (Growth-Hormone Releasing Hormone)
 - **Inhibition:** Somatostatin
 - **Actions:**
 - **Growth-Promoter** from *Early Childhood* → *Onwards*
 - Longitudinal Bone Growth & Remodelling
 - Skeletal Muscle Growth
 - Liver Growth
 - Stimulates IGF-Binding Protein Synthesis (Important carrier for IGF)
 - Stimulates IGF Synthesis
 - **Metabolic Effects:**
 - Stimulates:
 - Lipolysis
 - Ketogenesis
 - Gluconeogenesis
 - Protein Synthesis
 - Lactation
 - Inhibits Insulin Action.
 - Boosts Immune Function.
 - **Organs Stimulated by Growth Hormone:**
 - Liver - Hepatocyte Growth Factor
 - Chondrocyte - Fibroblast Growth Factor
 - Kidney - Epidermal Growth Factor
 - Pancreas - Nerve Growth Factor Receptor Production
 - Thymus - IL-1 (Interleukin 1) → Bone Marrow Proliferation
 - Uterus - Oestrogen Receptor Production
 - Breast - Oestrogen Receptor Production

Liver & Other Tissues:

- **Insulin-Like Growth Factors (IGF's):**
 - Both IGF-I & IGF-II are **Structurally Similar to Proinsulin** (The Insulin Precursor)
 - IGF-I - Chromosome 12
 - IGF-II - Chromosome 11
 - Circulates bound to **IGFBP (Insulin-like Growth Factor Binding Protein)**
 - Bind to **Specific Receptors**
 - **Stimulate Cell Division** together with other Growth Factors.
 - **Foetal Life:**
 - Act in **Paracrine** Fashion
 - IGF made by all foetal tissues (However, mainly by liver after birth)
 - **Absence of IGF-I in Foetal Life** → **Intra Uterine Growth Retardation**

The Growth Hormone Axis:



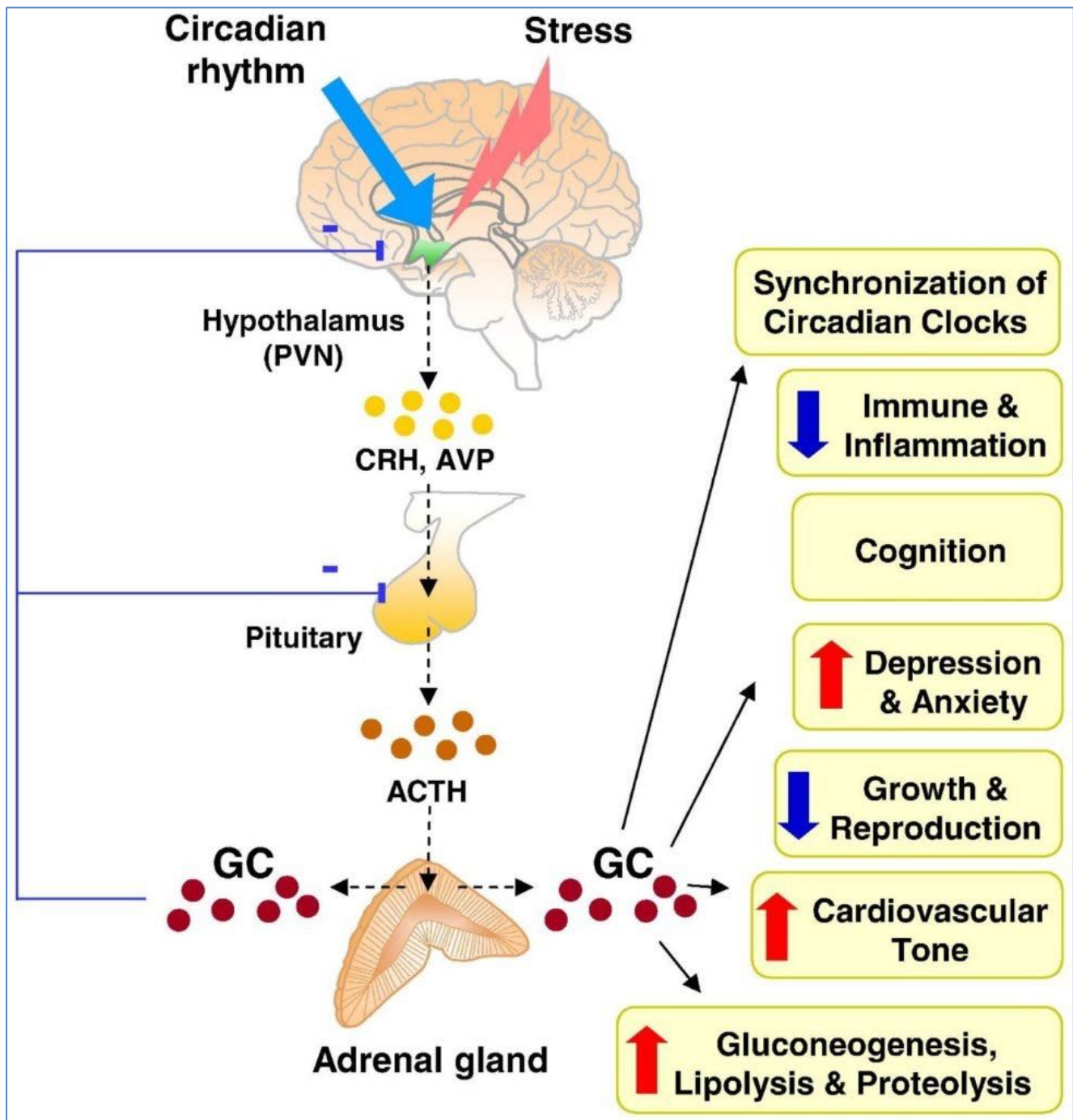
<https://www.austincc.edu/apreview/PhysText/Endocrine.html>

PHYSIOLOGICAL RESPONSE TO STRESS

PHYSIOLOGICAL RESPONSE TO STRESS

Stress & The Hypothalamo-Pituitary Axis:

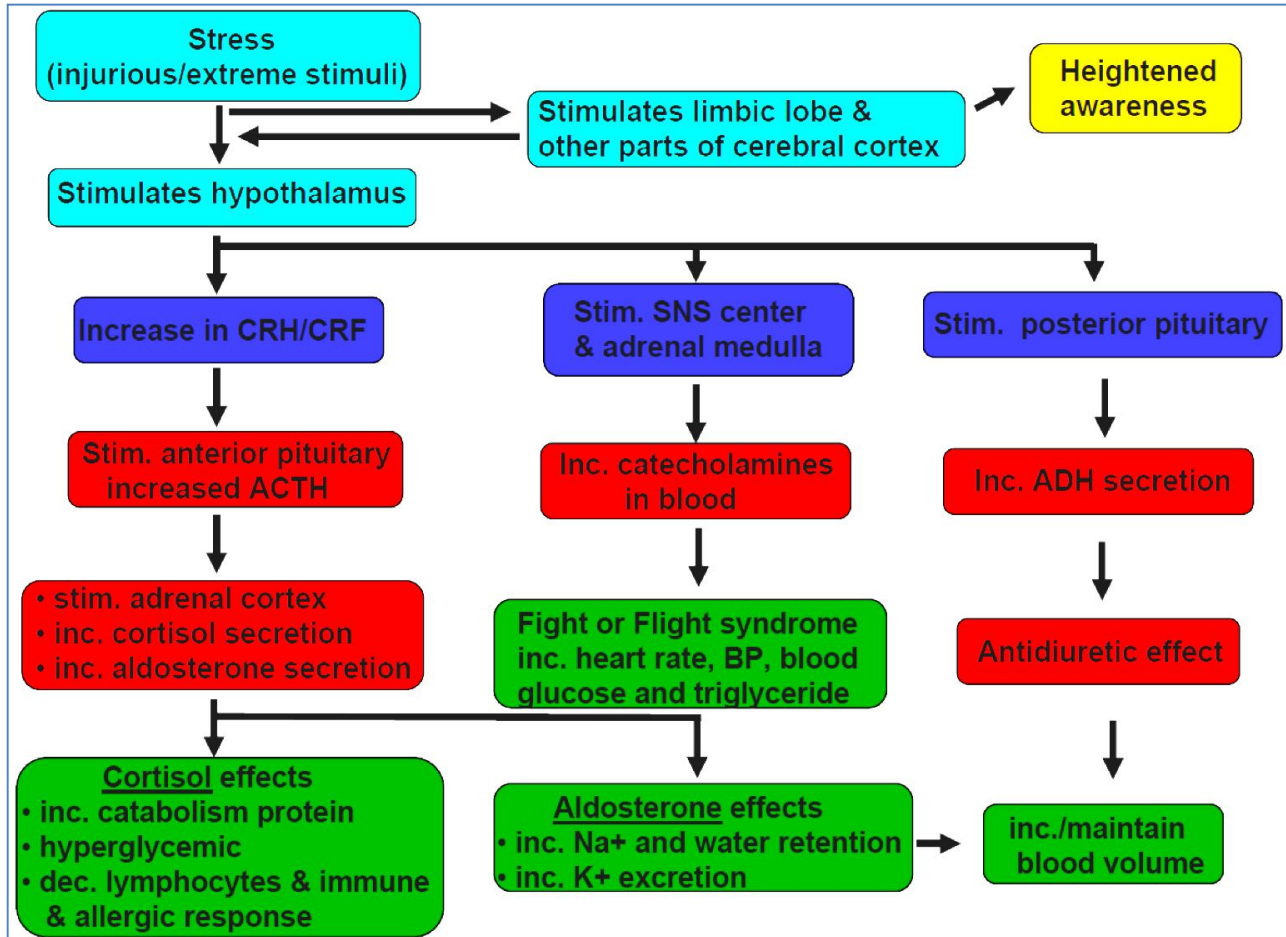
- 1- Stressors (Internal or External) trigger Receptors.
- 2- Receptors inform the Hypothalamus
- 3- Hypothalamus
 - Activates Sympathetic Pathways
 - Secretes Corticotrophin-Releasing Hormone → Ant. Pituitary releases ACTH.
- 4- Both Sympathetic Activation & ACTH Release → Stimulate the Adrenal Glands to produce Glucocorticoids.
- 5- Adrenal Glands
 - Secrete Catecholamines (Incl. Adrenaline)
 - Secrete Cortical Steroids (Incl. Cortisol)



Circadian rhythm of adrenal glucocorticoid Figure on ResearchGate;

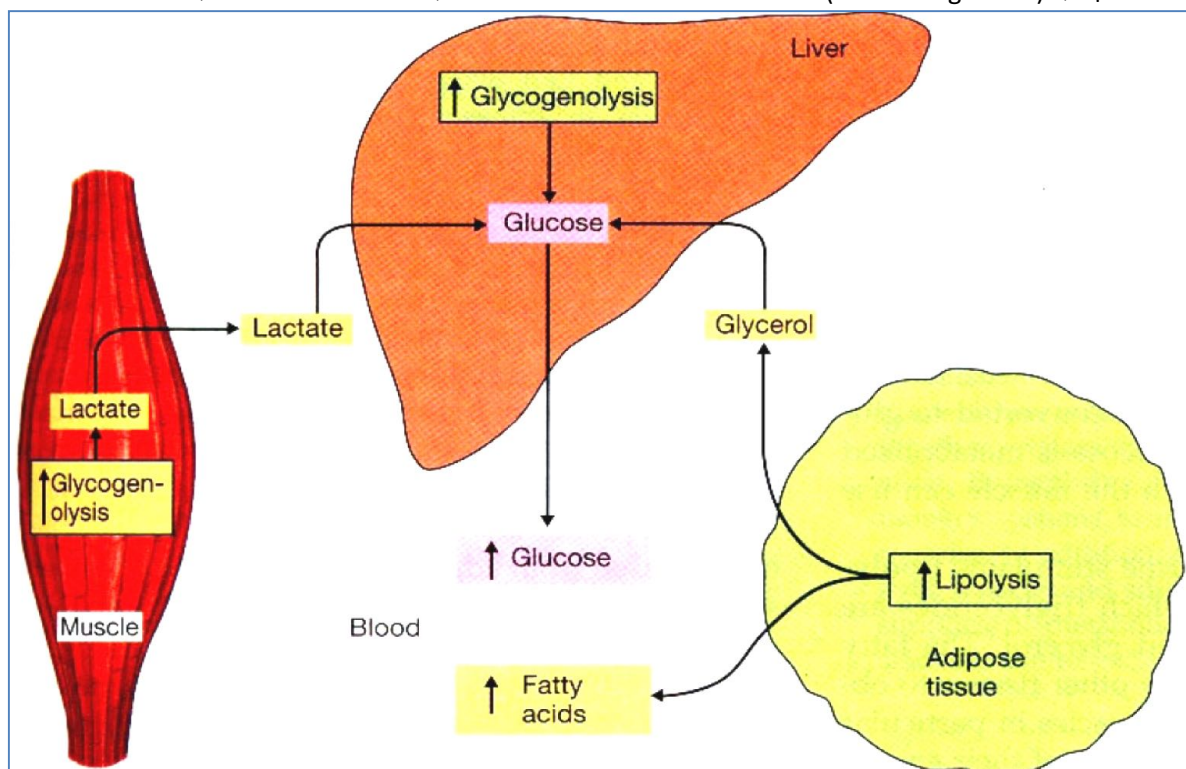
https://www.researchgate.net/figure/Neuroendocrine-regulation-of-adrenal-GC-and-its-physiological-roles-GC-is-primarily_fig3_49835615

Schematic of the Body's Response to Stress:



Metabolic Actions of Adrenaline/Epinephrine (Fight/Flight Response Hormone):

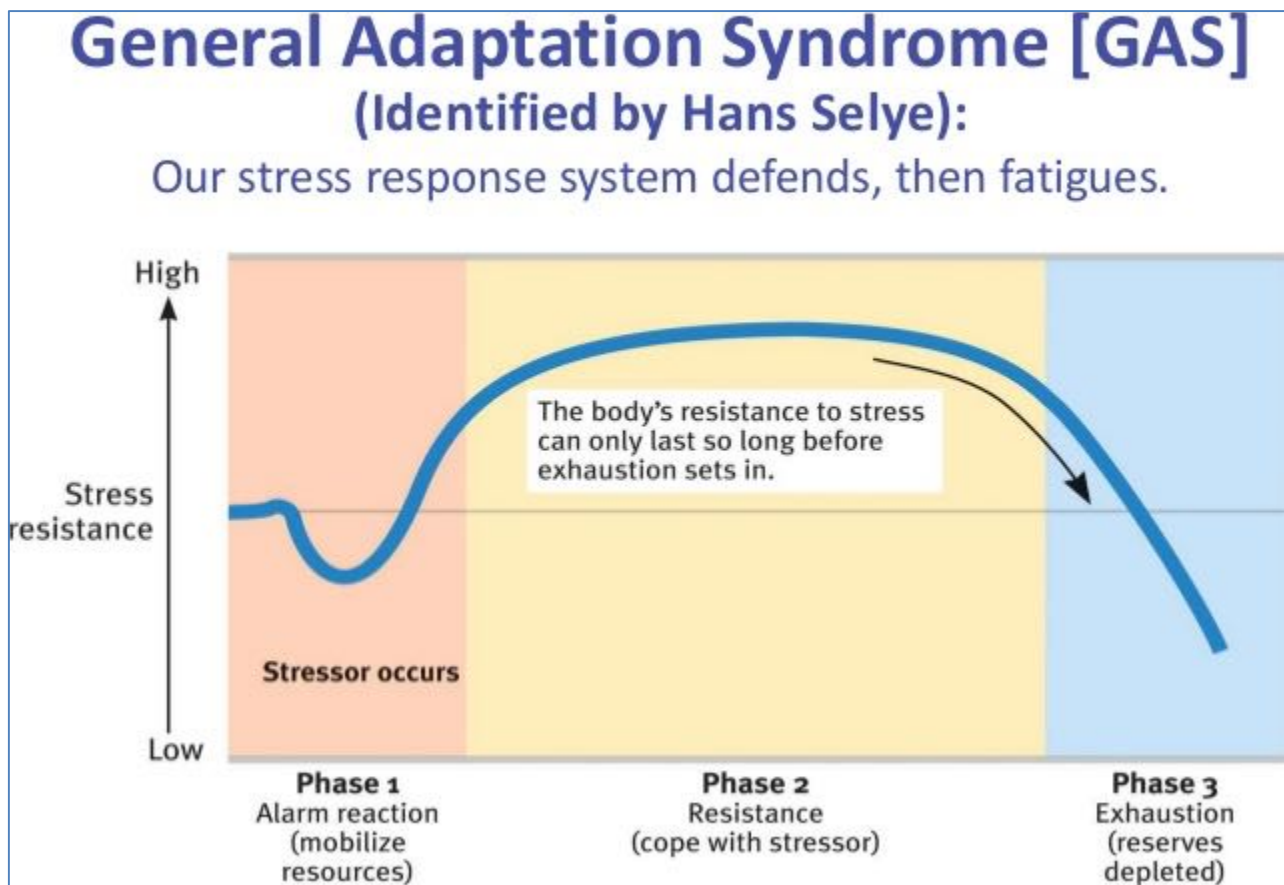
- ↑Glycogenolysis (Liver) → ↑Blood-Glucose. } ↑Energy Precursor
- ↑Lipolysis (Adipose) → ↑Free Fatty-Acids. } Levels in Blood
- ↑Glycogenolysis (Muscle)
 - → Fuels Muscle Cells
 - → Provides Lactate → Liver converts back to Glucose (Gluconeogenesis) → ↑Blood-Glucose



Source: Unattributable

The **General Adaptation Syndrome (GAS)** Theory:

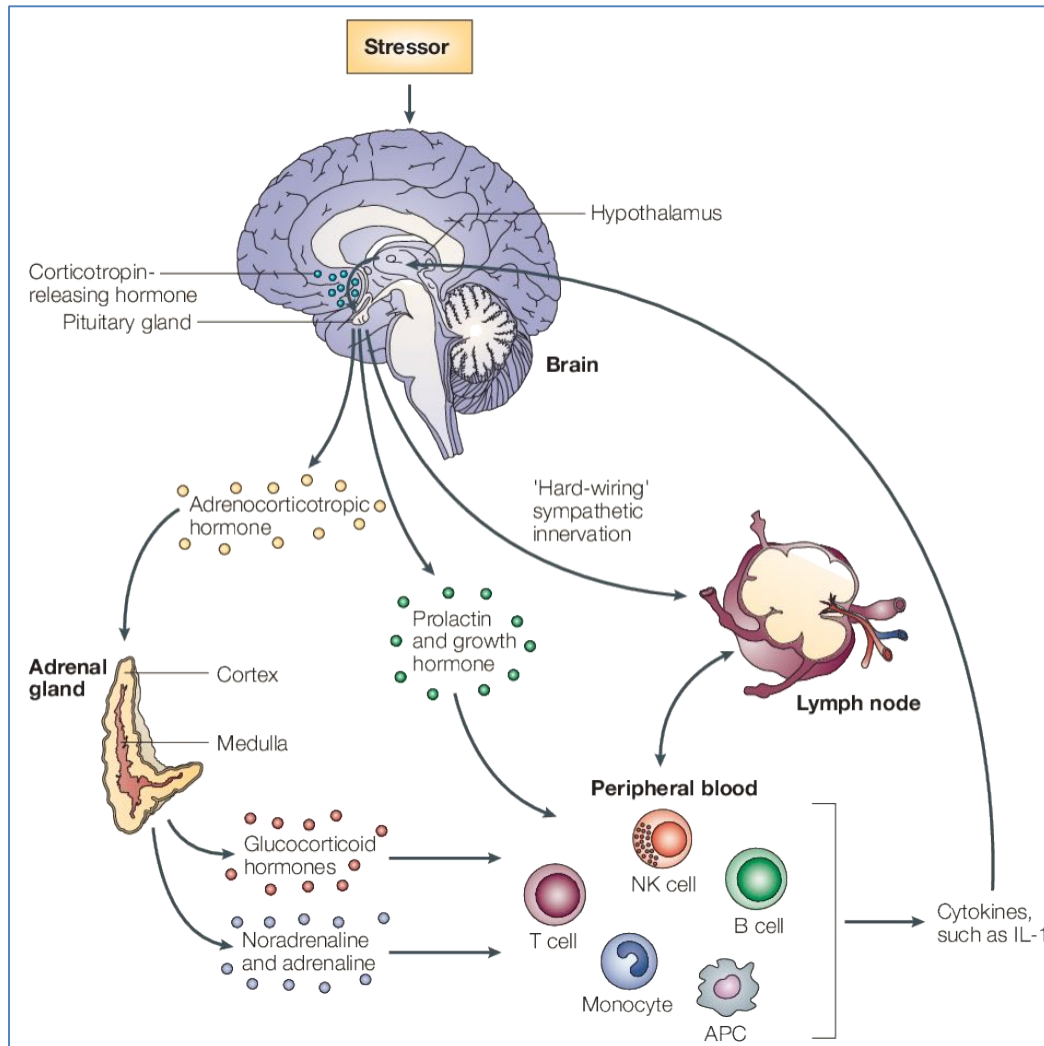
- Dr Hans Selye proposed the “**General Adaptation Syndrome**” as the Body’s Responses to Stress
- He also noticed 3 Universal Symptoms of Chronic Stress:
 - o Adrenal Cortex Enlargement
 - o Atrophy of Lymphoid Tissues
 - o Bleeding Ulcers in Stomach & GI Tract.
- **Overview:**
 - o Stress → Causes Physiological Changes → Causes Symptoms
 - o There are 3 stages. Note: If the stress is overcome during one of the stages, the ‘GAS’ will terminate in that stage
- **3 Stages of the General Adaptation Syndrome:**
 - o **Stage 1: ALARM REACTION:**
 - When we are surprised or threatened → Immediate Physical Reaction.
 - Fight or Flight Response
 - Prepares the body for life-threatening situations, channelling resources away from things like the Digestive & Immune Systems, to more immediate muscular needs.
 - **↑ Sympathetic Nervous System**
 - **↑ Catecholamines from Adrenal Medulla**
 - o **Stage 2: STAGE OF RESISTANCE:**
 - If stressors continue, the body enters the *Resistance Phase*, where we *feel* like we’ve adapted to the stressors, but the body is working at abnormally high levels to keep up with the ↑ demands.
 - **↑ Cortisol Secretion**
 - **Sustained Catecholamine Actions**
 - o **Stage 3: STAGE OF EXHAUSTION:**
 - Eventually, the body gives up on maintaining a high level of stress. Parts of the body literally start to break down → Sickness → Possible Death.
 - **↓ Adaptive Endocrine & Neuroendocrine Functions**



<https://sanescohealth.com/blog/general-adaptation-syndrome-stages/>

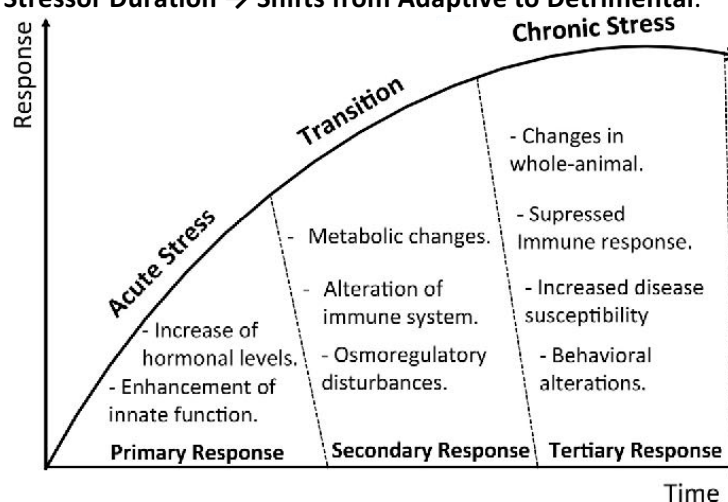
Effects of Stress on The Immune System:

- Similar to GAS theory, **Acute Stress ENHANCES** the Immune System, but **Chronic Stress SUPPRESSES** it
- **The Effect of Stress on the Immune System is 'BIPHASIC':**
 - **1- During Acute Stress** – There is a shift towards ↑ Innate Immune Responses.
 - (↑ Granulocyte/Macrophage/NK-Cell Activity + ↑ Complement & Acute-Phase Proteins)



Glaser, Ronald and Janice K. Kiecolt-Glaser. "Stress-induced immune dysfunction: implications for health." *Nature Reviews Immunology* 5 (2005): 243-251.

- **2- If Stress Continues** – There is a shift from *Cellular* Immunity to *Humoral* Immunity.
 - ↓ Type-1 Helper T-Cell Activity (→ Become Macrophages)
 - ↑ Type-2 Helper T-Cell Activity (→ Become Plasma Cells → Secrete Antibodies)
 - **3- If Chronic Stress** – There is a Decrease in almost all functional Immune Responses
- Hence: Increase in Stressor Duration → Shifts from Adaptive to Detrimental.**



Source: Unattributable

REPRODUCTIVE ENDOCRINOLOGY

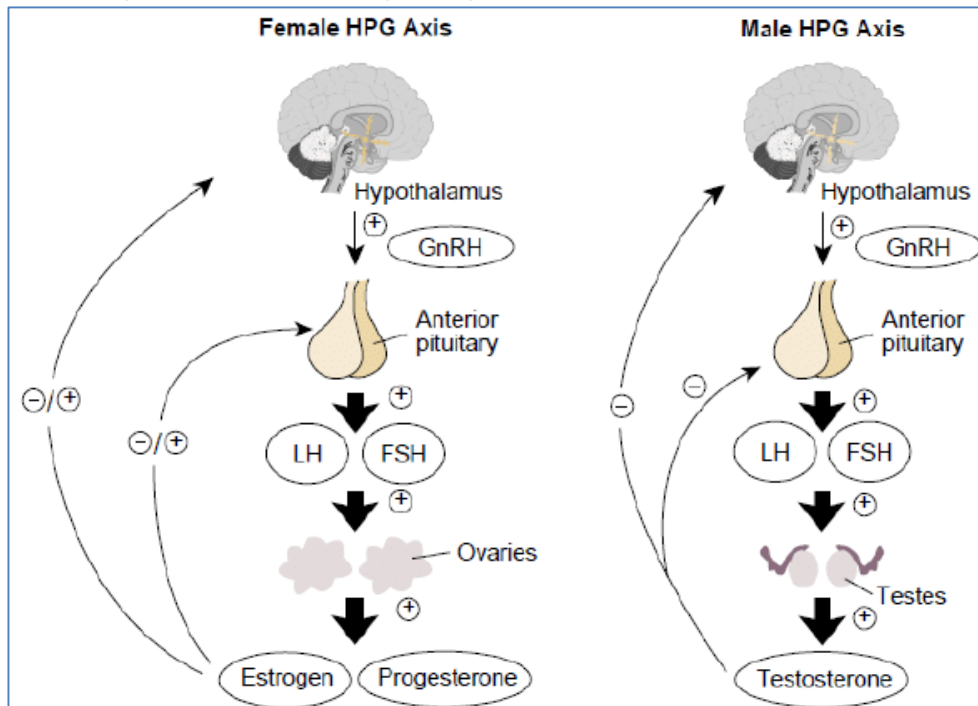
REPRODUCTIVE ENDOCRINOLOGY:

The Hypothalamic Sex Hormone:

- **Gonadotropin-Releasing Hormone (GnRH):**
 - Peptide Hormone
 - Pulsatile Release (≈ 90 mins)

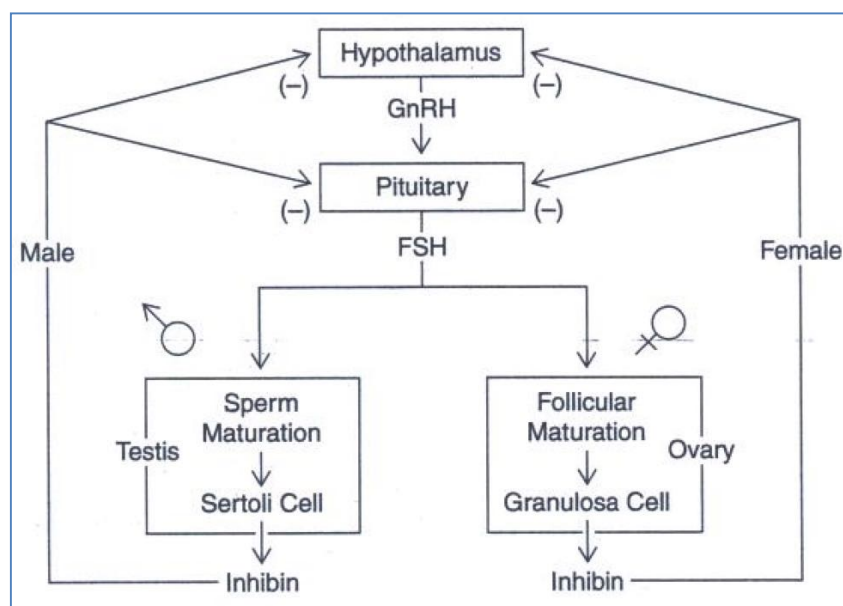
The Pituitary Sex Hormones:

- **Gonadotropins - FSH & LH:**
 - Are *Glycoproteins*
 - Released by the Anterior Pituitary in response to Pulsatile release (90mins) of **GnRH**.



Available from: https://www.researchgate.net/figure/Schematic-representation-of-the-hypothalamic-pituitary-gonadal-HPG-axes_fig5_268791949

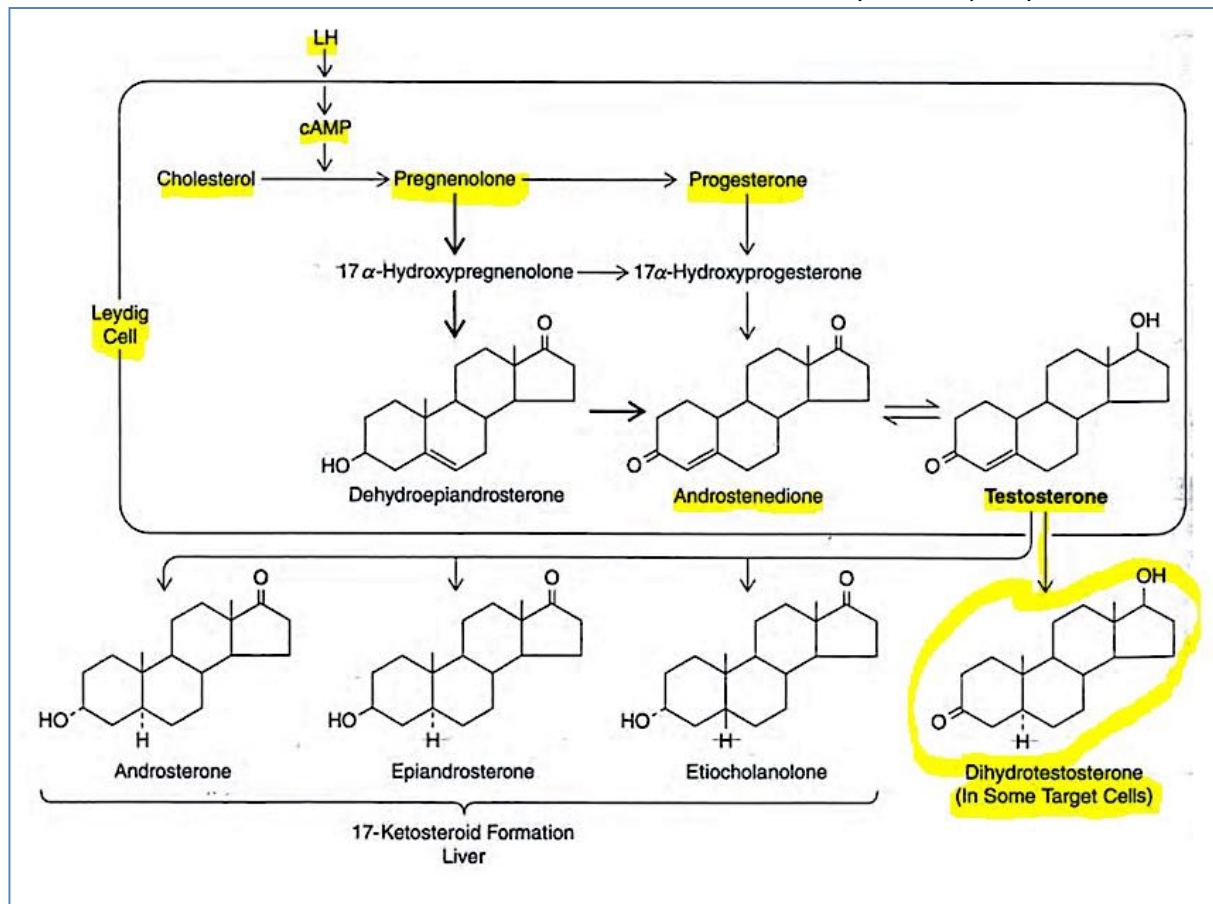
- **Inhibin:**
 - Produced by the Sustentacular/Sertoli Cells (Male) & Granulosa Cells (Female).
 - Released in response to high FSH.
 - Inhibits FSH release via Hypothalamic Inhibition.



- **Note: Sex Hormone Binding Globulin is an Important Transporter:**
 - Secreted by the Liver
 - Increased by ↑Oestrogen
 - Decreased by ↑Androgen
 - Constant in Males
 - Cyclical in Females – (but During Pregnancy, ↑↑Oestrogen → ↑SHBG)

The Important Androgens:

- **#1 – Testosterone** - Affects Mainly the Testes
 - o 40% - Bound to SHBG (Sex-Hormone Binding Globulin)
 - o 60% - Bound to Albumin
 - o 2% - Free (Active) – (Receptors are intracellular ∴ Must be able to enter the cell)
- **Dehydroepiandrosterone (Sulphate) – DHEA(S)** - Affects Mainly the Periphery
- **Androstenedione** - Affects Mainly the Periphery



- **Actions of Androgens:**
 - o **Primary Sex Characteristics:**
 - Growth & Maturation of Reproductive Tract @ Puberty
 - Maintenance of Reproductive Tract in Adulthood
 - Libido
 - Enhance Spermatogenesis
 - o **Secondary Sex Characteristics:**
 - Body Hair
 - Deep Voice
 - Thick, rough skin
 - Bone Growth
 - Androgen Binding Protein Synthesis (in Sertoli/Sustentacular Cells)
 - ↑Musculature
- **Measuring Androgen Levels – “The Free Androgen Index”:**
 - o Gives a measure of the “free” active fraction of Androgens.

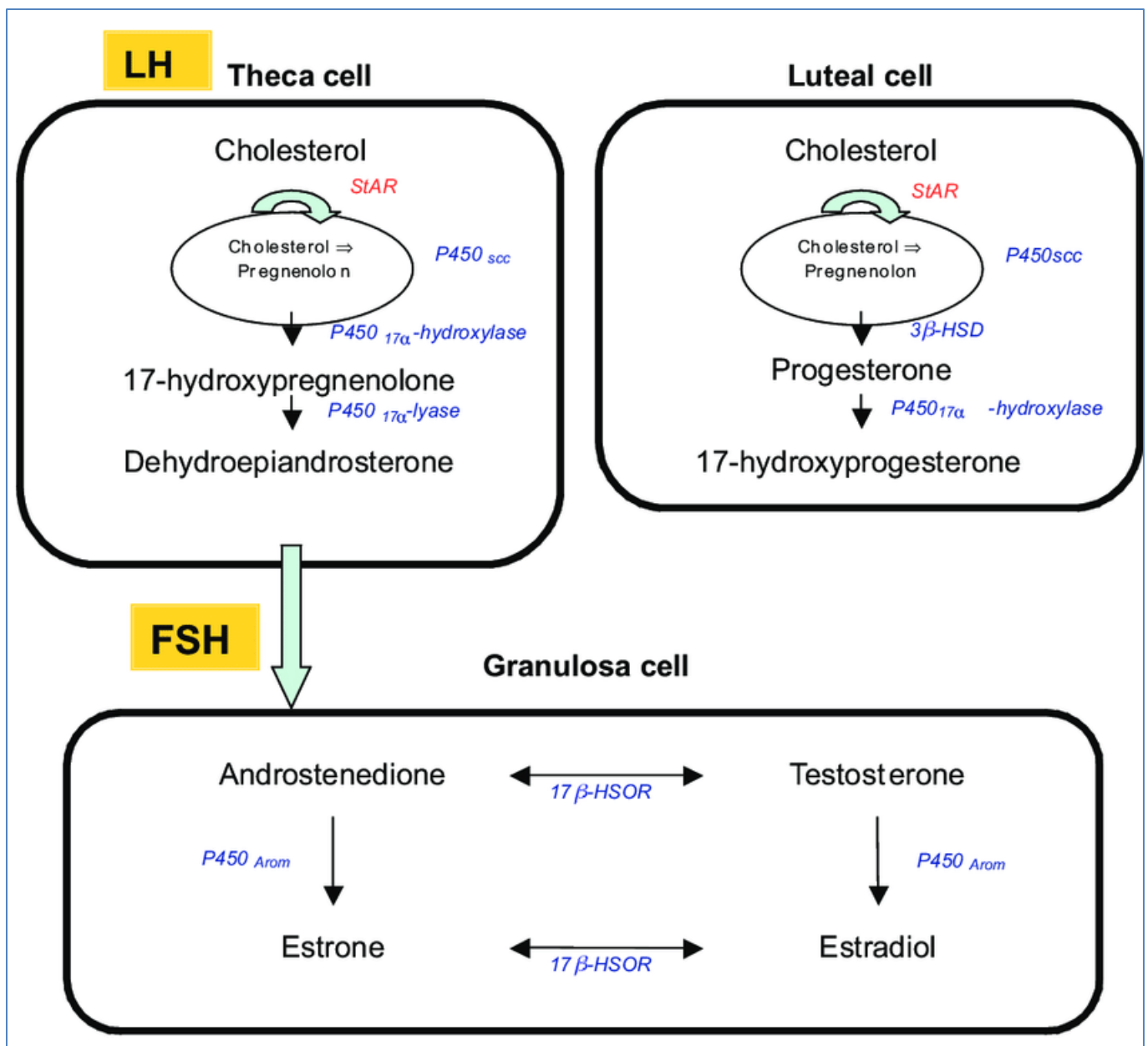
$$\frac{\text{Testosterone (nmol/l)}}{\text{SHBG (nmol/l)}} \times 100$$

Female 0 – 11
Male 25 – 190

SHBG = sex hormone binding globulin

The Important Female Reproductive Hormones:

- **What are they?**
 - Oestrogen/Estrogen (3 types: Estradiol, Estrone, Estriol)
 - Progesterone
- **Secreted By:**
 - Produced mainly by Ovaries (Corpus Luteum)
 - Some Estrogen is also produced in...
 - Adrenal Cortex
 - Adipose Tissue
 - Placenta (During Pregnancy)
- **Synthesis:**
 - **All Are Steroid Hormones...**
 - **Therefore, are produced from Cholesterol**
 - 1- **Theca Cells** (of ovarian follicle) convert Cholesterol to Pregnenolone
 - 2- Some Pregnenolone converted directly to → **Progesterone**
 - 3- Rest of Pregnenolone converted to → Dehydroepiandrosterone (DHEA) → Androstenedione
 - 4- Androstenedione diffuses to nearby **Granulosa Cells** which converts to → **Testosterone**
 - 5- **Testosterone** converted to → **Estradiol**
 - 6- **Estradiol** → Released into blood (bound to SHBG – *Sex Hormone Binding Globulin*)

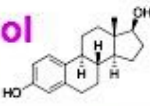
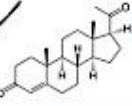


- Primary Actions of Estrogen/Oestrogen:

- **Puberty:**
 - Maturation of Female Reproductive Organs (uterus, fallopian tubes, vagina)
 - Secondary sexual characteristics (e.g. breast growth, fat distribution)
- **Menstruation:**
 - ↑ Estrogen (pre-ovulation) → prepares uterine epithelium for implantation (endometrial proliferation); endometrial secretion in collaboration with progesterone
 - Dominant hormone during the follicular phase of ovarian cycle; follicle maturation; initiates ovulation via FSH, LH surge
- **Pregnancy:**
 - Placenta secretes Estrogen to stimulate Myometrium development
 - Promotes cervical 'ripening' prior to labour.
 - Increases uterine responsiveness to oxytocin leading up to labour
 - Supports mammary duct growth
- **Systemic:**
 - Anabolic effect on bones
 - Promotes skin elasticity, blood vessel flexibility
 - Promotes fat deposition

- Primary Actions of Progesterone:

- **Menstruation:**
 - Dominant hormone during **Luteal/Secretory Phase** of ovarian cycle
- **Maintains Pregnancy:**
 - Reduces Myometrium irritability (Prevents pre-term labour)
 - Thickens Cervical Mucous Plug
 - Promotes breast gland development & milk production
 - Increases respiratory centre's sensitivity to CO₂ → Raises basal respiratory rate.
- **Systemic:**
 - Augments Estrogen's anabolic effect on bones → Increased bone density
 - Promotes skin elasticity

	Estradiol E2 	 Progesterone P4 
Cells and Tissues	Growth with Proliferation	Maturation with ↓ Proliferation
Uterus–Endometrium	Proliferative	Secretory
Uterus–Cervix	↑ Mucus Volume and Stretch	↓ Mucus Volume and Stretch
Breast–Epithelial	Proliferation	↓ Proliferation, Maturation
Breast–Lobular Alveolar (Areola and Nipple)	↑ Breast Volume	↑ Areolar Size
Bone Remodeling	↓ Resorption	↑ Formation
Cardiovascular—Vascular Endothelial Function	↑ FMD	↑↑ FMD
Cardiovascular—Electrical Function, QT	↑ QT Interval	↓ QT Interval
Brain	Excitation/Activation	Excitation/Calming
Sleep	?	↑ Deep Sleep
Central NE	↑	Likely ↓

- Estrogen/Progesterone Release is Stimulated By:

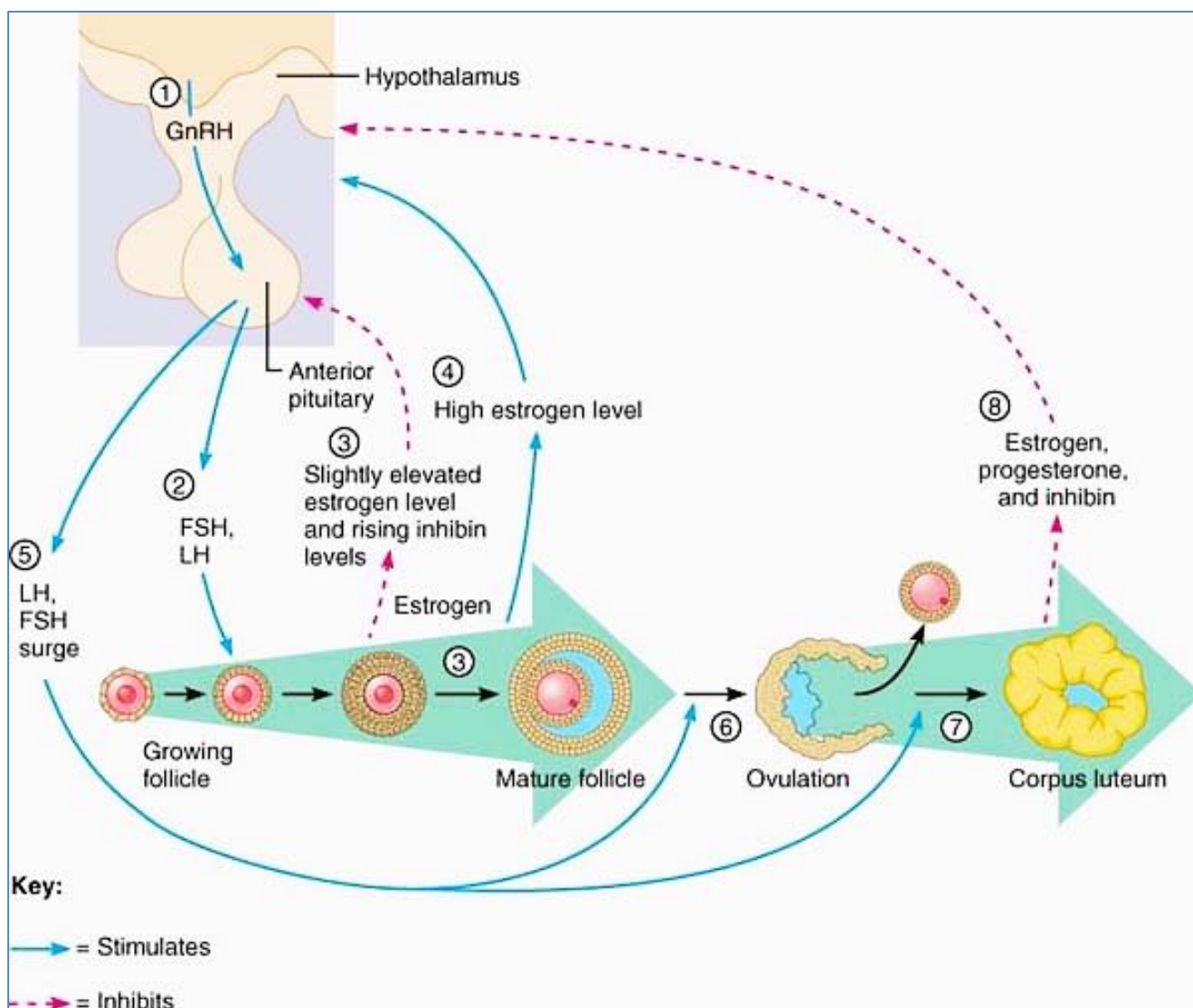
- Pulsatile releases of GnRH from Hypothalamus → FSH/LH Surges from Anterior Pituitary
- FSH/LH → Stimulates maturation of Ovarian Follicles → Secrete Estrogen & Progesterone

- Release is Regulated By:

- Hypothalamo-Pituitary-Gonadal Feedback Axis (See over the page)

Hormonal Regulation of the Ovarian Cycle:

- 1) **Day 1** – Hypothalamus increases levels of GnRH (Gonadotropin-Releasing Hormone) and stimulates the Anterior Pituitary to produce FSH (Follicle-Stimulating Hormone) & LH (Luteinising Hormone).
- 2) FSH & LH stimulate follicle growth, maturation & oestrogen secretion.
 - FSH targets follicle cells
 - LH targets the thecal cells – makes thecal cells produce androgen.
 - o Androgen diffuses through basement membrane, where the granulosa cells convert it to oestrogens.
- 3) Medium oestrogen levels exert **negative feedback** to the Ant. Pituitary, inhibiting FSH & LH release.
 - Inhibin released by granulosa cells also exerts **negative feedback** on FSH release.
- 4) At a critically high oestrogen level, **positive feedback** is exerted on the brain & Ant. Pituitary.
- 5) **Midcycle** – This positive feedback causes the Ant. Pituitary to release a sudden burst of LH (and also some FSH – role midcycle is currently unknown).
- 6) **LH surge** stimulates the primary oocyte of the dominant follicle to complete MEIOSIS I, forming a secondary oocyte + first polar body.
 - **LH surge** also triggers ovulation.
 - After ovulation, oestrogen levels decline due to the damaged dominant oestrogen secretor.
- 7) **LH Surge** also transforms ruptured follicle into corpus luteum – stimulates it to produce progesterone & oestrogen.
- 8) Corpus luteum secretes inhibin along with progesterone & oestrogen, exerting a strong negative feedback signal to the Ant. Pituitary – Stops the release of LH & FSH.
 - **End of cycle** – LH levels fall → corpus luteum degenerates → no oestrogen or progesterone production by Corpus Luteum → no negative feedback to the hypothalamus → hypothalamus increases FSH & LH levels → Back to square #1-



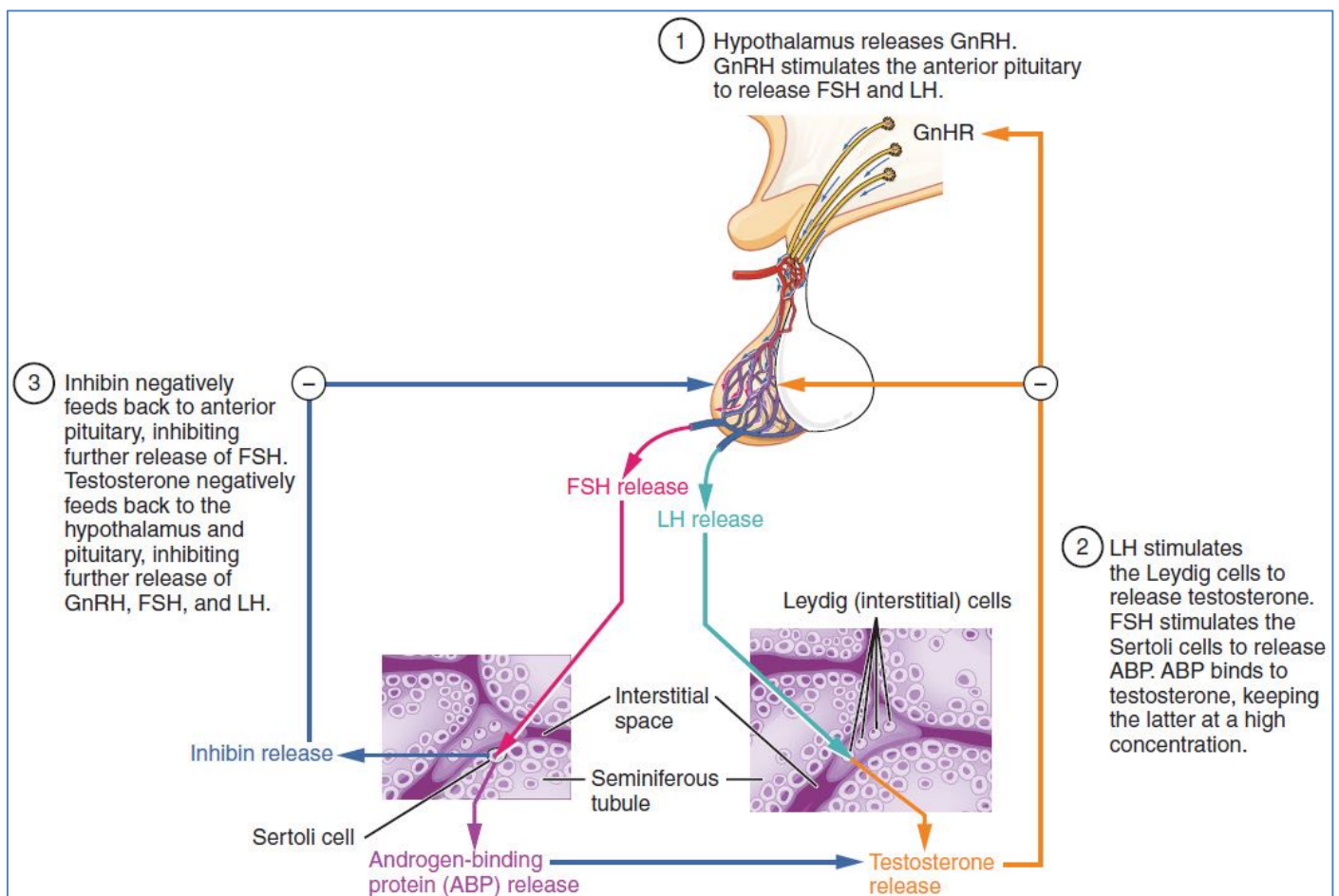
Source: Unattributable

Hormonal Regulation of Spermatogenesis:

- 1) **Hypothalamus** releases **GnRH (gonadotropin-releasing hormone)** which-
- 2) stimulates the release of gonadotropins: **FSH** (Follicle stimulating hormone) & **LH** (Luteinizing hormone).
- 3) **FSH**: stimulates sustentacular cells to release **Androgen-binding protein (ABP)** → Makes spermatagonium, spermatocytes, and spermatozoa **receptive to the androgen: Testosterone**.
- 4) **LH**: stimulates the **interstitial (Leydig) cells** [Basally external to Seminiferous tubules] to **produce testosterone** which **triggers & maintains spermatogenesis**.
- 5) **Testosterone** produced by Leydig (interstitial) cells **inhibits GnRH** production; as does **Inhibin**, produced by the sustentacular (sertoli) cells.

- When testosterone is at its peak → **sperm count is high (20Mil⁺)** → inhibin levels rise → GnRH decreases → FSH & LH levels decrease → Testosterone & ABP levels decrease → **spermatogenesis slows**.

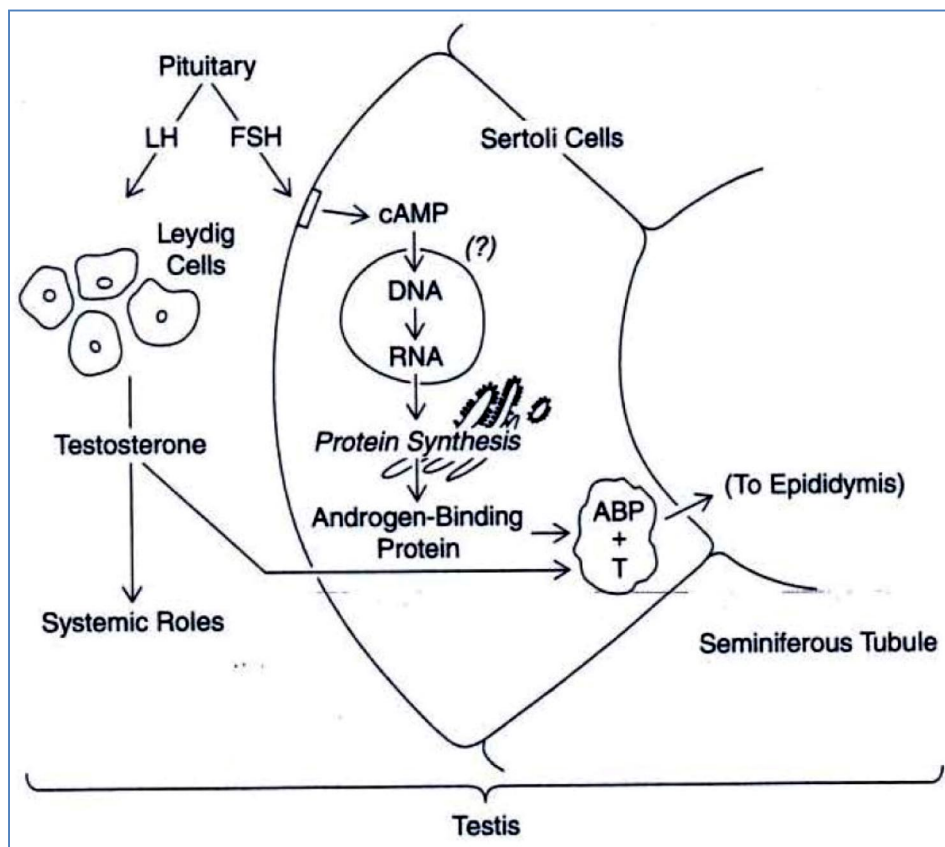
-When **sperm count is low (20Mil⁻)** → inhibin & testosterone levels are low → no negative feedback to hypothalamus → hypothalamus releases GnRH → Ant. Pituitary releases LH & FSH → FSH stimulates sustentacular (sertoli) cells to produce ABP; LH stimulates the interstitial (Leydig) cells to produce testosterone → Testosterone + ABP stimulates spermatogenic cells → **Spermatogenesis increases**.



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Functional Micro-Anatomy of the Testes:

- **Leydig Cells (In Interstitium of the Testes):**
 - #1 Function = Produce Testosterone (Stimulates Spermatogonia to enter Spermatogenesis)
 - Stimulated by **LH** (Luteinising Hormone)
- **Seminiferous Tubules (In Lobules of Testes):**
 - **Spermatogonia (Germ/Stem-Cells):**
 - In Basal Lamina of Seminiferous Tubules
 - #1 Function = Are the precursors for Spermatogenesis
 - Stimulated by **Testosterone**.
 - **Sertoli/Sustentacular Cells:**
 - Make up the Walls of the Seminiferous Tubules
 - Main Functions =
 - Endocrine – Production of **Androgen Binding Protein** (ABP)
 - – (Makes Spermatogenic Cells *receptive to Testosterone*)
 - Endocrine – Production of **Inhibin**
 - – (Provides negative feedback to the *Hypothalamus*)
 - Blood-Testes Barrier (because spermatids are genetically unique & require protection from autoimmunity)
 - Nourish Sperm
 - Phagocytosis – (mop up any dead/underdeveloped spermatids)
 - Produce Tubular Fluid – (Help transport the sperm)
 - Produce Plasminogen Activating Factor – (Help free the sperm from tubule wall)



GENERAL OVERVIEW OF ENDOCRINE DISORDERS

GENERAL OVERVIEW OF ENDOCRINE DISORDERS:

Level-Of-Function Disorders:

- **Hypofunction Disorders:**
 - Where the gland produces less than it should.
 - **Common Causes:**
 - Loss of reserve
 - Hypo-secretion
 - 'Agenesis' – failure to develop embryonically
 - Atrophy – Wasting away due to injury/disease/lack of use.
 - Active Destruction
 - Tumour
- **Hyperfunction Disorders:**
 - Where the gland produces more than it should.
 - **Common Causes:**
 - Hyper-secretion
 - Loss of suppression
 - Hyperplasia (↑Proliferation)
 - Neoplastic Change (Tumour)
 - Hyperstimulation
 - Ectopic Sites of Secretion (Some far-off tumours secreting hormone)

Hierarchical Classification of Hypothalamo-Pituitary Axis Disorders:

- **Note:** Endocrine disorders of the Hypothalamo-Pituitary Axis are often classified in a Hierarchical Fashion depending on the origin of the disorder:
- **Primary:**
 - Disorder of the Target Gland
 - (eg: *Primary Hypothyroidism* – the Thyroid Gland *itself* is under-responsive to TSH stimulation)
- **Secondary:**
 - Disorder of the Pituitary Gland
 - (eg: *Secondary Hypothyroidism* – the Pituitary Gland is under-producing TSH)
- **Tertiary:**
 - Disorder of the Hypothalamus
 - (eg: *Tertiary Hypothyroidism* – the Hypothalamus is under-producing TRH)

Testing for an Endocrine Disorder:

- **Basal Hormone Testing:**
 - A single 'snapshot' measurement of the concentrations of specific hormones.
 - Eg: High [Thyroid-Stimulating Hormone] → Therefore Primary Hypothyroidism.
 - Problem – Some secretions are ***Pulsatile***, meaning random measurements don't accurately diagnose a disorder of that gland. The Solution: Dynamic Hormone Testing.
- **Dynamic Hormone Testing:**
 - Using exogenous chemicals/hormones to Stimulate/Suppress activity of a target gland. This tests the responsiveness of a target gland to feedback stimuli.
 - **Suppression Tests:**
 - When Hyperfunction is suspected, an inhibitor is administered and then the hormone concentration is re-measured to see if it has decreased. If not, Hyperfunction is confirmed.
 - **Stimulation Tests:**
 - When Hypofunction is suspected, a stimulator is administered and then the hormone concentration is re-measured to see if it has increased. If not, Hypofunction is confirmed.

Overview of Common Endocrine Symptoms:

- **Diabetes (1 & 2):**
 - Weight Change
 - Polyuria, Nocturia & Thirst
 - Visual Disturbances
 - Infections & Immunosuppression
 - Constant Hunger
 - Nausea + Vomiting
 - Fatigue
 - DKA (Diabetic Ketoacidosis) = Emergency Presentation
- **Hyperthyroidism:**
 - Weight Loss
 - Fatigue
 - Suppressed TSH
 - Elevated T4
- **Hypothyroidism:**
 - Weight gain
 - Pretibial Myxoedema
 - Periorbital Oedema
 - Bradycardia
 - Bradypnoea
- **PolyCystic Ovarian Syndrome:**
 - Weight Gain
 - Hirsutism
 - Infertility
- **Cushings Syndrome:**
 - Caused by Excess Corticosteroids
 - Moon Facies (Fat, white, round faces)
 - Muscle Wasting + Weakness
 - Weight Gain (Truncal Obesity)
 - Stretch Marks due to Weight Gain
- **Pituitary Adenoma:**
 - Peripheral Vision Loss
 - Compression symptoms or Secretory Symptoms
 - Secretory → eg: Prolactin → Galactorrhoea + Gynecomastia
→ eg: GH → Gigantism (Pre-Puberty) → Acromegaly (Post Puberty)
→ eg: ACTH: → Cushing's Syndrome
- **Acromegaly:**
 - Soft-Tissue Swelling
 - Arthritis
 - Hyperhidrosis
 - Headache + Visual Field Defect
- **Addison's:**
 - Autoimmune
 - Weight Loss
 - Fatigue
 - Hypotension
 - Hyponatraemia
 - Hyperkalaemia
 - Hyperpigmentation
- **Anorexia:**
 - Weight Loss
 - Fatigue
 - ↓BMI
 - ↑FSH + LH (Due to no ovulation)
 - ↑GH
 - Hypokalaemia (often due to vomiting) → Arrhythmias

THYROID DYSFUNCTION:

THYROID DYSFUNCTION

HYPOTHYROIDISM (Most Common):

- **"Under-Secretion of Thyroid Hormone"**
- **Pathophysiologies:**
 - ****Dietary: - *Insufficient Iodine intake*** – (Worldwide greatest cause due to poverty)
 - ****Autoimmune: '*HASHIMOTO'S THYROIDITIS*'**
 - – **Anti-Thyroid Peroxidase Antibodies (Anti-TPO-Abs)** → ↓T3/T4 Production
 - - **or Anti-Thyroglobulin Antibodies** → Destroy T3/T4.
 - Classically Women >60yrs
 - ***Hypothalamic-Pituitary Disorder***
 - Either a failure of Hypothalamus to produce enough TRH
 - Or a failure of the Pituitary to produce enough TSH
- **Effects of Hypothyroidism:**
 - ↓Metabolic Rate
 - ↓Body Temperature & Cold Intolerance
 - ↓Sympathetic Sensitivity
 - If Extreme → "*Cretinism*" – Severely stunted physical growth & mental development.
- **Clinical Features – Affects virtually ALL Systems:**
 - **General:**
 - Fatigue
 - *Cold* Intolerance
 - Apathetic Face
 - Droopy Eyes
 - Hoarseness
 - Menstrual Irregularities
 - Muscle Weakness
 - **CVS:**
 - Bradycardia
 - Pericardial Effusion
 - ↓Cardiac Output
 - **GI:**
 - Weight Gain *Despite* Poor Appetite
 - Constipation
 - **Neuro:**
 - Paraesthesia
 - Slow Speech
 - Mental Sluggishness
 - **Skin:**
 - Pale, Cool, Dry (Due to ↓Blood Flow)
 - NON-Pitting Oedema (Due to Accumulation of Hyaluronic Acid & Glycosaminoglycans)
 - Face & Periorbital Oedema
 - **Hair:**
 - Dry, Coarse, Loss of Lateral 1/3 Eyebrow
 - **MSK:**
 - Muscle Cramps
 - "Hung Reflexes" – Delayed Relaxation in deep tendon reflexes.
 - **Haem:**
 - Macrocytic Anaemia
- **Diagnosis (See '*Diagnosing Thyroid Dysfunction*' section for more details):**
 - TSH Levels – To see if Secondary Hypothyroidism
 - T4/T3 Levels – To combine with TSH to confirm exact cause
 - **If Hashimotos:**
 - + Anti-Thyroid Peroxidase Antibodies (Anti-TPO-Abs)
 - + Anti-Thyroglobulin Antibodies

Hashimoto's Thyroiditis : *Pathogenesis and clinical findings*

Author:

Jaye Platnich

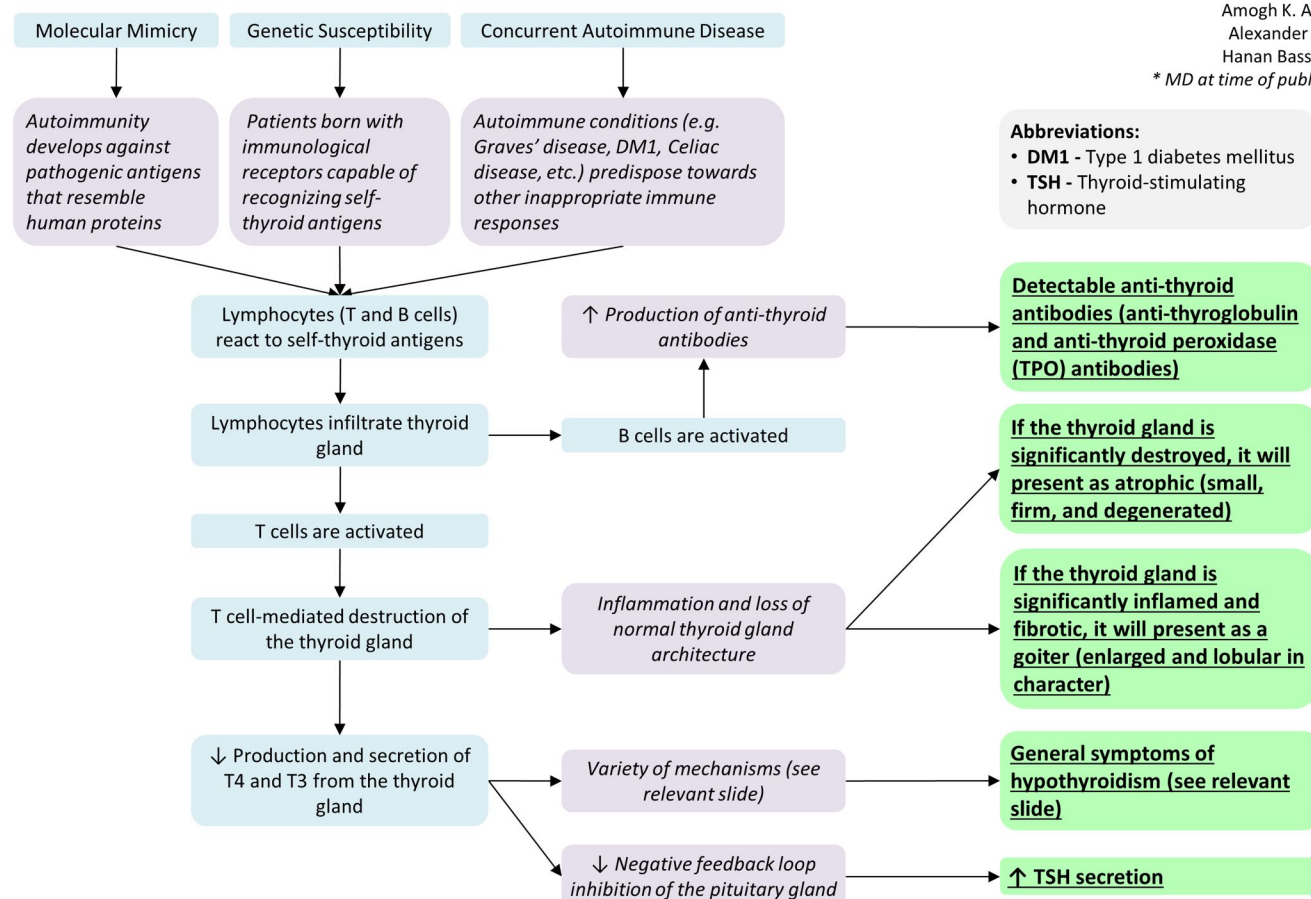
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Legend: Pathophysiology Mechanism Sign/Symptom/Lab Finding Complications Published August 8, 2013 on www.thecalgaryguide.com



<https://calgaryguide.ucalgary.ca/hashimotos-thyroiditis-pathogenesis-and-clinical-findings-2/>

Treatment of Hypothyroidism:

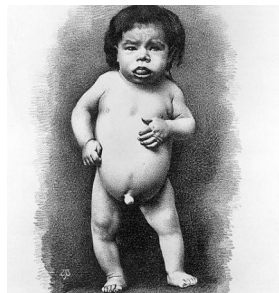
- Thyroid Hormone Replacement:

- **L-Thyroxine** (Thyroid Hormone Replacement)
 - ***Thyroxine**/levothyroxine (T4)
- Note: Thyroxine is the preferred agent as it is the least biologically active – (Longer Half-life), and can be Deiodinated by the body to T3 (Thyronine) when needed.

Complications of Hypothyroidism:

- ***CRETINISM*:**

- **Terminology:**
 - = “Hypothyroidism that develops in Infancy or Early Childhood → Severely stunted physical growth & mental development”
- **Aetiology:**
 - Maternal Hypothyroidism (Typically Iodine Deficiency)
- **Clinical Features:**
 - Short Stature
 - Severe Mental Retardation
 - Coarse Facial Features
 - Protruding Tongue
 - Umbilical Hernia
 - (Note: Severity of Disease depends on *When* Maternal Hypothyroidism occurred during Pregnancy)



Sir B. Bramwell, Sporadic cretinism. CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0>>, via Wikimedia Commons

- ***MYXOEDEMA COMA!:**

- – Most Severe Complication
- **Aetiology:**
 - *Longstanding Undiagnosed Hypothyroidism + Precipitant (Infection/Surgery/MI/CHF)*
- **Clinical Features:**
 - Hypothermia
 - Hypoventilation
 - Bradycardia
 - Hypertension
 - Hypoglycaemia
 - Stupor
- **Lab Findings:**
 - ↓↓↓ T3/T4
 - ↑↑↑ TSH
 - Hypoglycaemia
 - Check ACTH & Cortisol for ?Concomitant Adrenal Insufficiency?.
- **Treatment:**
 - Emergency management (ABCs)
 - Keep Pt Warm
 - Loading Dose Thyroxine
 - Adjuvant Corticosteroid (Hydrocortisone)
 - Assisted mechanical ventilation if respiratory acidosis.
 - IV Frusemide if Pulmonary Oedema
 - Treat Precipitant

HYPERTHYROIDISM:

- **"Excessive Secretion of Thyroid Hormone"**
- **Causes:**
 - ****Auto-Immune: 'GRAVE'S DISEASE'** – where **TsAb's (Thyroid-Stimulating Antibodies)** mimic TSH → Stimulate ↑T3/T4 Production
 - ****TOXIC MULTINODULAR GOITRE** – where a nodule/s in a non-toxic-goitre gain autonomy → Secrete excess thyroid hormones → Hyperthyroidism & Development of a 'Toxic' Multinodular Goitre.
 - ***Post-URTI Subacute ("De-Quervain's") Thyroiditis** (→ Transient ↑→↓→ Eu-Thyroid)
- **Effects of Hyperthyroidism:**
 - ↑Metabolic Rate
 - ↑Body Temperature (→ Heat Intolerance. Ie can't stand hot environments)
 - ↑Sympathetic Sensitivity
 - **If Grave's Disease:** Exophthalmos (Eye Protrusion) & Goitre (Thyroid Enlargement)
- **Clinical Features – Affects virtually ALL Systems:**
 - **General:**
 - Fatigue
 - Heat Intolerance
 - **CVS:**
 - Tachycardia
 - Palpitations
 - AF (Suspect Thyrotoxicosis if New-Onset AF in elderly)
 - Cardiomegaly
 - Congestive Heart Failure (In Elderly)
 - **GI:**
 - Weight Loss *Despite INCREASED Appetite*
 - Thirst
 - Hypermotility (↑Frequency of Bowel Movements, Diarrhoea)
 - **Neuro:**
 - Overactive Sympathetic NS
 - Fine Tremor
 - Irritability
 - Anxiety
 - Insomnia
 - Proximal Myopathy (Muscle Weakness) & Wasting.
 - **Eye:**
 - **Exophthalmos (Wide, Staring Gaze)**
 - Lid Lag
 - (Note: Proptosis only in Graves)
 - **Dermatology:**
 - **Acropachy (Digital Clubbing & Swelling; Fingers & Toes)**
 - **Hair:** Fine, Alopecia
 - **Skin:** Soft, Warm, Flushed, Sweaty.
 - Vitiligo (Pigmentation)
 - Soft Nails with Onycholysis (Plummer's Nails)
 - **MSK:**
 - ↑Bone Resorption → Osteoporosis
 - **Haem:**
 - Lymphadenopathy (esp. Graves' Disease)
 - **Others:**
 - Menorrhagia
 - **Pretibial Myxoedema**

Graves' Disease: Pathogenesis and Clinical Findings

Authors:

Ashley Hinthner

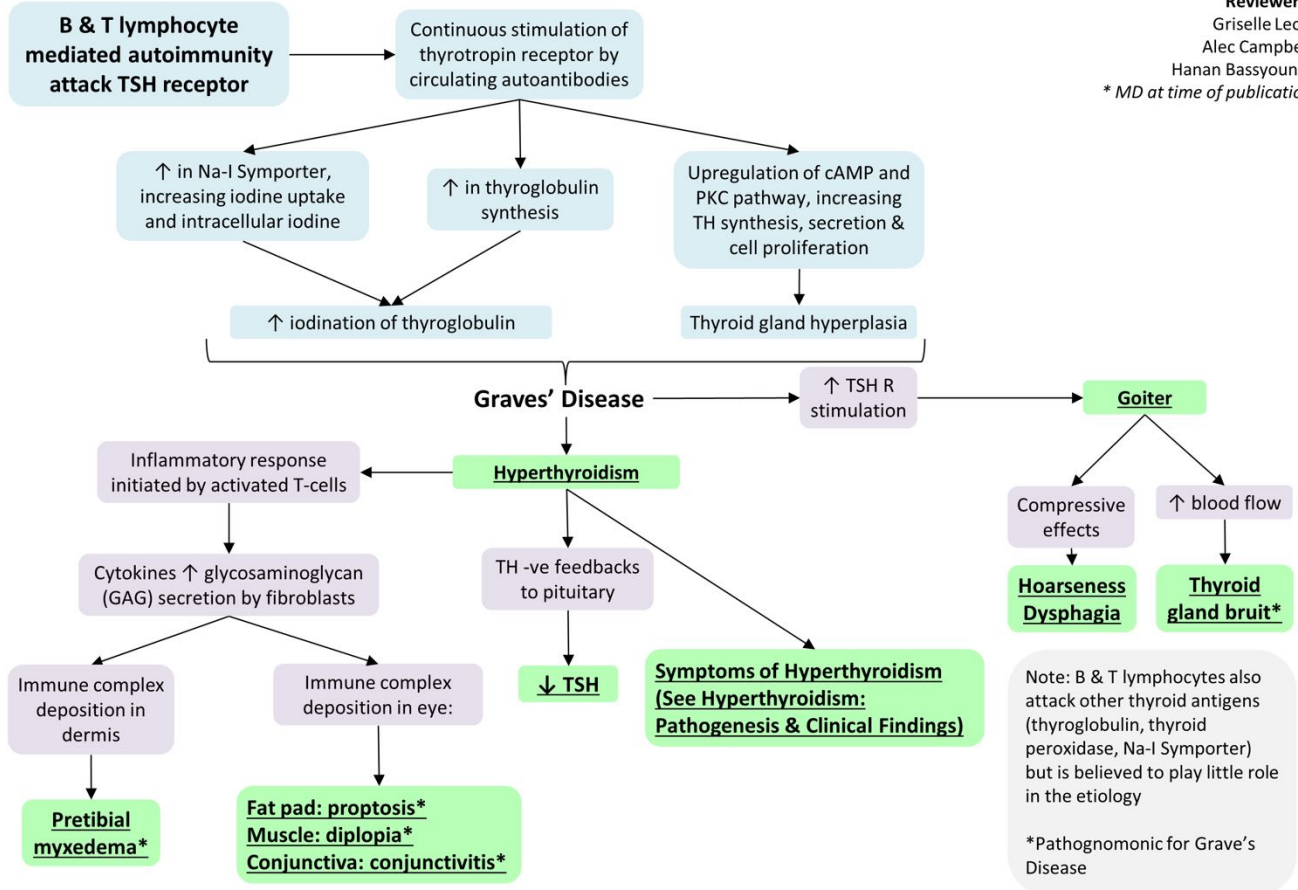
Reviewers:

Griselle Leon

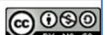
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Legend: Pathophysiology Mechanism Sign/Symptom/Lab Finding Complications Published MONTH, DAY, YEAR on www.thecalgaryguide.com



<https://calgaryguide.ucalgary.ca/graves-disease-pathogenesis-and-clinical-findings/>

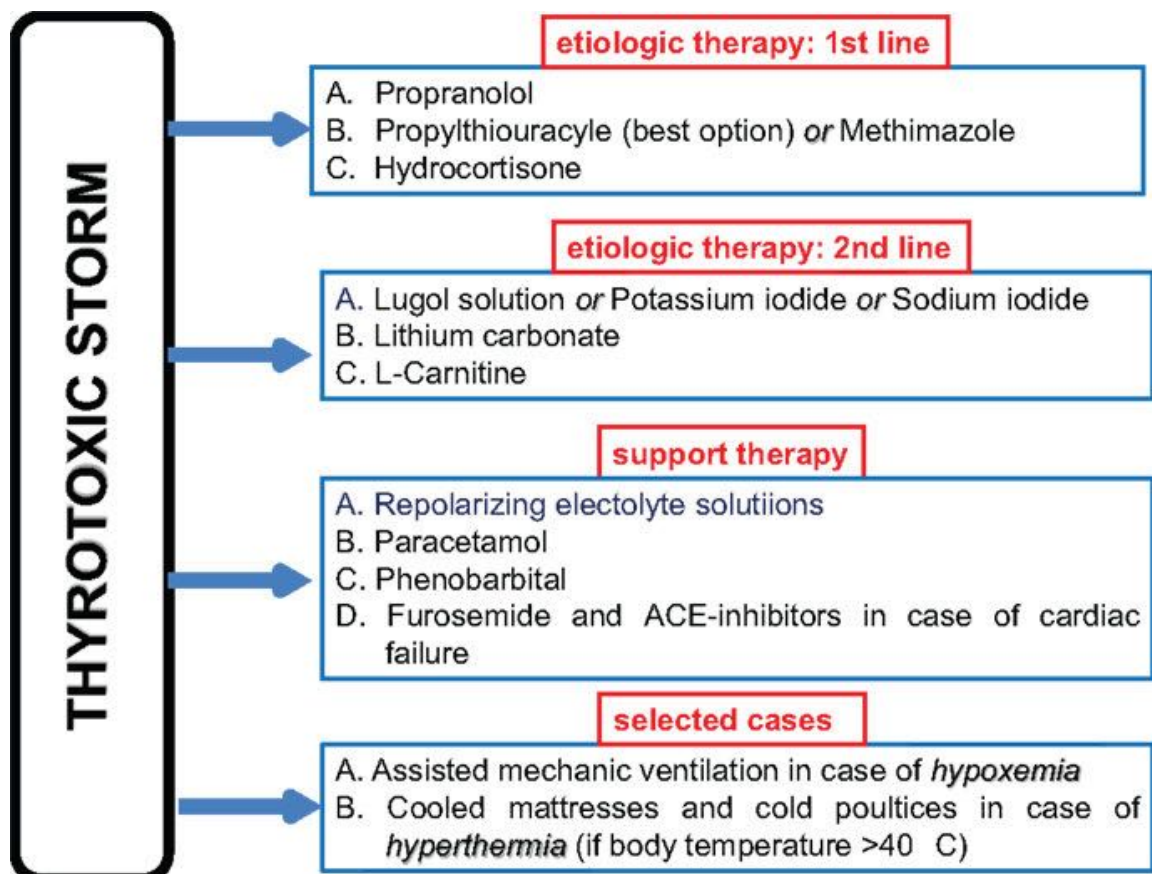
Treatment of Hyperthyroidism:

- **Surgery (Thyroidectomy):**
 - o **Total Vs Partial**
- **Radioactive Iodine - Destroy Thyroid Follicular Tissue:**
 - o **Radioactive Iodine (I^{125})**
 - o Note: Can kill too much thyroid tissue → Hypothyroidism.
- **Anti-Thyroid Agents:**
 - o ***Carbimazole**

Complications of Hyperthyroidism:

- **THYROTOXIC STORM:**

- **Aetiology:**
 - Precipitated by Infection/Trauma/Surgery/etc. In a Hyperthyroid Patient.
- **Pathogenesis:**
 - Pre-existing Hyperthyroidism → ↑ Sympathetic Sensitivity
 - + Precipitant → ↑ Catecholamine Levels → Sympathetic Symptoms.
- **→ SEVERE Clinical Features – 50% MORTALITY:**
 - Extreme Fever
 - Tachycardia/Arrhythmias
 - Vascular Collapse (Hypotension)
 - Congestive Heart Failure/Pulmonary Oedema
 - Vomiting/Diarrhoea
 - Confusion/Delirium/Coma
- **Differentials:**
 - Sepsis
 - Pheochromocytoma
 - Malignant Hyperthermia
- **Lab Findings:**
 - ↑↑↑ T3/T4
 - ↓↓↓ TSH
 - (Leukocytosis, Hypercalcaemia, ↑ LFTs)
- **Treatment:**
 - **Treat Precipitating Factor, Plus:**



DIAGNOSING THYROID DYSFUNCTION:

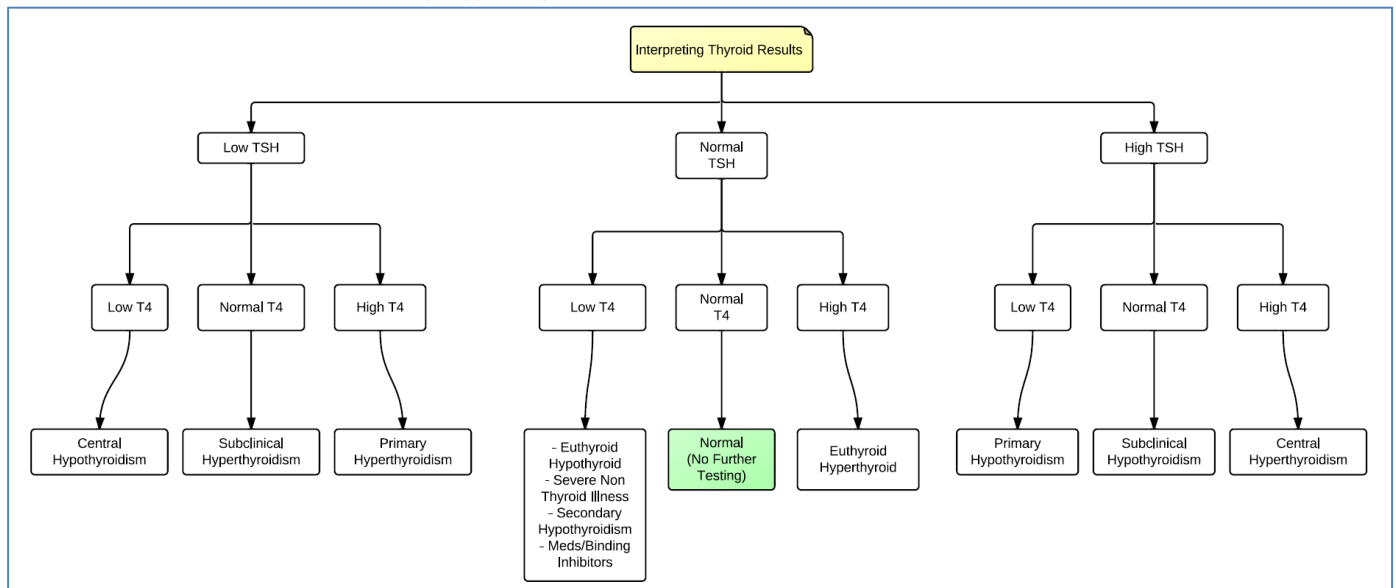
Step 1 – Think...What Level is the Problem?

- **Primary** ↓/↑-Thyroidism – (Glandular Level)
 - Eg: Primary Hyperthyroidism = Graves' Disease
 - Eg: Primary Hypothyroidism = Hashimoto's Disease
- **Secondary** ↓/↑-Thyroidism:
 - Problem with the Pituitary Gland or Hypothalamus (Ie: ↓TRH / TSH)

Step 2 – Test...Confirm Clinical Suspicions:

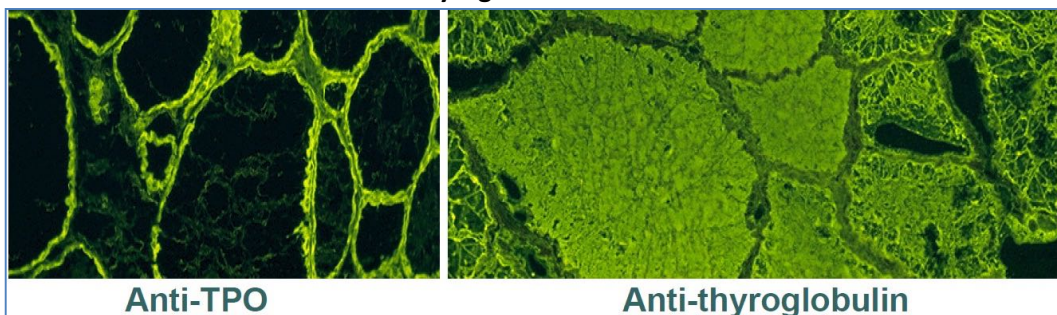
- **Thyroid Function Tests:**

- Test Plasma T₃, T₄ & TSH levels → To determine the 'level' of the dysfunction (Primary/Secondary)
 - If Primary Hypothyroidism: ↑TSH Levels, ↓T₃&T₄ Levels
 - If Primary Hyperthyroidism: ↓TSH Levels, ↑T₃&T₄ Levels
 - If Secondary Hypothyroidism: ↓TSH Levels, ↓T₃&T₄ Levels
 - If Secondary Hyperthyroidism: ↑TSH Levels, ↑T₃&T₄ Levels



- **Thyroid Autoantibody Assays:**

- **Grave's** – Presence of TsAb's (Thyroid-Stimulating Antibodies)
- **Hashimoto's** – Presence of Anti-Thyroid Peroxidase Antibodies (Anti-TPO-Abs)
 - or Anti-Thyroglobulin Antibodies



Note: Thyroid Peroxidase is contained within the Follicle Cells themselves, whereas Thyroglobulin resides within the Colloid – Hence the staining.

- **TRH Stimulation Test (Dynamic Testing)**

- Investigates Pituitary-TSH Deficiency

- **Imaging:**

- **Ultrasound** – for sizing.
- **Radioisotope Methods (Nuclear Medicine)** – (Radioactive Iodide) to measure activity of the gland.
 - Can distinguish Inactive (Cold) & Overactive (Hot) Nodules.

- **Histopathology:**

- Ie: Biopsy.

NON-TOXIC GOITRES – (Diffuse & Multinodular Goitres):

Terminology:

- **Goitre** = “Enlargement of the Thyroid”
- **Non-Toxic Goitre** = “Enlargement of the Thyroid Gland in a *EUTHYROID* Individual that is *NOT DUE TO* Inflammatory, or Neoplastic Changes”

(Simple) DIFFUSE GOITRE (Early):

- **Aetiologies:**
 - o **Normal Physiological Conditions** (Adolescence, Pregnancy, Lactation)
 - – Ie: Conditions requiring ↑ Metabolism
 - o **Iodine Deficiency – Most Common**
- **Pathogenesis:**
 - o **Note: Hyperplasia; NOT Neoplastic** – (Due to ↑ TSH Stimulation)
- **Morphology:**
 - o Goitre is Mild, Diffuse & Symmetrical
- **Clinical Features:**
 - o **Most Pts are Euthyroid**
- **Ix:**
 - o ↑↑ TSH
 - o Normal T3/T4 (Unless Severe - ↓ T3/T4)
- **Complications:**
 - o Mechanical (Dysphagia, Airway Obstruction)
 - o Endocrine (Toxic Nodule → Hyperthyroidism)



<https://www.ncbi.nlm.nih.gov/books/NBK285556/>

NON-TOXIC MULTINODULAR GOITRE (Late):

- **Aetiology:**
 - Prolonged Hyperplasia of a Diffuse Goitre due to Iodine Deficiency
- **Pathogenesis:**
 - 1- Simple *Diffuse Goitre* due to Iodine Deficiency
 - 2- Prolonged Hyperplasia of a Diffuse Goitre →→ **Multinodular Goitre.**
- **Morphology:**
 - Asymmetrical, Multinodular, Multilobulated, Goitres
 - Can be MASSIVE
 - Nodules are **Un-Encapsulated**, & Contain Variable amounts of Colloid (Brown, Gelatinous)
- **Clinical Features:**
 - Massive Goitre
 - **Most Pts are Euthyroid**
- **Complications:**
 - Toxic Adenoma/Toxic Nodule.



(Credit: "Almazi"/Wikimedia Commons)

THYROID NEOPLASMS – (Adenomas & Carcinomas):

Terminology:

- **Hot Nodules** - Those secreting Thyroid Hormones regardless of TSH Levels.
- **Cold Nodules** – Those that are Hypofunctioning regardless of TSH Levels

Clinical Features:

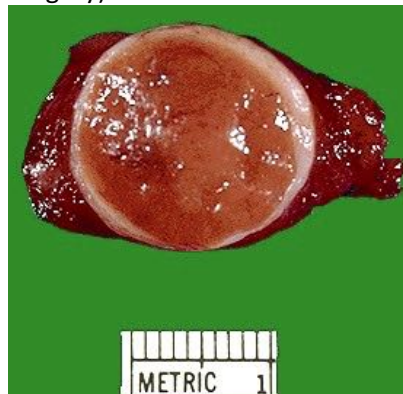
- **Mostly Asymptomatic**
- Palpable (sometimes visible) lump in throat
- Goitre (later sign)
- **Red Flags:**
 - o Rapid Growth
 - o Firm/Hard
 - o Immobile
 - o Voice Hoarseness
 - o Dyspnoea
 - o Dysphagia
 - o Lymphadenopathy
- **Green Flags:**
 - o Mobile, Painful & Inflammation – Non-Neoplastic.

Universal Investigations of Thyroid Neoplasms (Despite Patterns):

- **1- Imaging**
- **2- TFTs**
- **3- FNA + Biopsy**
- **4- Surgical Resection & Histology.**

THYROID ADENOMA: “FOLLICULAR ADENOMA” (Benign Neoplasms) – 90%:

- **Cold or Hot Adenomas**
- **Morphology:**
 - o Solitary, Spherical Mass
 - o Well-Defined, *Intact* Fibrous Capsule
 - o ≈3cm Diameter
 - o Colour:
 - **Cold Adenoma** = Grey-White colour (Due to ↓ Colloid)
 - **Hot (Toxic) Adenoma** = Red-Brown colour (Due to ↑ Colloid Content)
 - o Areas of Haemorrhage, Fibrosis & Calcification (Similar to MNG)
- **Clinical Features:**
 - o Unilateral Painless Mass
 - o **Cold Adenomas** = Euthyroid
 - o **Hot (Toxic) Adenomas** = Hyperthyroid
- **Investigations:**
 - o As above
- **Complications:**
 - o Excellent Prognosis (post-surgery) – No Recurrence or Metastasis.



THYROID CARCINOMAS (Malignant Neoplasms) – 10%:

- **4x Types – (Note: ALL begin as Follicular Cells *Except* Medullary Carcinomas, which are Parafollicular Cells):**

- **PAPILLARY CARCINOMA of the Thyroid – MOST COMMON:**

- **Clinical Features:**
 - Asymptomatic, Mobile Thyroid Nodule (Indistinguishable from Benign Nodule)
 - Note: Presenting Symptom is often Cervical Lymphadenopathy.
 - Symptoms of Severe Disease: Dysphagia, Hoarseness, Cough
- **Investigations:**
 - As above
- **Treatment:**
 - Surgical Excision
- **Prognosis:**
 - Malignant, but *Clinically Benign* – i.e. High survival rate (98% @ 10yrs).
 - Rarely extends outside the thyroid capsule or to other structures.



<https://webpath.med.utah.edu/ENDOHTML/ENDO063.html>

- **FOLLICULAR CARCINOMA of the Thyroid:**

- **Morphology:**
 - Single Nodules
 - May be Well-Demarcated (Similar to Follicular Adenoma) or Infiltrative
 - Grey-Tan Colour
 - Central Fibrosis & Calcification
- **Clinical Features:**
 - Slow-Growing, Painless Nodules
 - Follicular Carcinoma prefers Haematogenous Metastasis rather than Lymphatic
 - ∴ No Lymphadenopathy
 - Aggressive - Spreads Early to Bone (May present as a pathological fracture)
- **Treatment:**
 - Total Thyroidectomy
 - + Radioactive Iodine Ablation for ?Metastases.
 - + Supportive Thyroid Hormone Replacement



Follicular Neoplasms in the 4th Edition WHO Classification of Endocrine Organs;

https://www.researchgate.net/figure/Widely-invasive-follicular-thyroid-carcinoma-Multiple-invasions-by-tumor-nests-into_fig4_348663497

- **ANAPLASTIC CARCINOMA of the Thyroid:**

- **Pathogenesis:**
 - Typically arise due to **De-Differentiation** of a *Papillary* or *Follicular Carcinoma*
- **Morphology:**
 - Invasion out of the Thyroid Capsule & Into Adjacent Structures (Eg: Trachea & Jugular Vein)
- **Clinical Features:**
 - Typically Elderly
 - Rapid-Growing Nodule
 - Compressive Symptoms: Dysphagia, Dyspnoea, Hoarseness, Cough
- **Treatment/Prognosis:**
 - Highly Aggressive – Local Invasion & Metastasis @ Presentation
 - **No Treatments**
 - **100% Mortality @ 1yr.**

- **MEDULLARY CARCINOMA of the Thyroid:**

- **Aetiology:**
 - **70% Sporadic Proto-oncogene Mutation.**
 - **30% Occur in *Multiple Endocrine Neoplasia Syndrome Type 2 (MEN-2a & 2b)***
- **Morphology:**
 - Solitary if Sporadic; Multiple/Bilateral if MEN-2a/b.
 - Firm, Pale Grey-Tan, Areas of Haemorrhage & Necrosis
 - Invasion outside the Thyroid Capsule
- **Clinical Features:**
 - **Sporadic:**
 - Thyroid Nodule + Dysphagia or Hoarseness
 - Note: NO Hypocalcaemia, despite ↑Calcitonin.
 - **MEN-2 Also Involves:**
 - (Thyroid Gland – Medullary Carcinoma of the Thyroid)
 - Adrenal Medulla – Pheochromocytoma
 - Parathyroid Gland – Parathyroid Hyperplasia



Familial thyroid cancer: A review - Scientific Figure on ResearchGate. Available from:
https://www.researchgate.net/figure/Gross-pathologic-features-of-medullary-thyroid-carcinoma_fig13_50997813

Summary:



Carcinoma of Thyroid



Type	(%)	age	spread	Prognosis
Papillary	60-70	young adults 20-40 (<45y)	Lymphatic , to local nodes	Excellent
Follicular	20-25	Young-middle 40-50 (>45)	Blood stream, especially to bone	Good with radio-iodine therapy.
Anaplastic	10-15	Elderly	Aggressive local extension	Very poor
Medullary (C-cells)	5-10	Usually elderly, but familial cases occur	Local, lymphatic, blood stream	Variable. More aggressive in familial cases

GROWTH DYSFUNCTION:

GROWTH DYSFUNCTION

Defects in Endocrine Control of Growth:

- **Hyper:**
 - **Too Much Growth Hormone &/or Growth Factors (Rare):**
 - Eg: Childhood Gigantism
 - Eg: Adults - Acromegaly
 - **Non-GH Causes:**
 - Eg: Precocious Puberty
- **Hypo:**
 - **Defective Growth Hormone Axis:**
 - **GH-Deficiency:**
 - **Primary GH Deficiency:**
 - Hypothalamic Defect
 - And/Or Pituitary Defect
 - **Secondary Pituitary Deficiency:**
 - Eg: Tumour & other Destructive Diseases.
 - Eg: Psychosocial Deprivation (Ie: Kids in abusive/non-supportive environments → GH-Deficiency → exhibit slowed growth)
 - **Bio-Inactive GH:**
 - GH is produced by has little/no impact with receptors.
 - **Growth Hormone Insensitivity:**
 - **Primary GH Insensitivity:**
 - GH Receptor Defect
 - IGF Synthesis Defect
 - IGF Receptor Defect
 - Signal Transduction Defect (GH/IGF) (Ie: Secondary Messenger Defect)
 - **Secondary GH Insensitivity:**
 - GH-Inhibiting Antibodies
 - Malnutrition
 - Diabetes
 - Uraemia
 - **Deficient in *Other Hormones* essential for growth – Eg: Thyroid Hormone (Hypothyroidism)**

General Treatment Principles of Short Stature:

- **Treatment Depends on Origin of the Problem:**
 - Is it Endocrine?
 - What is the Defect?
 - GH Deficiency
 - Receptor Defects
 - IGF Deficiency
 - Other?
- **Common Treatments:**
 - Exogenous Growth Hormone
 - Exogenous IGF-I
- **Who Should Receive GH?**
 - GH Deficiency (Child/Adult)
 - Genetic Syndromes (Eg: Turners)
 - Chronic Kidney Failure (In Kids awaiting Transplant)
 - AIDS Wasting
 - Severe Burns

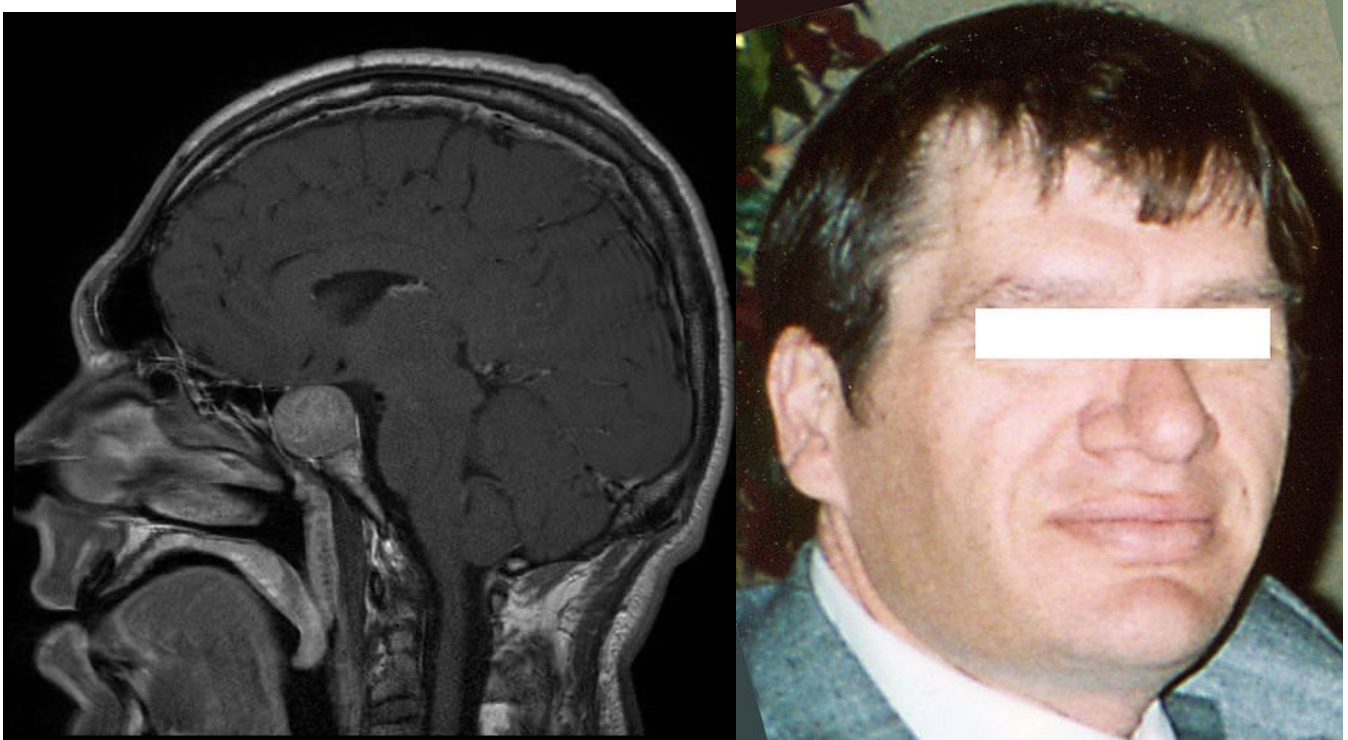
GROWTH HORMONE DEFICIENCY (AKA. PITUITARY DWARFISM):

- **Aetiology:**
 - o Insufficient production of Growth Hormone from the Pituitary
 - o Can be Congenital (Present at birth) – Eg: Genetic defects.
 - o Can be Acquired (develop later) – Eg: Severe brain injury
- **Pathogenesis:**
 - o Insufficient Growth Hormone → Abnormally short stature, but with NORMAL body proportions
- **Clinical Features:**
 - o Slow or absent growth
 - o Short stature (Below 5th percentile compared to children of same age/sex)
 - o Absent/delayed sexual development during puberty
 - o May have symptoms of other co-existent Pituitary hormone deficiencies:
 - Eg: Polyuria
 - Eg: Excessive thirst
 - Eg: Facial abnormalities
- **Diagnosis:**
 - o Clinical suspicion based on anthropometrics (patient's height, weight, arms/leg lengths)
 - o Blood tests – Low levels of Growth Hormone in the blood
 - o Imaging – Xrays + MRI-Brain
- **Treatment:**
 - o Synthetic Parenteral Growth Hormone Injections (Somatropin)
 - o Replace any other pituitary hormone deficiencies



GIGANTISM/ACROMEGALY:

- **Aetiology:**
 - Pituitary Adenoma
- **Pathogenesis:**
 - Pituitary Adenoma → Secretes Excess GH
- **Clinical Features:**
 - Insidious Onset
 - Mostly Middle-Aged Adults
 - **Symptoms:**
 - Severe Disfigurement
 - Soft-Tissue Swelling (Hands, Feet, Nose, Lips, Ears, Skin, Carpal Tunnel)
 - Prominent Jaw & Supra-Orbital Ridges
 - Hypertension
 - **Compressive Pituitary Adenoma** → Headache + Visual Field Defect
- **Diagnosis:**
 - IGF1 & GH Levels
 - Brain MRI
- **Treatment:**
 - Surgical Removal of Pituitary Adenoma
 - Somatostatin Analogues
- **Complications:**
 - Hypertrophic Cardiomyopathy & Heart Failure
 - Hypertension & Kidney Failure
 - Hyperglycaemia & Diabetes Mellitus
 - Accelerated Osteoarthritis
 - Possible Malignancy



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ADH DISORDERS:

ADH DISORDERS:

Disorders of Fluid/Electrolyte-Regulating Hormones:

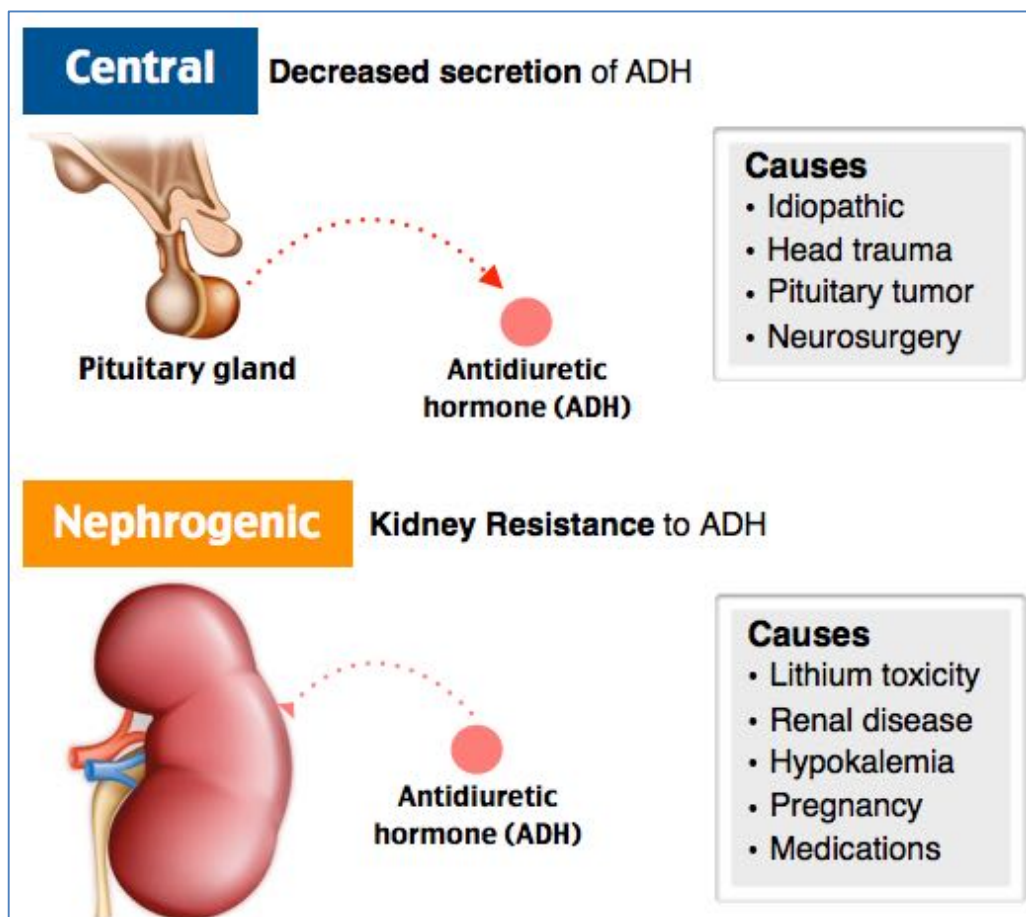
- Disorders of ADH:

○ **DIABETES INSIPIDUS (↓ADH):**

- Condition characterised by Excessive Thirst & the inability to Concentrate Urine
- **2 Types:**
 - **Neurogenic (Neuro)** - ADH Insufficiency
 - **Nephrogenic (Renal)** – Insensitivity of the kidneys to ADH
- **Signs/Symptoms:**
 - Extreme Thirst
 - Excessive Urination
 - Risk of Hypokalaemia
- **Diagnosis Criteria:**
 - Normal Blood Glucose
 - Normal Blood Bicarb
 - Normal Blood Calcium

} To Rule out other causes of Excess Urination.

 - Urinalysis – Low Osmolality, Electrolytes & Specific Gravity
 - Fluid Deprivation Test - No change in urine osmolality
 - Desmopressin Stimulation – Distinguishes between Neurogenic & Nephrogenic.
- **Treatment:**
 - Patients compensate by ↑H₂O Intake.
 - If Neurogenic – **Desmopressin** (Synthetic ADH) → ↓Urine Production.
 - If Nephrogenic – **Hydrochlorothiazide** Diuretic → ↓Urine Output in patients with DI.

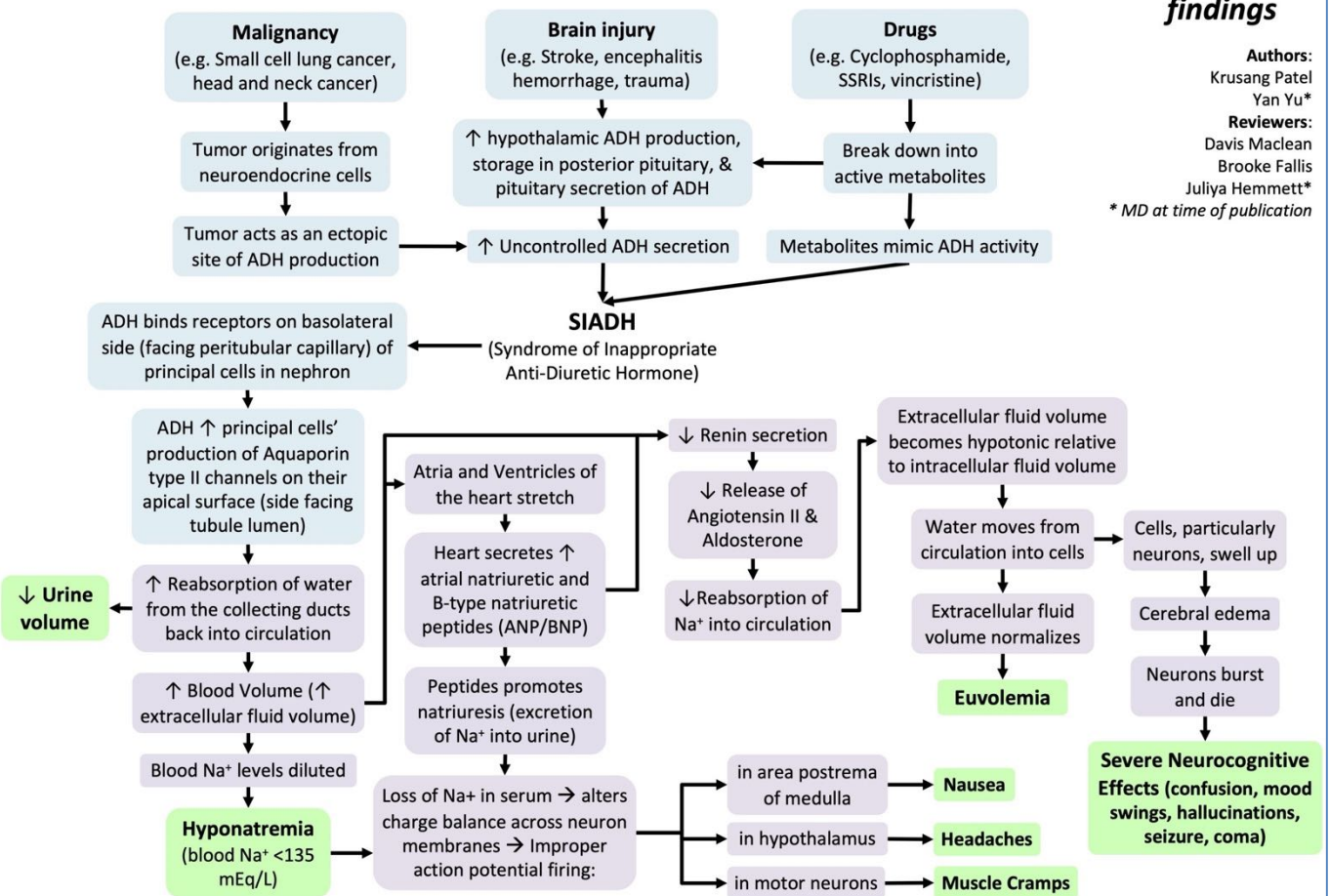


Source: Unattributable

SIADH (Syndrome of Inappropriate ADH secretion) (↑ADH):

- **Caused by:**
 - Insensitivity of Hypothalamic Osmoreceptors to ↓ Plasma Osmolality
 - Therefore, ADH release isn't inhibited by ↓ Plasma Osmolality
 - (Other causes: Malignancy, Drugs, Primary Brain Injury, Infection, Hypothyroidism)
- **Condition characterised by Excessive ADH Release from Post. Pituitary Or Ectopic Source.**
- **5 Cardinal Signs/Symptoms:**
 - 1- Fluid Overload (*Without* oedema or hypertension)
 - 2- Hyponatraemia (Dilutional) →
 - Headache
 - Nausea
 - Vomiting
 - Confusion
 - Convulsions (If Severe)
 - Coma (If Severe)
 - 3- Natriuresis (Excretion of Sodium in Urine – usually excessive)
 - 4- High Urine Osmolality relative to Plasma Osmolality.
 - 5- Normal Renal & Adrenal Function
- **Treatment:**
 - **Fluid Intake Restriction**
 - **Drugs – (ADH Inhibitors):**
 - Demeclocycline – Induces Nephrogenic Diabetes Insipidus as a Side Effect.
- Hence desensitises ADH receptors in the Nephron.
 - Conivaptan – Inhibits 2 of the 3 ADH Receptors.
 - Tolvaptan – Competitive inhibition of ADH Receptors

Syndrome of Inappropriate Anti-Diuretic Hormone (SIADH): Pathogenesis and clinical findings

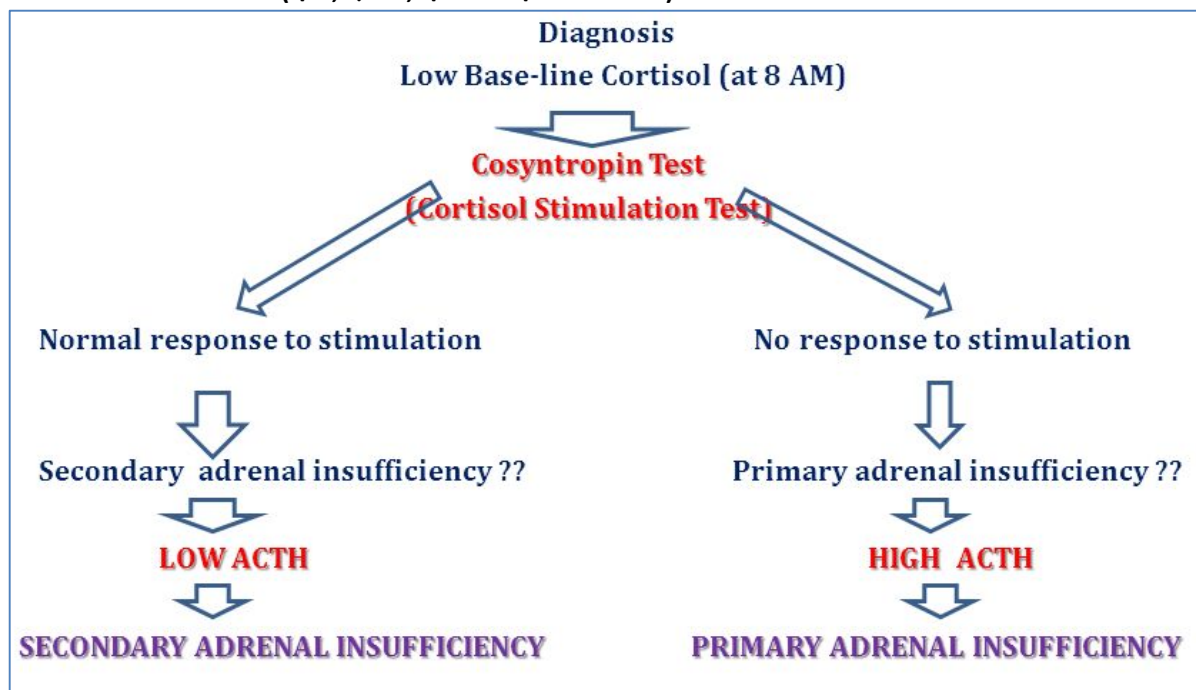


ADRENAL CORTEX DYSFUNCTION:

ADRENAL CORTEX DYSFUNCTION

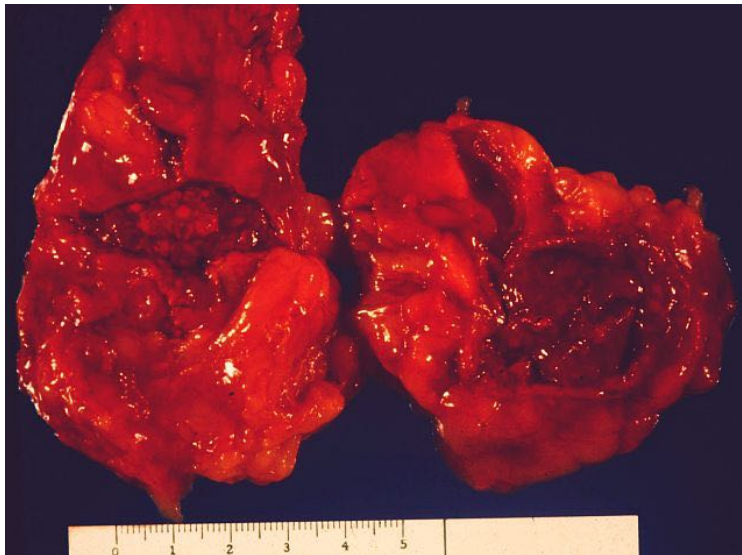
Adrenocortical Insufficiency (Hypoadrenal) Syndromes:

- **ADDISON'S DISEASE (Primary Chronic Adrenocortical Insufficiency):**
 - Aetiologies (Multiple Possible):
 - **Most Common = Autoimmune Adrenalitis (70%)**
 - Pathogenesis (Autoimmune Adrenalitis):
 - **↓↓Aldosterone → Hyponatraemia & Hyperkalaemia**
 - **↓↓Cortisol**
 - Clinical Features:
 - **Initially:** Progressive Weakness, Fatigue, Lethargy, Depression
 - **Later:**
 - **GI** - Anorexia, Weight Loss, Vomiting, Diarrhoea
 - **Skin** – Hyperpigmentation (Esp. Sun-Exposed & Pressure Point Areas)
 - **Electrolytes** (↓Aldosterone) – Hyponatraemia & Hyperkalaemia
 - Diagnosis:
 - **Synacthen (Synthetic ACTH) Test** → (Measure Cortisol and Aldosterone 30mins after)
 - **Adrenal-Autoantibodies**
 - **UECs** – (↑K, ↓Na, ↑Urea ↑Creatinine)



- Treatment:
 - **Cortisol Replacement (*Hydrocortisone*)**
 - **Correct Electrolytes**
- Complication - **Addisonian Crisis:**
 - **Why:** Stress/Acute disease → Adrenal Glands Cannot Respond → *Crisis*
 - **Clinical Features:**
 - Fever
 - Intractable Vomiting
 - Abdominal Pain
 - Hypotension
 - Coma
 - Shock (Vascular Collapse)

- **WATERHOUSE-FRIDERICHSEN SYNDROME (Acute Adrenocortical Insufficiency):**
 - **Aetiology:**
 - Overwhelming Sepsis
 - **Pathogenesis:**
 - Acute Haemorrhagic Infarction → Adrenal Necrosis → Acute Adrenal Hypofunction:
 - → ↓Aldosterone → Salt & Water Loss → Hypovolaemic Shock
 - **Morphology:**
 - **Macro:**
 - Haemorrhagic Mass (Blood Clot) Completely Obscures the Adrenal Gland
 - **Micro:**
 - Acute Haemorrhagic Necrosis (Starts in Medulla → Spreads to Cortex)
 - Islands of Recognizable Cortical Cells
 - **Clinical Features:**
 - Abrupt & Severe Clinical Course (Death in Hours-Days unless Treated)
 - Typically Meningococcal Septicaemia
 - ∴ Neck stiffness
 - ∴ DIC
 - Hypovolaemic Shock (Due to ↓Aldosterone)
 - **Treatment:**
 - Prompt Antibiotic Treatment
 - Fluids



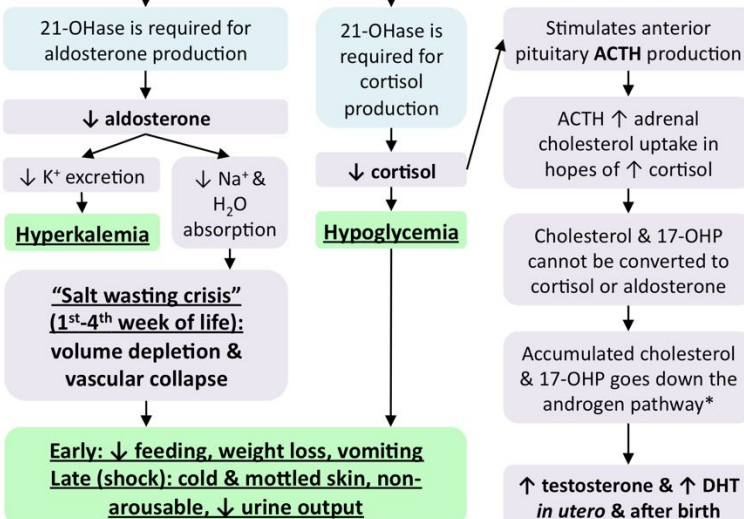
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- CONGENITAL ADRENAL HYPERPLASIA (CAH) - (Adrenogenital Syndromes/Virility Syndromes):

- **Aetiology:**
 - Autosomal Recessive 21-Hydroxylase Deficiency
- **Pathogenesis:**
 - 21-Hydroxylase Deficiency → ↓Cortisol/Aldosterone Synthesis → ↑ **Androgen Synthesis**
- **Clinical Features:**
 - **Androgen Excess:**
 - Masculinisation of Females (Clitoral Hypertrophy/Hirsutism/Oligomenorrhoea)
 - Masculinisation of Males (Penile Enlargement/Precocious Puberty/Oligospermia)
 - Neonate with Ambiguous Genitalia
 - **Mineralocorticoid (Aldosterone) Deficiency:**
 - Hypotension & Salt Wasting.

21-Hydroxylase Deficiency (21-OHD): Pathogenesis and clinical findings

Autosomal recessive mutation in *CYP21A2* coding for the enzyme 21-OHase causes varying degrees of 21-OHD in the adrenal glands



Note: Salt wasting crisis & hyperkalemia do not occur in utero as the placenta maintains appropriate ion concentrations.

Note: This is also known as "**congenital adrenal hyperplasia**". Females may be labeled as males at birth due to an enlarged clitoris. Severity depends on the amount of enzyme deficiency. Severe deficiencies present in infants with "salt wasting crisis" & virilization. Milder forms present with lesser virilization that may only become apparent at or before puberty.

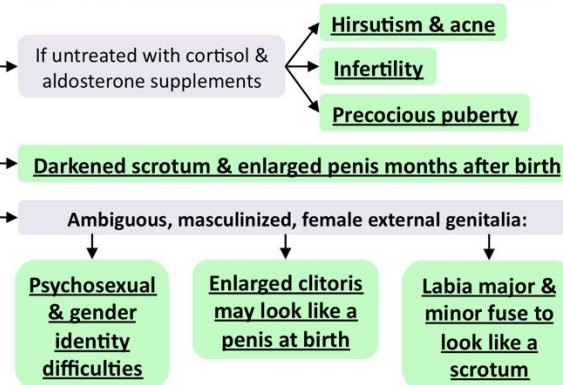
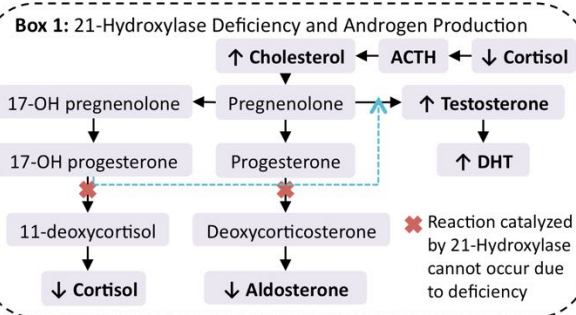
Abbreviations:

21-OHD: 21-hydroxylase deficiency

21-OHase: 21-hydroxylase

DHT: dihydrotestosterone

SR: sex determining region of Y which causes male pattern genitalia development



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Legend: Pathophysiology Mechanism Sign/Symptom/Lab Finding Complications

Published July 13, 2015 on www.thecalgaryguide.com

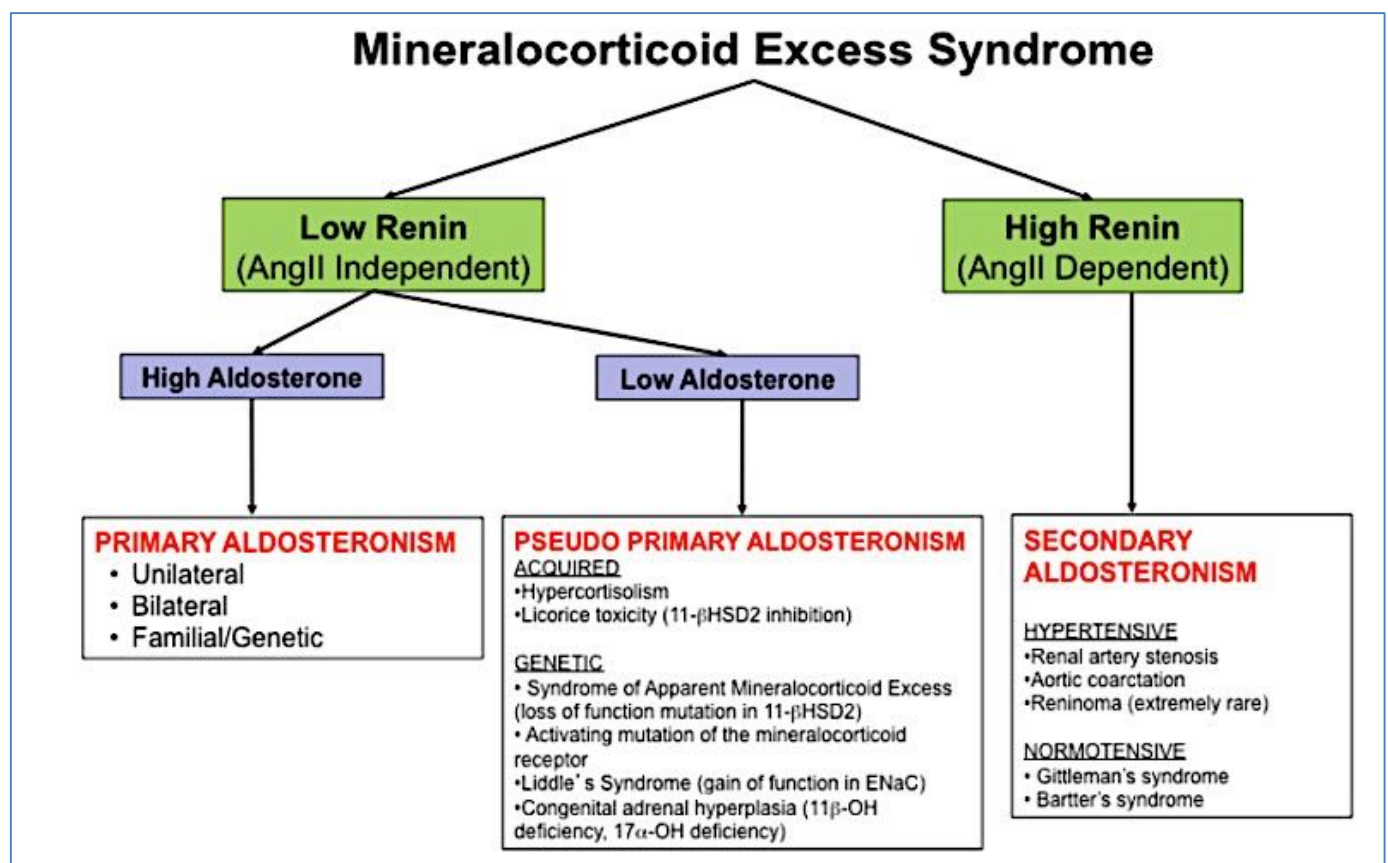


<https://calgaryguide.ucalgary.ca/21-hydroxylase-deficiency-21-ohd-pathogenesis-and-clinical-findings/>

Adrenocortical Hyperfunction (Hyperadrenal) Syndromes:

- **CONN'S SYNDROME (& other Primary Hyper-Aldosteronisms):**

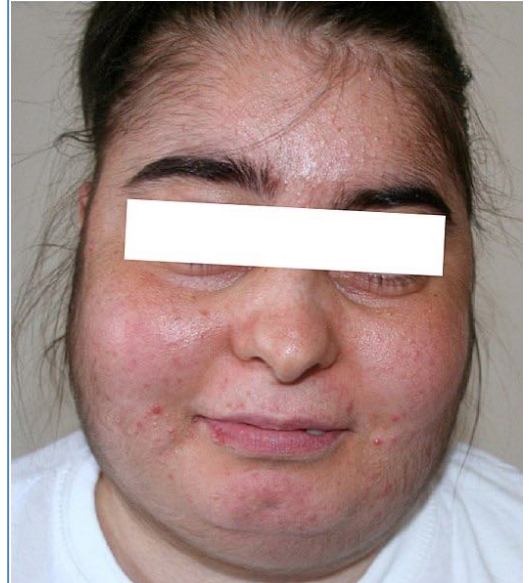
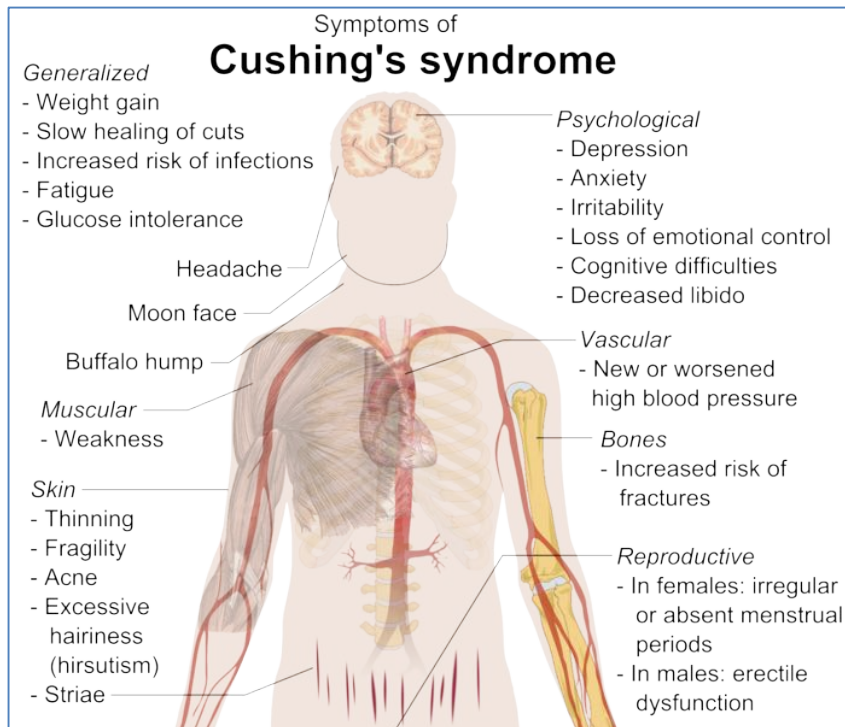
- Note: Aldosterone = Mineralocorticoid = Produced by the Zona Glomerulosa of the Adrenal Cortex.
- **Aetiologies:**
 - #1- Idiopathic Hyperplasia of Adrenal Glands
 - **#2- Aldosterone-Producing Adenoma (Conn's Syndrome)**
 - #3-(Rare) Aldosterone-Producing Carcinoma
- **Pathogenesis:**
 - → Chronic Excess ↑↑ Aldosterone Secretion → Na^+ Retention (& ↓ K^+) → Fluid Retention
 - → Hypertension
 - → Hypernatraemia
 - → Hypokalaemia
- **Clinical Features:**
 - ****Universal Sign = Hypertension**
 - Hypernatraemia (due to Renal Na Retention)
 - Hypokalaemia (due to Renal K Wasting):
- **Diagnosis:**
 - **Very HIGH Aldosterone**
 - Very Low Renin-Aldosterone Ratio (I.e: ↓Renin & ↑Aldosterone)
- **Treatment:**
 - **Idiopathic Hyperplasia:** Spironolactone (Aldosterone Antagonist)
 - **Adenomas (Conn's):** Surgical Resection



<https://www.ncbi.nlm.nih.gov/books/NBK279065/figure/hyperaldosteronism.F3/>

- **CUSHING'S DISEASE/SYNDROME (Hypercortisolism):**

- **Aetiology:**
 - **Cushing's Syndrome:** (Any cause of Excess Glucocorticoid Levels)
 - **Cushing's Disease:** (Central – ACTH-Secreting Pituitary Adenoma)
- **Pathogenesis (Cushing's Disease ONLY):**
 - **ACTH-Secreting Pituitary Adenoma** → ↑ACTH Levels → ↑Cortisol
- **Clinical Features:**
 - Slow onset
 - **Early Features** – (Hypertension & Weight Gain)



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Ozlem Celik, Mutlu Niyazoglu, Hikmet Soylu and Pinar Kadioglu, CC BY 2.5
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- **Diagnosis:**
 - **Dexamethasone Suppression Test (Central Vs. Primary)**
 - **ACTH Levels**
 - **Cortisol Levels**
 - **CT/MRI Brain (Pituitary Adenoma)**
- **Treatment – Depends on Aetiology:**
 - **If Exogenous Cortisol** – Wean Pt. off Cortisol.
 - **If Pituitary Tumour (Cushing's Disease)** – Surgical Removal + Temp Cortisol Replacement.
 - **If Adrenal Tumour** – Surgical Removal + Temporary Cortisol Replacement.

PHAEOCHROMOCYTOMA

PHAEOCHROMOCYTOMA

Aetiology:

- Idiopathic
- May be familial (in MEN2 Syndrome)

Pathogenesis:

- Tumour of the Medullary Chromaffin Cells (Which produce Catecholamines)
 - o → Increased Catecholamines (↑Adrenaline & Noradrenaline)
 - o → Secondary Hypertension → Hypertensive Crises

Clinical Features:

- Young Age
- 10% Are Malignant
- **Symptoms:**
 - o #1- Paroxysmal Hypertension
 - o Palpitations/Tachycardia
 - o Headache
 - o Sweating/Hot Flushes
 - o Tremor
 - o Anxiety
 - o Nausea/Vomiting
- (Note: Phaeos are a cause of *Surgically-Correctable Hypertension*)
- (Note: Phaeos May be associated with *MEN2 Syndromes*)

Diagnosis:

- Increased Urinary Catecholamines & VMA (Vanillylmandelic Acid – A Metabolite of Adrenaline & NA)

Treatment:

- Preoperative Sympatholytic Drugs (To prevent hypertensive crisis)
- Surgical Resection

Complications:

- **- of Hypertension:**
 - o Congestive Heart Failure
 - o Pulmonary Oedema
 - o Myocardial Infarction
 - o Ventricular Fibrillations
 - o CVAs



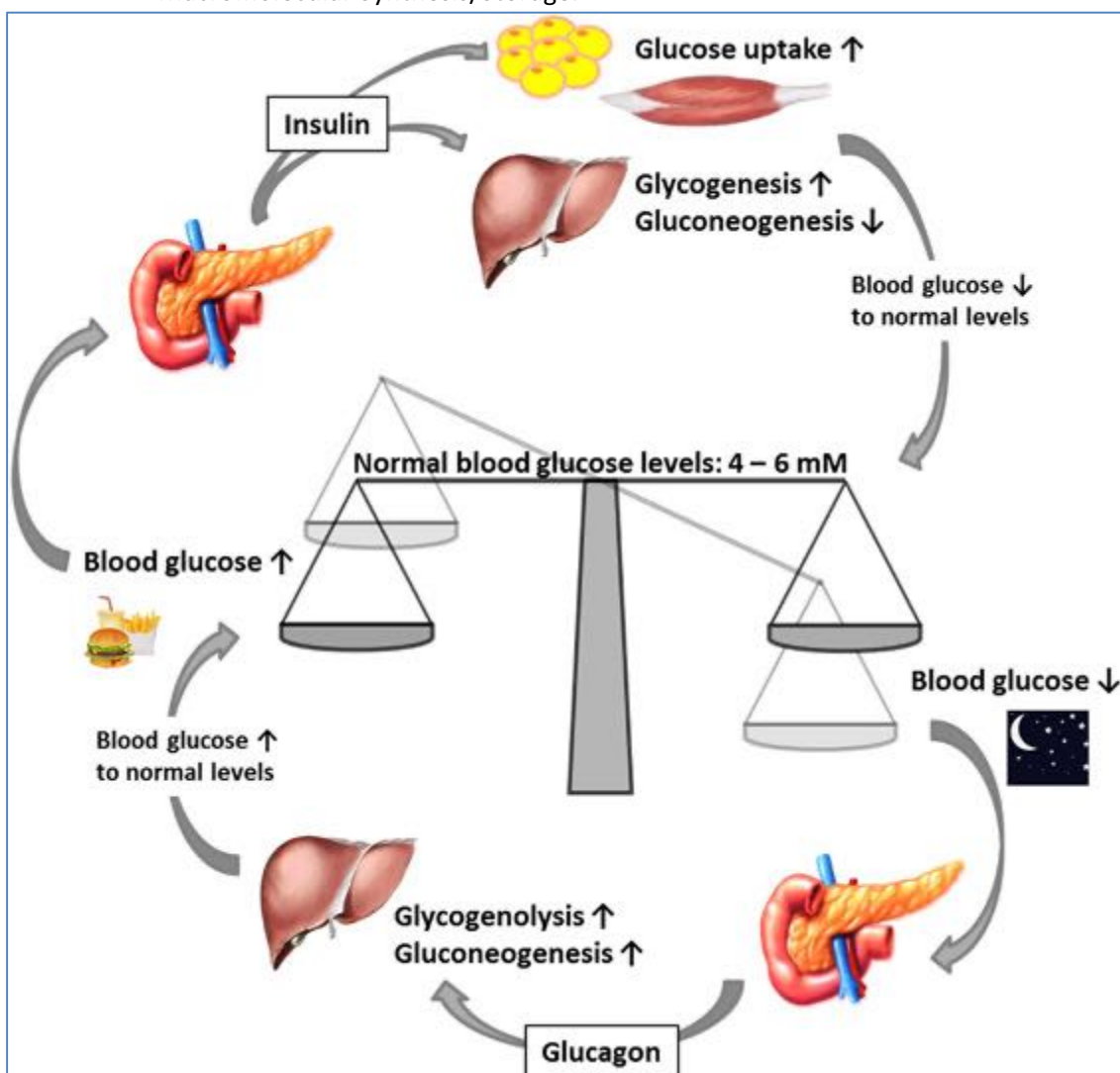
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DIABETES:

DIABETES

Review of Insulin & Glucagon:

- **Insulin:**
 - **Released Due to:**
 - ↑Blood Glucose
 - ↑Blood Amino Acids
 - **Stimulates:**
 - Glucose Uptake (Fat & Muscle)
 - Lipid Synthesis & Storage (Fat)
 - Protein Deposition (Muscle)
 - **Inhibits:**
 - Ketogenesis
 - Macromolecular Breakdown
- **Glucagon:**
 - **Released Due to:**
 - ↓Blood Glucose
 - ↓Blood Amino Acids (le: Fasting)
 - **Stimulates:**
 - Glycogenolysis
 - Gluconeogenesis
 - Lipolysis
 - Ketogenesis
 - **Inhibits:**
 - Macromolecular Synthesis/Storage.



Pancreatic regulation of glucose homeostasis - https://www.researchgate.net/figure/Maintenance-of-blood-glucose-levels-by-glucagon-and-insulin-When-blood-glucose-levels_fig3_297896821

Diabetes: General Information:

- **Diagnostic Criteria (The “7-11 Rule”):**
 - **Fasting BSL ≥ 7.0 mmol/L** (Note: For Non-Pregnant)
 - **Random BSL of >11** (Note: If Fasting BSL = 5.5-7.0 mmol/L → Perform OGTT)
 - **OGTT – Oral Glucose Tolerance Test** (Fasting) **>11 @ 2hrs**
 - **Autoantibodies (If Type 1 Diabetes):**
 - **+ Anti-Islet-Cell Antibodies (Anti-ICAs)**
 - **+ Anti-Glutamic Acid Decarboxylase Antibodies (Anti-GADs)**
 - (Note: **HbA1c** for monitoring only)
 - Note: People who have **Impaired Glucose Tolerance** and/or **Impaired Fasting Glucose** have slightly raised fasting & Post Prandial BSL's, but not high enough for diagnosis of diabetes.
- **Initial Presentation:**
 - **PPP – Polyuria, Polydipsia, Polyphagia**
 - Unexplained Weight Loss/Fatigue/Lethargy
 - Recurrent/Persistent Infections, Delayed Healing & Immunosuppression (Eg: Genital Thrush)
- **Emergency Presentations:**
 - **HYPERs:**
 - **DKA** - Diabetic Ketoacidosis
 - **HONC** - Hyperosmolar Non-Ketotic Coma
 - **HYPOs:**
 - **Eg: Insulin Overdose/Over exercise/Missed Meal**
- **Treatment:**
 - Lifestyle (Diet + Exercise + Weight Loss)
 - Medications:
 - **Insulins** – (Broad range of Rapid to Long-Acting)
 - Oral Hypoglycaemic Agents:
 - **“Insulin Secretagogues” – (*Sulfonylureas):**
 - **Biguanides – (*Metformin)**
 - Incretin Mimetics:
 - **Incretin Analogues – (*Exenatide):**
 - **DPP-4 Inhibitors – (*Sitagliptin)**

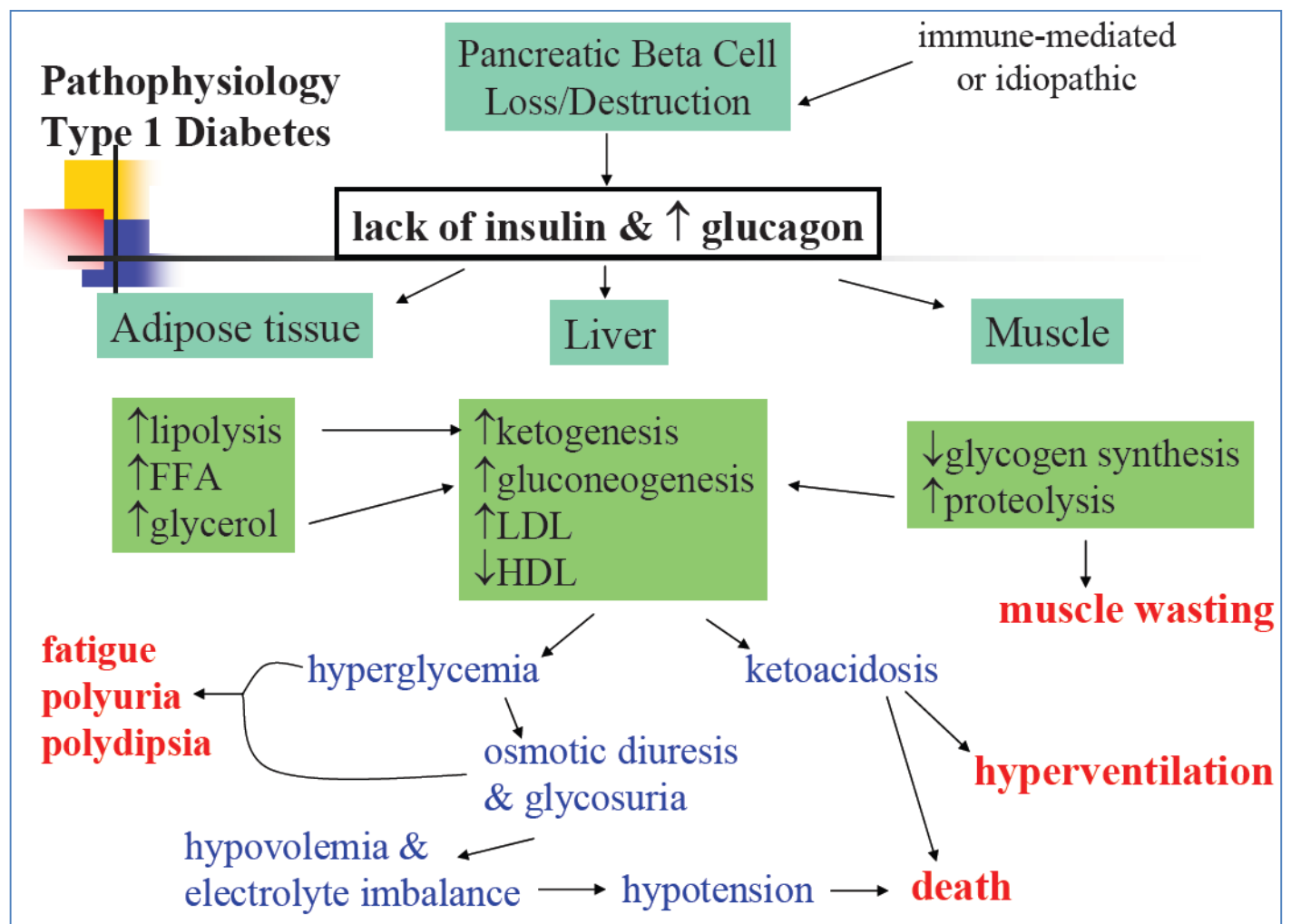
Snapshot Overview of Types of Diabetes:

- **Type 1 Diabetes – Insulin Deficiency, Juvenile, Rapid Onset:**
 - Aetiology – (Autoimmune Destruction of the β -Cells of the Pancreatic Islets)
 - Clinical Features – (Juvenile Disease, Rapid Onset)
 - Diagnosis – (+ Anti-GADs, Anti-Islet-Cell Antibodies (Anti-ICAs) & Insulin Auto Antibodies (IAAs))
 - Treatment: - (**Exogenous Insulin**)
 - Complications – (**Diabetic Ketoacidosis**)
- **LADA – Latent Autoimmune Diabetes of Adults:**
 - Aetiology – (Delayed Autoimmune Type I in Adults)
 - Clinical Features – (Slim Adults with Diabetes Symptoms)
 - Complications – (HONC, DKA)
 - Diagnosis – (Hyperglycaemia, + Anti-GADs, Anti-ICAs & Insulin Auto Antibodies)
 - Treatment – (Insulin)
- **Type 2 Diabetes – Insulin Resistance, Adults, Insidious Onset:**
 - Aetiology – (Insulin Resistance And/Or Relative Insulin Deficiency)
 - Clinical Features – (Adults, Slow Onset +/- ‘Pre-Diabetic State’)
 - Diagnosis – (Random BSL >11 , OGTT >11 @2hrs)
 - Treatment – (1- Diet & Lifestyle, 2- Orals [Metformin/Sulfonylureas/Incretins], 3- Insulin)
- **MODY – Maturity Onset Diabetes of Youth:**
 - Aetiology – (Autosomal Dominant)
 - Pathogenesis – (Essentially a Type II DM in a Child)
 - Clinical Features – (Young, Non-Obese, Autosomal Dominant ∴ FamHx)
 - Treatment – (1- Orals [Metformin/Sulfonylureas/Incretins], 2- Insulin)
 - Complications – (Like Type II Diabetes (Ie: HONC rather than DKA))

Different Types of Diabetes:

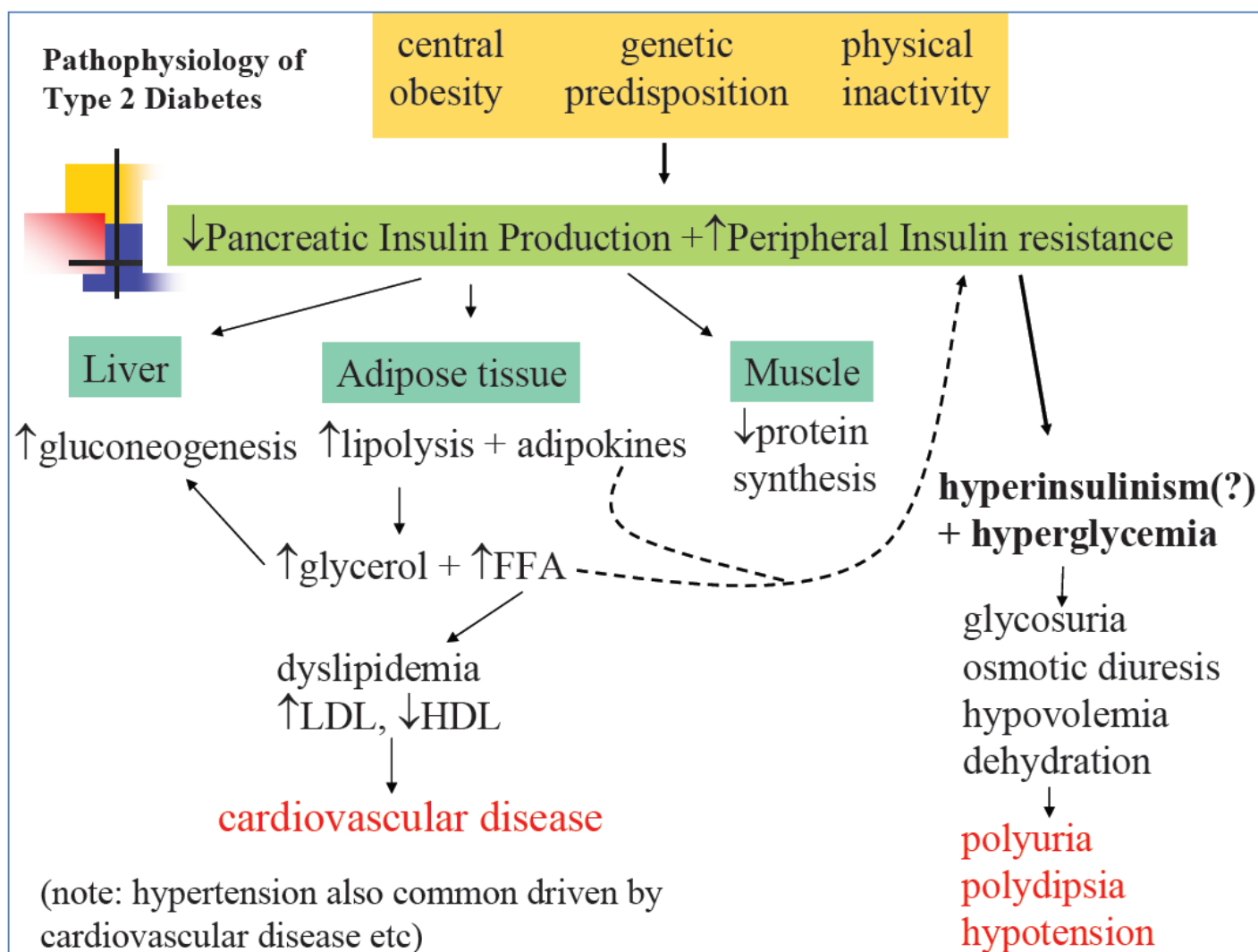
- **Type 1 Diabetes – Insulin Deficient:**

- Autoimmune attack on the β -Cells of the Pancreatic Islets.
- Results in a **Physical Lack of Insulin Production**
- Aetiology can be Genetic & Environmental
- Rapid Onset → Therefore Fewer Complications @ Diagnosis.
- Presentation:
 - Hyperglycaemia
 - Ketonuria (Ketoacidosis) →
 - Hyperventilation
 - Nausea
 - Vomiting
 - Abdo Pain
 - Rapid Significant Weight Loss
 - Excessive Hunger (Polyphagia)
 - Mental Fatigue
- Treated with Exogenous Insulin



- **Type 2 Diabetes – Insulin Resistance:**

- Results from **Insulin Resistance** sometimes combined with **Relative Insulin Deficiency**.
- **Note: Obesity** is the #1 Predisposer of Insulin Resistance:
 - Due to Change in Adipose-Release of '**Adipokines**' – Hormones that Mediate Insulin Resistance.
 - Incl: Resistin/Leptin/Adipopectin.
- Peak onset @ ~50yrs
- Gradual Onset → Therefore ~1/4 have Complications @ Diagnosis (Eg: Vascular)
 - Note: Many people spend years in a '**Pre-Diabetic State**' – where BSL is higher than normal but not high enough for a diagnosis of Type 2 Diabetes
- Presentation:
 - Same as Type 1, Except:
 - No Ketonuria
 - Usually Overweight (Central Obesity)
 - **Metabolic Syndrome** (Due to Insulin Resistance):
 - ↑Circulating FFA's
 - ↑Insulin
 - ↑Glucose
 - ↓HDL's
 - ↑BP
- Treated with Diet / Tablets / Exogenous Insulin
 - Note: Over time Insulin Resistance Increases & β-Cell Function Decreases →
 - Therefore the later stages require ↑Amount of Treatment.



Complications of Diabetes:

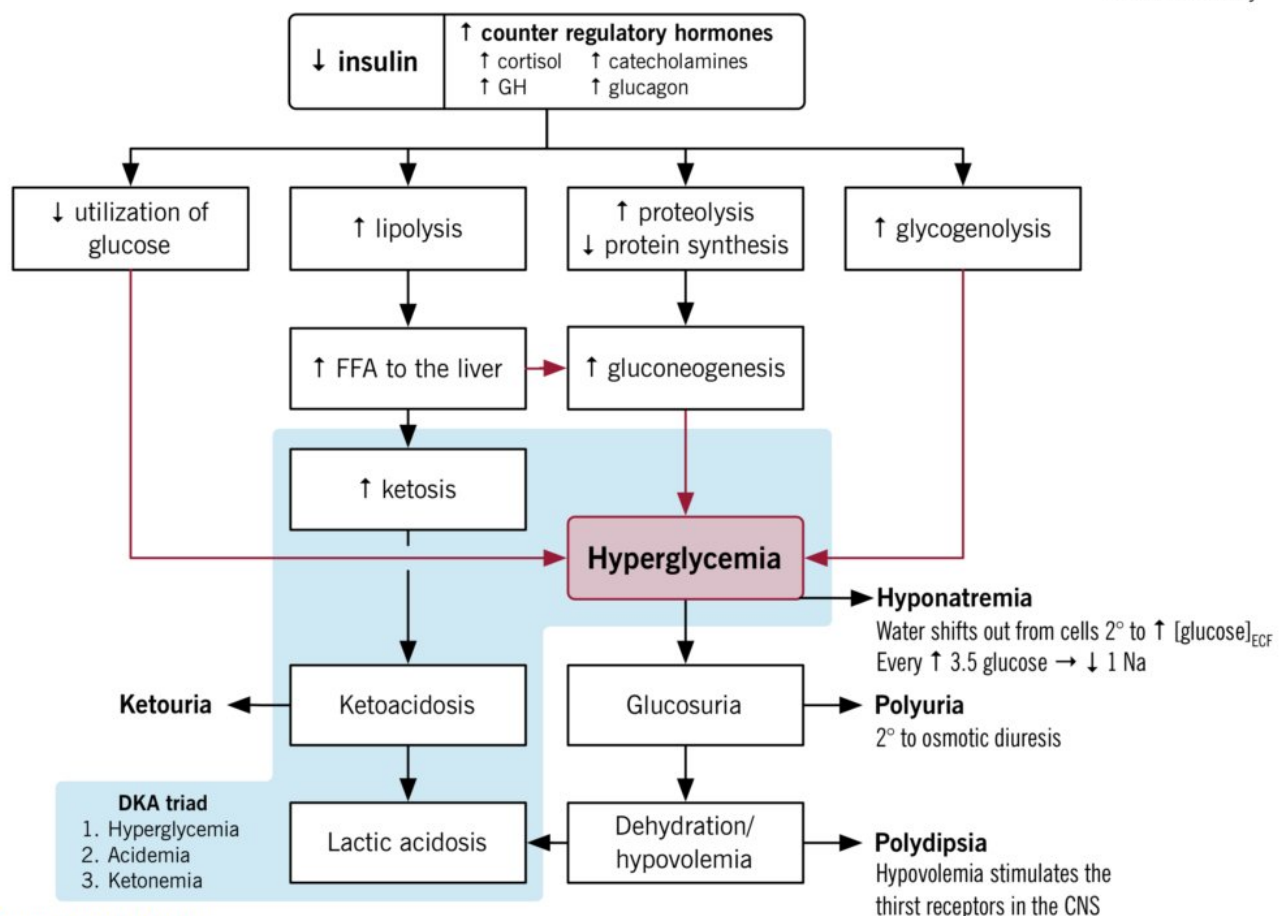
Acute Complications:

- Diabetic Keto-Acidosis (DKA):

- **Acute life threatening**
- **Caused By** – (Type I Diabetes - Lack of Insulin (Eg: Forgotten to take insulin))
- **Diagnosis:**
 - **Hyperglycaemia:** High Glucose (>15 mmol/L)
 - **Ketoacidosis:** Low pH, Low Bicarbonate (< 15 mmol/L), Sweet Breath, Ketonuria
- **Symptoms:**
 - – of Underlying Diabetes – (Polyuria, Polydipsia, Weight loss)
 - – of Hyperglycaemia – (Glycosuria/Osmotic diuresis, **Severe Dehydration**)
 - – of Hyperketonaemia → KetoAcidosis – (Vomiting, Acetone Breath, Hyperventilation)
 - - of Electrolyte Disturbances [↓Na & ↓K] – (Cardiac Arrhythmia / Bradycardia)
- **Treatment:**
 - 1- IV access → Correct Dehydration
 - 2- Insulin Infusion → Correct Hyperglycaemia
 - 3- Monitor/Correct Electrolytes – Particularly Potassium
- **Complications:**
 - 40% mortality – medical emergency
 - Severe Dehydration

Pathogenesis and pathophysiology of diabetic ketoacidosis

Sultan Chaudhry

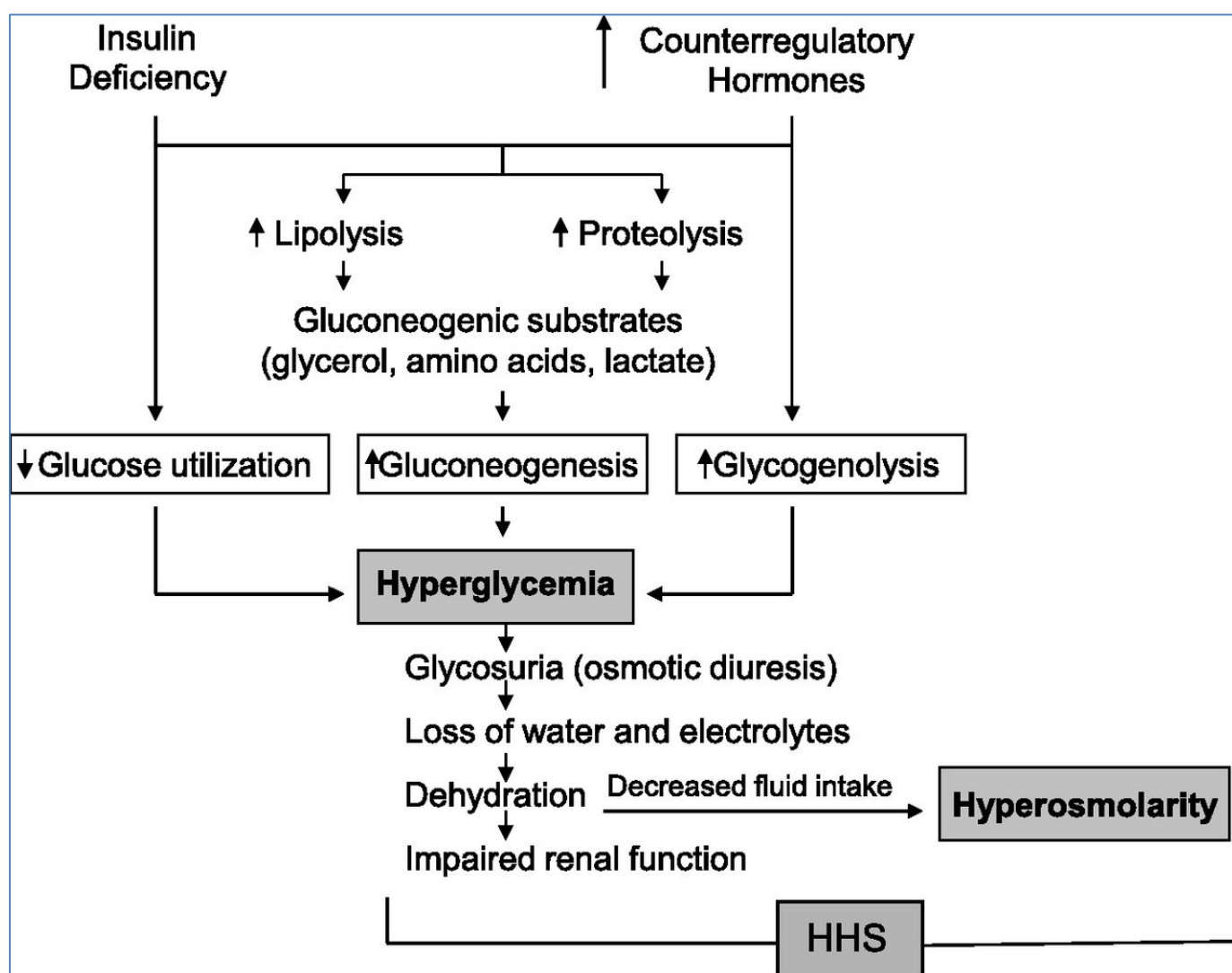


Source : www.pathophys.org

Source: Sultan Chaudhry; www.pathophys.org

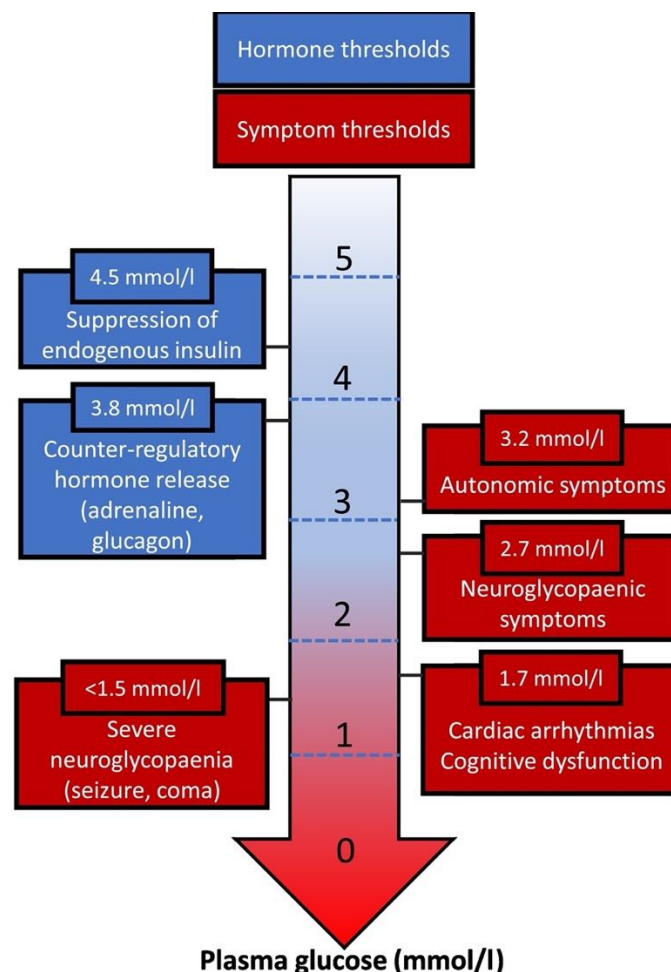
- **HONC – Hyperosmolar Non-ketotic Coma:**

- **Caused By** – (*Type II Diabetes* - Relatively Low Insulin/Insulin Insensitivity + Precipitant)
- **Diagnosis:**
 - **Hyperglycaemia**
 - **NO KetoAcidosis**
 - Osmolality > 330 mOsm/Kg
- **Symptoms:**
 - Confusion/Coma
 - Marked Dehydration
 - Polyuria (Osmotic Diuresis)
 - Neurology – (Sensory/Motor Impairment, Focal Seizures, Hyporeflexia, Tremors)
- **Treatment:**
 - IV Fluids
 - Insulin + Potassium (Since Insulin causes K^+ Shift *Into Cells*)
 - Electrolyte Replacement (Esp. Potassium)
- **Complications:**
 - Fatal if Untreated



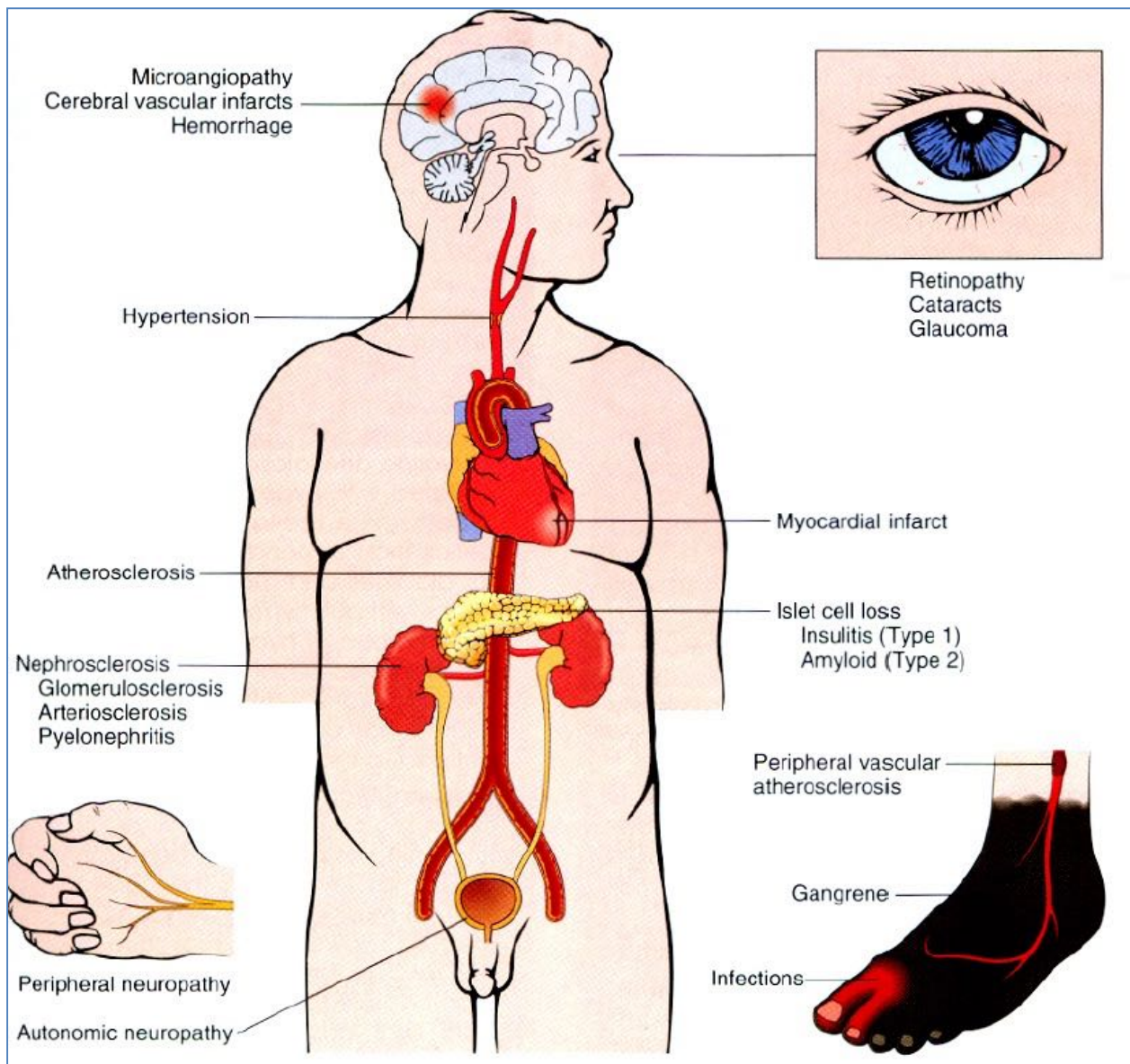
- **Hypoglycaemia (BSL < 6.0mmol/L):**

- **Aetiology – (**Diabetes + Insulin Overdose / Alcohol / Sepsis)**
- **Mild (BSL 3.5 - 6.0 mmol/L):**
 - ↓ Insulin Secretion
 - ↑ Counter Regulatory Hormones (Glucagon/Catecholamines/Cortisol/GH)
- **Severe (BSL < 3.5 mmol/L):**
 - **Neurogenic Symptoms:**
 - Adrenergic:
 - Palpitations
 - Anxiety
 - Tremor
 - Cholinergic:
 - Hunger
 - Sweating
 - Paraesthesia
 - **Neuroglycopaenic Symptoms:**
 - Behaviour Change
 - Cognitive Seizures
 - Coma
- **Symptoms (Summary):**
 - **Autonomic –** (Sweating, Anxiety, Hunger, Tremor, Palpitations, Dizziness)
 - **CNS –** (Confusion, Drowsiness, Visual Disturbances, Seizures, Coma)
- **Treatment:**
 - #1 – Oral/IV Glucose (Jellybeans/Juice/Biscuits/etc.)
 - OR – IM/IV Glucagon



Chronic Complications:

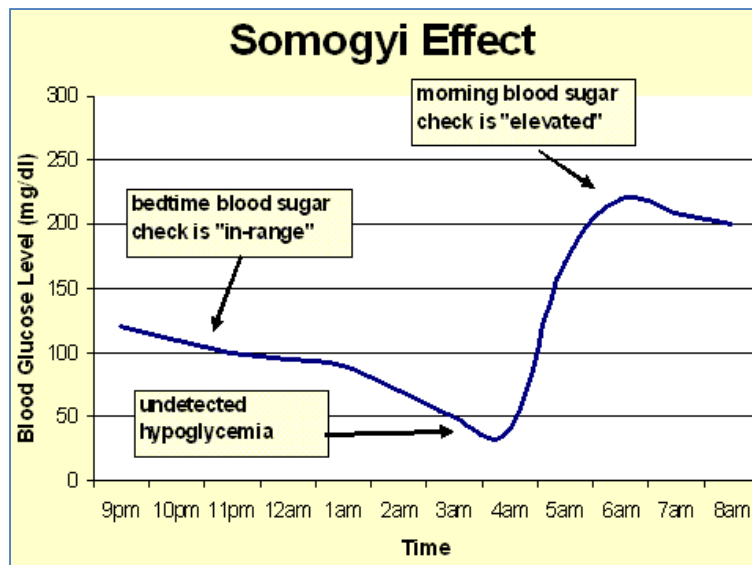
- Caused by chronic exposure to Hyperglycaemia & Dyslipidaemia in the Insulin *Insensitive Tissues*
- Vascular:
 - Macrovascular:
 - Heart Disease/Coronary Artery Disease/Atherosclerosis
 - Stroke
 - Peripheral Vascular Disease
 - Microvascular:
 - Retinopathy
 - Neuropathy
 - Nephropathy



Source: Unattributable

- **The 'Somogyi' Effect:**

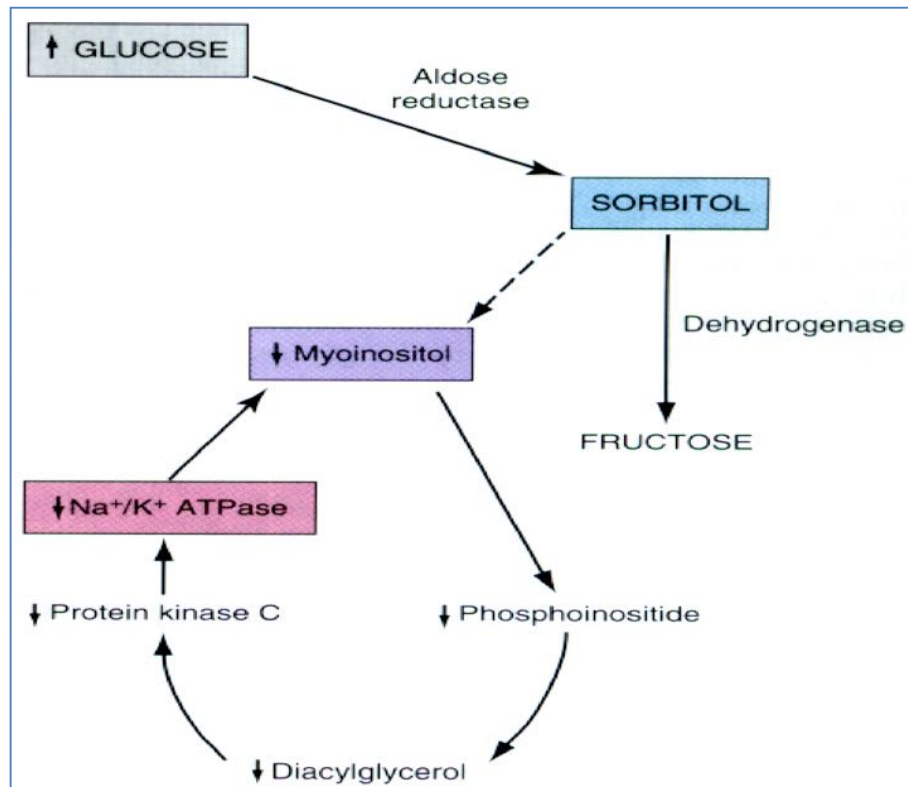
- Where Circadian Elevations in Counter-Regulatory hormones (GH & Cortisol) cause Hyperglycaemia early in the morning → Patient Increases Insulin Dose @ Bedtime → Sustained Hypo all night → Body releases Glucagon, Adrenaline & Cortisol → Hyper @ Dawn.



Possible Causes of Chronic Complications:

- The 'Polyol Pathway':

- Hyperglycaemia → ↑ Glucose Accumulation in Insulin *Independent* Tissues
- Much Glucose → Converted to **Sorbitol** → Slowly converted to Fructose.
- However, **Sorbitol** is **Osmotically Very Active** → Sorbitol Accumulation →
 - Osmotic Cell Injury
 - Affects Ion Pumps
 - Affects Some Cellular Secondary Messenger Pathways



- Hyperglycosylation of Proteins:

- ↑ Glucose → 'Glycosylation' of proteins (ie: Linkage of Glucose to Free Amino Groups in Proteins)
 - Eg: HbA_{1c} (Glycosylated Haemoglobin)
- Affects proteins in Blood Vessel Walls & Basement Membranes → Can stimulate Inflammation.
 - Includes Collagen, Fibronectin, etc.
- Leads to formation of '**Advanced Glycosylated End-Products**' (**AGE's**):
 - Can form cross-linkages between peptides → Forms a "Mesh" → Traps Other Molecules:
 - Eg: LDL Trapping → Cholesterol Deposition → Atherosclerosis
 - Eg: Immunoglobulins & Complement → Inflammation
 - Can form binding-sites for other proteins (eg: Albumin)
 - Inactivate Nitric Oxide → Reduce Vasodilation
 - Stimulate Growth-Factor & Cytokine Secretion
 - ↑ Vascular Permeability
 - ↑ Clotting Activity → Stroke/Embolus/etc.
 - ↑ Extra-Cellular Matrix Deposition.

- Reactive Oxygen Species:

- ↑ Hyperglycaemia → ↑ O₂ Free-Radical Production.
- Compounded by ↑ Immune Cell Activation

CALCIUM & PHOSPHATE BALANCE DISORDERS:

CALCIUM & PHOSPHATE BALANCE DISORDERS

Disorders of Calcium & Phosphate Regulation:

- **Hypercalcaemia:**
 - **Caused By:**
 - Hyperparathyroidism
 - Malignancy (Eg: BRCA, Multiple Myeloma)
 - Vit D Excess
 - Secondary Renal Hyperparathyroidism
- **Hypocalcaemia:**
 - **Caused by:**
 - Vit-D Deficiency/Disorders of Vit-D Metabolism (Activation) → ↓Active Vit-D → **Rickets**
 - Eg: Lack of sunlight
 - Eg: Lack of Dietary Vit-D
 - Eg: Chronic Kidney Failure
 - Hypoparathyroidism – (Because PTH is required for Vit-D Activation in the Kidneys)
 - Acquired/Congenital
- **Rickets:**
 - **What is it?:**
 - A Vit-D Deficiency → Resulting in a Calcium/Phosphate Deficiency.
 - Note: Clinical Signs occur after a few months (Once the Bone's Ca/P Reservoirs are Depleted)
 - **Effects:**
 - Marked ↑PTH Secretion → Extreme Osteoclastic Activity:
 - → ↑↑Plasma Calcium
 - → ↓↓Plasma Phosphate (Due to ↑Renal Excretion)
 - Tetany – Once the Bone's Ca^+ Reservoir is Depleted, Plasma Ca^+ falls to dangerous levels.
 - **Treatment:**
 - Dietary Calcium Supplements
 - Exogenous Vit-D Administration.
- **Hypoparathyroidism:**
 - **What is it?:**
 - When the Parathyroid Glands don't secrete sufficient PTH.
 - **Effects:**
 - ↓Resorption of exchangeable Calcium → Hypocalcaemia
 - When Ca^+ falls too low, Tetany can develop (Can occur in larynx → obstructs respiration)
- **Hyperparathyroidism:**
 - **What is it?:**
 - When the Parathyroid Glands secrete an inappropriate excess of PTH.
 - **Effects:**
 - ↑↑Extreme Osteoclastic activity in bones.
 - → Hypercalcaemia
 - → Hypophosphataemia (Due to ↑Renal Excretion)
- **Osteoporosis:**
 - **What is it?:**
 - Decreased Bone **Matrix** (Not decreased bone *calcification*)
 - **Possible Causes:**
 - Usually due to poor **Osteoblastic Activity** → ↓Osteoid Deposition.
 - Can be due to ↑↑**Osteoclastic Activity** → ↑Osteoid Resorption.
 - Inactivity → Lack of physical stress on bones
 - Malnutrition
 - Postmenopausal Lack of Oestrogen (Oestrogen normally ↓Osteoclast Activity)
 - Cushing's Syndrome - ↑↑Glucocorticoids cause ↓Protein deposition throughout the body.

PARATHYROID DISORDERS:

HYPERPARATHYROIDISM:

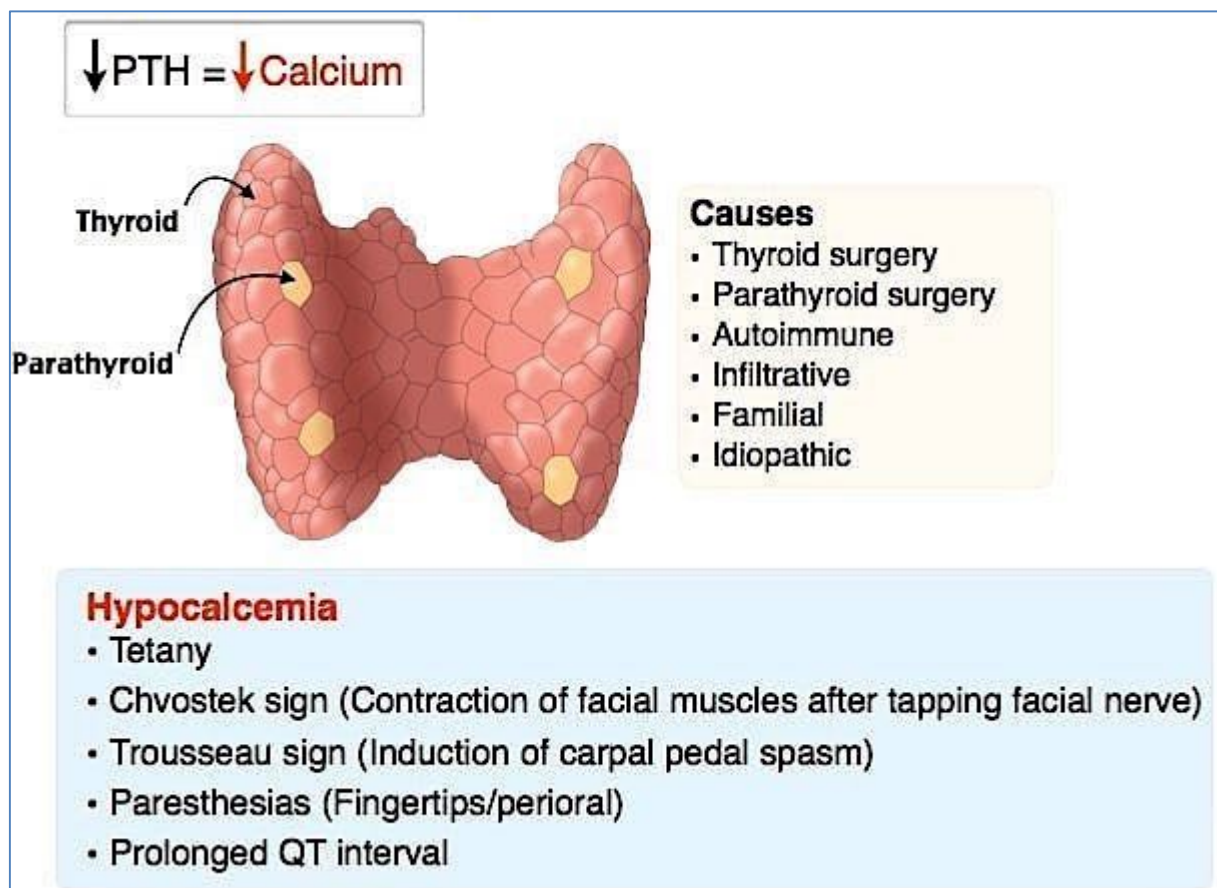
- **What is it?:**
 - When the Parathyroid Glands secrete ↑↑PTH.
- **Effects of ↑↑PTH:**
 - ↑↑Extreme Osteoclastic activity in bones → Hypercalcaemia
 - ↑↑Renal Phosphate Excretion → Hypophosphataemia
- **Types:**
 - **Primary Hyperparathyroidism** – **Autonomous, Spontaneous Overproduction of PTH:**
 - **Aetiologies/Pathogeneses:**
 - **Adenoma(Sporadic or MEN)**/Hyperplasia/Carcinoma → ↑↑PTH
 - **Clinical Features:**
 - **F>>M**
 - **Hypercalcaemia Triad** – **“Bones, Moans & Abdominal Groans”:**
 - **1- Bone:** Pain/Osteoporosis/Fractures
 - **2- Moans:** Depression/Lethargy/Seizures
 - **3- Abdo:** Constipation/Nausea/Ulcers/Gallstones
 - **+ (Renal:** Renal Stones)
 - **+ (Heart:** Aortic/Mitral Calcification)
 - **Diagnosis:**
 - ↑PTH
 - ↑Serum Calcium
 - ↓Serum Phosphate
 - **Treatment:**
 - Surgical Excision
 - **Secondary Hyperparathyroidism** – **Secondary to Chronic Renal Insufficiency:**
 - **Aetiology:**
 - Secondary to Renal Failure → Hypocalcaemia
 - (Others incl. Dietary Calcium Deficiency, Vit-D Deficiency)
 - **Pathogenesis:**
 - **Renal Failure** → Hypocalcaemia → ↑PTH to Compensate → Hyperplasia
 - **Clinical Features:**
 - Symptoms of Chronic Renal Failure
 - **Osteoporosis**
 - **Treatment:**
 - Vitamin D + Calcium Supplementation
 - Partial Parathyroidectomy

hyperparathyroidism			
	Primary hyperparathyroidism	Secondary hyperparathyroidism	Tertiary hyperparathyroidism
PTH	↑	↑	↑↑↑
Ca ²⁺	↑ *	↓/N	↑
P04 ⁺	↓	↑/N	↑
Vitamin D	N/↓	↓↓↓	↓/N
Comments	In 85% a solitary parathyroid adenoma is present Important differential diagnosis is FHH (see Table 3)	Causes include: • chronic renal failure • vitamin D deficiency	Caused by: • end stage renal failure
* Differential diagnosis of hypercalcaemia includes drug induced (eg. lithium, thiazides) and hypercalcaemia of malignancy			

Blackburn, Miriam and Terry H Diamond. "Primary hyperparathyroidism and familial hyperparathyroid syndromes." *Australian family physician* 36 12 (2007): 1029-33 .

HYPOPARATHYROIDISM:

- **Effects:**
 - ↓ Resorption of exchangeable Calcium → Hypocalcaemia
- **Aetiologies:**
 - **Iatrogenic** – Surgery (Eg: Thyroidectomy/Lymphadenectomy/Over-resection in 1°HyperPT)
 - **Genetic** – (Autoimmune/Familial/Congenital Absence of Gland)
- **Clinical Features:**
 - ***Hypocalcaemia**
 - ***Hallmark = Tetany:**
 - → Neuromuscular Irritability
 - → Distal Paraesthesias
 - → Carpopedal Spasm
 - → *Laryngospasm (Life-Threatening)
 - → Seizures
 - **CNS:** Confusion/Depression/Hallucinations/Psychosis
 - **Eyes:** Cataracts (Calcification of Lenses)
 - **CVS:** Characteristic Prolonged QT-Interval



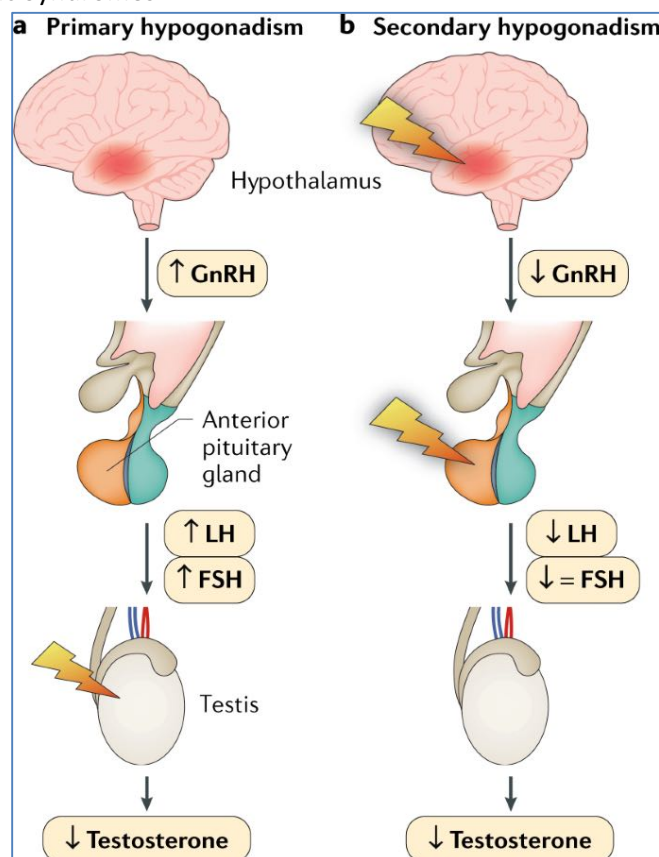
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GONADAL DYSFUNCTION:

GONADAL DYSFUNCTION

MALE HYPOGONADISM:

- **What is it?**
 - o A deficiency in Testosterone due to problems with either:
 - 1) Testes, or - Primary
 - 2) Hypothalamus/Pituitary - Secondary
- **Primary Hypogonadism (Hypergonadotrophic)**
 - o Ie: Problem with the Leydig Cells in the Testes → ↓↓Testosterone Production → ↑↑Hypothalamo-Pituitary release of Gonadotropins (FSH/LH).
 - o **Causes:**
 - Trauma/Irradiation of Testes.
 - Mumps
 - Klinefelter's Syndrome (XXY)
 - Androgen Resistance
 - Autoimmune
 - Congenital
- **Secondary Hypogonadism (Hypogonadotrophic)**
 - o Ie: Problem with the Hypothalamo-Pituitary Axis → ↓↓Gonadotropin Release (FSH/LH) → ↓↓Testosterone Production
 - o **Causes:**
 - Developmental
 - Pituitary Tumour/Trauma/Pituitary Irradiation/Autoimmune
 - Genetic Syndromes



Salonia, Andrea et al. "Paediatric and adult-onset male hypogonadism." *Nature Reviews Disease Primers* 5 (2019)

- **Effects of ↓↓Testosterone:**
 - o Infertility (Low Sperm Count), ↓Libido, Erectile Dysfunction
 - o ↓Muscle Mass, ↓Bone Mass
 - o ↓Beard/Body Hair
 - o ↑Breast Tissue
 - o ↑Body Fat
- **Treatment – Testosterone Replacement Therapy**

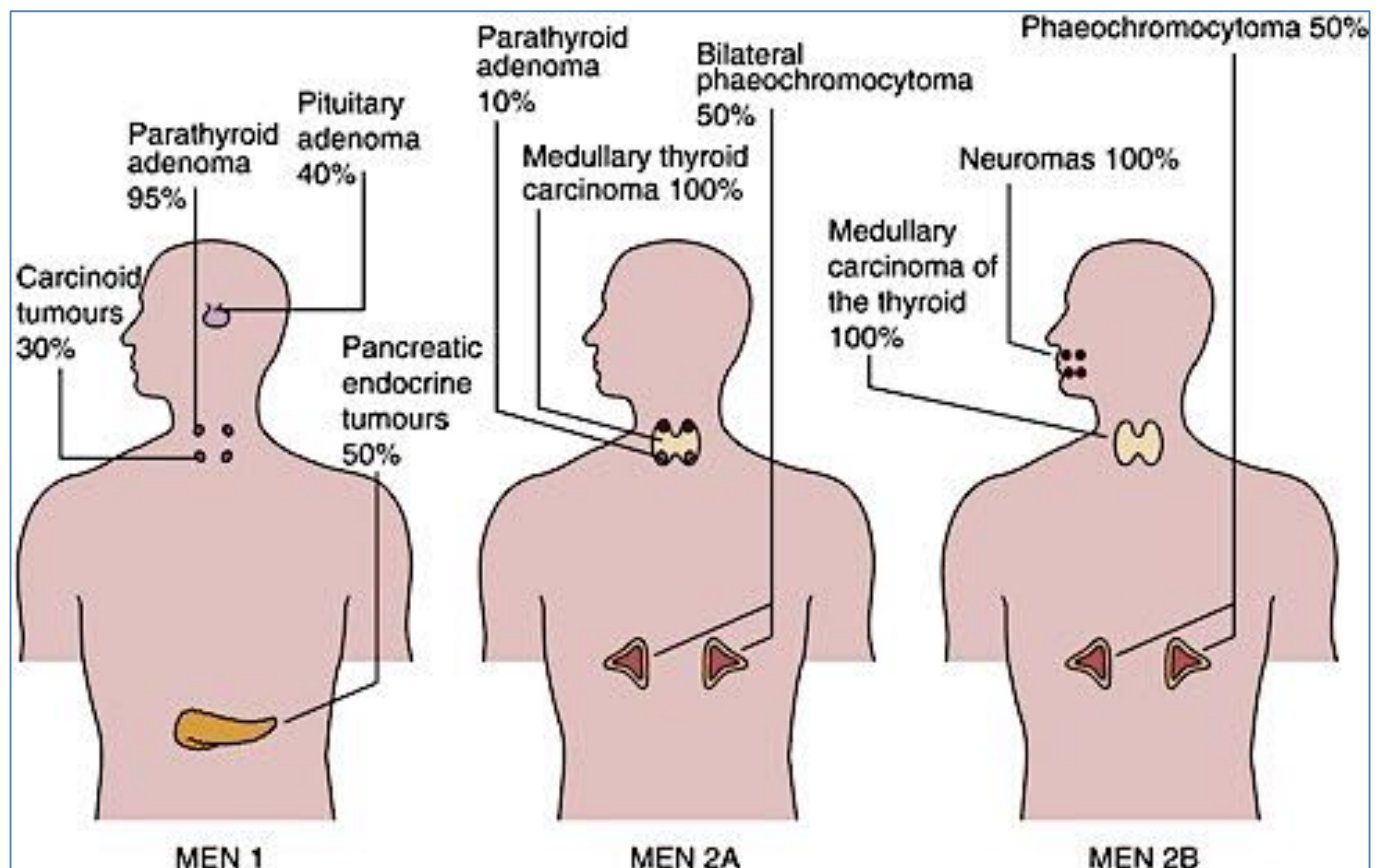
MULTIPLE ENDOCRINE NEOPLASIA SYNDROME:

MULTIPLE ENDOCRINE NEOPLASIA SYNDROME

Familial Endocrine Neoplasias – MEN Syndromes:

- **Multiple Endocrine Neoplasia** = Genetic Disorders where 2/More Tumours are found in Endocrine Glands (Parathyroid, Pituitary, Thyroid, & Adrenal Medulla).
- MEN Disorders → Greatly ↑ the risk of developing multiple cancerous and noncancerous tumours in glands such as the parathyroid, pituitary, and pancreas.

Multiple Endocrine Neoplasm Classifications			
Type	Aetiology	Organs Involved	Clinical Features
MEN1: <u>Wermer's Syndrome</u>	Autosomal Dominant Genetic Mutation in MEN1 Tumour Suppressor Gene on Chromosome 11	1. Pituitary 2. Parathyroid 3. Pancreas (Top 2/3 of Body)	60% of Wermer's Pts have >2x of the Following: -Pituitary Adenoma → Eg: Prolactinomas/GH - Can also → Bilateral Hemianopia -Parathyroid Adenomas → Hyperparathyroidism -Pancreatic Gastrinoma → Peptic Ulcer Disease -Pancreatic Insulinoma → Hypoglycaemia -Pancreatic VIPomas → ↑VIP→ Secretory Diarrhoea
MEN2a: <u>Sipple's Syndrome</u>	Autosomal Dominant Genetic Mutation in MEN2 Tumour Suppressor Gene on Chromosome 10	1. Thyroid 2. Parathyroid 3. Adrenal Medulla (Lower 2/3 of Body)	100% of Pts → Medullary Thyroid Cancer 50% of Pts → Pheochromocytoma → ↑Adrenaline 30% of Pts → Parathyroid Hyperplasia → ↑PTH
MEN2b:	Autosomal Dominant Genetic Mutation in MEN2 Tumour Suppressor Gene on Chromosome 10	1. Thyroid 2. Adrenal Medulla (Lower 2/3 of Body)	100% of Pts → Medullary Thyroid Cancer 50% of Pts → Pheochromocytoma → ↑Adrenaline (Other: Mucosal Neuromas, Marfanoid Features)



Source: Unattributable



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